ENERGY Harvesting for AUTONOMOUS SYSTEMS

STEPHEN BEEBY
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Energy Harvesting for Autonomous Systems

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Contents

CHAPTER 1

Introduction

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1.1 Background and Motivation

The concept of energy harvesting generally relates to the process of using ambient energy, which is converted, primarily (but not exclusively) into electrical energy in order to power small and autonomous electronic devices. The phrases "power harvesting" or "energy scavenging" are also used to describe the same process. The concept is not new and has wider applications that are more common. We are all familiar with larger-scale deployments such as electrical and thermal power generation for buildings and the presence of large-scale solar panels and wind turbines is evident in many of today's rural and urban environments across the globe. For the purpose of this text, however, we will not be covering these larger-scale systems but will examine the issues of powering miniature, stand-alone electronics systems. As we will see, there are many examples of such systems where the main focus of such devices is that of a sensing node that has the ability to monitor its surrounding environment as well as using a variety of forms of ambient energy to power itself. For those that are new to this field, you may now be taking a sharp intake of breath and wondering about issues such perpetual motion, but we can assure you that within these pages you will hear no mention of over unity devices or free energy. All of the systems described here are designed to exploit excess or wasted energy within an environment for powering the sensor node. The adopted transduction processes are often quite inefficient but can, nevertheless, produce sufficient electrical energy to a node to take a measurement and sometimes to transmit data via a radio frequency link to a remote base station. The reader of this text will become comfortable with the prefix of a lowercase m before watts to denote acceptable power levels, as opposed to the uppercase M that would be expected for domestic energy generation. Further, we will find that it is not out of the question to exploit levels of ambient energy that only generate micro (μ) watts of electrical power.

There is a broad range of different energy domains within any typical environment, whether internal or external. Solar energy (light), thermal energy (heat) and kinetic energy (motion) are three examples that we can envisage as possible sources for harvesting electrical energy from a typical outdoor environment. As we gaze out of our office windows on a dull, wintry day in Southampton there is not much scope for exploiting the former two sources but the latter looks promising as the trees sway slowly in the breeze. No doubt our colleagues in California will have a somewhat different opinion about using solar and thermal sources, as these are much more abundant over there! This simple observation raises some interesting technical questions and challenges for the field energy harvesting:

- Is the source always available?
- Does it vary in intensity?
- How many different sources can be exploited?
- Is it a cost-effective solution?
- To what extent does the harvesting process affect the primary energy source?

It is difficult to generalize regarding the typical power levels that are available from the three types of energy source, or which source is most suitable. Table 1.1 provides an indication of typical power levels along with the conditions assumed. It is clear that in terms of power density solar power in outdoor conditions is hard to beat. However, it becomes comparable with the other sources if used indoors and is not suitable for embedded applications or dirty environments where the cells can become obscured. At the end of the day, the choice of energy source and method of implementation is largely governed by the application. There can be a fundamental link between the energy source and the design of the harvester. In the case of kinetic energy harvesting exploiting vibrations, the source vibration spectra will vary enormously for different applications. For example, generating power from human movement requires a totally different solution to the design of a generator for harvesting machinery vibrations. In every case, clear and precise data of the energy source is required at the outset.

One of the key questions that often get asked at conferences and technical seminars is: Why choose energy harvesting? This is a very important question, the implications of which should be thought through in detail before committing to a particular approach. Wireless sensors offer many obvious advantages such as ease of installation, flexibility, suitability for retrofitting and avoidance of the added cost, weight, and unreliability of wired connections. In some scenarios it might not even be possible to get access to mains electricity supply. If a sensor node was to

Table 1.1 Typical Data for Various Energy Harvesting Sources

	Conditions	*Power Density*	*Area or Volume*	*Energy/Day*
Vibration	1 m/s^2	100 μW/cm^3	1 cm^3	8.64J (assuming continuous vibration)
Solar	Outdoors	7,500 μW/cm^2	1 cm^2	324J (assuming light is available for 50% of the time)
Solar	Indoors	100 μW/cm^2	1 cm^2	4.32J (assuming light is available for 50% of the time)
Thermal	$\Delta T = 5°C$	60 μW/cm^2	1 cm^2	2.59J (assuming heat is available for 50% of the time)

be used for monitoring the environment on a glacier or in the desert, the nearest power socket could be tens of miles away. Batteries would seem to be an obvious source of electrical power, but they have a limited lifetime. For applications requiring several hundreds (or even thousands) of sensor nodes scattered over a wide area, it might not be realistic to expect the batteries to be changed as soon as the source is depleted. Furthermore, some applications require the electronics to be embedded where access to replace batteries is inconvenient (e.g., implanted medical devices) if not impossible. A solution for powering the sensor nodes that exploits the availability of ambient energy therefore has clear benefits, providing of course that sufficient levels of electrical power can be generated. Energy harvesting devices should naturally be designed to operate for the lifetime of the system thereby enabling a long term, self powered, wireless sensing solution.

1.2 Typical System Architecture

Throughout this text, there are several specific examples of a variety of architectures for an energy harvesting system. A simple block diagram of the key elements covered in this book is presented in Figure 1.1. This serves to illustrate some of the important design considerations that need to be addressed when one considers adopting an energy harvesting strategy to solve a particular problem.

The output of an energy harvester takes the form of the electrical variables, voltage, and current. Depending upon the nature of the harvester, the characteristics of these parameters can vary considerably; in particular the phase, frequency, and amplitude of the AC waveforms and/or the magnitude of the DC level. In order to power an electronic subsystem such as a sensor or a microcontroller, it is a usual requirement to modify the output of the harvester in order to supply the desired excitation for the subsystem. For example, a vibration energy harvester might produce an AC voltage having a frequency of 50 Hz and magnitude of 1V and this would need to be converted to, say, a DC voltage of 3V to power a solid state accelerometer. It cannot always be assumed that the output power from the energy harvester will be continuous. Take the case of a solar cell and consider the variability of the incident sunlight that it is exposed to during a typical 24-hour

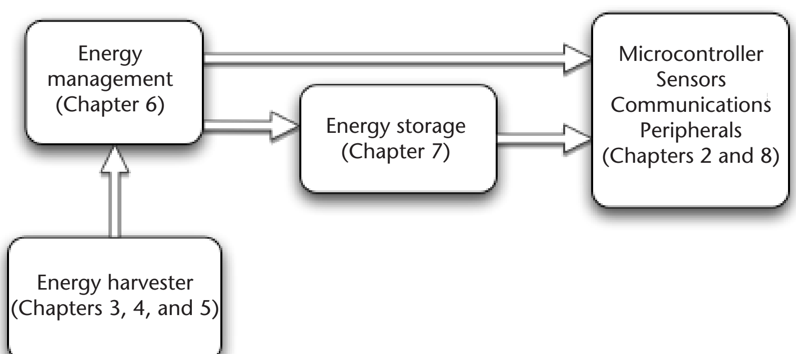

Figure 1.1 Generalized block diagram of an energy harvesting sensor node.

period. The intensity of the light will change in accordance with overhead cloud cover and the position of the sun in the sky and, of course, at night there will be no output from the harvester at all. In the case of vibration energy harvesting, the power generated will depend upon the characteristics of the source vibrations (such as amplitude and frequency). These source vibrations will often vary and the resulting fluctuations in harvested power could be quite significant, especially in the case of a tuned resonant generator. It is therefore often necessary to store the energy on a temporary basis so that it can be delivered in a controlled manner to the required electronic subsystem. The form of energy storage element might be a supercapacitor or a rechargeable battery. Naturally there are some scenarios where the energy supply is constant and there is no need to use a storage component. The sensor node will typically comprise a low-power microcontroller, a variety of sensors, a communications module and other peripherals specific to the task. A majority of the communications modules described in the literature are wireless (radio frequency) systems, which are capable of analog or digital transmission over distances of around 100m. The recent increase in interest in energy harvesting has been driven in the main by the growth in wireless sensors that typically use the IEEE 802.15.4 or ZigBee communications protocols.

1.3 Intended Readership for This Book

This book is primarily intended for graduate researchers and practicing engineers/scientists. Wherever possible, we have tried not to assume that the reader has specialist knowledge of the field, but has sufficient technical background to follow the core underlying engineering and scientific principles. In view of this, the book will also therefore be suitable for senior undergraduates and master's level students who wish to get an insight into the multi- and crossdisciplinary area of energy harvesting systems. As with any textbook that addresses a new and rapidly expanding area, it is difficult to ensure that the content is up-to-date at the time of publication. For the latest advances within the field, the interested reader is advised to refer to some of the many journals and conferences that now have energy harvesting as either a theme or their main focus.

Conferences

- PowerMEMS (held annually in Japan, North America, or Europe);
- Energy Harvesting and Storage (held annually in both Europe and the United States);
- NanoPower Forum (held annually in the United States);
- Transducers—the International Conference on Solid-State Sensors and Actuators (held biennially in either Asia, North America, or Europe);
- Eurosensors (held annually in Europe);
- IEEE Sensors Conference (held annually in North America);
- Energy Harvesting and Storage Conference (run by IDTechEx; held biannually in Europe and the United States).

Journals

- *Sensors and Actuators A-Physical*
- *Measurement Science and Technology*
- *Journal of Micromechanics and Microengineering*
- *IEEE Sensors Journal*
- *IEEE Pervasive Computing*
- *Sensor Review*
- *Measurement and Control*
- *Microsystems Technology*
- *Smart Materials and Structures*
- *Energy Harvesting Journal (online journal published by IDTechEx)*
- *Journal of Sound and Vibration*
- *Journal of Intelligent Material Systems and Structures*
- *Solar Energy Materials and Solar Cells*

Reference

[1] Ó Mathúna, C., et al., “2008 Energy Scavenging for Long-Term Deployable Wireless Sensor Networks,” *Talanta,* Vol. 75, No.3, May 15, 2008, pp. 613–623.

CHAPTER 2

Wireless Devices and Sensor Networks

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2.1 Introduction

This book focuses on the use of a range of technologies to power wireless autonomous systems. Although this term may include devices as diverse as mobile phones, entertainment devices, and other consumer electronics, it is most often used to describe devices used in the industrial, building, or environmental setting. This term is often used to describe wireless sensor nodes, devices that can be deployed to monitor parameters of interest and to report these observations back, often to a central data collector (known as a *sink*). Collections of wireless sensor nodes, known as wireless sensor networks (WSNs) can communicate and cooperate with one another. In the industrial setting, they may be used to monitor the vibration of machines in order to detect the development of faults and prompt remedial action [1]; in the environment, they may be used to monitor geological processes, habitats, or wildlife [2]; in buildings, wireless sensors may be used as part of a system to control lighting and heating [3]. Wireless devices enable parameters to be monitored that would be too expensive or impractical with conventional wired systems (a study has estimated the cost of installing wiring to each sensor in a commercial building at $200 [4]), allowing sensors to be embedded in complex machinery or in remote locations. These devices are distinct from data loggers as they can continuously communicate their data wirelessly, thus enabling data to be monitored remotely without the need to continually revisit the deployment site.

Wireless sensor nodes are devices that are able to interface with physical sensing hardware, perform some processing on the sampled data, and communicate this information wirelessly (typically via a radio transceiver) to another device. The basic components of a typical wireless sensor node are shown in Figure 2.1. Additionally, some devices may have off-processor nonvolatile memory to provide a data storage capability. Timing facilities are normally provided by low-power oscillators or crystals, with the communications subsystem normally including a stable high-frequency crystal. As a minimum, a node will consist of an 8-bit microcontroller, transceiver, sensor and power supply, along with some additional passive circuitry. As these devices are wireless, they cannot depend on a wired

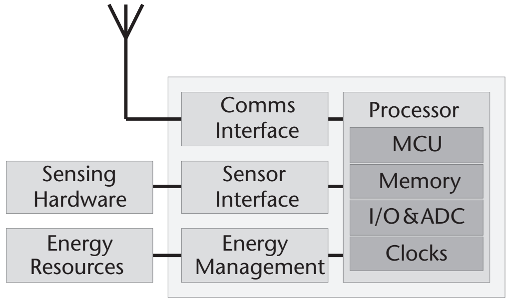

Figure 2.1 Basic components of a wireless sensor node. Modern sensor nodes may feature an integrated processor and radio transceiver.

power supply; nodes have conventionally been battery powered, but this places limitations on their useful lifetime and the activities they can undertake. Indeed, the powering of wireless sensor nodes was recently identified as a "critical barrier to the uptake of this technology" [5]. For this reason, energy harvesting technologies are particularly attractive as they offer the potential to sustain the operation of sensor nodes indefinitely, provided that there is sufficient exploitable environmental energy (such as light, vibration, or temperature difference) available. Energy harvesting has a number of other benefits: it can eliminate the cost and inconvenience of replacing batteries on sensor nodes, reduce waste, and potentially enhance the energy-awareness of sensor nodes—meaning they are able to manage their energy resources intelligently—in order to deliver sustainable operation. Research interest in the use of energy harvesting in wireless sensor networks has increased rapidly in recent years. This trend is visible in Figure 2.2, which shows both the number of papers published on energy harvesting sensor networks, along with the percentage of papers on sensor networks that feature energy harvesting.

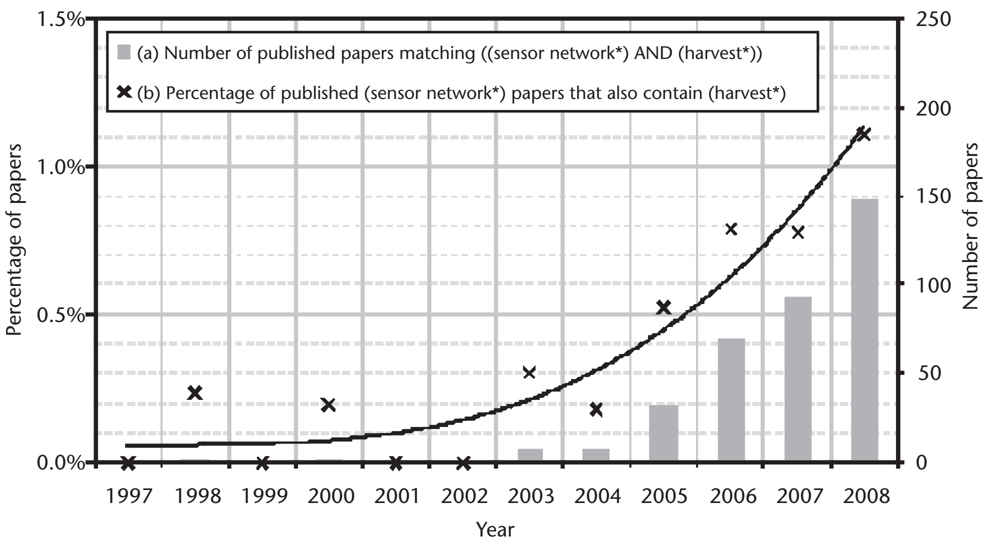

Figure 2.2 The increase in research into energy-harvesting WSNs, shown by (a) the number of published papers matching the topic [(sensor network*) AND (harvest*)], and (b) the percentage of published papers containing the topic (sensor network*) that also feature the topic (harvest*). (*Source:* ISI Web of Knowledge.)

The complexities of interfacing with energy harvesting devices mean that they cannot be treated as simple drop-in replacements for batteries. In order to obtain optimal efficiency, it is normally necessary for the output of the energy harvester to be put through a maximum power point tracking circuit or other type of power conditioning arrangement. The raw power output from many devices is insufficient to directly power a sensor node in its active mode, so it is normally necessary to buffer energy in a large capacitor or a battery to cope with peak current demands (nodes typically spend most of their time in a low-current sleep mode, waking up periodically to execute tasks). A system-level approach to the selection and design of the energy hardware of the sensor node is required. Indeed, in some situations it may be necessary to incorporate multiple energy harvesting sources; an extension of this is that energy harvesters can potentially be used in combination with nonrenewable energy sources such as nonrechargeable batteries or wireless power transfer technologies. Ultimately a flexible approach will allow system designers to select the appropriate energy resources for the sensor node, considering the power requirements of the deployment, its cost, the required endurance, and the environmental energy available in each location.

In this chapter, we first look at the energy requirements, capabilities and applications of autonomous devices—from wireless sensor nodes through to mobile phones and MP3 players, to radio frequency identification (RFID) tags—along with a broader consideration of trends and likely future developments. Later in the chapter, we look at the enabling technologies for autonomous devices: microcontrollers and transceivers (including system-on-chip devices), sensors and peripherals, and interface methods. Communication methods, protocols, and technologies are covered, including energy-aware protocols, routing, and network topologies.

2.2 Energy Requirements of Autonomous Devices

Ongoing advances in low-power electronics and energy harvesting are making the powering of wearable, handheld or pervasive devices from ambient energy a distinct and real proposition. The power consumption of various computing platforms is shown in Figure 2.3.

2.2.1 From Mobile Phones to MP3 Players

Aside from pervasive systems such as WSNs and RFID, the multibillion dollar portable electronics market—from mobile phones to MP3 players to digital cameras—will be an attractive application for micro- and macro-scale energy harvesting when the power requirements can be met. The average mobile phone has a power consumption in the order of 1W during a call and 10 mW in standby. Clearly, where

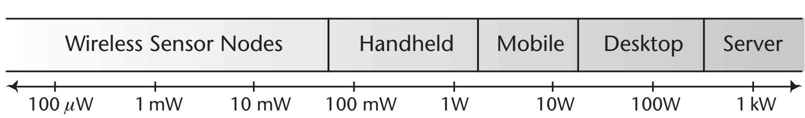

Figure 2.3 Hierarchy of computing based on power consumption. (*After:* [6].)

energy harvesting is incapable of delivering watts of power, it may permit a near-indefinite standby lifetime or even recharge the phone between calls.

Advances in low-power electronics are reducing the energy requirements of other mobile consumer devices including MP3 players (as shown in Figure 2.4) that currently consume less than 50 mW during playback. There is a clear possibility for energy harvesting to be used to extend the battery life of these devices significantly or even indefinitely. The possibility of using energy harvesting to recharge the battery in a mobile phone, MP3 player, digital camera, or other mobile device is certainly realizable, and a number of different USB chargers powered from solar cells have recently appeared on the market.

2.2.2 Radio Frequency Identification (RFID)

RFID devices enable the identification of an object without the requirement for line-of-sight (LOS) or physical contact. Originally, RFID's greatest competitor was the simple barcode, with potential end-users viewing RFID as an unnecessary electronic replacement. The barcode remains widespread, and instead of being a replacement technology, RFID's additional capabilities—including the ability to simultaneously read multiple tags, store additional data, and read from greater separation distances—have resulted in it creating a distinct market for applications with more demanding identification requirements. Typical applications often associated with RFID include asset management (in retail, healthcare, baggage handling, supply chains, and commercial services) and personal identification (using credit card size smartcards). In 2002, the annual global market for RFID systems was £550m [7]. It is predicted that, by 2015, 900 billion food items will be tagged and stricter livestock legislation will require around 824 million more sophisticated and expensive tags (with an annual market of billions of Euros for livestock applications alone) [8].

An RFID system consists of a host computer, one or more readers, and a number of transponder devices commonly referred to as *tags*. The reader emits a radio frequency (RF) signal that is detected by the tags which, in turn, respond by

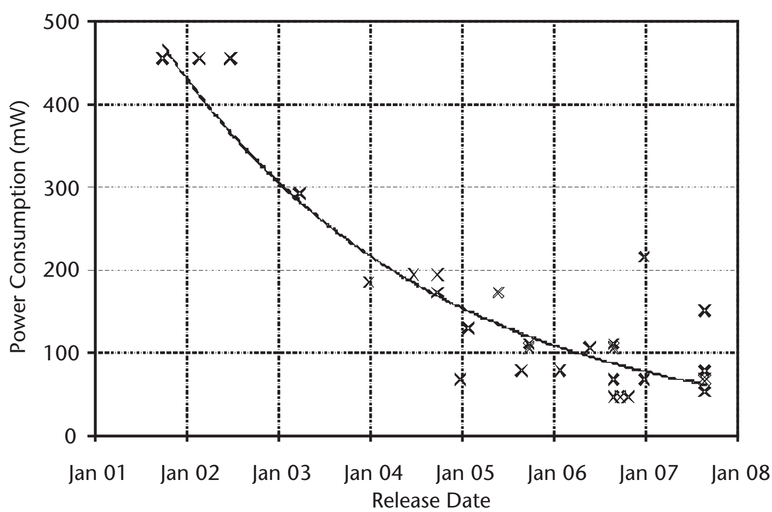

Figure 2.4 The decreasing trend of power consumption for the Apple iPod (*From:* www.ipodbatteryfaq.com).

transmitting their stored data. Anticollision algorithms are utilized to allow multiple tags to respond at the same time (for example, through a multitier TDMA system) [9]. RFID tags can be classified by their power resources [10, 11]:

- *Passive tags:* Transponders consisting of an encapsulated aerial and IC, but no on-board power source. Passive tags rely on the reader to provide power via the interrogation field that they emit. The harvesting of energy from this field is performed using either near-field inductive coupling or far-field electromagnetic coupling (using a technique referred to as *backscatter*). Passive tags have a limited operable range, but are cheap to produce and do not have a battery-limited operational lifetime.
- *Active tags:* Transponders that contain an integrated power source or are wired to a powered infrastructure. Active tags have higher performance, resources, and range (up to several kilometers) but are larger, more expensive, and have an operational lifetime limited by the battery.
- *Semipassive tags:* Transponders that are a hybrid of active and passive tags, using passive techniques for communication and active circuitry to power on-board devices.

Further classification of RFID devices is provided by the EPC global tag class definitions [12], which places devices into one of four categories. Most notably, the semiactive and active classes (three and four, respectively) include the capability of performing advanced tasks that are not traditionally associated with RFID (including sensing, data logging, and ad hoc networking). Furthermore, research advances are producing passive RFID tags that are able to deliver this functionality while retaining the benefits associated with passive transponders [13, 14]. The expansion of RFID from wireless communication (as a wireless ID device) into sensing and networking can be seen in Figure 2.5.

There is a degree of overlap between the capabilities of advanced RFID devices, described earlier, and of wireless sensor nodes. A useful distinction between the two technologies (neglecting the required cost distinction, see Figure 2.6) is perhaps that wireless sensor nodes have *sensing* (typically of environmental parameters, and

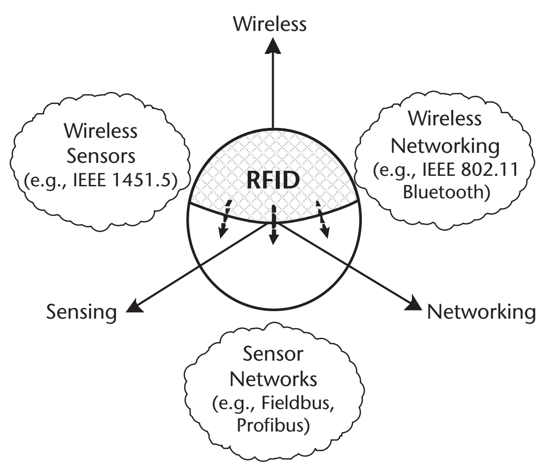

Figure 2.5 The capabilities of RFID compared with related technologies.

commonly the location of themselves or other devices) as their main task, whereas RFID devices are mainly concerned with *identification* of the objects to which they are attached. An example of a device that sits on the boundary between the two device classifications is the Intel Wireless Identification and Sensing Platform (WISP) [16], which is interrogated by conventional UHF RFID readers and incorporates a microcontroller and sensor suite, enabling parameters such as temperature or the orientation of the device to be measured.

2.2.3 Wireless Sensor Networks

Radio-based communication technologies have achieved dominance for wireless sensor nodes, but some other communication methods (such as sound or light) are also used by some devices. A network of wireless sensor nodes can be formed, which offers the potential for nodes to cooperate and participate in routing data from one node to another node in the network (possibly over several hops). Wireless sensor networks may incorporate many hundreds or thousands of nodes spread over a wide area. The capabilities and power requirements of common communications protocols and node technologies are described later in this chapter.

The concept of smart dust [17] was outlined in the late 1990s by Kris Pister from the University of California, Berkeley (UCB), and is the origin for the use of the term *mote* (which means a small particle) to describe wireless sensor nodes. The challenging vision of smart dust was for cubic millimeter-sized sensor nodes (incorporating sensing, power, computation, and communication hardware) distributed liberally throughout the environment, providing intelligence to everyday objects. While the realization of the extremely compact volumes proposed may be many years away and of questionable benefit, a number of wireless sensor node platforms have been developed. A particularly small example is the Eco mote [18] from University of California, Irvine (UCI), which measures 1 cm^3 including its battery. In general though, to provide adequate sensing capabilities and a suitable power supply, sensor nodes are much larger. For example, the Mica mote (developed at UCB in 2001) was the first design to be commercialized by Crossbow Technology and had an overall volume of approximately 13 cm^3 excluding its battery pack [19]. The device had a stacking connector structure that allowed additional boards to be attached to the mote providing sensors or an alternative power supply. The development of this device was followed by the Telos mote in 2004 [20], which has a lower power consumption than the Mica and a built-in USB programming interface. These motes incorporate a low-power 8- or 16-bit microcontroller, sensor interface circuitry, and a radio transceiver for communication with other devices. Examples of wireless sensor nodes are shown in Figure 2.7.

The embedded software for sensor nodes is normally written in a variant of ANSI C. The communication-related functions of the node (which form the communications stack) are normally provided by the device manufacturer either in full

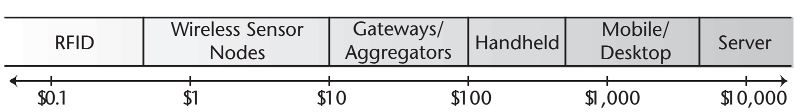

Figure 2.6 The cost spectrum of various computing platforms. (*After:* [15].)

(a) (b)

Figure 2.7 Examples of commercially-available wireless sensor nodes. (a) A MEMSIC TelosB. (*From:* [21]. Courtesy of MEMSIC Inc.) (b) A Texas Instruments eZ430-RF2500-SEH. (*From:* [22]. Courtesy of Texas Instruments.)

or as precompiled libraries. A number of embedded operating systems for sensor nodes have been developed. TinyOS [23] is a popular example, and is written in nesC. The nesC language [24] is component-based and event-driven, and components (which represent functions including communications, routing, sensing, and storage) are wired together to create TinyOS applications. Criticisms of the original TinyOS 1.x system were that it was unnecessarily restrictive and difficult to adapt to new platforms; these concerns have been addressed by the new hardware abstraction architecture of TinyOS 2.x, which uses a three-layer abstraction for hardware components. The result of the new architecture is a hardware-independent interface to the application running on the sensor node [25]. TinyOS is perhaps the most widespread operating system for WSNs, being supported by UCB motes including the Imote2, and has been ported to a number of other platforms. Other operating systems, such as Contiki [26] and MANTIS [27], allow coding of modules in C and have some advantages over TinyOS such as multithreading and over-the-air programming. On the other hand, the Sun SPOT platform executes programs written in Java, and Gumstix devices natively run a compact version of Linux.

While motes are conventionally highly resource-constrained, a number of more computationally capable nodes with 32-bit processors have also been developed. The Imote2 [28] is a result of the partnership between Intel and UCB and uses a similar stacked-PCB architecture to the Mica, but supports devices such as cameras and USB peripherals in addition to the standard sensor board. The Imote2 has a significant amount more memory than the Mica or Telos mote series which, along with its enhanced processor, makes it suitable for intensive applications such as image processing. Similarly capable nodes include the SunSPOT [29] (developed by Sun Microsystems), and the Gumstix [30] family of single-board computers. In general, these types of node have a higher active and sleep current draw than typical motes, and take substantially longer to transition between power modes; however, their overall power draw may be lower as mathematical tasks and other computationally intensive applications can be completed in a much shorter time period. These more capable motes will typically run programs written for Java or .NET, with operating systems including cut-down versions of Linux.

The capabilities and operating characteristics of a range of WSN platforms are shown in Table 2.1. The active current draw of sensor nodes is normally several orders of magnitude larger than the sleep current. For this reason, the overall power

Table 2.1 Features and Characteristics of a Selection of Popular Wireless Sensor Nodes

Node	*Mica2*	*TelosB*	*Imote2*	*SunSPOT*
Freq. band	868/915 MHz	2.45 GHz	2.45 GHz	2.45 GHz
Processor	Atmel 8-bit ATMega128L	TI 16-bit MSP430	Intel 32-bit PXA271 XScale	ARM 32-bit ARM920T
Transceiver	FSK	802.15.4 DSSS	802.15.4 DSSS	802.15.4 DSSS
Max. data rate	38.4 kbps	250 kbps	250 kbps	250 kbps
Flash memory	512 kB	1 MB	32 MB	4 MB
Voltage	2.7–3.3V	1.8–3.6V	3.2–4.5V	3.7V
Current Draw				
Sleep	<16 μA	6.1 μA	390 μA	33 μA
Active	8 mA	1.8 mA	31 mA	86 mA
Transmit	27 mA	18 mA	18 mA	18 mA
Receive	10 mA	23 mA	23 mA	23 mA

requirement of sensor nodes is highly dependent on their duty cycle. For example, a TelosB mote will draw 6.1 μA when sleeping, and around 25 mA when active and communicating (which is around 4,000 times higher). With a supply voltage of 2.0V and a 1% duty cycle, the node will consume an average power of approximately 510 μW; in contrast, with a 0.1% duty cycle it would consume only 62 μW. The energy saving from duty-cycled operation is given by (2.1).

$$\text{Energy Reduction} = (1 - DC)\left(1 - \frac{P_{sleep}}{P_{active}}\right) \tag{2.1}$$

These submilliwatt average operating powers mean that sensor nodes can potentially be powered for very long periods by batteries, or indefinitely from harvested energy. The rate of power generation from energy harvesting devices is typically insufficient to directly power the node in its active mode, so it is generally necessary to buffer energy in supercapacitors or rechargeable batteries.

A number of experimental devices have been developed that exploit environmental energy to power wireless sensor nodes. These include the Heliomote platform [31] from the University of California, Los Angeles (UCLA), which harvests energy from outdoor light and buffers the energy in rechargeable batteries, powering a Mica mote. By directly buffering energy in the batteries, they were exposed to repeated charge/discharge cycling that resulted in premature aging of the cells. This problem was addressed by Prometheus [32] (from UCB), which has a two-stage energy storage arrangement incorporating a battery and supercapacitors that reduces the stress of repeated charge/discharge cycling on the battery. Commercial energy-harvesting nodes have been released by MEMSIC in the form of their eKo system [33] (which powers the sensor node from outdoor light). Other sensor nodes that incorporate energy harvesting include the eZ430-RF2500-SEH from

Texas Instruments [34] (powered from indoor light), and a range of devices from enOcean [35] (harvesting energy from light, temperature difference, or even the action of pressing a switch).

Other projects have used multiple energy resources to power a sensor node. For example, AmbiMax [36] from UCI is a modular architecture that allows the connection of multiple energy harvesting devices (demonstrated with light and wind harvesting, each with their own maximum power point tracking circuit), buffering the harvested energy in supercapacitors to power a sensor node. Along similar lines, researchers from the University of Trás-os-Montes and Alto Douro, Portugal have developed MPWiNodeX [37], which is intended for precision agriculture applications and harvests energy from light, wind, and the flow of fluid through an irrigation pipe; their system also permits the node to monitor the rate of energy harvesting. A system incorporating a modular architecture for the energy subsystems of wireless sensor nodes [38] has been developed at the University of Southampton, United Kingdom; this allows energy harvesting devices to be connected to the system in a plug-and-play manner.

2.2.3.1 Future Trends in WSNs

WSNs are a relatively new phenomena, and because each application has different requirements, it is the application that drives the development of the area. Applications of wireless sensors networks include bodywide sensor networks for health parameter monitoring, monitoring of elderly people for quality of life improvement, precision agriculture, personnel or asset location monitoring, environmental control in buildings, and home automation, all of which are expanding areas. In this section, some of the current key drivers are briefly discussed, together with the relevance of energy harvesting.

A key advantage for wireless sensors is that they are flexible and easy to install and, if they can be made energy autonomous, then this flexibility is enhanced as they have the added advantage of being easier to maintain. These features make them attractive in both situations where the sensors can be installed for long-term use, and also for rapid deployment into areas for short-term use, where if necessary the sensors can organize themselves into networks.

Condition Monitoring Condition monitoring is an important area for wireless autonomous systems with some applications only becoming realistic with the advent of energy harvesting.

Process control equipment. The process control industry is embracing industrial networking through Fieldbus, which is a collection of IEC wired standards for the connection of digitally controlled equipment in industrial environments, and work is progressing towards developing a wireless Fieldbus. Industrial equipment is a prime candidate for condition monitoring, as this allows maintenance to be planned with a resulting minimizing of downtime of equipment. Frost and Sullivan's 2008 report on "World Condition Monitoring Equipment and Services Markets" states that the demand for wireless solutions is driving the condition monitoring market [158]. Of particular relevance for self-powered wireless sensors is their application to rotating equipment, as this is usually synchronized to the local power frequency and so is well-defined. Such equipment lends itself particularly well to vibration powered systems, with research active in the area [39, 40], and with companies now selling vibration powered systems for use at fixed frequency, such as GE Energy, who are offering vibration powered wireless sensor condition monitoring networks [41] such as the 185300 Essential Insight.mesh product. Such a harvester is capable of generating a few milliwatts to a few tens of milliwatts from typical levels of vibration seen on rotating machinery, while the vibration is present.

Health and usage monitoring (HUMS). The term HUMS is at present confined to the aerospace industry, with systems originally developed for monitoring offshore helicopter usage, but this is now extending to the fixed-wing sector. The European Aeronautics Science Network Association [42] defines HUMS as research and technology associated with a network of sensors tasked with monitoring the health, usage, fatigue, and performance of various aircraft systems and subsystems. HUMS systems monitor life consumption of life-limited components by monitoring actual damage exposure due to combinations of loads, speeds, temperature and so forth, allowing timely replacement, as well as allowing data logging activities for such parameters as flying-time and airframe loads. The concept has proved to be successful with evidence showing that in the military environment, HUMS equipped helicopter units were able to fly 27% more missions that equivalent units without HUMS, but in the more general aviation world, since the introduction of HUMS systems in the early 1990s, helicopter accidents caused by mechanical failure have reduced by over half [43]. One area of current research is to incorporate energy harvesters into HUMS sensors, with a view to allowing the sensors to be installed into components that otherwise would not be possible. Recent work in the United States by MicroStrain Inc. [44] has demonstrated a wireless helicopter pitch link sensor, powered by PZT elements located on the rotor head, and the European Union-funded project TRIADE [45] is developing self-powered wireless sensors with a view to developing a credit card sized self-powered sensor that will be embedded into the structure of a rotor blade at the time of manufacture, with a target harvesting energy budget of 200 μW.

Structural monitoring. Civil infrastructure health monitoring is a major market. As an example, the Federal Highway Administration has a 2-year cycle for inspecting its current bridge inventory, consisting of over 500,000 significant bridges [46]. The optimum solution would be to monitor these structures in real time, rather than discretely every 2 years, so that catastrophic failures can be predicted and prevented. Wireless sensor solutions are being trialed, notably on the Golden Gate Bridge, where a 46-node battery-powered network was deployed and used to measure the dynamics of the bridge [47]. As well as monitoring, dynamically controlling structures is also of interest, using wirelessly networked sensors to provide control inputs for actuators [48]. Although buildings and other structures usually have power supplies available, energy scavenging systems allow increased flexibility of installation, particularly in retrofitting. Research is progressing on producing self-powered structural health monitoring systems [49] and this will be a significant growth area in the future.

Tracking and Border Monitoring One of the obvious applications of WSNs is in tracking, both for military and civilian use. In military applications, it is desirable to be able to track enemy movements and an attractive approach is to be able to air-drop a network of sensors that can organize themselves once distributed and report back on events within their sphere of detection [50]. In terms of energy usage, short-range communication is attractive as it is less prone to interference and uses less energy for transmission than equivalent systems over a longer range. However, to cover longer distances, nodes can be utilized as relays. This, results in the energy budget being distributed across the network, which is advantageous for energy constrained systems. An example of this approach is the Line in the Sand project [51].

Border monitoring is also cited as a potential application for wireless sensor networks, and here the potential to self-power the sensor improves the practicality of the scenario, with the vision of being able to monitor the U.S.-Mexico border driving the results reported in [52], which indicates that 1,000+ node networks are feasible for monitoring. Less ambitious, but equally valid applications with shorter perimeters are attractive, such as placing a sensor network around a building or installation. Wireless, self-powered sensors have attractions in such a scenario as they are flexible and easy to install and augment.

Environmental Monitoring Environmental monitoring is possibly the most advanced application area for WSNs, and many examples of deployments can be found, with Zebranet [53] and Glacsweb [54] being just two examples. Research is also active in applications that monitor remote environments over long periods for infrequent events. Forest fire monitoring is one active area, with many areas of the world susceptible to forest fires [55]. WSNs have applications in high risk areas, which can be monitored by a network of self powered sensors with occasional nodes configured for satellite communication [56].

Rapid deployment for detection of short-term events is also a promising application area, currently receiving significant research interest. In [57], a sensor network is dropped by helicopter to monitor a volcano, and in [58], an application for a network of buoys for oil-slick monitoring is described.

Urban Monitoring Urban monitoring is a growth area, with the prevalence of the cellular network allowing a ready-made infrastructure to be exploited, with major social projects being developed such as those under the Urban Sensing Theme [59] at the Center for Embedded Network Sensing (CENS). Cell-phones are capable of either communicating with local phones using Bluetooth or by transmitting back to the mobile network, allowing potentially significant data capture to occur routinely. Mobile phones have many different peripherals, such as GPS, and Nokia is extending their use significantly [60] with many exciting visions under its Eco Sensor concept. Although a resource-constrained node, in so much that cell phones are battery powered, work is progressing at enabling the recharging of mobile phones by energy scavenging, with a demonstration by Orange at the 2008 Glastonbury Festival of a system that harnessed the power of dancers to recharge a cell phone battery [61].

Wireless Multimedia Sensor Networks The development of low-cost sensors over recent years has enabled increased capability to be included in sensor networks. In particular, CMOS camera sensors (now a standard item on mobile phones) and microphones have become available for integration into low-power nodes. Research is now addressing the concept of utilizing resource (including energy) constrained wireless sensor networks for multimedia applications [62] including video. Having the hardware is, however, only half the story. Processing the large amounts of data that these sensors can generate is equally important and there is a choice of either processing the data on the sensor node or within the network, or transmitting the data to some central processor for analysis. This is all linked to the power budget of the sensor node—if power is plentiful, then high-power processing nodes can be utilized, but if power is scarce, then more care needs to be taken in sensibly using the power budget.

The attractions of video enabling sensor networks are obvious. CCTV networks are traditionally mains powered, with cabled links to a control center. A wealth of raw data can then be monitored and recorded, with the capability of being analyzed later in less than real time to track an event. Increasingly, it is becoming possible to automate this process, using various biometrics to identify the target and significant processing power to track this. An example of this is the Polymnia project [63], an EU Framework 6 project that allows visitors to a theme park to be identified and tracked throughout their visit by cameras in the vicinity. At the end of the day a DVD of their visit is available. Such impressive technology requires significant processing power and is not currently suitable for power-constrained WSNs. Nevertheless, research is active in the area and, with the development of technology (particularly data processing, compression techniques, and the relatively wide bandwidth that is available with UWB systems, together with solar and wind powered systems), such applications will become more feasible [62, 64, 65].

Distributing Resources An alternative strategy is to process data on the node itself, only transmitting when something interesting is happening, or even just transmitting part of the image. An interesting halfway house is beginning to emerge, with data being processed within the network itself, utilizing the power of local nodes to assist in the processing. Thus data is kept local and only the information contained within that data needs to be forwarded. An example of this approach is given in [66]. This approach is extended in the idea of distributed processing, with a particular subset of this, known as agent-based computing, being particularly suited to WSN, where no central controlling node is used, but nodes extract information from the network to use in local decision making. In particular, multiagent systems have the ability to negotiate, make decisions, and be proactive in terms of achieving their goals, which leads to the notion of a social level that sits above the actual hardware level. The agent-based approach finds particular application in a dynamic environment, which is typical for a wireless sensor network [67].

In conclusion, it can be seen that there are many potential applications for WSNs, but it is power that is one of the main drivers that both constrains and encourages their application. Energy harvesting is rightly generating a lot of interest and the combination of energy scavenging and wireless sensor networks is a powerful enabler, with self-powered sensor nodes being able to be installed in environments where nonrechargeable devices cannot be considered.

2.3 Enabling Technologies: Devices and Peripherals

Wireless sensor networks are complex systems, comprised of a number of sensor nodes that incorporate microcontrollers and peripherals. These devices communicate, typically using radio transceivers. At all times, there is a requirement for systems to be low-power (as these systems are wireless, they will normally operate from batteries or harvested energy), robust, low cost, and interoperable. This section addresses the technologies that enable these systems. Communication issues are discussed in Section 2.4.

2.3.1 Low-Power Microcontrollers and Transceivers

The operations of wireless sensor nodes are typically coordinated by microcontrollers. A number of low-power microcontrollers have been developed for use in resource-constrained wireless sensing applications. Examples include the Texas Instruments (TI) MSP430 [68], a 16-bit RISC mixed-signal microcontroller, which is used in some motes and has achieved dominance for use in wireless sensor network applications. Generally, microcontrollers used in wireless sensor networks will feature very low sleep currents of around 1 μA, fast wake-up times, and the ability to interface with a range of peripherals including radio transceivers and analog sensors; other options for low-power microcontrollers include the Atmel AVR series that is used in some MEMSIC motes. Some higher-power node platforms are based on more powerful processors (for example, the Imote2 platform is based on the Marvell PXA271 XScale processor). The sleep current for these more capable processors is typically around 100 times that of the low-power microcontrollers (although their active-mode power consumption is comparable), which makes these devices more suited to less energy-constrained applications where higher processing power is required.

In wireless sensor nodes, the radio transceiver provides the interface between the microcontroller and the physical radio channel. At present, the most widely adopted protocol for wireless sensor networks is IEEE 802.15.4 [69], and a typical transceiver is the TI CC2520 [70]. Many of the requirements of the protocol are implemented in hardware (such as data handling and buffering, burst transmission, encryption and authentication, clear channel assessment and quality indication). It is a requirement of 802.15.4 transceivers that they must have an adjustable output power (which allows the transmit power to be optimized to deliver the required range); settings for the transmit power, along with various other parameters, are accessed via memory registers on the transceiver. A number of single-chip solutions, which incorporate a microcontroller and radio transceiver (thus delivering a reduced footprint and potentially lower component cost), have been developed. For example, the TI CC2530 [71] integrates an enhanced 8051 microcontroller and 802.15.4-compliant transceiver onto a single chip. The TI CC430 [72], shown in Figure 2.8, integrates an MSP430 and subgigahertz transceiver into a 9.1 mm x

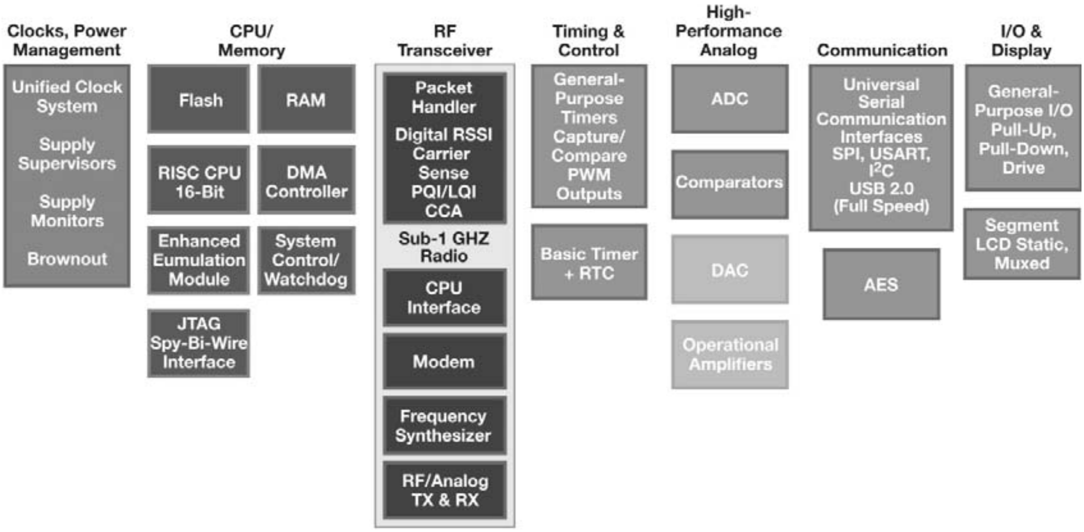

Figure 2.8 Outline of the Texas Instruments CC430, a system-on-chip device incorporating an MSP430 processor and subgigahertz radio transceiver. (Image reproduced from [72].)

9.1 mm package; this device potentially offers high levels of performance and very low power consumption (in common with other MSP430 devices) along with the good RF performance offered in the subgigahertz frequency range.

As mentioned earlier, microcontrollers on sensor nodes will normally spend most of their time in a low-power sleep mode in order to conserve energy, waking up periodically to take measurements and transmit data. However, most devices offer a range of power modes; for example, the TI MSP430 has five low-power modes in addition to its active mode. In the deepest low-power mode, the CPU and all clocks and peripherals are disabled, meaning that the system can only be woken by an external interrupt. In shallower sleep modes, certain peripherals such as clocks and DC-DC converters are disabled. The depth of sleep mode has implications for the length of time in transitioning to the active mode, and so the overall power consumption and responsiveness of the processor. A comparison of the current draw of the various power modes is shown in Figure 2.9. For devices with separate transceivers, the power consumption of the radio in receive mode is often equal to or higher than its consumption in transmit mode. For this reason, it is normally essential to “sleep” the radio, rather than leaving it in receive mode when idle.

2.3.2 Sensors, Peripherals, and Interfaces

Although fundamentally the microcontroller and transceiver are the heart of a sensor node, they cannot perform all the requirements of the sensor node on their own—at the very least a sensor is required. Advancing system-on-chip technology is allowing such peripheral capability to be included within the chip design to produce highly integrated devices. As more functionality is included, the chip becomes more complex but offers the promise of small size and good value for money. For example there are well over 3,000 variations on the Microchip PIC family of

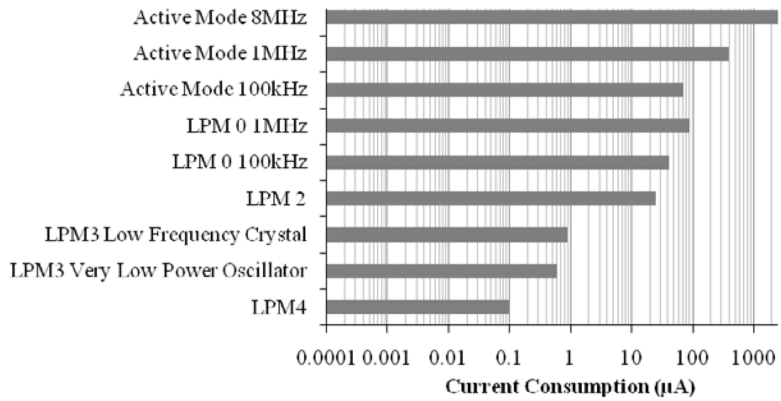

Figure 2.9 Current consumption of MSP430F2274 microcontroller, in various active and low-power modes, with a 3.0-V supply. (Figures obtained from MSP430 datasheet [73].)

microcontrollers. The difficulty comes when the specific requirements of a sensor node require a capability that is not available within a single package, or conversely a chip is used that contains excess capacity, resulting in a solution that is physically larger, more power-hungry, and more expensive than it needs to be.

Although a custom design is feasible, it is common to take a systems approach and add the required functionality to the basic microprocessor. This has the advantage that improved sensing performance can be added to a basic design as technology improves. For example, analog-to-digital converters can be replaced with higher-resolution or faster devices as they become available without having to completely redesign the node architecture. If we pursue this example further, it becomes apparent that this is only possible if there are standard interfaces between the analog-to-digital device and the microprocessor.

More complex nodes require more thought about the sensor interface, and can result in a flexible hardware solution, where several standard functions can be assembled together. Good examples of this approach are described in [74, 75], where a modular assembly of sensor interfaces can be used, with [75] in particular demonstrating the significant amount of work necessary to produce an efficient node. Such complex systems use proprietary methods for connecting, but at a more fundamental level, there are several standard methods that exist for peripheral interconnection and are particularly applicable for microprocessor-based systems.

2.3.2.1 Standard Interfaces

There are two main interfaces used for communicating between chips: inter-integrated circuit (I^2C) and serial peripheral interconnect (SPI). Both are serial interfaces but I^2C trades software complexity against required microcontroller input/output resources, and SPI requires more hardware resources but allows simpler software as each SPI device is enabled individually.

I^2C was developed by Philips (now NXP) in the early 1980s as a flexible way for ICs to communicate with each other. This system only requires two signal lines, irrespective of how many I^2C devices are attached and is in the form of a bus structure. Devices make use of a clock line and a data line and the specification [76] allows for multiple bus masters and therefore implements bus contention management. This is achieved by the use of open collector interface stages that require pull-up resistors, and it is this feature that can have an impact on energy constrained systems. The specification calls for a minimum bus current of 3 mA when active, which can cause significant power consumption if the bus is heavily used. It is apparent that this flexibility gives added complexity over the SPI interface as devices need individual addresses and need to check that the bus is available. However, such features can enable elegant hardware designs in complex systems that have several devices.

Although there is no official standard for SPI (Motorola is credited with developing it), it has proved to be a very popular interface standard. This interface requires up to four input/output lines from the microprocessor to control a device and allows full duplex communication. It is a single master system and its structure resembles a shift register. Data is clocked into or out of the master under the control of the master, with a chip select signal indicating the start or finish of a transaction. This synchronous structure means that clock rates can be very flexible and the simple hardware means that no pull-up resistors are required, reducing the power consumption. If more than one SPI device is required then each subsequent device requires an extra I/O line, used as a chip select. Thus a system with two SPI devices could potentially require five I/O lines from the microprocessor. Consequently, systems with multiple devices can consume I/O capability and board space very quickly. In addition, the lack of a formal standard can make controlling multiple device solutions complicated, and it is necessary to take care to check the requirements of each device from its data sheet.

2.3.2.2 Typical Sensors and Peripherals

Devices that are available with serial interfaces include analog-to-digital converters, digital-to-analog converters, memory devices (RAM and FLASH), display units, and certain sensor blocks (temperature and pressure). For example the Analog Devices ADT7516 [77] contains a temperature sensor, four channels of D/A and four channels of A/D in a single package, which can be controlled via SPI or I^2C, allowing a relatively simple microcontroller to be used.

Accelerometers such as the ADXL345 [78] three-axis device are available with both SPI and I^2C, but if such digital interfaces are not suitable then the output can be provided as an analog signal for more custom processing (through a device such as the ADT7516 mentioned above), or as a pseudodigital signal such as pulse width modulation (PWM), which potentially allows direct connection to a microcontroller equipped with an accurate timer, to time the pulse width.

2.3.2.3 Reducing Power Requirements

In general, sensors consume power, and the nature of the sensing technique dictates how much power is required. If it is necessary to conserve power then fundamentally the first requirement is to only sense when required, and turn off as much of the system as possible in between measurements. This strategy applies to all aspects of the node architecture. Thus the entire node architecture needs to be designed with the specific application in mind to maximize the efficiency. It is a common assumption that the radio aspects of a sensor node are the most power hungry, but it may be that both the sensor and the processing are major consumers. In terms of the processing capability, it may be unavoidable to use high speed, wide bandwidth processing at times, but other functionality may better be served by a slower processor. Some work has been done in trying to establish whether high-speed, high-power processors are more applicable than slower, low-power devices. For example, does it take more energy to carry out a calculation at 8-bit resolution at 8 MHz compared with doing the same calculation at 32 bits with 400 MHz? This does depend on the nature of the calculations being performed, but such assessments are taken into account in [79] where identical tasks were run on an 8-bit ATMEL ATmega128L and a 32-bit Intel PXA255 and indicate that the high-speed processor is more efficient, but has a much higher peak power usage. However, this comparison was unable to determine the degree that the processor architecture affected the result, but a later paper [80] demonstrated that energy efficiency of the high-power processor was increased when it was run at a lower clock frequency, showing that it is important to match the computing power to the application in hand. This approach is illustrated in [81] where a two-processor solution is implemented—a small low power processor is used for low-intensity tasks, whereby a larger processor is turned on to perform processor-intense functions and communication. An example of the trend to optimize sensor subsystems is given in [74], although the system illustrated still has a basic consumption of 182 μW.

2.3.2.4 Sensors

Sensors can range from relatively simple devices such as thermocouple temperature sensors to complex systems such as lab-on-a-chip biological sensors, and are very much defined by the application. However, where there is more than one technique for a measurand, it is necessary to assess which technique offers acceptable performance at the lowest power consumption. It is beyond the scope of this book to discuss sensor options in detail, but some general examples will be presented.

In general terms, individual sensors can be active or passive. Active sensors are those that interact with their environment and measure the response. Such sensors include sonar range finders and radars, and tend to be high peak energy users so are used sparingly in resource-constrained networks. The Dnet DNS-000 series motion detector, for example, typically uses 150 mW while active (www.dnoco.kr).

Passive sensors can be viewed as those that listen to their environment and can have very low power consumption, and in certain cases, can even be self-powered. For example, piezoelectric sensors generate a charge when they are subject to mechanical deformation and this charge can be used to sense, for example, acceleration. Another example of a passive sensor is the thermopile, commonly used in motion detectors, which generates a voltage from infrared radiation. However, even passive sensors can have high power demands. For example, the SirfStar III GPS chipset, commonly used in GPS enabled devices, consumes 23 mW in TrickleMode and even the new SirfStar IV chipset, with much improved power capability, requires 8 mW in the same conditions, making their incorporation into resource constrained, self-powered systems difficult (www.sirf.com).

In all cases the output from the sensor needs to be processed, and this uses power. The amount of power used will vary from device to device and is usually related to speed of response, bandwidth, sensitivity, and accuracy. Nonself-generating passive sensors include such devices as piezoresistive strain gauges, where the resistance of the gauge changes with strain. This resistance then needs to be measured and a common way of achieving this is to use a resistive bridge where the sense resistor is compared with a fixed resistor in a potential divider, with the bridge potential varying with the strain. Being resistive in nature, power is dissipated in the resistances, which makes the choice of the operating voltage and the value of the resistance important. Capacitive sensors are nondissipative and so are in principle low power, but again the signal processing will use some energy. A sensor whose capacitance varies with the measurand can be interrogated by a capacitance bridge, which operates in a similar way to a resistive bridge except that the drive signal needs to be AC, and this adds to the complexity of the circuitry for reading the bridge. Alternatively the capacitance can be used to define a time constant that can be then measured, or it can be used to define the frequency of an oscillating circuit, however both options will incur some power usage.

2.4 Wireless Communication

Autonomous, self-powered sensor nodes are generally required to communicate sampled data wirelessly. Whether this is through radio, optical, acoustic, or other means, the node's transceiver (the component that sends and receives data) has no appreciation of a network, or even the concept of communicating with other devices. These tasks are the role of the communication protocol stack, which provides a structure for the networking and communication subsystem with multiple layers. The number, content, and function of layers differ between protocols [82]. Each layer of the protocol stack implements distinct communication tasks, and offers services to the layer above while masking the details and complexity of its implementation. Indeed, by utilizing protocol stacks, different implementations of the same layer become possible. The Open Standards Interconnection—Basic Reference Model (OSI-BRM) [83] was introduced in the mid-1970s, proposing a basic layered structure for communication protocols, shown in Figure 2.10. The majority of modern communication protocols, stacks, and models (for example the Internet Reference Model [84] shown in Figure 2.10) have added, removed or merged layers from the OSI-BRM, tailoring the model to their specific requirements. Additionally, many protocols of relevance to WSNs have absorbed higher layers of the OSI-BRM into their application layer.

2.4.1 Communication Protocols and Power Requirements

Communication in wireless sensor networks is generally performed with a low data rate, short transmission range (typically 10m–100m) and a low duty cycle. Short-range wireless communications can be delivered via a range of media, including

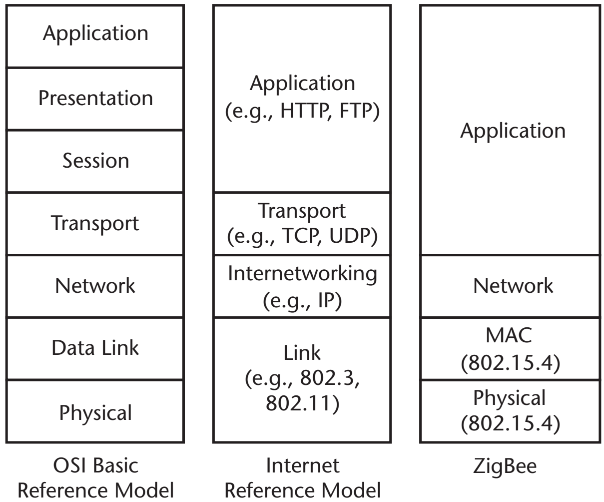

Figure 2.10 The OSI Basic Reference Model [83], Internet Reference Model [84], and ZigBee Stack [85].

infrared, radio frequency (RF) communications, magnetism, and acoustics. RF communications have achieved dominance for wireless sensor networking, mainly due to their low component cost, early standardization, and applicability to a wide range of deployment environments. Dependent on the complexity of the radio protocol and the capability and/or heterogeneity of the hardware, they permit star, tree, and mesh network topologies (as shown in Figure 2.11). Other technologies, however, have niche applications where RF communications are impractical.

As mentioned previously, the dominant basic standard for short-range RF wireless sensor communications is IEEE 802.15.4 (which defines only the Medium Access Control and Physical layers of the communications stack). The standard defines the radio channels and transmission schemes to be used in wireless personal area networks (WPANs) and facilitates schemes that allow data to be routed through a number of sensor nodes. The standard defines physical channels in the industrial, scientific, and medical (ISM) bands at 868 MHz in Europe, 915 MHz in North America, and 2.45 GHz worldwide. The 2.45-GHz band has been widely

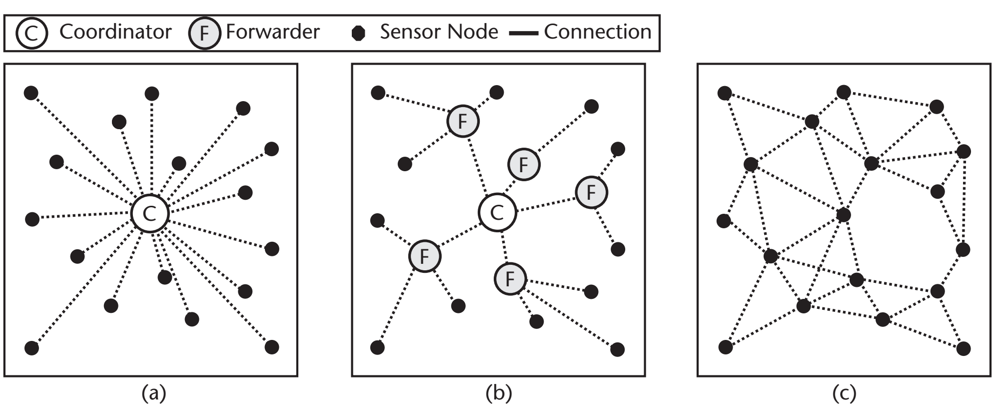

Figure 2.11 Common network topologies for sensor networks (a) star, (b) tree, with a mix of node types, and (c) mesh with a number of homogenous nodes.

adopted, and offers data rates up to 250 kbit/s and a transmission range of up to 100 meters.

Additional functionality has to be added to 802.15.4 to implement the higher layers of the protocol stack, providing routing, network management, and application support. The most established 802.15.4-based protocol is ZigBee, which is managed by the ZigBee Alliance [86]; compliant devices are capable of forming star, tree, or mesh network topologies. Additionally, the 6LoWPAN standard [87] is being developed to allow IPv6 packets to be routed over IEEE 802.15.4 networks. This allows sensor nodes to communicate with other devices over the internet, thus enabling the "Internet of Things" concept, where embedded devices are connected to the Internet, which was popularized by the Auto-ID Labs at Massachusetts Institute of Technology and has now been adopted by the International Telecommunication Union [88].

The ZigBee stack is intended to be based on cheaper hardware than conventional Bluetooth [89], with a simpler software stack and a much lower energy requirement; however, the code size and power consumption of ZigBee remains too high for some applications. Many of the energy savings are derived from much faster handshaking, with ZigBee devices able to join a network and make a transmission in a very short time period. A standard for RF-based remote controls for consumer electronics, RF4CE [90], is based on ZigBee and is set to challenge the dominance of infra-red remote controls with a nondirectional, power-efficient, two-way protocol. Other 2.45-GHz based protocols, such as MiWi [91] and SimpliciTI [92], have been developed by device manufacturers to get around the limitations of ZigBee by providing less able, but typically more device-specific, protocols. There are some moves to establish more energy-efficient but nonhardware-specific protocols: two schemes worthy of mention are ZigBee Green Power [93] and the EnOcean Alliance [94].

Bluetooth 2.1, another WPAN protocol, supports data rates up to 3 Mbit/s at a range of up to 10m. Bluetooth Low Energy [95], originally developed by Nokia as WibRee and known for a short time as Low Power Bluetooth, has the aim of allowing ultralow-power end devices to operate for over a year from a single coin cell. Future Bluetooth transceivers in mobile telephones are likely to incorporate Bluetooth Low Energy capabilities, meaning that the protocol is likely to become popular for body area networks. The Bluetooth protocols are only capable of star network operation (i.e., all end devices must communicate directly with their host), which means they are less suited to medium-to-large WSNs.

Wireless local area networks (WLANs) are defined by the IEEE 802.11 WiFi standard (first released in 1997) and its amendments. 802.11g [96] devices operate at 2.45GHz with data rates of up to 54 Mbit/s and a range of up to 140 meters. Wireless LAN has a much higher power requirement, and higher data rate, than wireless personal area network (WPAN) radio protocols, meaning it can place costly demands on battery-powered devices. WLAN protocols are capable of implementing star or tree network topologies.

Work is progressing on wireless body area networks (WBANs) under the IEEE 802.15.6 [97] task group, which will facilitate the wireless connection of implanted or worn body sensors. The sensed parameters can be received by a base station which can then forward the information to a healthcare professional. At the time of writing, no standard has been released. An important distinction between WBAN and other wireless standards is that the 802.15.6 standard is likely to use frequencies in the subgigahertz range, due to the attenuating effect of the human body on 2.45-GHz transmissions.

Some alternatives to conventional RF communications exist. In very short range (less than a meter) line-of-sight applications, infrared may be a viable solution. IrDA [98] offers data rates of between 115.2 kbit/s and 16 Mbit/s. Devices must be aligned accurately as transmission cones can be as little as 15°, which means that, in practice, devices would have to be permanently fixed in position to guarantee a reliable data link. This property does have some advantages for security of transmissions, but this is rarely of interest in wireless sensor networks. Some wireless sensor nodes use sound and radio transmissions in combination, so that the difference between the times of arrival of the two transmissions can be measured to determine the distance between transmitter and receiver [99]. In underwater environments, communications are most frequently acoustic: for example, the CORAL system uses piezoelectric acoustic transducers [100].

To facilitate long-range backhaul from the sensor network, directional antennae or cellular communication networks are frequently used. Cellular data communications are typically carried out using GPRS [101], which supports data rates between 8 and 48 kbps (dependent on the GPRS class) and is a development to the original 2G standards. Clearly a GPRS modem must be used for this purpose, which considerably increases the power consumption of the network. Alternatively, directional antennae and power amplifiers can extend the range of WPAN transmissions in excess of 5 km with a minimal impact on the overall power consumption of the network.

2.4.2 Energy-Aware Communication Protocols

The design of a sensor network for energy efficiency and energy awareness must consider far more than the selection and management of low-power hardware. The software that a node executes—including the communication protocol, application, and operating system—must also be energy-aware, and should be designed to maximize the useful life of the node and of the network as a whole. Energy-aware design affects every layer of the communication stack, from the modulation scheme used in the physical layer to the high-level protocols in the application layer. Further, energy-efficient operation is often considered as a key parameter in the design and optimization of crosslayer protocols (these are protocols that break the rules of a protocol stack by, for example, sharing variables between the layers of the stack, providing interfaces between nonadjacent layers, and merging adjacent layers) [102]. A complete overview of specific energy-aware communication protocols is outside the remit of this book, and interested readers are instead referred to the considerable body of documentation that exists on this area [103–108]. However, this chapter describes properties of routing and medium access control (MAC) algorithms that significantly contribute to and influence a node's energy use and energy-aware operation.

2.4.2.1 The Routing (Network/NET) Layer

In WSNs, routing is the process of delivering a packet from a source node to a sink node. The routing scheme can generally be classified into one of three categories: direct, flat multihop or clustered multihop [103, 104]. In direct routing, both the source and sink node communicate over a single hop (where a hop is defined as a direct transmission between two nodes). In flat multihop routing (also referred to as peer-to-peer or mesh routing) schemes, the source node uses intermediate nodes to relay messages to the sink. The route taken (and indeed the number of routes taken) depends on the protocol, but could take a large number of short hops, or a smaller number of longer hops. Finally, clustered multihop routing schemes subdivide the network into a number of clusters inside each of which nodes route packets to a cluster head (clusters and their cluster heads can be designed into a network using heterogeneous nodes or dynamically allocated using in-network runtime algorithms) that relays messages onto the sink. These three routing topologies are depicted graphically in Figure 2.12.

The process of direct routing is largely trivial: as the diameter of the network increases, so does the mean transmission power that the source requires in order to successfully communicate the message to the sink node. In order to demonstrate the theoretical benefits of multihop routing, (2.2) and (2.3) show the end-to-end power consumed routing a packet with single-hop and multihop schemes, using a generalized version of the Friis free-space model [109]. In these equations, P_{tx} [W] is the transmission power consumption, P_{rx} [W] is the receive power consumption, d [m] is the separation distance between the source and sink nodes, λ [m] is the wavelength, η is the path loss exponent (representing the rate at which the path loss increases with distance), and H is the number of hops.

$$P_{tx}^{direct} \propto P_{rx}\left(\frac{d}{\lambda}\right)^{\eta} \tag{2.2}$$

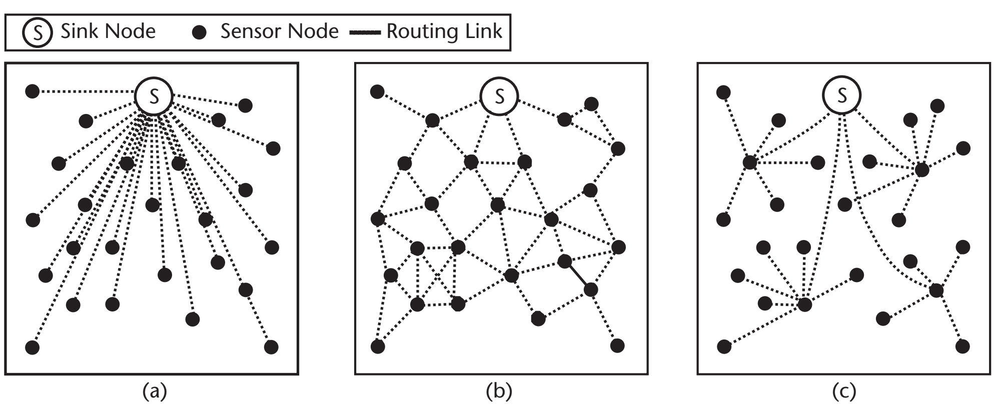

Figure 2.12 Different communication topologies: (a) direct, (b) flat multihop, and (c) clustered multihop.

$$P_{tx}^{multihop} \propto HP_{rx}\left(\frac{d}{H\lambda}\right)^{\eta} \tag{2.3}$$

By combining (2.1) and (2.2), the benefit of multihop routing can be obtained, providing the factor by which the transmit power consumption is reduced (2.3).

$$\frac{P_{tx}^{direct}}{P_{tx}^{multihop}} = H^{\eta-1} \tag{2.4}$$

It can be observed from (2.3) that a transmit power saving of $H^{\eta-1}$ is achievable through multihop routing; for example, routing over five hops with a path loss exponent of four (feasible for an indoor environment [110]) results in a reduction in the transmit power consumption of two orders of magnitude compared to single-hop routing. A graphical representation of (2.3) is shown in Figure 2.13.

From Figure 2.13, the benefit of multihop routing appears to be clear. In practice however, the relative benefit is not fully realized due to both implementation challenges and the overheads of such a scheme. These include the power required for initializing, discovering, and maintaining a complex routing scheme, the difficulty in accurately matching the transmission power to the separation distance, and a number of energy losses in the MAC layer (discussed later), particularly with regard to the energy consumed to listen for and receive data (not accounted for in the theoretical model presented above). Further, in practice oscillators and bias circuitry often dominate the power consumption of transmissions; this means that a

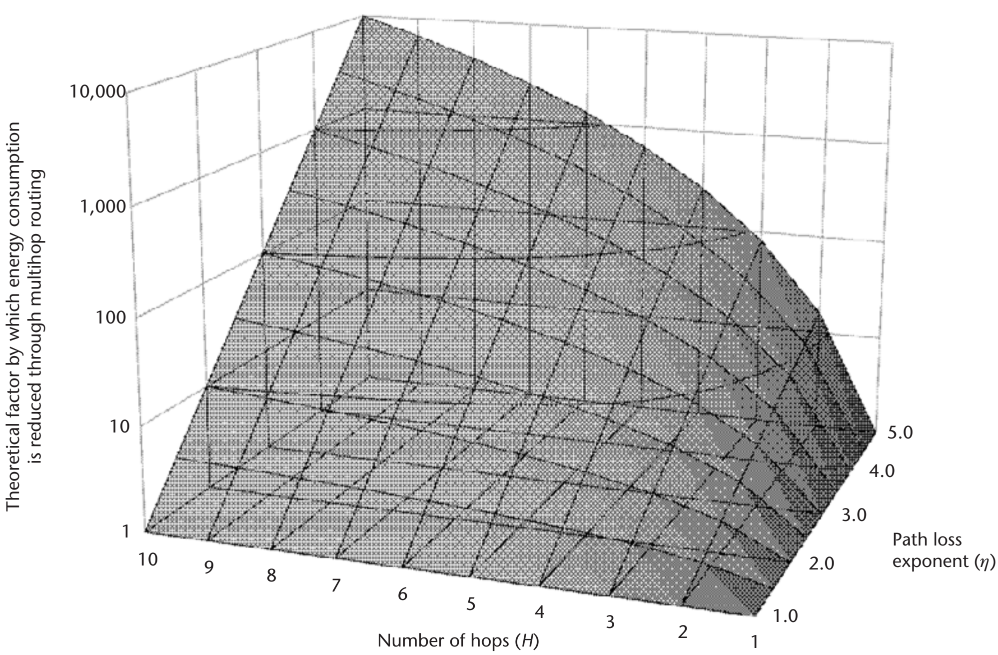

Figure 2.13 The theoretical factor by which the energy consumption is reduced through multihop routing, and its dependence on the number of hops and path loss exponent.

decrease in the separation distance does not necessarily result in a proportional reduction in the power consumption [111]. Aside from these energy efficiency issues, multihop routing can also give rise to problems including traffic accumulation, latency, and end-to-end reliability [112]. Multihop routing is however particularly useful when operating in hostile environments (the high path loss results in low transmission distances) or when a large separation distance would necessitate significant transmission power levels (which may not even be greater than the maximum permissible by radio communication regulations). It is therefore generally recognized that there is no clear winner between single-hop and multihop routing, instead the optimum solution depends heavily upon the particular environment, hardware, network configuration and application constraints [113, 114].

Reactive routing algorithms dynamically compute routes as they are required, whereas proactive algorithms periodically compute and maintain routes whether they are needed or not [104, 115, 116]. Reactive algorithms are generally favorable in networks with infrequent communication, or with a network that is rapidly changing (for example in a network supplied by energy harvesting, where individual nodes' routing involvement may change rapidly due to fluctuations in the availability of environmental energy). The simplest flat multihop routing algorithm is packet flooding, a reactive algorithm where nodes broadcast received packets to all their neighbors unless they have already forwarded the packet, a maximum number of hops have been reached, or the node is the packet's destination. Packet flooding is simple, scalable, robust, and efficient for one-to-many communication but inefficient for one-to-one communication due to the energy wasted in transmitting the packet to every node in the network [117, 118]. A number of modified flooding algorithms have been presented in the literature, which constrain the epidemic nature of flooding [119–123].

Geographical (or location-aware) routing is a truly reactive process, with algorithms usually operating by routing packets via nodes that are progressively closer to the sink node (though this is not always the case [124]). This simplistic approach to routing encounters problems when the packet is routed via a communication hole; that is, a local maxima is encountered when the current node is closer to the sink node than any of its neighbors. Different location-aware algorithms tend to differ by their method of routing around communication holes [125–129].

Routing algorithms designed for energy harvesting sensor nodes have received comparatively limited research. Voigt et al. [130] proposed a modified version of diffusion that takes the current state and availability of energy harvesting into account when selecting the next hop. The authors also proposed a similar modification to LEACH (a clustered multihop routing algorithm [131]) that takes into account the availability of energy harvesting when selecting a node to become the next cluster head [132]. Research by Kansal et al. [133] into predicting energy-harvesting availability has demonstrated the possibility of implementing harvesting-aware routing by not only considering how much energy is currently being harvested, but also how much is likely to be harvested in the future. For example, the authors propose a routing cost metric that considers both the residual energy on the node and the harvesting opportunity of the node (the amount of energy it is likely to harvest in the future).

2.4.2.2 The Medium Access Control (MAC) Layer

Medium access control (MAC), the regulation of nodes' access to a shared medium, is traditionally a sublayer of the data link layer (DLL) [110]. In WSNs however, the workload of the DLL is considerably reduced while that of the MAC is heavily increased. As a result, the MAC is often considered to absorb the tasks of the DLL and is referred to a distinct layer in its own right: the MAC layer. In WSNs, the primary performance metric is often energy efficiency, while classical metrics such as scalability, adaptability to changes, latency, throughput, fairness, and bandwidth utilization are of secondary importance. Energy-efficient MAC design is performed by addressing the following areas of energy waste [106, 134]:

- *Overheads:* Energy is consumed in overheads such as MAC headers and footers, control packets, and acknowledgments. As packets in WSNs are usually much shorter than those in traditional networks, the proportion of overheads can be considerable; hence simplistic protocols are essential in reducing the energy consumption.
- *Collisions:* Packets that collide are corrupted, and require retransmission by the source node (incurring an energy cost). Due to the low channel utilization in WSNs, the chance of collisions is not as great as with other networks.
- *Overemitting:* Energy is wasted in a node when a packet is transmitted but the sink node is not listening (thus requiring a retransmission).
- *Idle listening:* Radio transceivers for wireless sensor networks typically have four different levels of power consumption that correspond to the following states: transmitting, receiving, listening (waiting to receive), and sleeping [135]. As listening often consumes as much energy as receiving (as discussed above—and which is often more than is consumed for transmitting), a node that is listening but not receiving wastes significant energy.
- *Overhearing:* If a unicast packet (a packet with a single destination) is broadcast by a node, neighboring nodes waste energy in receiving enough of the packet to realize that it is not destined for them. This is not a problem for broadcast packets where every neighbor is supposed to receive the packet.

Many of these problems can be mitigated using fixed-assignment MAC protocols (heavily used in traditional wireless networks), which proactively allocate discrete sections of time (TDMA), spectrum (FDMA), or coding (CDMA) to different nodes for communication [106, 110]. These protocols are not particularly suited to WSNs as they generally require a central authority to assign the discrete section, with good time synchronization, additional circuitry, and/or reasonably heavy computation.

In random access MAC protocols, nodes transmit when they need to as opposed to having to use assigned sections of frequency or time. As random access protocols all communicate over the same channel, nodes inspect the channel before transmitting to see if they are going to interfere with other nodes [136]. In order to mitigate the problem of idle listening, nodes' receivers spend most of their time in a low-power sleep state, and only turn on briefly to listen for nodes wishing to communicate with them. This can create further problems as accurate time synchronization is rarely possible in WSNs (due to the absence of accurate timing hardware or intensive network algorithms).

To reduce energy wasted through idle listening, a scheme referred to as low power listening (LPL) shifts the energy cost from the receiver to the transmitter; this is particularly suited to networks with very infrequent transmissions. LPL forms the basis of the Berkeley MAC (B-MAC) protocol, which is the default MAC in TinyOS [137]. The concept of B-MAC is that a node wishing to transmit first performs CSMA and, if clear, transmits a long preamble for a time Δt_p [26]. After this has elapsed, the packet is transmitted. A node that is not transmitting wakes up periodically (with period Δt_p) and briefly samples the channel to see if a node is transmitting a long preamble. If it is, it stays on long enough to receive the packet. In this way, the receiver node only has to wake up for a very short period of time to see if a node wishes to communicate with it, as shown in Figure 2.14. However, the use of LPL decreases the effective channel capacity as the transmitting node utilizes the channel for > Δt_p per packet. The X-MAC protocol attempts to overcome many of the shortcomings of B-MAC (including overhearing, excessive waiting at the receiver for the preamble to finish, and latency) by modifying the preamble sequence [138]. Instead of transmitting a single nondescript preamble for the duration Δt_p, X-MAC repeatedly transmits the destination address. Between repeats, X-MAC also waits for a period of time to allow the receiver to acknowledge and stop the preamble, initiating data transfer. This reduces the latency (as there is not a delay of Δt_p for every packet), reduces the idle listening energy consumption (as the receiver does not have to listen to the preamble even after it has detected it), and reduces overhearing (receiver nodes that hear the preamble but with a different node's address simply go back to sleep).

An attractive concept for low-energy reactive networks is that of the wakeup radio, where a receiving node's primary receiver is awoken by a low-power secondary receiver in response to a trigger from the transmitting node [110]. This virtually eliminates idle listening on the primary radio (presuming that only the desired node wakes up) and reduces latency (as receivers are woken up when they are needed) [135]. In the MAC proposed for Pico-Radio [139], the authors assume the presence of a secondary wakeup radio (consuming less than 1 μW at full duty cycle [140]) that is able to encode a destination address into the wakeup beacon, hence reducing energy wastage through overhearing. Zero-power wakeup radios that, using techniques established for passive RFID, are powered from ambient energy are very

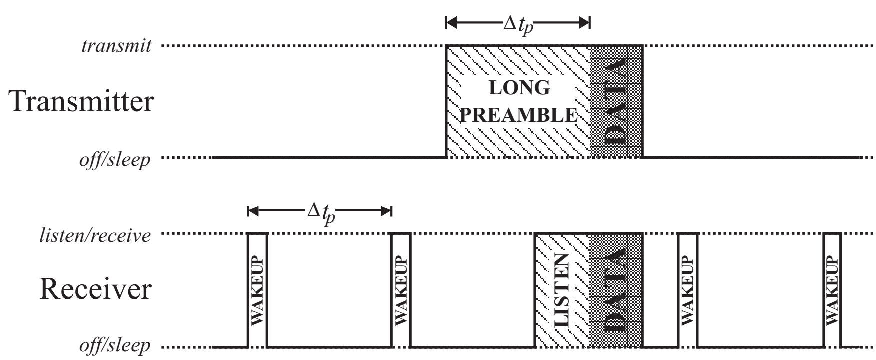

Figure 2.14 The operation of low power listening the B-MAC protocol.

attractive in accomplishing the goals of wakeup radio MACs [141, 142]. Gu et al. [143] proposed that such a virtually zero-power (typically <1 μA) wakeup radio is a realistic proposition. The wakeup detection circuitry creates a voltage directly from energy in the received electromagnetic signal, which drives a hardware interrupt on the microprocessor. The authors propose both simple wakeup beacons (with no modulation or encoded data), and also the possibility of selecting only certain nodes to wake up by using multiple frequency transmitters (for example, transceivers that are able to concurrently receive or transmit at multiple frequencies) to encode a wakeup ID beacon.

2.5 Energy-Awareness in Embedded Software

Energy-adaptive behavior or energy-aware algorithms are useful as they permit the sensor node to use information on its energy status to adjust its participation in the network (generally increasing its activity when energy is plentiful, or decreasing it when energy is restricted). Communications tasks are normally taken to be the most energy-intensive operation for sensor nodes, but energy-aware algorithms may also control sensing or processing tasks, which also consume large amounts of power. Perhaps the simplest form of energy-aware algorithm is one proposed by Delin et al. [109] in which nodes enter a sleep state when their energy stored drops below a threshold value, and wakes up again when it has been recharged sufficiently by the solar cell. Cianci et al. [144] have used threshold rules to balance the workload between a group of nodes in a network. Other algorithms use distributed processes to permit redundant nodes (i.e., those nodes whose communications or sensing coverage is also covered by other nodes) to sleep to conserve energy [145]. Merrett et al. [146] have developed schemes to discard messages of low importance in the interests of ensuring that high-importance messages can traverse the network.

These energy-adaptive algorithms must be able to monitor the energy status of the node. At its very simplest, the system should monitor the amount of energy stored by the system at a given time; this may be as straightforward as determining the energy stored on a capacitor by measuring its voltage and using the standard capacitor equation. In systems with multiple energy sources and stores, the situation can be complex and the monitoring of the energy subsystem should report with sufficient detail to enable appropriate decisions to be made. For example, a system may interface with multiple energy devices. These may include energy stores (such as supercapacitors and primary or secondary batteries); and energy sources (including mains adapters and energy scavenging devices).

2.5.1 Operating Systems and Software Architectures

The algorithms and applications operating in a node's embedded software play an important part in its energy-aware operation. A number of different operating systems (OSs) exist for WSNs [26, 27], with TinyOS being widely used internationally in both research and commercial WSNs [147]. TinyOS does not have the features of a conventional OS (for example scheduling, multithreading or memory management), but is a lightweight OS for embedded microcontrollers that provides a library of services and abstractions to assist in tasks such as clock synchronization, power management, and radio communication [148]. TinyOS is designed for ultra low-power operation, with the default operation being to power-off peripheral devices when they are not in use, and providing the programmer with abstractions to change the microcontroller and transceiver power states.

However, the use of an OS can itself restrict the functionality, capability, and efficiency of a WSN; hence, many researchers and end-users of wireless sensor nodes develop the software for the embedded microcontroller without the use of an operating system. For these users, the design process and structure is largely orientated around communications, with energy-efficiency considered in the communication stack (for example, in the routing and MAC algorithms presented above) and energy-awareness (for example, the monitoring of hardware and the node's energy-reactive operation) implemented at the top of the stack, usually in an application or user layer (depending on the particular protocol being used). For the former, research into the crosslayer design and optimization of protocols (merging some or all of the layers of the OSI-BRM) has been performed with the aim of reducing the power [149]. While crosslayer design can increase the energy-efficiency of the communication subsystem, it reduces the transferability and modularity of developed layers as, by not following the "standard" rules of the protocol stack, different implementations of the same layer now rely on different interfaces, variables, and provided functionality.

In [150] the authors propose a stacked architecture for embedded software that structures all of the major node functions (other than just communication); including a stack for energy-management. The architecture uses a basic template stack, based upon the principles of the OSI-BRM, from which different interface stacks can be derived. A number of these stacks, each implementing distinct tasks, are combined via a shared application layer. This forms a unified stack, such as that shown in Figure 2.15, which manages the complete functionality of the node.

This establishes a structured platform for the formal design, specification, and implementation of modern sensor, and wireless sensor, nodes. Energy management is provided with the benefits of a structured, layered model including the modular specification, design, and structuring of the node's energy subsystem and the

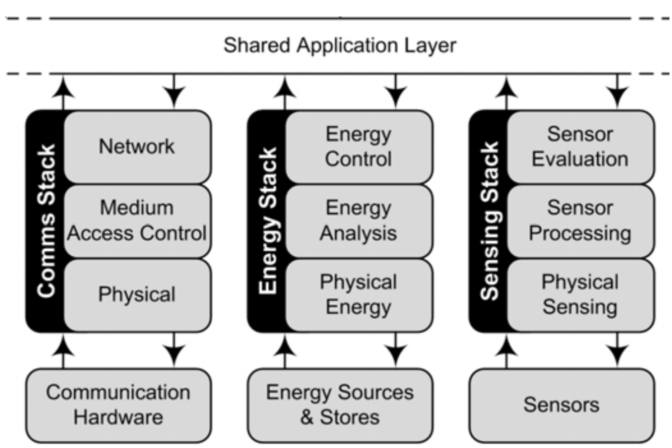

Figure 2.15 A unified stack structuring communications, energy management, and intelligent sensing.

promotion of standardized and interchangeable energy protocols to permit efficient code reuse.

2.6 Alternative Nonrenewable Power Sources

There are some interesting alternative technologies for powering autonomous sensors, and two approaches will be discussed in this section. The technique of powering devices by radio signals has long been an ideal, and can be viewed as a form of energy harvesting, although the energy is being transmitted to it for the express purpose of being used, rather than scavenging it from the environment. Another power source of interest is the fuel cell that converts chemical energy to electricity and so in principle is a nonrenewable supply.

2.6.1 Direct Transmission

Direct remote powering of devices was first demonstrated towards the end of the nineteenth century by Tesla. Ever since, there has been interest in remotely powering devices, but it is only recently that technology to achieve this has been commercialized.

2.6.1.1 WiTricity

WiTricity (wireless electricity) uses ideas similar to Tesla's to allow devices to be remotely powered over a distance of up to a few meters [151]. The company is a spinoff from MIT and builds on work developed there to use strongly coupled magnetic fields [152] to transfer power. Any devices fitted with a receiver can benefit from this and it enables the powering of devices such as laptops and mobile phones. The technique is fundamentally different to the familiar inductive recharger, and allows transfer of power over distances greater than the diameter of the coils used. The total efficiency currently available is of the order of 15% when powering a 60-W lightbulb at a distance of 2m, with the power transfer part of the system offering between 40% and 50% efficiency [152], and it is shown that although efficiency falls off with distance, it is a relatively linear characteristic. Potentially kilowatts of power can be transmitted through most common objects (although it is affected by metal) and so the technology has a range of applications.

2.6.1.2 Powercast

A similar system has been developed by the Powercast company [153], although this system concentrates on delivered power requirements of microwatts to milliwatts. It offers a received power conversion efficiency of up to 70% for its P1200 Powerharvester module, which is designed for lithium-ion single cell recharging. However, the total system efficiency is less impressive, and depends fundamentally on distance from the transmitter to the receiver and the directionality of the transmitter antenna as defined by the Friis equation, and is about 1% at distances of the order of a meter. Fundamentally the power falls off in an inverse square relationship. However, with directional antennas the system has been demonstrated to power sensors at a distance of 20 feet using a 3W transmitter.

2.6.1.3 Fuel Cells

Fuel cells are electrochemical converters that turn fuel directly into electricity (i.e., no combustion is involved). Hydrogen fuel cells have received significant publicity as in principle they offer an alternative to the internal combustion engine, using hydrogen (and oxygen from the atmosphere) as the fuel. Potentially there are carbon-free methods for producing hydrogen, such as hydrogen generation from solar cells (or other sources of carbon free electricity such as nuclear, wind, etc.) or from thermal or biological biomass reformation. The world's first operational fuel cells were developed for the Gemini space program and subsequently developed for other missions such as Apollo and the Space Shuttle and were hydrogen-powered, but other fuels are available. Of particular interest for smaller system are liquid-fuel fuel cells, utilizing fuels such as methanol or ethanol, and methanol fuel cells have successfully been used to power laptops and mobiles. Methanol is not an ideal fuel, due to its toxic nature, which has lead to developments in ethanol fuel cells, but the efficiency of ethanol devices is currently less than that of methanol (quoted as being up to 40%). However, fuel cells for smaller devices have less efficiency, at about 35% [154] with the remainder being converted to heat—potentially an issue for smaller devices. Research is active in the area and several microengineered devices have been demonstrated [155].

Another branch of fuel cell development is that of microbe powered fuel cells [156]. An interesting development in fuel cell technology that could be used by environmental sensor networks, particularly if they are mobile, is that of using foraged material as a source of fuel to power systems known as gastrobots. Such biologically inspired systems can either bring food back to a central point (or digester), and then top up on the electricity produced (literally energy scavenging), or can utilize biomass that they find directly, similar to a digestive system. A mobile system that is powered on such items as insects, fruit, and pieces of vegetables has been described [157]. A system that can run on flies has a certain attraction as the fuel is inherently mobile, so it may well be possible to attract it to the sensor—a sort of electronic self-powered fly-trap.

2.7 Discussion

This chapter has provided an overview of the energy requirements of autonomous devices, ranging from mobile consumer devices through the wireless sensor nodes. This has highlighted the wide-ranging assortment of different autonomous devices and applications, and as a result of this, the diversity of energy requirements that may span several orders of magnitude. Any application that is considering using energy harvesting will need to find a balance between the energy required for it to operate and the energy available from the environment. If a suitable balance is not found within the application requirements (as it often is not), energy harvesting may not be a viable option.

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CHAPTER 3

Photovoltaic Energy Harvesting

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3.1 Introduction

Any stand-alone electronic systems positioned outdoors, in rooms with windows, or frequently used artificial light sources are likely to find that photovoltaic (PV) technologies can provide the primary power source. Outdoors, the Sun can provide around 100 mW/cm^2 of optical power, a cloudy day will provide around 10 mW/cm^2, and around 0.5 mW/cm^2 will be incident on most surfaces within a well-lit room. Typical solar cells have efficiency values in the range of 5% to 20% under standard conditions; they will often be much less efficient under low illumination levels. The very best devices, typically very expensive "concentrator" cells, are designed to operate under the power of many suns and are up to 40% efficient. The power density available from solar cells operating outdoors can exceed that available with other energy harvesting technologies by several orders of magnitude (see Table 3.1). The value is much less for indoor operation; nevertheless, even indoor light energy harvesting can provide sufficient power densities for low power technologies such as wireless sensor nodes [1–3].

There is no doubt that plenty of optical power is available for many applications that require modest levels of energy. However, very careful considerations of the nature and frequency of illumination conditions and the total power usage of the device are required, and the area of the solar cell used must be chosen accordingly. Furthermore, devices must have energy management and storage systems that ensure that essential features (such as time keeping or critical monitoring) can be maintained throughout the longest likely periods of darkness.

Solar energy is commonly used within commercial devices, particularly low-power consumable electronics such as calculators. Solar energy is also often employed for isolated noncritical outdoor systems such as parking meters, weather stations, telephone boxes, and traffic information systems. It is less likely to be used for alarm systems or any portable high-power systems such as mobile phones or laptop computers and even less likely to power electric vehicles. Systems based on solar energy will nearly always require the end user of the equipment, be it

Table 3.1 Power Densities of Various Energy Harvesting Technologies

Energy Harvesting Technology	*Power Density Per Volume of Total System ($\mu W/cm^3$)*
Photovoltaics (outdoors, η = 15% cell, 100 mW/cm^2 incident irradiance)	15,000
Photovoltaics (indoors, η = 6% cell, 0.5 mW/cm^2 incident irradiance)	30
Piezoelectric (shoe inserts)	330
Vibration (small microwave oven)	116
Thermoelectric (10°C gradient)	40
Acoustic noise (100 dB)	0.96

Source: [3].

stationary or portable, to diligently place the device in an appropriate location, and this is often a limiting constraint.

The surface area of a photovoltaic module required for a desired power is perhaps the most limiting constraint. The size of an array required to power a house would ideally be no more than the area of one side of a roof; the size required for a laptop should be that of an A4 sheet of paper. Ideally, a single-chip sensing/transceiver system would require a solar cell no greater than its own area and ideally we would use the same piece of silicon to provide the base material for the solar cell. In many cases applications require device efficiencies higher than those currently available at a reasonable cost.

In this chapter we review the basic physics of photovoltaic devices and consider the optical power available under different conditions. We then review the main photovoltaic device technologies in order to explore and understand the limitations on efficiency for the mainstream commercial devices and then discuss PV energy harvesting system considerations. We conclude the chapter with an exploration of how PV technologies and systems might evolve to provide high-efficiency and low-cost integrated solutions in the future.

3.2 Background

3.2.1 Semiconductor Basics

A semiconducting material has an arrangement of electronic states that provide the opportunity for electrons to move and thereby provide the possibility of a current flow. For a semiconductor at absolute zero, all electrons completely occupy all the available electron states up to and including a band of energy states known as the *valence band*. Above the valence band, there lies a region of forbidden energies known as the *bandgap*. Above this, there lies a further band of allowed energies known as the *conduction band*. At absolute zero a semiconductor will have no electrons in the conduction band. At absolute zero, electrons in the valence band cannot move, as there are no available states in the valance band for them to move into, and since there are no electrons in the conduction band, there is no possibility of electron flow there either. At absolute zero there is no possibility of current flow, but things improve greatly at higher temperatures. Even modest increases of temperature will allow some electrons in the valence band to attain enough energy (an energy greater than the bandgap energy) to be thermally excited from the valence band into the conduction band. This process leaves electrons that are free to move among all the available energy states in the conduction band. It also leaves empty electron states or *holes* in the valence band. The holes in the valence band can be filled by the neighboring electrons and can thereby move around the valence band. Free electrons in the conduction band and free holes in the valence band provide two fundamental mechanisms for the current flow in a semiconductor.

When a potential difference is applied to a semiconductor at room temperature, it will be able to support a current as free electrons move from a negative charge to a positive charge and as free holes move (in the direction of a conventional current flow) from a positive charge to a negative charge.

Reliance on relatively low concentrations of thermally generated "intrinsic" free electrons and free holes to carry the current will result in a highly resistive material that is not much use. However, we can artificially (extrinsically) change the free electron or free hole concentrations by adding suitable *dopant* impurities to the semiconductor. Adding trace quantities of *donor* atoms from group V of the periodic table such as phosphorous, to a group IV semiconductor such as silicon will effectively donate free electrons to the semiconductor. The phosphorous atom has five valence electrons and it will find itself occupying a site within the silicon lattice meant for a silicon atom with four valence electrons. The extra electron will be readily excited from the phosphorous impurity into the conduction band. The semiconductor now contains a large concentration of free electrons (n) in the conduction band and positively charged ionized impurities (N_d^+) that are fixed within the silicon lattice. The semiconductor will be a much better conductor because it will have a free electron concentration typically many orders of magnitude larger. It will be known as an *n-type* semiconductor and will have *n-type conductivity.*

We can make semiconductors *p-type* by adding *acceptor* impurities with too few valence electrons, boron in silicon for example. Acceptor impurities (N_a) accept electrons from the valance band and thereby provide large concentrations of positively charged free holes (p) and fixed negatively charged ionized acceptor impurities (N_a^-).

3.2.1.1 Photoconductivity

When a photon with energy greater than the bandgap energy of a semiconductor is absorbed by that semiconductor, an electron in the valence band can be excited from the valence band and into the conduction band and an *electron-hole pair* is formed. The extra free hole and extra free electron will momentarily increase the conductivity of the semiconductor as these extra carriers allow an increased current, but after a short time the electron will lose its energy by emitting a phonon or a lower energy photon and will *recombine* with a hole. Photoconductivity cannot be used to generate useful electricity, but it can be used to rather inefficiently detect light. To make even better detectors and to use light to generate electricity, we need to physically separate electrons in the conduction band from holes in the valence band before they have time to recombine; we can do this by forming a junction between n-type and p-type semiconductors.

3.2.1.2 The p-n Junction

To envisage the equilibrium established by a p-n junction, it is most convenient to imagine n- and p-type materials being magically adjoined. In practice this is not possible; the junction between the materials has to be continuous, with no break in the regular lattice of the semiconductor, and the only thing to change from one side of the junction to the other would be the majority impurity (donor or acceptor) that is present. In reality this can only be achieved by implanting or diffusing n- or p-type impurities into a p- or n-type semiconductor to form a *homojunction* or by growing an n- or p-type layer of semiconductor on top of a p- or n-type semiconductor to form a homojunction within a single semiconductor type or a *heterojunction* between two suitably matched semiconductors.

When our n- and p-type semiconductors are magically pushed together to form a junction, we immediately create a nonequilibrium set of conditions that cannot be maintained. The n-type region will contain a massive concentration of electrons that have thermal energy and are therefore able to move. Likewise, the p-type semiconductor contains energetic and mobile holes. The concentration gradient formed at the junction causes a net diffusion of electrons from the n-type material into the p-type material and a net diffusion of holes from the p-type material to the n-type material (see Figure 3.1). As electrons leave the n-type material, a region of ionized donor impurities is left behind, and as the holes leave the p-type material, a region of ionized acceptors is left behind. These ionized impurities form charged regions on either side of the junction that are depleted of free carriers. The p-type region will have a net negative charge and the n-type region will have a net positive charge. These charged *depletion regions* will lead to the formation of an electrostatic field

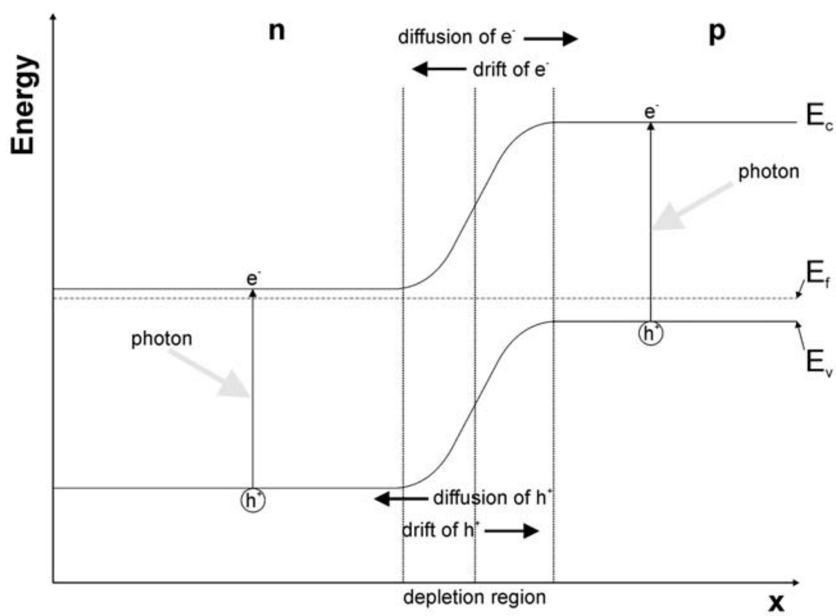

Figure 3.1 The band diagram of a p-n junction showing the directions of diffusion and the drift of electrons and holes and the generation of electron-hole pairs in both the n and p sides of the junction. E_V marks the tops of the valence band, E_C is the energy at the bottom of the conduction band, and E_f is the Fermi level.

and a built-in voltage across the junction. Eventually an equilibrium will be found in which the diffusion of electrons from n- to p-type and the diffusion of holes from p- to n-type are balanced by the drift of electrons from p- to n-type and the drift of holes from n- to p-type caused by the field across the junction. In this equilibrium the probability of finding an electron at any given energy in the system is constant (as indicated by the Fermi level, E_f) as a result of the evolution of a built-in potential difference between the n- and p-type materials.

If we allow light to fall on a p-n junction, we can use the built-in potential to separate the photo-generated electrons and holes and provide a net current flow and a useful voltage with which we can do work. To understand this, we must first consider what happens when a p-n junction is connected to a circuit.

3.3 Solar Cell Characteristics

If electrical contacts are added to the p-n junction and a voltage, V, is applied, the device exhibits rectifying behavior and the current, I_d, passing through the device can be described by the ideal diode equation:

$$I_d = I_0\left(e^{qV/kT} - 1\right) \tag{3.1}$$

where I_0 is the reverse saturation current, k is the Boltzmann constant, and T is the temperature. The form of this equation is plotted as the upper curve in Figure 3.2. Applying a forward bias (positive voltage) opposes the built-in field and so lowers the diffusion barrier, resulting in an exponential increase in the diffusion current with applied voltage. Conversely, applying a reverse bias (negative voltage) adds to the built-in field and so increases the diffusion barrier. The diffusion current is reduced exponentially with applied voltage, leaving only the drift current. Under dark conditions, the drift current is very small as it is limited by the diffusion of minority carriers, of which there are few, into the depletion region.

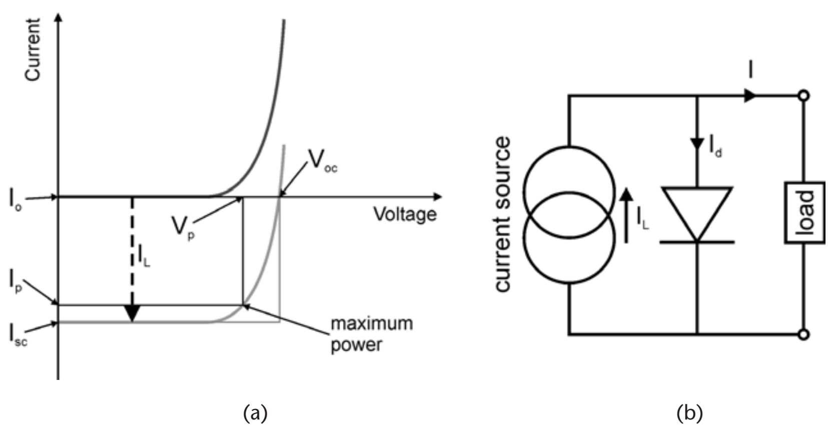

Figure 3.2 (a) Graph showing the I-V characteristics of an ideal solar cell in darkness (darker, upper curve) and under illumination (lighter, lower curve). The fill factor is the ratio of the areas of the rectangles. (b) Equivalent circuit for an ideal solar cell, consisting of a current source producing a current I_L and a diode through which a current I_d flows. The remaining current, I, flows through the load, driven by the voltage created by the separation of carriers at the junction.

A solar cell is effectively an unbiased diode that is exposed to light. Light is absorbed in the device, creating mobile electron-hole pairs. The minority carriers (i.e., holes created in n-type material and electrons created in p-type material) diffuse to the depletion region where they experience the built-in field which sweeps them to the opposite side of the junction. Under open circuit conditions, the separation of carriers leads to the buildup of a voltage across the junction, the open-circuit voltage (V_{oc}). If the n and p regions are connected by a resistance-free current path, a current will flow, the short-circuit current, I_{sc}, to balance the flow of minority carriers across the junction. If a load is added to the circuit, power can be extracted from the device.

In effect, the injection of minority carriers due to the absorption of photons adds to the drift current and this can be incorporated into the diode equation as an illumination current, I_L. A solar cell can then be represented by an equivalent circuit as a current generator and a diode [Figure 3.2(b)], with I-V characteristics described by

$$\begin{aligned} I &= I_d - I_L \\ &= I_0\left(e^{qV/kT} - 1\right) - I_L \end{aligned} \tag{3.2}$$

The illumination current shifts the I-V curve downwards, creating a region in the bottom right quadrant of the graph from which power can be obtained [see Figure 3.2(a)]. The IV graph is often flipped vertically so that the maximum power point is in the positive upper right quadrant. The V_{oc} and I_{sc} are the intercepts of the I-V curve with the voltage and current axes, respectively. The maximum power rectangle is defined by the voltage and current values (V_p and I_p) of the maximum power point (MPP), which is reached by optimizing the resistance of the load to draw the maximum power from the circuit. The ratio of the area of the maximum power rectangle to the area of the V_{oc}/I_{sc} rectangle is the fill factor (f), and the closer this value is to unity, the better the quality of the solar cell.

$$f = \frac{V_p I_p}{V_{oc} I_{sc}} \tag{3.3}$$

The energy conversion efficiency, η, is the ratio of the maximum electrical power obtained from the cell, P_p, to the incident light power, P_i, that is,

$$\eta = \frac{P_p}{P_i} = \frac{V_p I_p}{P_i} = \frac{f I_{sc} V_{oc}}{P_i} \tag{3.4}$$

3.4 Module Characteristics

A standard silicon solar cell will typically consist of a pn-junction formed in a wafer of silicon, with top and bottom contacts to allow power to be extracted from the device and a thin-film antireflective coating on the front surface to minimize the amount of light lost due to reflection [see Figure 3.3(a)]. The back contact is usually a continuous metal layer, while the front metal contacts are usually in the form of separated metal fingers to allow light into the cell. Many possible variations on this simple design have been developed to improve efficiency, including texturing the top or rear surfaces to reduce reflectance, and to improve light trapping [4–6] and locally contacting the rear of the cell to reduce back surface recombination [7].

When operating in the field, solar cells are encapsulated in the form of modules [Figure 3.3(b)]. This serves three main purposes:

- To protect the cells from mechanical damage;
- To prevent water corroding the metal contacts;
- To allow the connecting together of individual cells in a series, thereby increasing the voltage and power output to a useful level.

The number and size of solar cells to be incorporated into a module are determined by the requirements of the system needing power. For example, a module consisting of 36 crystalline silicon solar cells, each with an area of 100 cm^2 and each contributing around 0.6V, would give a maximum open circuit voltage of 21V and so should be sufficient to charge a 12-V battery when taking into account parasitic voltage drops, lower light intensities, and general suboptimum operation. Lower powered devices are less demanding, so the module can be much smaller and be made with cheaper and less efficient technologies (e.g., amorphous silicon cells for pocket calculators).

3.5 Irradiance Standards

3.5.1 Outdoor Operation

In photovoltaic device design, it is important to consider the spectral properties of the available light which for outdoor operation comes from the Sun. The center of the Sun is a fusion reactor that reaches temperatures of around 15,710,000K [8], but the light we see from the Sun comes from the Sun's surface, or photosphere, which can be modeled as a black body emitter at a temperature of 5,760K [9]. In its journey through space from the surface of the Sun to Earth, the spectral distribution

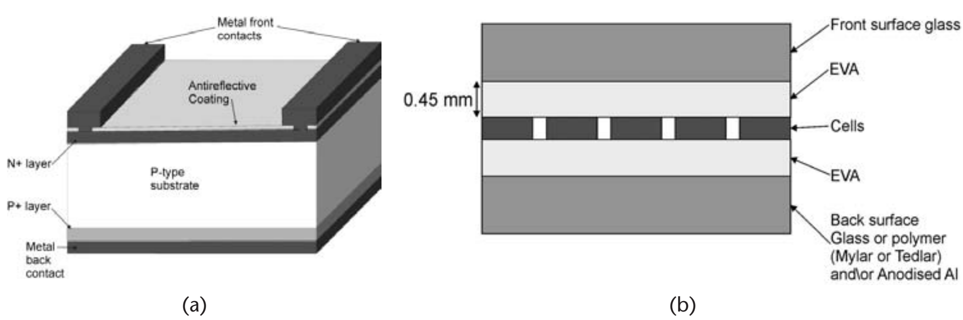

Figure 3.3 (a) Typical design of a single crystal silicon solar cell; and (b) arrangement of cells encapsulated to form a module.

of light remains unchanged, but the power density reduces as the square of the distance. Solar radiation that impinges on the outer layer of the Earth has a power density of approximately 1.37 kW/m [10]. This is called the solar constant, although the actual value varies during the year with solar activity and the variation in the Earth-Sun distance during orbit [10, 11]. However, these variations are small compared to the variations in the solar spectrum received on the Earth's surface. As sunlight passes through the Earth's atmosphere, absorption and scattering significantly alter its spectral characteristics. These alterations depend heavily on the quantity and nature of atmosphere through which the light has to pass. These factors vary considerably with latitude, season, time of day, and local weather conditions. To allow direct comparisons of solar cell performance, various standard solar spectra have been defined. These are named according to the principle of air mass that describes the path length of light through the atmosphere relative to the shortest possible path length, which is when the Sun is directly overhead (air mass, AM = 1). As the angle of the Sun from directly overhead, θ, increases, the AM value is calculated by (see Figure 3.4):

$$AM = \frac{1}{\cos\theta} \tag{3.5}$$

For devices operating outside the Earth's atmosphere (e.g., satellites), there is no atmospheric absorption to take into account and the standard spectrum AM0 is used with an integrated spectral irradiance of 1.3661 kW/m^2 (ASTM E-490-00 [12]). For terrestrial solar cells, the standard spectrum AM1.5 Global, with an integrated spectral irradiance of 1 kW/m^2, is normally used (ASTM G-173-03 [13]). This includes the direct and diffuse components of light for a surface on Earth tilted by 37° towards the equator (chosen as these parameters represent averages in the 48 contiguous states of the United States). There is a second terrestrial standard spectrum called AM1.5 Direct (+circumsolar), which is often used for concentrator systems. This has an integrated spectral irradiance of 0.9 kW/m^2 and does not include diffuse light caused by scattering in the atmosphere. These three standards are plotted in Figure 3.5.

It is important to note that the actual spectral irradiance experienced by a solar cell can be very different to the standards. The latitudinal position of the cell on Earth will determine the amount of atmosphere through which light has to pass before it reaches the cell. The air mass will also change throughout the day as the

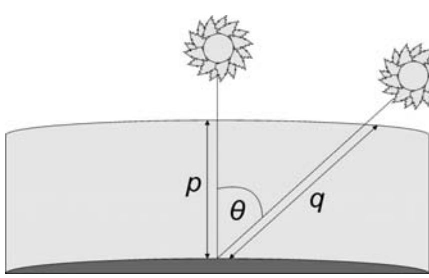

Figure 3.4 Derivation of air mass (AM) showing that as the Sun's position varies from directly overhead, the amount of atmosphere through which sunlight must travel increases.

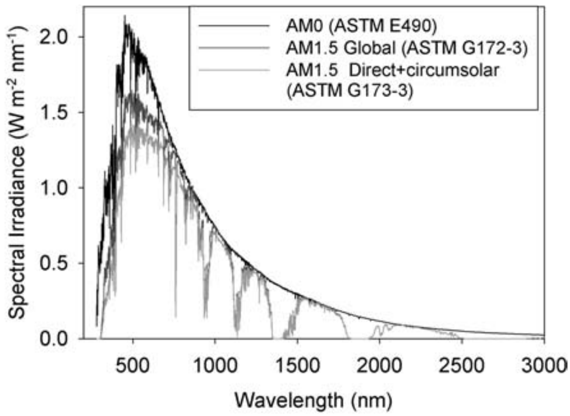

Figure 3.5 Plots of standard solar irradiance spectra from ASTM [12, 13].

Sun precesses across the sky: In the early morning and the late evening, the Sun is lower in the sky, so sunlight has to pass through more atmosphere (the AM is greater). This means that more light is scattered and absorbed before reaching the cell, an effect that more strongly affects the lower wavelengths. Therefore, the light incident on the cell is of a lower intensity and is spectrally shifted towards longer wavelengths (which is also why the sky appears redder at these times of day). The extent and characteristics of cloud cover also have a strong effect on the incident light. Clouds scatter light very effectively and can block most of the direct light, causing only diffuse light to reach the cell. Hence, on cloudy days, the light intensity on a cell surface can be reduced to approximately 10% of that on a sunny day [14].

Models are available to simulate nonstandard solar spectra data for a cell placed at a particular longitude, latitude, slope, date, and time and with specific atmospheric conditions [15, 16]. Figure 3.6 shows an example of spectral data acquired using the SPCTRL2 model [15] from dawn to midday for a horizontal surface at the equator. These models, together with historical typical meteorological year (TMY) [17] data, may be used to predict the performance of solar cells placed in specific locations and predict whether the output from the cell is sufficient for the application in mind. If a solar cell is to be located in a position whereby the

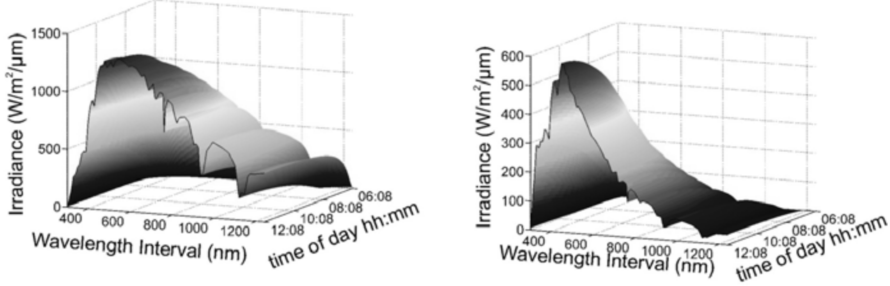

Figure 3.6 An example of the spectral variation solar irradiance over half a day for a horizontal cell placed on a horizontal surface at the equator on March 20, compiled from SPCTRL2 [15]: (a) direct irradiance and (b) diffuse irradiance.

incident spectrum is likely to differ considerably from the AM1.5 standard, this consideration can be incorporated into the choice of device technology and design of the cell (e.g., tuning the antireflective coating for spectral characteristics of the light that the cell is likely to experience [18]).

3.5.2 Indoor Operation

Many low-power PV powered devices are required to operate indoors where the spectral characteristics of the available light can be considerably different to the standard spectra described in the previous section. In general, the intensity of available light is much lower indoors and it can originate from the Sun (through windows) or from an artificial light source such as an incandescent lightbulb, fluorescent tube, or halogen lamp. Measured and simulated spectra for indoor light sources, assuming an intensity of 500 lux, are plotted in Figure 3.7 [19].

These spectra can be combined with quantum efficiency data for different solar cell technologies in order to select the optimum technology for a particular light source. For example, an amorphous silicon absorbs more in the visible wavelength region and so would be more suitable for operation under a fluorescent or halogen lamp with a cold filter, for which most of the emission is in the visible. Likewise, a cell technology that absorbs more in the infrared, for example, crystalline silicon, would be more suitable for a device that uses an incandescent bulb as its light source. It is also possible to tune the device characteristics, such as the antireflective coating thickness, to optimize performance under the light source that will be available to the device.

3.6 Efficiency Losses

The maximum theoretically possible efficiency of a solar cell can be estimated from a thermodynamic viewpoint, by considering the conversion of heat from the Sun, modeled as a black body at a temperature of ~6,000K, to the cell at 300K. This results in an efficiency of 95% (the Carnot efficiency) or, by a more detailed analysis, 93.3% (the Landsberg efficiency) [20]. Most commercial single-junction solar cells,

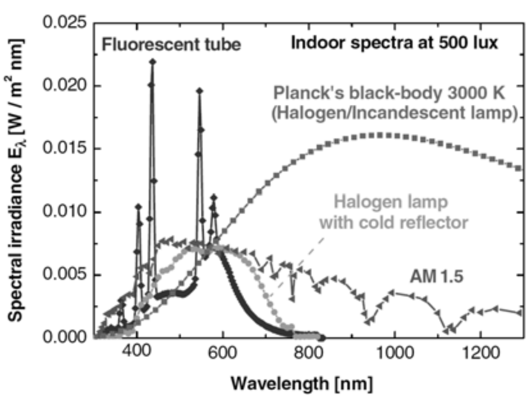

Figure 3.7 Measured or simulated spectral distributions of indoor light sources. (*From:* [19]. Reprinted with permission.)

however, have efficiencies in the range of 5% to 22%, that is, only 5% to 22% of the optical power incident on the device is converted into electrical power. These values are somewhat disappointing to those who hear this for the first time, but most people who work in photovoltaics understand that efficiencies of around 20% are remarkably high considering the fundamental and technological challenges that have to be faced. The losses of power for a high-efficiency silicon solar cell are summarized in Figure 3.8.

3.6.1 Intrinsic Losses

The solar spectrum spans from ultraviolet light with photon energies of around 4 eV to infrared photons with photon energies of less than 0.4 eV. The peak intensity,

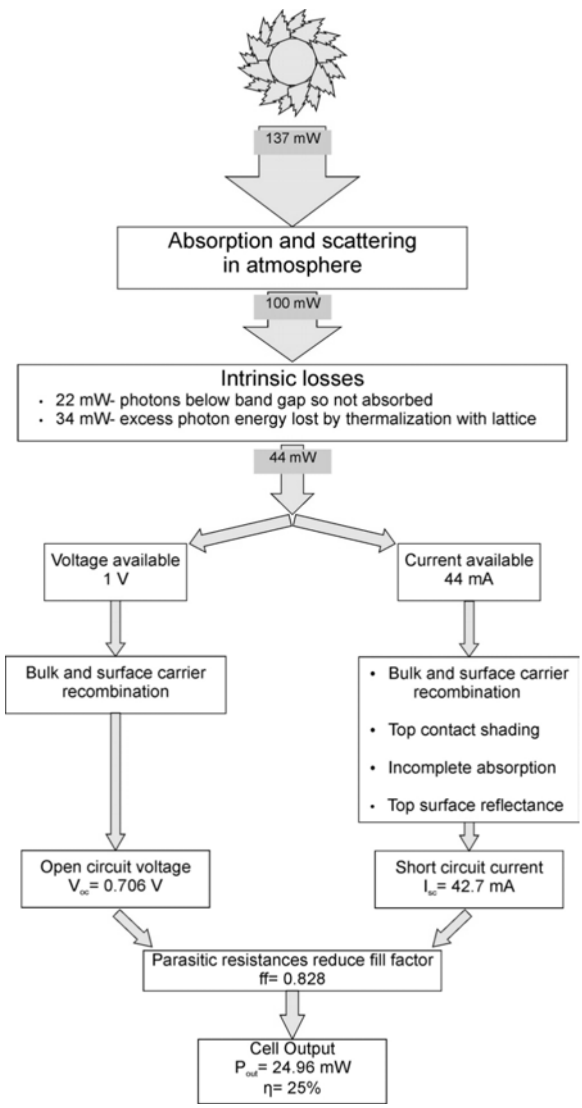

Figure 3.8 Summary of losses in a solar cell. Values are for high-efficiency crystalline silicon laboratory cells. Further losses result from encapsulation in a module structure. (Values for V_{oc}, I_{sc}, and *ff* are taken from [21].)

in terms of energy and photon number, lies in the visible range; this is why we evolved sight in this range. If we choose silicon with a bandgap of 1.12 eV, it is clear that all photons with energy less than 1.12 eV do not have sufficient energy to create an electron-hole pair; meanwhile, although any photon with energy greater than 1.12 eV has the ability to generate an electron-hole pair, all of those photons can only generate a single electron-hole pair. The extra energy is lost to the lattice through thermalization so that the useful energy contributed by each photon is only equal to the bandgap (see Figure 3.9).

Shockley and Queisser showed that this mismatch between the solar spectrum experienced on Earth and a single bandgap solar cell limits the efficiency to a 44% [22].

An additional intrinsic loss is that of *radiative recombination*, which is the relaxation of an electron back into the valence band, with the excess energy released as a photon. This radiative current subtracts from the current available for useful work. When this is accounted for, together with a consideration of the size of the solar disk as experienced from Earth, the efficiency limit for an ideal solar cell at 1 sun intensity (i.e., no concentration) is calculated to be approximately 31% [22, 23].

3.6.2 Extrinsic Losses

In practice, conventional single bandgap solar cells suffer from other losses that prevent them from even reaching the Shockley-Queisser efficiency limit of 31%. These extrinsic losses fall into two categories: electrical losses and optical losses.

In addition, the connection of an array of cells in a series and the encapsulation of this array into a module introduce further losses. These loss mechanisms can be minimized by careful design, but often the mitigation of one source of loss will inevitably increase another, and ultimately a compromise has to be found.

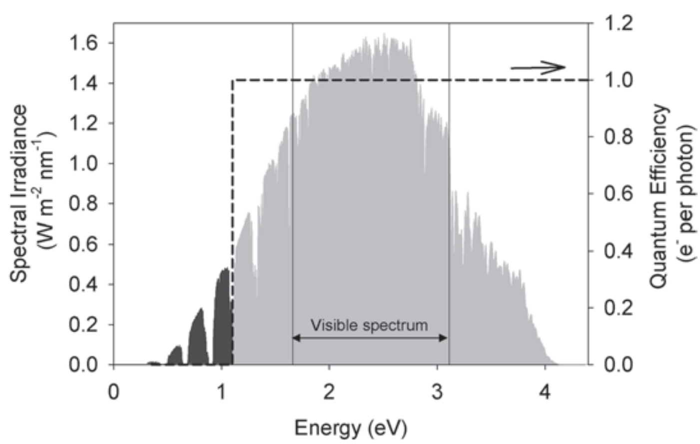

Figure 3.9 Illustration of intrinsic reductions in efficiency due to the system having a bandgap. The AM1.5 solar spectrum in terms of photon energy is shown. The dark gray shaded area represents photons that cannot contribute to current as they have energies less than the bandgap of 1.1 eV. The light gray shaded area represents photons that have a sufficiently high energy to excite a valence electron into the conduction band but any excess energy lost due to thermalization. The number of electrons generated per photon, also known as the quantum efficiency, is represented by the dashed line.

3.6.2.1 Electrical Losses

Bulk Recombination As we have already discussed, minority electrons or holes generated within the active thickness of a solar cell (i.e., within an average diffusion length of the depletion region) will be swept across the junction. Any absorption that takes place within the semiconductor but outside the active region will generate carriers that are not collected, and this will reduce the quantum efficiency and the current produced by the device. The average carrier lifetimes and diffusion constants and therefore average diffusion lengths for electrons in the p-type material and for holes in the n-type material must be as long as possible. Short carrier lifetimes reduce the average diffusion lengths (and the active thickness) and increase recombination in the device. As a consequence of all this, the design of the device must take into account the likely range of carrier diffusion lengths; these will be a strong function of the semiconductor growth or deposition conditions. High-quality materials with long diffusion lengths are expensive to produce and control accurately. The total thickness of a solar cell absorption region should not greatly exceed the active thickness; successful device technologies can absorb sufficient photons within this thickness (either with or without light trapping). Perhaps the biggest trade-off in a solar cell is this decision over thickness. Carrier collection is best for thin devices, photon collection is best for thick devices, and, to complicate this matter, average diffusion lengths are hard to control or predict.

Recombination is often caused by defects within the semiconductor. These defects include: structural defects, where there is imperfect alignment of the atomic lattice; intrinsic defects, where atoms sit on the wrong lattice sites or where atoms are absent from a lattice site; and extrinsic defects, which are caused by the presence of impurity atoms that replace native atoms or sit on interstitial lattice sites. In order to minimize defects, the source materials that are used must be of very high purity, particularly with regard to the worst extrinsic impurities (no gold jewelry in a silicon solar cell manufacturing plant). Impurities can also emanate from manufacturing tools, particularly those that require high temperatures and the materials on which (or in which) the semiconductor is grown. A glass substrate, for instance, is a massive source of extrinsic impurities if not properly specified or treated and often barrier layers are required between the glass and the transparent conducting oxide (TCO)/semiconductor.

Equally, metal contacting to the solar cell can act as a source of carrier loss, as with shadowing (see Section 3.6.2.2); small contact areas help as long as resistive losses do not increase significantly.

If the semiconductor bulk comprises many small crystallites rather than one single crystal, it is described as *multicrystalline* or *microcrystalline*. The *grain boundaries* between individual crystallites in these materials represent another significant source of recombination, and particular attention has to be paid to the reduced diffusion lengths. Growth techniques that encourage large crystallites that extend all the way from the bottom to the top of the solar cell and thereby provide a path for electrons and holes that is free of grain boundaries are beneficial. Similarly, schemes based on the use of extrinsic impurities such as hydrogen that passivate grain boundaries can also improve matters.

Surface Recombination Recombination at the surface is of particular concern to solar cell designers. The presence of a surface causes a departure from the symmetry of the crystalline lattice, leading to nonterminated "dangling" bonds. These form defect states in the bandgap that act as trapping centers and so aid the recombination of electrons and holes [24]. Carriers generated by ultraviolet (and other) photons absorbed at the very surface of a solar cell are often lost to surface recombination. Carriers that recombine before they are collected do not contribute to the current produced by the cell and so minimizing the surface recombination is essential in the solar cell design. It can be shown, using the Shockley-Read-Hall theory [25], that the surface recombination rate is proportional to the defect density at the surface and the concentration of free carriers at the surface.

Therefore, efforts to reduce surface recombination tackle one or both of these factors. For example, the addition of a thin layer of a passivating material such as SiO_2 will act to saturate the dangling bonds and reduce the surface defect density. Positive charges in the oxide layer will also repel positive carriers, decreasing the concentration of holes at the surface in a process known as *field-effect passivation* [26]. For the back surface of a cell, an anneal is often performed to allow the diffusion of aluminum from the back contact into the silicon, creating a doping profile and an electric field (called a back-surface field), which forces electrons away from the surface and hence reduces surface recombination [27].

Nonideal Diode Behavior A good solar cell requires a good dark current/voltage characteristic as close to an ideal diode as possible. Nonuniform acceptor and donor impurity profiles, thickness variations, and lifetime variations across the area of a device will cause deviations from the ideal and the saturation current and ideality factor that characterize the diode increase. Large saturation currents and ideality factors lead to a poor fill factor and a lower efficiency.

Series Resistance The series resistance, R_s, should be as small as possible, as this results in a reduction in the short-circuit current and the fill factor of a cell [Figure 3.10(a)]. The series resistance arises due to the resistance of the bulk semiconductor material, the contact resistance at the semiconductor-metal interfaces, and the resistance of the metal contacts. The metal interconnects and bus-bars will present a resistance that increases with temperature and is a source of voltage loss. Losses will be large if the interconnects are made too thin and there has to be a trade-off with

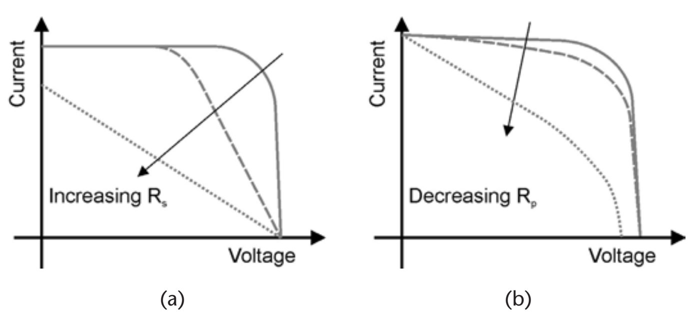

Figure 3.10 (a) Effect of parasitic resistance on the IV characteristics of a typical solar cell: (a) series resistance (R_s) and (b) shunt resistance (R_p).

shadowing. In addition, any TCO between the metal and absorber or on the surface of the absorber also presents a volume through which electrons or holes must travel and has an associated series resistance.

The actual metal-semiconductor contacts can occupy a smaller area than interconnects and bus-bars (though they often are the same area). The contacts must be ohmic or else they will form Schottky junctions that may lead to carrier, current, or voltage losses. If an ohmic contact is formed, the resistance that it presents is another part of the total series resistance of the device.

Shunt Resistance Any short-circuit paths for carriers between the n- and p-type regions or contacts represent a form of loss quantified by the parallel or shunt resistance, R_p, of a device. A low shunt resistance causes a decrease in the open-circuit voltage and fill factor [Figure 3.10(b)], and thus the shunt resistance should be as high as possible for good solar cell performance. Typical shunt resistances are caused by problems at the edges of devices, pin holes that break through thin film pn-junctions, and conductive paths that can be formed through grain boundaries. Suitable fabrication techniques and procedures can ensure that shunt resistances are suitably high, though this often requires additional processing steps such as edge deletion.

Equivalent Circuit Treatment of Electrical Losses The change in the properties of a solar cell due to electrical losses can be incorporated into an equivalent circuit using a two-diode model and adding series and shunt resistors (see Figure 3.11). Using two diodes allows us to separate the contributions from recombination in the neutral regions (proportional to $e^{qV/kT}$) and recombination in the depletion region (proportional to $e^{qV/2kT}$) [28]. The cell equation (3.2) is modified to:

$$\begin{aligned} I &= I_{d1} + I_{d2} + I_r - I_L \\ &= I_{01}\left(e^{\frac{q(V+IR_S)}{kT}} - 1\right) + I_{02}\left(e^{\frac{q(V+IR_S)}{2kT}} - 1\right) + \frac{V+IR_S}{R_p} - I_L \end{aligned} \tag{3.7}$$

Alternatively, a single diode can be used with a nonideality factor, n, of between 1 and 2 in the denominator of the exponential term:

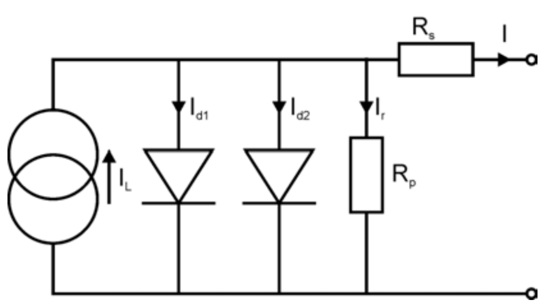

Figure 3.11 Equivalent circuit for a nonideal solar cell. (*After:* [29].)

$$I = I_0\left(e^{\frac{q(V+IR_S)}{nkT}} - 1\right) + \frac{V + IR_S}{R_p} - I_L \tag{3.8}$$

3.6.2.2 Optical Losses

Solar cell design has to take account of every photon capable of generating an electron-hole pair. Any process that leads to photon loss will lead to a decrease in the current that can be generated. There are several sources of optical loss in a solar cell.

Shading by Top Contacts A solar cell must have electrical contacts and often conductive bus-bars on both the top and bottom. Ultimately, these contacts need to be formed by metals, but any metal will reflect light and shadow the underlying device. Using TCOs or thin metals in an effort to reduce shading comes at the expense of an increase in series resistance. Contacts and bus-bars are often arranged as interdigitated fingers with a carefully designed thickness to optimize the trade-off between resistance and shadowing [30]. The laser-grooved, buried contact cell design was developed to allow a large contact area (and small contact resistance) together with a small shadowing effect by forming the contacts in the trenches carved into the surface of the cell [31]. In another example, the interdigitated back contact solar cell design eliminates the need for top contacts all together by forming both contacts to both the n and p regions at the rear of the cell structure [32, 33].

Incomplete Absorption Although high-energy photons will be absorbed within the first few tens of nanometers of a semiconductor, near-bandgap photons are rather more difficult to absorb because there are fewer energy states available at the top of the valence band and the bottom of the conduction band. This is especially important for indirect bandgap materials such as silicon because these tend to be poor absorbers of light due to the requirement of a change in momentum as well as energy in the absorption process [34]. Consequently, the amount of light absorbed depends heavily on the amount of material through which the light is forced to pass. A 1-cm-thick piece of silicon will absorb 99% of all light of wavelength 1.1 μm. Reduce the silicon to a thickness of 35 μm, and only 2% of the light is absorbed [35]. Materials costs associated with thick cells are higher both because of the greater amount of material required and because the quality of the material must be higher as much of the light will be absorbed far away from the junction and so efficiency would be limited by the carrier recombination if the carrier lifetimes are insufficient. Moving to cheaper, thin-film designs necessitates some form of a light-trapping scheme to increase the effective optical path length of light through the cell.

The simplest form of a light-trapping scheme is a back-reflector at the rear of the device, which increases the optical path length by 2 [Figure 3.12(a)]. If light can be scattered laterally at either the front surface or rear surface, then the effective optical path length can be further enhanced via two mechanisms:

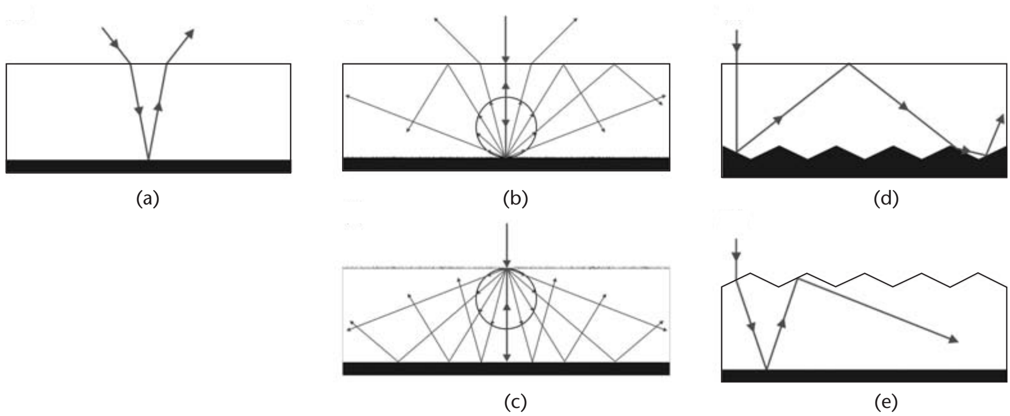

Figure 3.12 Possible light-trapping mechanisms: (a) planar reflector on the back of device; (b, c) Lambertian diffuse interface on the front or back of the device; (d, e) texturing with geometric features on the front or back of the device.

1. Light deflected away from the normal to the cell may undergo a total internal reflection and so will be forced to pass through the cell multiple times.
2. The distance across a cell for light traveling at a more oblique angle will be greater than that of light traveling normal to the cell.

An ideal randomizing surface, called a Lambertian surface, scatters light according to the Lambertian cosine rule [36]:

$$I(\theta_S) = I_0 \cos\theta_S \tag{3.9}$$

where θ_s is the angle between the viewer's line of sight and the surface normal and I_0 is the incident light intensity.

Figure 3.12(b, c) shows two ways to use a Lambertian diffuse surface for light trapping. Light will enter the device from the top and be transmitted diffusely through the front surface or reflected diffusely from the back surface. A proportion of this light will now be traveling at a sufficiently oblique angle to undergo a total internal reflection at the top surface. Goetzberger has shown that the fraction of light that undergoes a total internal reflection at the front surface in the setup shown in Figure 3.12(c) is $1 - 1/n^2$, where n is the refractive index of the cell material [37]. The scattering of light by a Lambertian reflector in all directions also leads to the average path length of light per pass through the cell being twice the width of the cell. The combination of these two effects allows a Lambertian reflector light-trapping scheme to achieve a path length enhancement factor of $4n^2$, which is about 50 for silicon [37–39]. Furthermore, Yablonovitch claimed that this is not limited to ideal Lambertian surfaces, but that any combination of illumination conditions and surface texturing, including regular geometric texturing [Figure 3.12(d, e)], that is sufficient to lead to the angular randomization of light within the cell would lead to values of path length enhancement factors approaching 50 [38]. This has led to the development of many techniques to practically realize near-Lambertian and geometric texturing on silicon. Etching can be used to create arrays of geometric features (pyramids and grooves) [4, 40, 41] or other scattering textures [5, 7, 42, 43] on the top surface of cells. In amorphous silicon technologies, the glass or TCO substrate materials are often produced with dramatic textures to scatter the light and enhance confinement in the active layers that are grown on top [44, 45].

Recently, there has been considerable interest in light-trapping schemes that exploit the strong interaction with light that metal nanoparticles exhibit due to the excitation of localized surface plasmons [46–48]. Arrays of such particles can scatter light over a range of wavelengths, and with careful choice of particle material, size, and distribution to avoid excessive absorption, this technology could enhance the confinement of light in thin film solar cells, leading to higher device efficiencies.

These light-trapping schemes are essential to some types of devices, but are often costly to other aspects of device performance, if not to the manufacturing process.

Top Surface Reflectance The semiconductor surface itself will reflect a component of light depending on the refractive index of the semiconductor, the angle of incidence, and the wavelength. It can be shown using the Fresnel equations that, for normal incidence, the reflectance, R, from an interface between two materials with refractive indices of n_1 and n_2, respectively, is given by [21]:

$$R = \left| \frac{n_2 - n_1}{n_2 + n_1} \right|^2 \tag{3.10}$$

For absorbing materials, the reflectance is given by replacing the n terms in (3.10) by a complex refractive index, $\breve{n}$, where

$$\breve{n} = n - ik \tag{3.11}$$

The imaginary component, k, describes the absorption in the material and is often referred to as the *extinction coefficient*. The reflectance is then given by:

$$R = \left| \frac{\breve{n}_2 - \breve{n}_1}{\breve{n}_2 + \breve{n}_1} \right|^2 = \frac{\left(n_2 - n_1\right)^2 + \left(k_2 + k_1\right)^2}{\left(n_2 + n_1\right)^2 + \left(k_2 + k_1\right)^2} \tag{3.12}$$

Equation (3.12) shows that the larger the difference between the refractive indices of the two materials, the greater the reflectance at the interface between them. Taking silicon as an example, the real part of the refractive index of this material, n, ranges from 3.5 to 6.9 over wavelengths in the range of 300–1,200 nm. This leads to a normal incidence reflectance for an air-silicon interface, which is important for laboratory cells, between 31% and 61% and for an EVA-silicon interface, found in encapsulated cells, between 16% and 46%. The reduction in the short-circuit current due to reflectance losses if no antireflection (AR) scheme is employed is approximately 36.2% for a laboratory cell and 19.5% for an encapsulated cell [49]. These high values highlight the need for effective methods of reducing reflectance losses, a need that has led to the development of AR techniques falling into three main categories:

- *Thin film coatings:* Destructive interference between light reflected from the interfaces created by adding one or more thin films to a surface minimizes reflectance at certain wavelengths (Figure 3.13). The refractive index and thicknesses of the layers must be carefully chosen for optimum reflection reduction over the required wavelength range. Common materials for single-layer thin-film ARCs on silicon include SiO_2 [50], TiO_2 [51, 52], and SiN_x [53, 54]. More layers can be added to broaden the wavelength range over which the coating is effective. One example of this is a double layer of MgF_2/ZnS coating used on the highest efficiency crystalline silicon solar cells [7]; however, this approach is ultimately limited by thermal and mechanical properties of available materials and the extra costs involved. For solar cells, the reflectance weighted with the solar spectrum is often used as a figure of merit for the optimization of thin film AR coatings [55, 56]. Single-layer coatings can lead to predicted encapsulated cell performances only 4.6% lower than that of a cell with no surface reflectance (i.e., an ideal antireflective surface). This can be further improved to only 3.5% of the ideal using a double-layer coating [55]. The application of a thin-film coating can also help with passivating the surface and so reducing the surface recombination (see Section 3.6.2.1).
- *Micron-scale texturing*: The techniques described in Section 3.6.2.2 for light trapping also confer an antireflective effect. Texturing with features of dimensions above the wavelength of the incident light reduces overall reflectance by forcing the light to undergo multiple reflections from the inclined walls of the features, with a portion of this light being coupled into the substrate at each reflection (Figure 3.14). Micron-scale antireflective surface textures are often coated with a thin-film antireflective coating to achieve a further reflectance reduction and to act as a passivating layer.
- *Subwavelength-scale texturing:* Incident light cannot resolve individual texture features of dimensions less than the wavelength divided by the refractive index of the textured material and so instead sees a weighted spatial average of the surface's optical properties. Consequently, a grading in the width of the features in the direction normal to the surface leads to a grading in the refractive index, effectively blurring out the interface and resulting in a low

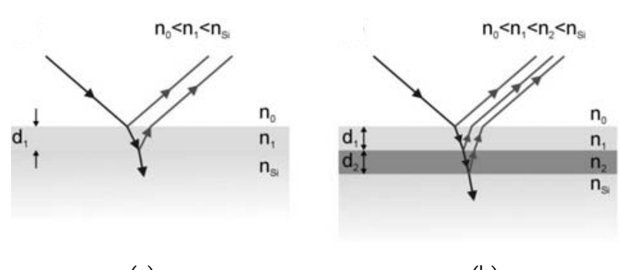

Figure 3.13 Destructive interference between light reflected from the interfaces of (a) a single-layer or (b) a double-layer antireflective coating can reduce the amount of light reflected from a surface.

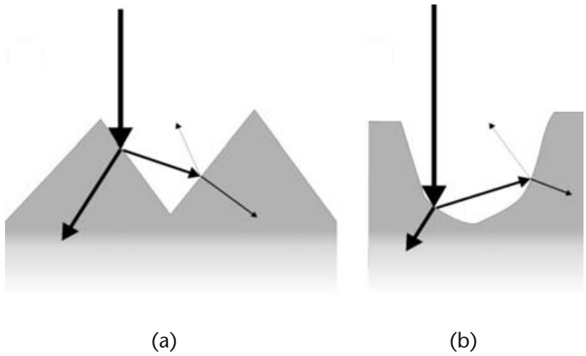

Figure 3.14 Light reflected from one part of a textured surface is directed to a different part of the surface and so is incident more than once onto the surface of the solar cell, resulting in more light being coupled into the cell: (a) for pyramid-type texturing schemes and (b) for well-type texturing schemes.

reflectance for a broad range of wavelengths. Random arrays of sufficiently small features also produce this effect and it is the mechanism behind the very low reflectance achieved by nature as a form of camouflage on the corneal surfaces (and wings) of some species of moth [57, 58] [Figure 3.15(b)]. Researchers have textured quartz [59, 60], polymer [61], GaSb [62], and silicon [63–66] surfaces with subwavelength features [for example, see Figure 3.15(b)] and demonstrated very low reflectance across a broad range of wavelengths and angles of incidence. Fabrication techniques include dry etching through an electron beam patterned mask [64] or an anodic porous alumina mask [67, 68]. The latter was used in one example by Sai et al. to provide subwavelength texturing in a crystalline silicon solar cell, achieving a 38% increase in cell efficiency compared to an untextured cell [68]. Catalytically enhanced wet etching techniques can also be used to form such textures [69, 70] leading to very low surface reflectance and higher cell efficiency. In one example, Koynov et al. used this technique during the fabrication of multicrystalline silicon solar cells and reported a 36%–42% increase in photocurrent compared to untextured cells [71]. The resulting reflectance spectrum is dependent on the feature spacing, shape, and height of the subwavelength-scale textured surface, and optimization of the design is important to achieve

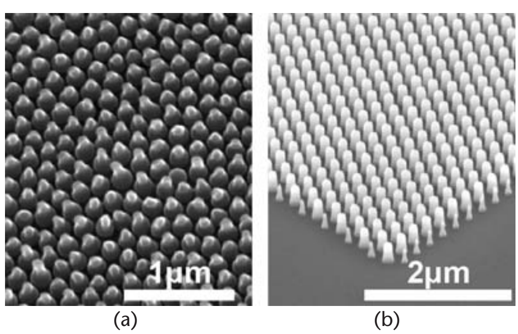

Figure 3.15 SEM images of (a) antireflective subwavelength features found on the transparent section of the wing of the *Cryptotympana Aquila* and (b) a biomimetic silicon moth-eye surface. (*From:* [66]. Reprinted with permission.)

good performance. In one such optimization, it was shown that using this type of texturing can lead to predicted encapsulated cell performances only 0.6% lower than that of an encapsulated cell with no surface reflectance [18]. This represents the most effective technique for achieving antireflection for photovoltaic devices, but the costs involved in texturing down on this length scale often outweigh the benefits of lower reflectance.

3.6.3 Module Losses

Further losses result when a cell is encapsulated into a module. Reflectance and absorption losses in the encapsulant materials (e.g., glass and EVA) are low because they tend to have low refractive indices and are chosen for their transparency. Antireflective coatings and texturing can help to reduce encapsulant interface reflectance too, but it is important to consider the optical properties of the media either side of each interface when designing AR schemes.

If there is any surface region of a photovoltaic module or a solar cell where there is no semiconductor or appropriate mirror, then immediately the headline efficiency of that device is reduced. The arrangement of single crystal silicon solar cells that are generally made from circular wafers presents an immediate problem for module and submodule manufacturers. Cleaving or sawing to make square or hexagonal wafers is both expensive and wasteful, and therefore some loss of efficiency is inevitable.

Connecting together individual cells to form a module inevitably introduces losses because the series configuration required to increase the voltage causes the current to be limited by the worst performing cell in the array. Careful placement of the module is required to ensure that all cells receive the same level of irradiance. Shading of one cell by, for example, an overhanging tree will reduce the current generated by this cell and so limit the current available from the module. In extreme cases, this can cause power dissipation in the poorly performing cell, which can lead to overheating and cracking of the module. The use of bias diodes in the circuit design can prevent this, but the more complicated circuitry requirements lead to an increase in the cost of the module.

Encapsulation also effects heat dissipation, raising the operating temperature of the cells and reducing the open-circuit voltage and fill factor. Severe overheating can cause thermal expansion–related cracking, so careful design of the module to ensure that the heat generated by the cells can be effectively dissipated and thermal expansion can be tolerated is essential.

3.7 Device Technologies

There is a large diversity of photovoltaic device types in commercial production and even more devices being developed in research labs. The diversity of devices reflects the diversity of applications including satellite arrays (where expense is no object but weight and lifetime really matter), residential and commercial installations (where cost needs to be as low as possible and efficiency needs to be as high as possible), and semidisposable consumer toys (where only cost matters).

The most important commercial devices are the module technologies that might be around 1 m^2 in area and produce 60–200W at peak power. These modules are designed to provide power to homes and businesses via rooftop installation or be arrayed within solar energy farms. Often solar energy farms will employ mechanical tracking systems so that the full area of PV modules is presented to the Sun as it moves through the sky.

There are also markets for expensive high-efficiency devices that can be used within concentrator systems that use mirrors or lenses to focus sunlight onto photovoltaic devices. These devices are designed to cope with high temperature and generally have to be installed in systems that track the Sun.

A large number of small area devices are on the market that are designed to power a myriad of low-power consumer products including pocket calculators, radios, garden lamps, USB charging devices, and many other low-power applications. These devices include everything from versatile low-specification module devices that can be bought from electronics component suppliers to highly bespoke designs that integrate into the faces of wristwatches. As the availability of high-efficiency and low-cost photovoltaics begins to increase, we can expect solar PV to be increasingly used to provide power in more consumer applications.

Device technologies are often divided into three major categories [72]. First generation devices are the most established and are high-efficiency, expensive devices based on silicon wafers. Second generation devices are devices constructed from semiconductor thin films deposited onto glass, metal, or plastic substrates. In principle, though not always in practice, these devices should be much less expensive than first generation devices, but will most likely have lower efficiencies. Third generation devices are devices that have efficiencies greater than the single-junction Shockley-Queisser limit. Although several schemes for such devices have been suggested [73], only multijunction devices have as yet been shown to work. Multijunction devices have thin top cells that absorb the short-wavelength photons and thicker bottom cells that absorb the long-wavelength infrared light. Any number of cells can be inserted in between, but the current from each cell must be matched to extract the best efficiency out of the devices. These multijunction designs are used in tracking concentrator systems, as this offsets the cost of the cells.

3.7.1 Silicon Wafers

The predominant commercial photovoltaic technologies are those based on single crystal silicon (C-Si) and multicrystalline silicon (mC-Si) wafers. The cost of silicon wafer preparation accounts for roughly half of the total cost of a C-Si-based PV module. Since the success of PV is critically dependent on module cost and PV cost is typically two to three times too expensive, it is clear that wafer cost cannot be ignored.

Single crystal silicon is typically produced by the same processes used to produce wafers for the microelectronics industry. The Czochralski method involves slowly drawing an oriented seed from molten silicon in a pure quartz crucible [74]. The float zone method involves passing a molten zone of silicon along a silicon rod to produce a purified single crystal ingot [75].

Multicrystalline silicon ingots can be produced in large furnaces that allow molten silicon to crystallize from the bottom of huge crucibles [76]. Furnaces and processes are carefully designed to produce large columns of silicon and good uniformity within ingots, providing a cheaper alternative to single crystal that can produce useful but less efficient solar cells.

Once single crystal or multicrystalline ingots are produced, they are sliced into wafers by wire-sawing, and then a number of shaping, cleaning, polishing, and gettering stages are required. All processes required to produce these wafers are expensive and are continuously evolving to reduce energy requirements, processing time, and waste silicon. Other techniques are emerging based on casting and ribbon growth, and over time we can expect significant progress in these areas and commensurate price reductions as a result of technological improvements. In the meantime, most wafer-based technologies will rely upon industrial scaling to bring in cost reductions.

3.7.2 Single Crystal and Multicrystalline Devices

Commercial C-Si wafer module efficiencies have reached 22.9% [77], but are more typically in the range of 14%–17%. Meanwhile, mC-Si modules have reached up to 15.5% [78], but are more typically sold as modules in the 10%–14% range.

There is a large diversity of C-Si and mC-Si device types; Figure 3.16 illustrates a typical commercial device. Nearly all commercial C-Si and mC-Si devices have a shallow n-type region formed by the diffusion of phosphorous from $POCl_3$ into a lightly doped p-type wafer to form the junction. The polished front surface of a C-Si solar cell is typically textured by a KOH etch to produce a randomized array of micron-scale inverted pyramids [40]. This provides a cost-effective antireflection scheme. The textured surface is then covered by a single layer of a dielectric material, typically silicon oxide or silicon nitride deposited at low temperature by Plasma-Enhanced Chemical Vapor Deposition (PECVD) [53]. This layer provides an antireflective coating and a passivation layer that reduces surface recombination. The polycrystalline form of the mC-Si wafers will not facilitate the formation of the inverted pyramid structure, so mC-Si devices rely solely upon 1/4 wavelength AR coatings. A large number of contacting materials and deposition schemes are employed. On the front surface a finger grid of titanium (protected by nickel or palladium) is deposited by evaporation through a shadow mask or by screen printing to form the top electrical contacts. Typically aluminum is screen-printed or evaporated onto the entire rear surface of the wafer to form the back electrical contact.

The devices described above will reach efficiencies in the range 16%–18%; however, increased sophistication (and cost) will provide efficiency enhancements.

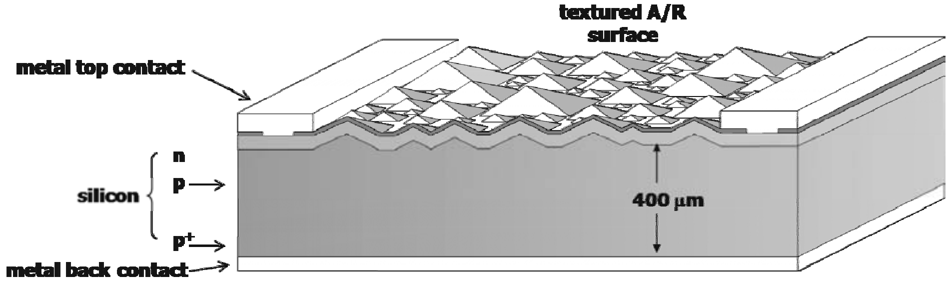

Figure 3.16 Cross-sectional schematic of a typical commercial crystalline silicon device.

The Passivated Emitter and Rear Contacts (PERC) (see Figure 3.17) device uses a thinner wafer, uses an oxide to passivate much of the rear surface of the device, and selectively heavily dopes the regions where the metal is allowed to contact the silicon [79]. In this design, light is reflected back from the rear surface and carrier recombination is reduced.

The passivated emitter, rear locally diffused (PERL) cell is the most efficient single-junction silicon solar cell reported [7, 80]. The most striking feature of the PERL cell (Figure 3.18) is the lithographically defined "inverted pyramid" structure on the top surface that is covered with a thin passivating oxide and a double-layer antireflection coating. This structure not only provides low reflectance, but it also increases absorption lengths by ensuring that most of the absorbed light is directed obliquely into the device. Careful optical design, sometimes refined with the asymmetric tiling of the inverted pyramids to minimize the out-coupling of internally reflected light that has otherwise taken a symmetric route [4], reduces optical losses to only 6%–7% of the incident light. The remaining optical losses are due mostly to shadowing and reflection at both the front and top contact regions.

These devices reach efficiencies up to 25% [7], but the extra efficiency comes at some considerable cost due to the need for double polished silicon wafers, a number of careful lithographic stages, and a number of vacuum depositions. These devices are generally only manufactured for high-value applications such as powering satellites and for best-practice research purposes.

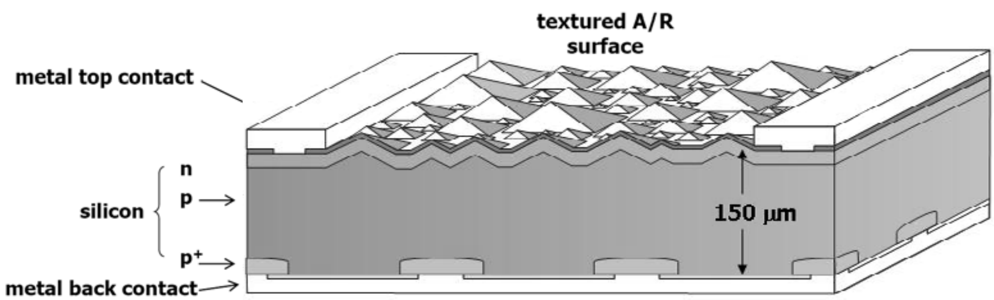

Figure 3.17 Schematic cross-section of a PERC cell [79].

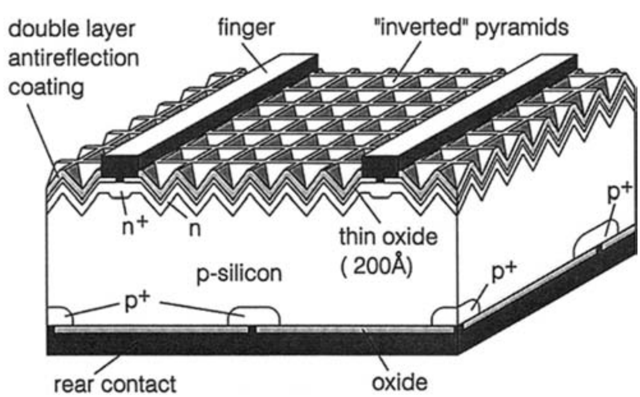

Figure 3.18 Schematic cross-section of a PERL cell. (*From:* [80]. Reprinted with permission.)

Other design features used within high-efficiency C-Si devices illustrate the extent of research efforts. The use of buried contacts has been shown to provide a more reliable contacting scheme than screen printing and at the same time reduce shadowing and resistance [50]. Heterojunction with intrinsic layer (HIT) cells combine a-Si technology and C-Si technology. Unusually, these devices start with an n-type substrate, but then surround the C-Si with p- and n-type a-Si layers on the top and bottom of the device. These layers provide excellent passivation and low resistances, ease contact formation, and allow large open-circuit voltages. Although absorption in the a-Si cannot contribute minority carriers and device currents are reduced, an impressive 22.8% of efficient devices have been demonstrated [81].

3.7.3 Amorphous Silicon

Amorphous hydrogenated silicon (a-Si:H) is a disordered network of silicon and hydrogen atoms that is usually deposited from a silane precursor onto a heated substrate [82]. The effective bandgap of a-Si:H is near 1.7 eV, so it cannot usefully absorb photons with wavelengths much longer than 700 nm. However, the absorption of short-wavelength light in a-Si:H is strong, so solar cells built around the a-Si system are suitable candidates for indoor and outdoor applications.

In 1977, Staebler and Wronski [83] discovered that the photocurrent of an a-Si:H cell would slowly decrease with exposure to light. The original photocurrent of a cell was found to be restorable with an anneal at 150°C or sometimes a large reverse bias. Called the Staebler-Wronski (SW) effect, this degradation was attributed to the breaking of Si-Si or Si-H bonds within the random amorphous network. This increases the midgap defect density and hence the recombination current. The Staebler-Wronski effect is a serious limitation of basic a-Si:H technology as cells can lose as much as 10% of their initial efficiency after a few months of use before reaching a stabilized efficiency.

The deposition process is very flexible, and by varying process conditions, the grain size of the material can be altered from truly amorphous to nano, micro, or polycrystalline. A-Si:H is often deposited at temperatures of 200°C–300°C or lower which allows deposition onto flexible polymer substrates that can not readily support CIGS or CdTe (see Sections 3.7.6 and 3.7.7) deposition. The record efficiency for a-Si:H laboratory scale devices is 9.5% (stable) [84]; the best modules are usually 8%–10% efficient (initial). The devices typically used in pocket calculators are generally around 5% efficient.

Amorphous silicon solar cells have been in development for over 3 decades. From the original basic pn-junction structure, they evolved quickly to incorporate p-type/intrinsic/n-type (pin) designs (Figure 3.19). A region of low-impurity concentration placed between heavily doped p and n regions will sit within an electric field. Within this region carriers will form drift rather than diffusion currents between the n- and p-type regions. This design successfully combats poor diffusion mobility and ensures effective minority carrier collection over a thickness that is dominated by the thickness of the intrinsic region and depends much less upon the carrier diffusion lengths.

It is unlikely that a-Si:H will ever be able to challenge CIGS on the basis of lab-scale device efficiency or CdTe in module efficiency, but considering cost-per-watt, a-Si:H remains an important technology, and although it will be displaced

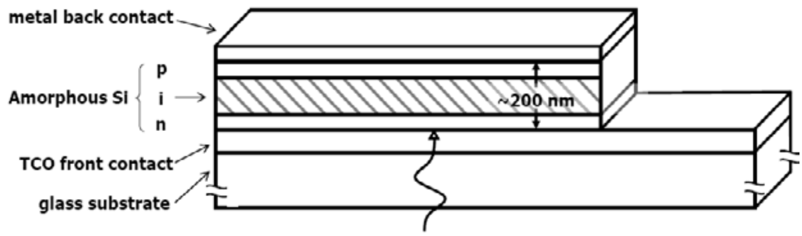

Figure 3.19 Cross-section of a generic a-Si:H device (typically the TCO front contact and all subsequent surfaces are heavily textured for light-trapping; texturing is omitted from this figure for clarity).

by micromorph modules in large-scale production, it will maintain a niche in low-power indoor applications.

3.7.4 Thin Film Polycrystalline Silicon

Polycrystalline silicon (polySi or p-Si) on glass offers an enticing system for photovoltaics. It has a near optimum bandgap at 1.12 eV and a much better utilization of the spectral bandwidth than a-Si. It is also immune from the SW effect. Furthermore, thanks to the success of display industries and the need for reliable thin film transistors (TFTs), there is a highly evolved symbiotic technology with established large-area uniform deposition techniques developed for an already highly profitable commercial device.

Unfortunately, polycrystalline silicon has two challenging disadvantages. First, just like crystalline silicon, it has very low absorption levels across a broad spectral range: 50 μm of crystalline silicon struggles to absorb enough incident IR light to produce efficient devices, so a 3-μm thin film of polysilicon cannot under normal circumstances produce a viable device. Second, the grain boundaries that exist between crystallites in a poly-Si thin film act as significant recombination centers that cause considerable carrier loss as well as channels for the preferential diffusion of dopant impurities during fabrication. These have presented significant challenges to overcome, but now 10.5% cell efficiencies [85] and 8.2% module efficiencies have been reported [86].

The best-known p-Si devices are Kaneka's "naturally Surface Texture and enhanced absorption with a back Reflector" STAR devices [87] [Figure 3.20(a, b)] and the "Crystalline Silicon on Glass" CSG devices developed by UNSW and commercially produced by CSG solar [Figure 3.20(c)] [85].

The key to the success with these devices is, first and foremost, the use of texturing schemes to provide light-trapping that increase effective absorption lengths by as much as 20 times the actual silicon thickness. The second key to success is the careful optimization of deposition conditions to produce suitable material and device characteristics.

In each of the devices light-trapping is achieved by texturing at least one of the interfaces. The effectiveness of this scheme is strongly dependent on the dimensions of the texturing and the thickness of the p-Si layer, and these optical optimizations have to be carried out with due concern for the quality of the resultant p-Si layers and the electronic performance of the device.

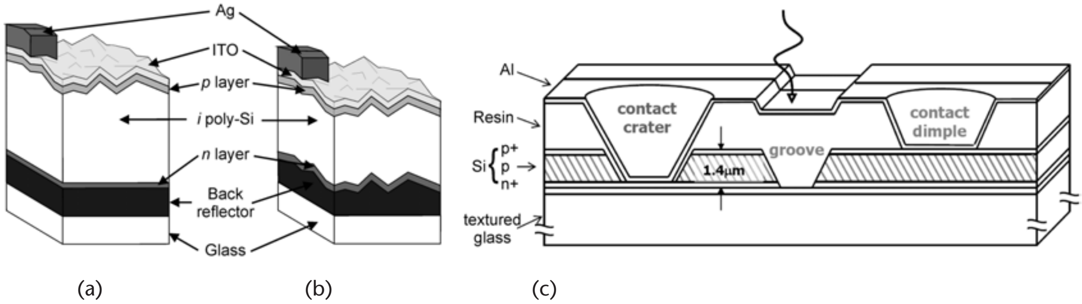

Figure 3.20 (a, b) Schematic views of thin film poly-Si solar cell with STAR (naturally Surface Texture and enhanced Absorption with back Reflector) structure [87]. (a) First generation of poly-Si cell with a flat back reflector. (b) Second generation of poly-Si cell with a rough back reflector for a thinner cell. (c) Schematic cross-section of a CSG thin-film polysilicon device [85].

Deposition at low temperature, using PECVD, often in separate chambers for n- and p-type regions, can produce device structures with appropriate electronic properties. Here there is a great deal of detailed research; it now appears that the best layers consist of relatively small crystalline grains surrounded by hydrogenated a-Si grain boundaries that provide adequate passivation. In this way the distinction between a-Si and p-Si devices becomes a continuum; now the best devices are constructed from nano- or proto-silicon layers that share p-Si and a-Si properties.

3.7.5 Multijunction Silicon

Over the last few years there has been a huge increase in the fabrication of multijunction silicon devices. Using a combination of a-Si and p-Si in a tandem "micromorph" (microcrystalline-amorphous) arrangement, devices with higher efficiencies (~12%) that are less susceptible to the SW effect can be produced [88]. A schematic diagram of a generic tandem cell is shown in Figure 3.21.

The top device is a thin a-Si solar cell that absorbs most of the high-energy photons to produce a modest current at a relatively large voltage. The bottom p-Si device absorbs the red and infrared light, producing the same current and a smaller voltage. The current through the whole device will be limited by the device that produces the least current, so the thickness of each layer has to be carefully optimized for the prevalent lighting conditions. Overall a high-voltage, low-current device is produced that, when carefully optimized, can provide lab efficiencies of 12% and

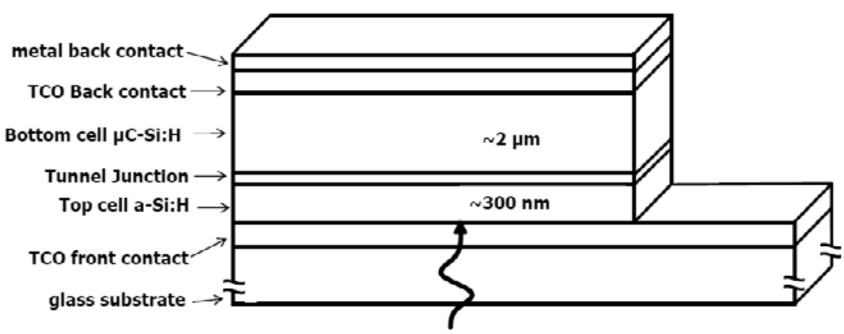

Figure 3.21 Schematic cross-section of a typical micromorph device [88].

module efficiencies of 9% under standard conditions [89]. The two devices are separated (connected) by two heavily doped n- and p-type regions that provide a tunnel junction under the appropriate conditions. These tunnel junctions will be a source of loss, but overall the device offers a sufficiently large improvement to ensure that all new thin-film silicon plants will produce micromorph devices.

Triple-junction silicon devices have also been produced, generally by adding a-SiGe or p-SiGe bottom cell to the tandem structure (Figure 3.22). Again, each subdevice must be current matched. Lab cells with initial efficiencies as high as 15.4% have been reported [90], and modules produced with an a-Si, a-SiGe, a-SiGe combination have reached 10.4% [91]. However, some have predicted that efficiencies over 20% might be achieved based on these concepts [92].

A key consideration if these devices are to be considered for indoor or any non-standard (or even nontracking) applications is the business of current matching. If any spectral component of the incident light is reduced out of proportion with the rest of the spectrum (for instance, if a red sky at night or a window removes short wavelengths), or if room lights are to be the primary source of light, then the overall efficiency will decrease considerably since the current produced by the worst performing subcell will determine the device current and significantly reduce the efficiency.

3.7.6 Cadmium Telluride/Cadmium Sulphide

Cadmium telluride (CdTe) has a direct bandgap of 1.45 eV and typically forms with a p-type conductivity. It forms a good heterojunction with cadmium sulfide (CdS) that has a wide bandgap of 2.4 eV and typically forms with a n-type conductivity. Laboratory CdTe/CdS devices have reached efficiencies of 16.5% [93] and modules have reached 10.9% [94]. Typically, modules are 7%–10% efficient.

A typical device process begins with the deposition of a TCO, typically indium oxide, tin oxide, or indium tin oxide (ITO), onto a glass substrate. A 100-nm layer of an *n*-type CdS is deposited on the TCO by any number of techniques, but typically by an inexpensive chemical bath deposition. After the CdS, a thick (>5 μm) layer of p-type CdTe is deposited, typically by a vacuum deposition with a substrate

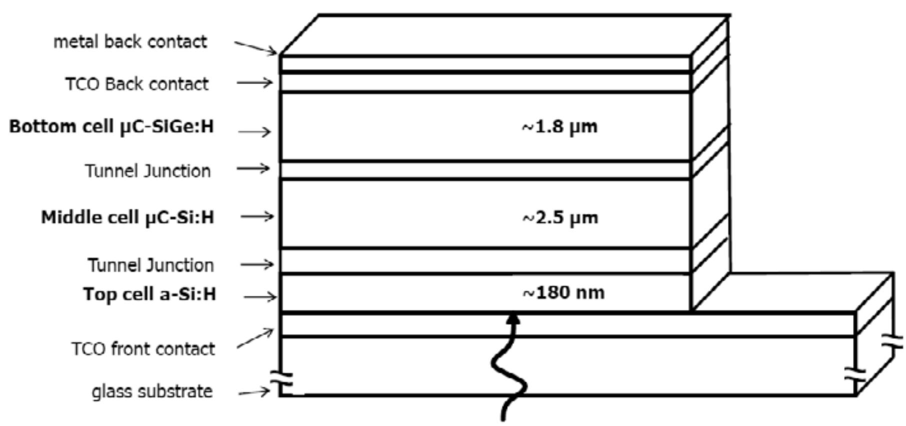

Figure 3.22 a-Si:H/μC-Si:H/μC-SiGe:H triple-junction thin film silicon solar cell (AIST). Similarly, a-Si:H/a-SiGe:H/a-SiGe:H have also been manufactured.

temperature of around 400°C. The final back contact is formed by the deposition of a suitable metal, typically copper. A typical final device structure is shown in Figure 3.23.

In operation light enters the device through the glass substrate, photons pass through the TCO and CdS layers, and some inevitable absorption in either of these two layers becomes a source of loss as any carriers generated will generally not reach the heterojunction. Most of the incident light is absorbed in the first two microns of the CdTe absorber layer, very close to the heterojunction. Much of the 5 μm of CdTe is not required to absorb the light and much of the 5 μm typically deposited is there to ensure good electrical device properties.

Unlike $CuInSe_2$, both CdTe and CdS have strong tendencies to form suitable stoichiometric layers of p- and n-type materials as required, and it appears that one of the key strengths of this technology is the ability to form suitable layers across large substrate areas in commercial module production. Devices are efficient and stable and can be produced cheaply.

In 2009 First Solar announced the first <1$/W PV module production based on CdTe technology, so at this time it is possible to argue that CdTe is now the leading PV technology. It is certainly expected to make significant inroads into the world PV market over the next few years, displacing C-Si and mC-Si technologies. CdTe and to a lesser extent cheaper thin film silicon modules are finally realizing the true cost-reduction potential long expected of second generation (thin-film) technologies.

In spite of these commercial successes, there still remains some potential for increased efficiency and reduced cost of CdTe-based modules. In particular, reducing losses associated with absorption in the TCO and CdS or deep within the CdTe by reducing layer thicknesses. Improving the quality and, to some extent, improving the stability of the back contact are complex challenges that could also improve efficiency. Meanwhile, further cost reductions could be realized by reducing thickness, using less expensive (rare) materials such as indium, and developing a roll-to-roll process for use with flexible substrates.

3.7.7 Copper Indium (Gallium) Disselenide

The chalcopyrite semiconductor, $CuInSe_2$ (CIS), has the highest absorption coefficients across the broadest spectral range and a slightly less than optimum energy bandgap of 1.04 eV that can be increased by the addition of gallium to form $CuIn_{1-x}Ga_xSe_2$ (CIGS).

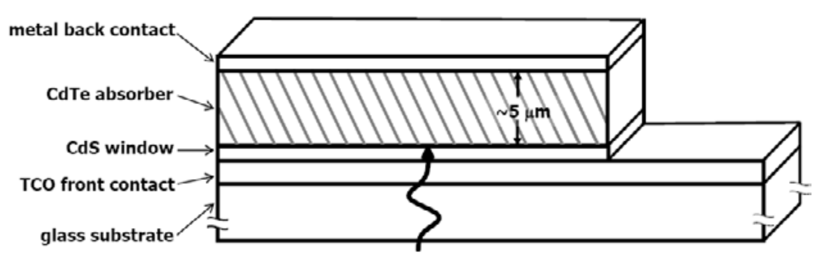

Figure 3.23 Schematic cross-section of a generic CdTe/CdS solar cell.

Solar cells based on CIS and CIGS have been under continuous development for a number of years and are now mature device technologies. Of all the thin film technologies, it is the CIGS laboratory cells that have the highest efficiency record of 19.4% [95]. Unfortunately, the scalability of this technology into large area modules has proved a significant challenge, and although submodule efficiencies have been reported up to 16.7% [96], the best full module efficiency is 13.5% [97]. Commercialization has been hampered by low yields and high production costs.

These difficulties can, at least in part, be attributed to the complex nature of CIGS, which has multiple phases and complex defect chemistries with strong dependence on temperature and film composition. Relatively small variations in the In(+Ga)/Cu ratio have significant effects on resistivity, for example. A generic device is shown in Figure 3.24.

Typically, a ~1-μm layer of molybdenum is deposited on a soda-lime glass substrate to form the back contact. The 1–2-μm p-type CIGS absorber layer is formed on the Mo by either coevaporation or by the controlled selenization (using H_2Se) of metal layers. The heterojunction is formed by a 50-nm n-type CdS layer that is usually deposited by chemical bath deposition and a sputter-deposited ~70-nm layer of undoped but intrinsically n-type ZnO, which provides the top contact TCO.

Light is incident on the ZnO window layer, so called because almost no light is absorbed as the light passes through the top of the device. Some of the short-wavelength photons are then absorbed within the thin layer of CdS and will generate electron-hole pairs. Here holes will be swept across the junction to form part of the external current. However, in this configuration most of the available light is absorbed in the CIGS very close to the junction. This ingenious configuration ensures that there is a minimum requirement on long carrier diffusion lengths (as these can be short in the regions close to the substrate). It also exploits the relatively high mobility of electrons when compared to holes. In any case n-type window materials are not commonly found.

CIGS devices have been extensively researched, and it seems unlikely that laboratory cells will increase significantly beyond the 20% already demonstrated. Successful large area fabrication, higher module efficiencies, and a reduction in the volumes of indium required to make devices are three important research objectives.

3.7.8 Single and Multijunction III-V Cells

Efficient but expensive solar cells can be made out of a range of III-V materials in single-junction and multijunction designs. Single-junction concentrator cells (232

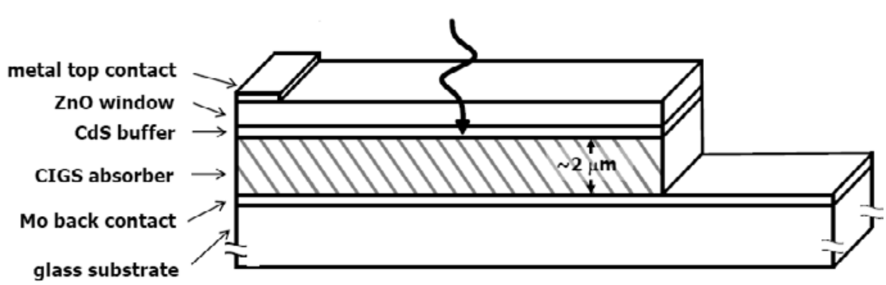

Figure 3.24 Schematic cross-section of a generic CIGS solar cell.

suns) based on GaAs wafers have demonstrated 28.8% [21] efficiency, and lab-scale thin film GaAs devices have reached 26.1% efficiency [98]. High-efficiency, high-temperature and radiation-tolerant III-V-based devices have been commonly used to provide solar power for satellites, an application for which device cost is of little importance.

Single-junction device efficiencies are only a little better than those of the best silicon devices and in general are less easy to produce. On the other hand, multi-junction devices based on the diversity of direct bandgap III-V materials can produce very efficient devices.

For many years a great deal of work was focused on mechanical stacking of devices, as this provides the most flexibility of design. However, over recent years, the impressive development of III-V light-emitting devices has provided deposition technologies capable of producing complex lattice-matched multilayered structures that now allow monolithic integration of multiple current matched cells. At the same time there has been an increase in interest in concentrator technologies, and, as a consequence, high-efficiency multijunction III-V cells have emerged as an important technology.

Triple-junction GaInP/GaAs/Ge devices similar to the generic device shown in Figure 3.25 have been produced with efficiencies of up to 32% (under AM1.5) and 41.1% (under 454 suns) by Spectrolab and the Fraunhofer, respectively [21]. These devices must be designed with two key constraints. First, each cell in the device must produce the same current (for the most part, this can be tuned by cell thickness). Second, every layer in the device must be lattice matched (or strain compensated) at room temperature and be at least tolerant to differences in lattice expansion at high temperatures. Fortunately, germanium provides an excellent infrared absorbing bottom cell and provides a suitable large area substrate upon which lattice matched semiconductors in the GaAs/InP/GaP system can be found with a large range of possible bandgap values in the 1.4–2.2-eV range. Ironically,

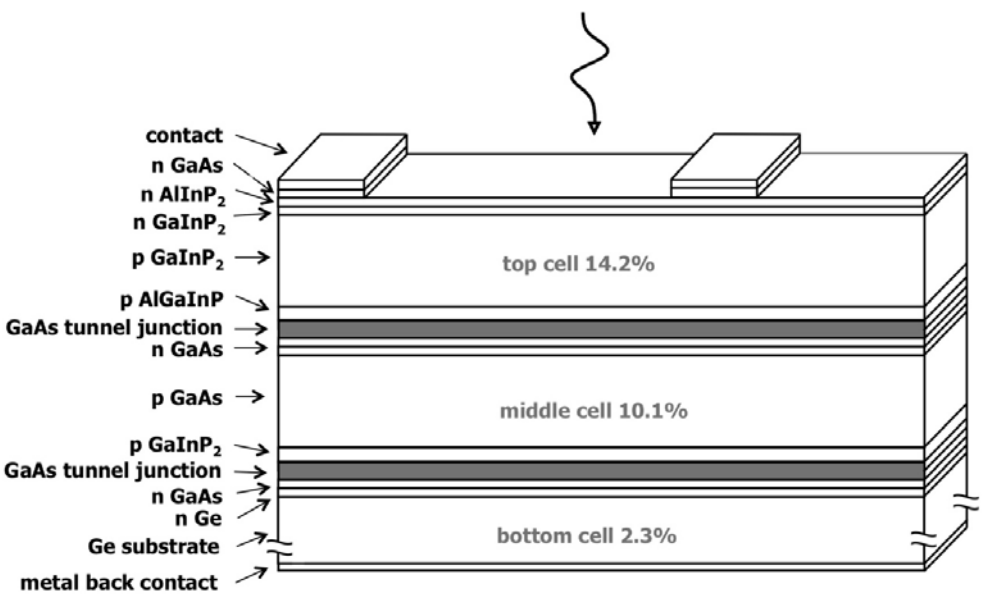

Figure 3.25 Triple-junction GaInP/GaAs/Ge device.

the biggest weakness in the system, preventing the formation of >50% efficient quadruple junction devices is the absence of a suitable semiconductor with a bandgap similar to that of silicon but lattice matched to Ge.

Even with 1 sun efficiencies greater than 30%, triple-junction III-V solar cells are still much too expensive for use in anything other than concentrator systems with mechanical tracking. Equally, although the 30% efficiency seems like a tempting option for high-value nonstandard applications, as with the triple-junction thin film silicon devices, the problem is current matching. These devices will perform rather poorly under nonstandard illumination and when not facing the Sun.

3.7.9 Emergent Technologies

Single-junction devices based on silicon wafers and a-Si, CdTe, and CIGS thin films are all relatively mature device technologies. There has been significant and sustained research on each and the efficiency values for these systems seem to be saturating. Although it is impossible to rule out future step changes in efficiency, it is hard to imagine any laboratory scale devices enjoying substantial increases above current records. In most cases improving module efficiencies in line with lab efficiencies and reducing module cost are the most important challenges.

The most recently developed of the technologies discussed so far are the single-, double-, and triple-junction devices based around the use of polycrystalline thin-film silicon. These are relatively new technologies, and, although we might expect some significant improvements in efficiency over the next few years, it remains unlikely that thin-film efficiencies greater than the CIGS lab record of 20% will be surpassed without a great deal of effort.

Meanwhile, a number of new technologies are currently being researched that have the potential for substantial cost reductions.

3.7.9.1 Dye Sensitized Solar Cells

The first dye sensitized solar cells (DSSCs), or Grätzel cells, were produced by O'Regan and Grätzel in 1991 [99]. Impressive progress has been made over the last 20 years, and now DSSCs have demonstrated laboratory cell efficiencies of 10.4% [100] and small submodule efficiencies of 8.4% [101].

DSSCs are bulk-heterojunction devices. Instead of having planar junctions formed by n-type material being deposited on p-type material (typical of all devices discussed so far), bulk heterojunctions are formed by the complete intermixing of the two materials that provide the anode (n-type) and cathode (p-type). In the DSSC, the anode is typically formed by nanospheres of titania (TiO_2) photosensitized by dye, and the cathode is formed by an electrolyte (liquid or solid) that completely surrounds every part of the anode.

Figure 3.26 shows a cross-section of a typical DSSC device. Light enters through the TCO of the anode and is absorbed by the dye anywhere in the bulk of the cell. This absorption creates an excited state in the dye, which injects an electron into the conduction band of the TiO_2. These electrons then diffuse through successive nanospheres until they reach the top contact. During this process the dye molecule has lost an electron, so in order to maintain neutrality, the dye molecule takes an

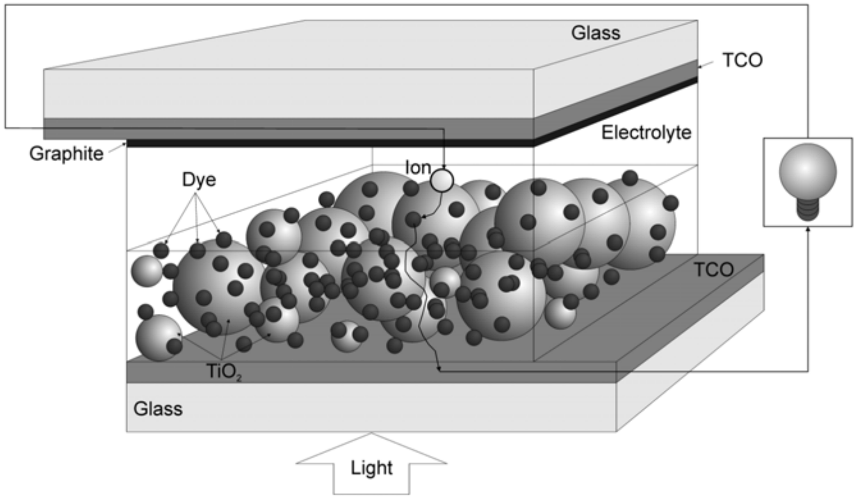

Figure 3.26 Cross-sectional schematic of a typical DSSC device.

electron from the electrolyte, forming a positively charged ion that diffuses through the electrolyte to the cathode.

The DSSC is a remarkably efficient system. Much of the efficiency loss (when compared to C-Si) is due to its inability to usefully absorb red and infrared photons, so one key target for increasing efficiency is extension of quantum efficiency into the lower energy region of the spectrum. Otherwise, quantum efficiency is very high, typically >0.9 for the blue-green part of the spectrum, with modest transport losses and losses associated with absorption in the TCOs. Use of nanowires and nanotubes for improved electron transport and dielectrics or metal scattering for enhanced light-trapping and therefore thinner devices are currently under investigation.

The efficiency values already achieved for DSSCs compare very favorably with inorganic thin-film efficiencies, particularly those based on a-Si and have one key advantage. DSSC devices can be manufactured using what are essentially inexpensive printing technologies rather than expensive semiconductor technologies. Devices are so cheap to manufacture that the first markets have been for use within semidisposable consumer electronics. As confidence in long-term stability and manufacturing maturity increases, it seems likely that DSSC modules will become a significant player that may displace a-Si and CdTe devices where DSSC could certainly compete on efficiency and cost, if not C-Si, where the proven stability and high efficiency will be hard to beat in the medium term.

3.7.9.2 Organic Polymer Solar Cells

Many fundamentally different types of n-type and p-type organic semiconductors are available and each of these can have their properties (such as bandgap) tuned by chemical modification to the molecules. It is easy to imagine the construction of a basic organic pn-junction solar cell since all the processes are analogous to those within inorganic devices. Unfortunately, diffusion lengths for electrons and holes in organic polymers are rather short and the electrons and holes exist in excitonic form and can be hard to separate. In fact, the best organic polymer solar cells are also bulk-heterojunction devices in which n- and p-type materials are mixed. Here, the reliance on diffusion is reduced, but the separation of electrons and holes becomes more of a problem. The current best lab and module efficiencies, 5.15% and 2.05%, respectively [102, 103], are relatively low because a significant volume of red and infrared photons are not utilized and there are carrier collection challenges. There are also serious stability issues as light soaking significantly degrades efficiency over relatively short timescales.

Organic polymer solar cells are not yet commercially viable, but have many promising properties that offer considerable hope for the future. If suitable polymers can be developed, organic polymer devices could be painted or printed onto the devices or buildings and could conform to the shape of any surface, but for now much more development is required.

3.7.9.3 Inorganic Bulk-Heterojunction Solar Cells

Researchers are currently exploring the possibility of inorganic bulk-heterojunction systems that use three-dimensional junctions instead of planar junctions. The most advanced example of this device concept is perhaps the core-shell nanowire device developed by the Atwater group (Figure 3.27) [104].

In this device, self-organized p-type nanowires are grown by physical deposition processes using a patterned silicon template. A second deposition regime deposits an n-type layer around the p-type wires creating an array of core-shell nanodevices that each form classic single crystal solar cells. These nanodevices can act as stand-alone energy harvesting units, possibly to individually power single CMOS gates, for example, or, to make a large area device, they can be surrounded by a flexible polymer and then peeled off the substrate (the substrate can then be reused) before electrical contacts are made to the top and bottom of the flexible array.

These devices are still in the early stages of development and have relatively low efficiency, but they embody a number of key ideas for future device technologies. They are based on nanotechnology and the use of self-organization, they form bulk heterojunctions and thereby relax the constraints placed by the need for long carrier diffusions lengths, and they greatly reduce the volume of expensive semiconductor material required. It is likely that any future device technologies will make use of these generic concepts in order to compete with the established technologies.

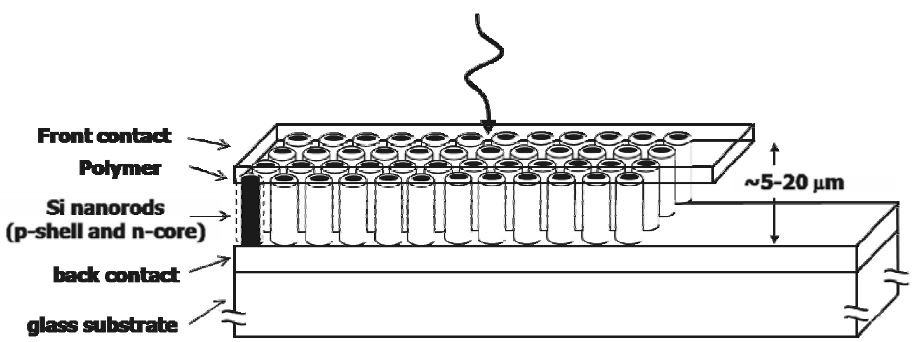

Figure 3.27 Schematic cross-section of a nanowire solar cell.

3.8 Photovoltaic Systems

Photovoltaic systems include MW power stations, grid-connected domestic systems, isolated high-power systems, and self-powered electronic devices (everything from pocket calculators to wireless sensors). There are a number of excellent texts that provide detailed reviews of PV systems; here we will focus our attention on the small self-powered devices.

Since outdoor or indoor lighting will rarely be continuous and will nearly always be rather unpredictable, PV arrays must act in conjunction with energy storage, especially battery storage.

Outdoor systems will experience a daily cycle, a climatic cycle (days to weeks) and a yearly cycle. Systems that wholly depend upon room lights will experience very different daily and weekly cycles and might also have to contend with holiday periods. Energy balance is a key concern. The energy generated by a PV system must be greater than the energy required by the application over any given time period. Otherwise, systems have to be designed to recover from complete power loss or else have a fail-safe low-power mode that keeps time or at the very least allows system recovery.

In many low-power systems, or in circumstances where there is no limit on the size of the PV array and the power generated is always much greater than the power consumed, relatively simple systems can be built that simply charge a battery. System design is more challenging when there is a space (or financial) limit on the size of the array and the maximum energy has to be extracted. To build the most efficient system, a detailed model of the patterns of illumination, the corresponding performance of the photovoltaic device, and a detailed description of the load are required. The solar cell needs to be operated at the maximum power point—where the external resistance matches the internal resistance of the PV cell—and maximum power point tracking (MPPT) needs to be used.

3.8.1 Basic System

The simplest system can be constructed using a solar cell, a diode, and a battery. When the solar cell is illuminated, the current passes through the diode and charges the battery. When the solar cell is in darkness, the diode prevents the battery from discharging through the solar cell. The solar cell or a number of cells in series need to generate enough potential to overcome the turn-on voltage of the diode and then charge the battery. The application (load) is attached to the battery terminal and is continuously powered by the battery or the solar cell or both. With some limitations the battery acts to ensure that the supply voltage is more or less constant. This simple system does not control the discharge of the battery or prevent the overcharging of the battery. The system will provide a fluctuating supply voltage and does not extract the maximum power from the solar cell. For systems requiring long-term reliability (and minimal battery replacement), bespoke or off-the-shelf charge controllers should be used.

3.8.2 Charge Controllers

In general, there is a limit to the current that should be used to charge a battery (in the bulk charge phase) and when the battery approaches full charging the rate of charging and the voltage should be reduced (tapered charge phase). Most batteries will discharge slowly when not in use and should be trickle-charged with a low voltage.

Two basic methods are used to control the current/voltage from a PV cell series regulation and shunt regulation (Figure 3.28) [105]. The series regulator reduces the voltage presented to the battery by varying a resistor placed in series, and the shunt regulator removes the current by varying a resistance placed across the PV cell terminals. In each of these instances the variable resistor (typically a transistor) is governed by a controller unit that in the past was often an analog circuit, but is now nearly always a digital system.

In fact, using the transistors as resistive components can require considerable heat dissipation within the transistor, so as an alternative, the transistors can be switched to ensure that excess power is dissipated in the array. In these shunt switching and series switching designs, the duty cycle (on/off) ratio controls the average currents through the regulators [106].

3.8.3 DC-DC Converters and Maximum Power Point Tracking

The optimum load resistance required by a solar cell will change with both illumination and temperature. In order to maximize the transfer of energy from the cell to the load (battery), controlled DC-DC converters are required. These systems decouple the solar cell from the array and can also form the basis maximum power point tracking (MPPT).

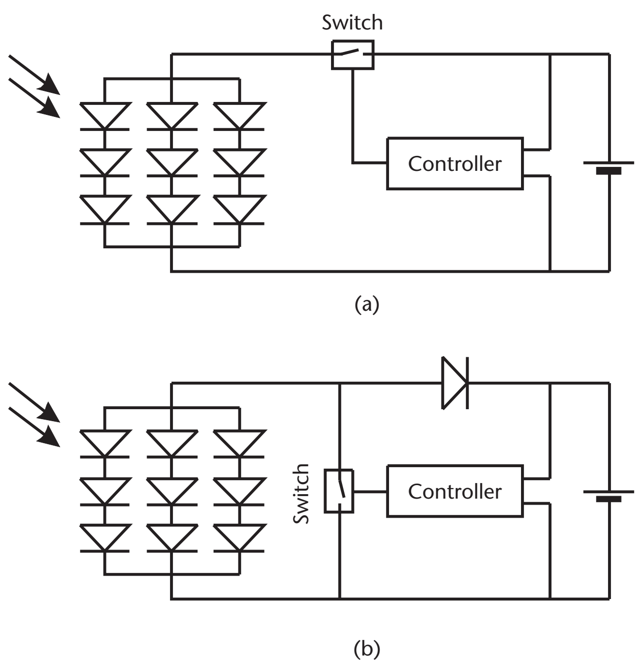

Figure 3.28 (a) Series switching and (b) shunt switching regulators [105].

Buck and boost are two simple DC-DC designs (Figure 3.29) [105]. The buck design [Figure 3.29(a)] uses a switching device in series with an inductor and in parallel with a capacitor to smooth the output voltage as the switch is cycled. Typically, the switching component is an MOS field effect transistor. This is cycled rapidly (typically 100 kHz) and when in a continuous current mode (the switching rate is such that the current does not decay to zero), the load voltage to source voltage ratio is the same as the on/off duty cycle ratio (D):

$$\frac{V_L}{V_S} = D \tag{3.13}$$

For a fixed frequency, modulating the pulse width controls the load voltage. For low currents the load voltage rises to the source voltage and a degree of control is lost.

The boost design [Figure 3.29(b)] places the switching device parallel to the load. In this scheme, switching acts to increase rather than decrease the voltage. Now increasing the on/off duty cycle decreases the load voltage:

$$\frac{V_L}{V_S} = \frac{1}{1 - D} \tag{3.14}$$

In a continuous current mode this design draws a continuous current from the source and therefore does not require filtering between the solar cell and the converter.

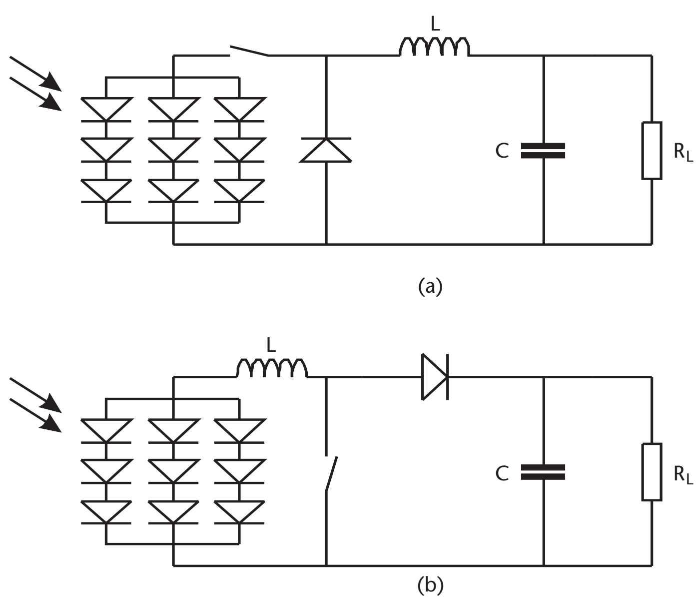

Figure 3.29 (a) Buck and (b) boost DC-DC designs [105].

Although the boost design is preferred, either of these DC-DC converters can be used to control the maximum power point by controlling the duty cycle and thereby controlling the load voltage.

Controlling devices set the duty cycle, but in order to do this, the devices need a suitable algorithm and sufficient information. Indirect trackers can be based around certain assumptions concerning the intensity of illumination at certain points of the day or time of year. Direct trackers will calculate the required voltage based on the continuously measured values of the currents or voltages of the solar cell or of the converter output. As long as this information is provided, suitable algorithms can be applied to find the optimum point. "Mountain climbing," "fractional open circuit voltage," and "perturb and observe" algorithms are popular, and "artificial neural networks" and "fuzzy logic" have been tried [107].

In fact, the voltage corresponding to the maximum power point is never so far away from any fixed value and in systems where the balance of energy is reasonably favorable, heroic MPPT is not necessarily a worthwhile exercise, especially considering that fixed or indirect schemes are much cheaper to implement and much more robust.

3.8.4 Miniaturization and Low-Power Systems

Most new self-powered applications use photovoltaic devices and systems that are already commercially available in a modular form. Development has concentrated on circuit design for efficient power management rather than implementation of more efficient PV devices and systems. This may begin to change as sensors become smaller and higher device and system efficiencies are required in order to maintain energy balance for a decreasing solar cell area.

3.8.5 Device Technology

Currently, most PV devices are designed and optimized for outdoor use; very few are designed specifically for high-efficiency indoor use. It is easy to understand why. There is a vast range of different indoor lighting scenarios with many different types of illumination and often very different emission spectra. It is clear that all of the device technologies discussed will behave differently in various situations. In particular, we know that DSSC devices and a-Si devices will perform better in blue-green conditions, but perform very poorly in conditions where red light is dominant. We know that any multijunction device will be extremely susceptible to significant spectral differences from those for which the device was designed. In the future it is likely that photovoltaics devices will be designed for some very bespoke applications, but for now it is worth considering how the efficiency of a range of devices designed for AM1.5 might change when moved inside.

Table 3.2 compares the performance of various solar cell technologies for both indoor and outdoor operations. The outdoor efficiencies were measured using a solar simulator to simulate an AM1.5 spectrum at an irradiance level of 100 mW/cm^2. Gray filters were then used to reduce the irradiance to one typically experienced indoors of 0.5 mW/cm^2.

Table 3.2 Efficiencies of Various Photovoltaic Technologies Measured with Solar Simulators for Operation Outdoors (100 mW/cm^2) and Indoors (0.5 mW/cm^2)

Reference	*Supplier*	*Technology*	*Outdoor Efficiency (%)*	*Indoor Efficiency (%)*	*Cell Efficiency Indoors Relative to Outdoors (%)*
[108]	BP Solar	c-Si	15.7	7.8	50
[109]	Unknown	c-Si	15.2	6.8	45
[108]	MAIN TESSAG	mc-Si	13.5	1.4	10
[109]	Unknown	mc-Si	14.4	3.4	24
[108]	Tessag	a-Si	7.2	6.3	88
[108]	Solems	a-Si	4.3	3.7	86
[109]	Unknown	a-Si	—	—	94
[108]	NREL	GaAs	12.9	9.9	77
[109]	Unknown	GaAs	—	—	84
[108]	NREL	GaInP	11.0	9.3	85
[108]	NREL	GaAs-GaInP tandem	14.2	10.5	74
[108]	ZSW, University of Stuttgart	CIGS	12.9	6.0	47
[108]	Parma University	CdTe	7.5	6.5	87

It is clear that a-Si (and CdTe) cell efficiencies suffer much less in poor lighting conditions than the highly considered C-Si or mC-Si devices. This is because in low light conditions, the low shunt resistance typical of wafer technologies acts to reduce the fill factor. A-Si cells naturally have a high shunt resistance and generally exhibit lower efficiencies in bright conditions compared to the other silicon technologies and thus show a lower degradation in cell efficiency as the light level is decreased [109].

Of course, in reality, indoor illumination has different spectral characteristics to those experienced outdoors because of the different light sources available (see Section 3.5.2), but from this simple analysis, it appears that a-Si would be a good first choice for low-power, indoor applications.

3.8.6 Systems Considerations

In Section 3.8.3 we briefly considered the most popular systems used to regulate PV power supply and track maximum power points. Again, decreasing low power levels and increasing miniaturization will require new solutions. Brunelli et al. have provided a detailed analysis of a photovoltaic scavenging system that could be used to power a small wireless sensor node [110] (see Figure 3.30).

This scheme employs a buck DC-DC converter, controlled by a maximum power point tracker that is based on a direct measurement from a 0.1 cm^2 pilot cell and a fractional open circuit voltage (FOC) approach that assumes a nearly linear relationship between the operating voltage at the maximum power point and the open circuit voltage.

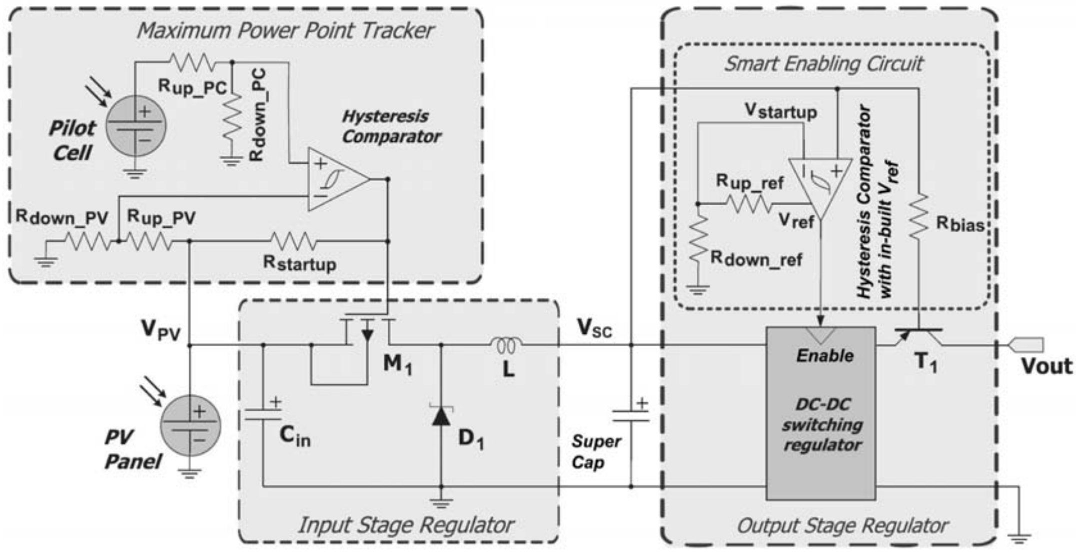

Figure 3.30 Schematic diagram of the harvester platform proposed by Brunelli [110]. Reprinted with permission.

Instead of charging a battery, an ultracapacitor (or supercapacitor) is used. The output stage regulator completes the design. The optimization process [assuming 400 mW of power generated (~700 W/m^2) for this design], detailed in [110], illustrates many of the key concerns that would feature in any similar exercise. Careful consideration is given to the choice of components, the losses associated with MOSFET switching and the power consumption of the comparator (including parasitic resistance, capacitance losses, and inductor losses), and careful selection of the best operating frequency. The final optimized system is found to have an overall efficiency in the range of 50%–70% depending on the charging level of VSC. Considering the challenge of miniaturization, this low-power system compares favorably with the larger systems that typically operate at ~90% efficiency. Yet it is also clear that further miniaturization and decreasing illumination will rapidly challenge the energy balance for similar systems, with more than 50% of energy being lost in transfer from solar cell to appliance.

Other challenges arise as available volumes are reduced. Cantatore and Ouwerkerk [106] set a target volume of 100 mm^3 for a self-powered sensor. A 1 cm^2 solar cell is allowed, which in turn means that the largest permissible inductor is typically smaller than 10H and the largest permissible capacitance is 10 μF. The battery or supercapacitor will also have to be small. All of these restrictions place considerable constraints on the design of control circuitry (the small capacitance and inductance values ensure that a high frequency is required) and the ability to store energy. In this analysis PV compares very favorably when compared to electromagnetic generators as the AC-DC conversion required for those systems requires transformers that cannot be built in the volume made available, but it is also clear that the efficiency of the solar cell will be of increasing importance as soon as area and volume restraints are imposed.

Taking miniaturization one step further, it is interesting to consider the monolithic integration of an entire system. It is clear that all the required components—the

PV device, any digital (CMOS) circuitry, capacitors, and inductors—can all be produced using silicon microfabrication techniques.

C-Si devices with 14% efficiency (AM1.5) have been integrated with CMOS and MEMS devices on a single SOI wafer [111]. The challenge faced here is protecting the solar cell while CMOS and MEMS processing steps are carried out. An alternative approach would be to deposit an a-Si device on top of the silicon chip at the end of the process. This could be carried out at a low temperature, would maximize the surface area of the photovoltaic device, and would provide a device that maintains its efficiency under artificial lights and under low illumination conditions. It is likely that systems used to manage such devices would have to be relatively simple in order to minimize the energy loss at the system level.

3.9 Summary

Outdoors and often indoors, the illumination level is often sufficient to allow photovoltaic devices to provide significant energy to small, medium, and large electronic systems. Current device technologies are largely aimed at optimal performance under standard (AM1.5) outdoor conditions and can operate with efficiencies of up to 30%. Typically devices are between 5% and 15% efficient, and most mature technologies are unlikely to rise significantly above current levels. Newer device technologies based on the use of multiple junctions yield higher efficiencies at AM1.5 but very reduced efficiencies under nonstandard or low illumination.

For most isolated systems the size of the photovoltaic device can be scaled to ensure that the energy supply outstrips energy demand and the primary concern is system stability and obtaining the best possible performance from the battery. With diminishing illumination levels and size restrictions, the efficiency of the solar cell, the efficiency of the system, and a careful energy management of the total device must all be carefully optimized in the context of a detailed understanding of lighting conditions and how intensity and spectra vary during daily, climatic, and yearly cycles.

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CHAPTER 4

Kinetic Energy Harvesting

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This chapter gives an overview of the field of vibration energy harvesting for autonomous, self-powered microsystems. Vibration-powered generators are typically, although not exclusively, inertial spring and mass systems. The characteristic equations for inertial-based generators will be presented, together with a theoretical overview of the mechanisms that show how three main transduction methods can be employed to extract mechanical energy from the system and translate this into electrical energy. The transduction mechanisms are piezoelectric, electromagnetic, and electrostatic techniques. Piezoelectric generators use active materials that generate an electrical charge when stressed mechanically. A review of existing piezoelectric generators will be given, which includes impact-coupled, resonant, and human-based devices. Electromagnetic generators operate on the principle of electromagnetic induction, which arises from the relative motion of a conductor moving through a magnetic flux gradient. Several electromagnetic generators described in the literature will be reviewed, including large-scale discrete devices and wafer-scale integrated versions. Electrostatic generators utilize the relative movements between electrically isolated charged capacitor plates to generate energy. The work done against the electrostatic force between the plates provides the harvested energy. Electrostatic-based generators are reviewed under the classifications of in-plane overlap varying, in-plane gap closing, and out-of-plane gap closing; the coulomb force parametric generator and electret-based generators are also covered. The coupling factor of each transduction mechanism is discussed, all the devices presented in the literature are summarized in tables classified by transduction type, and conclusions are drawn as to the suitability of the various techniques.

4.1 Introduction

This chapter will discuss kinetic energy generators, which convert ambient mechanical energy into electrical energy. Kinetic energy is typically present within the environment as vibrations, random displacements, or forces and can be converted into electrical energy using electromagnetic, piezoelectric, or electrostatic mechanisms. Useful vibration levels are to be found in numerous applications including common household goods (refrigerators, washing machines, and microwave ovens), industrial plant equipment, moving structures such as automobiles and airplanes, and civil structures such as buildings and bridges [1]. Human movement tends to be characterized by low-frequency, high-amplitude displacements [2, 3]. The amount of electrical energy that is attainable by these approaches is dependent upon the quantity and form of kinetic energy available in the environment and also on the efficiency of both the generator and the power conversion electronics.

This chapter will discuss the fundamental principles of kinetic energy harvesting and a variety of transduction mechanisms that can be employed. Issues relating to miniaturizing the generator and the operating frequency bandwidth of the device will also be discussed in detail. These will be illustrated by a comprehensive review of generators that have been developed commercially and within the laboratory.

4.2 Kinetic Energy-Harvesting Applications

Kinetic energy can be harvested from a range of applications. These have been categorized as human-, industrial-, transport-, and structural-based and the following sections provide example data describing the form that the kinetic energy can take in each case. It is apparent that the characteristics associated with each case are very different. This highlights a particular challenge with kinetic energy harvesting in that generators for various applications will often be very different. It is essential that generators are designed from the outset with a prior knowledge of the application and the characteristics of the kinetic energy targeted.

4.2.1 Human

Human motion is characterized by large amplitude movements at low frequencies and some degree of impact on the heel of the foot during walking. These impacts send shock waves through the human body, but these are rapidly absorbed by the joints. The average gait of a walking human of a weight of 68 kg produces 67W of energy at the heel of the shoe [4]. A study carried out during a European-funded research project, Vibration Energy Scavenging (VIBES), measured the vibrations at various locations on the body while walking. The subject was 1.7m tall, weighed 76 kg, and was wearing standard running shoes; measurements were taken at a walking speed of 5 km/h from the ankle, wrist, chest, upper arm, and head. The maximum accelerations were found at the ankle in the direction of walking with a peak acceleration of over 100 m/s^2 and a frequency of 1.2 Hz. The acceleration in the vertical direction was 20 m/s^2. At all other locations on the body, the frequency remains constant, although the magnitude of the accelerations in the vertical and walking axes was less than 7 m/s^2. These findings are similar to the results published by von Büren et al. [5].

The low frequencies and high amplitudes of displacement mean that human applications are difficult to address using inertial-based generators described in Section 4.3. The large forces generated, for example, heel strikes while walking or during breathing, and angular displacements at the joints also present possibilities for harvesting energy. One further point to note is that great care must be taken with regard to the amount of energy being extracted and by what method, in order to ensure that the subject does not directly experience the effect of energy harvesting (i.e., the process does not noticeably affect their movement or effort). This is discussed further in Section 4.7 when real devices are described.

4.2.2 Industrial

Industrial applications include plant and a variety of equipment found in the manufacturing and process industries. Where such equipment is powered by mains electricity, it is common for the frequency of the supply to be present in the vibration spectra of the machine. Figure 4.1 shows the frequency spectrum of the vibrations found on a U.K. mains-powered air compressor. The peak vibration at 50 Hz is clearly visible. The level of vibration on the plot is shown in units of g where 1g represents an acceleration of 9.81 m/s^2; the peak corresponds to 0.25 m/s^2. The amplitude of the corresponding displacements of these vibrations is 2.5 μm, which is very low compared to those associated with human applications. In many application scenarios greater levels of power are available at the higher-order harmonics of the mains frequency (typically the second-order harmonic) and, especially on new machines, vibration levels tend to be lower. For equipment not powered from the mains, the vibration frequency can often vary, but will typically be in the range of 20–200 Hz with similar vibration amplitudes to those of mains-powered equipment.

4.2.3 Transport

Transport applications cover a wide range of vehicles including cars, trains, aircraft, and ships. The vibrations present in each area can often be quite different, as illustrated by looking at two sources. In the case of cars, the vibration levels will vary depending upon the location of the generator on the vehicle (i.e., on the wheel or

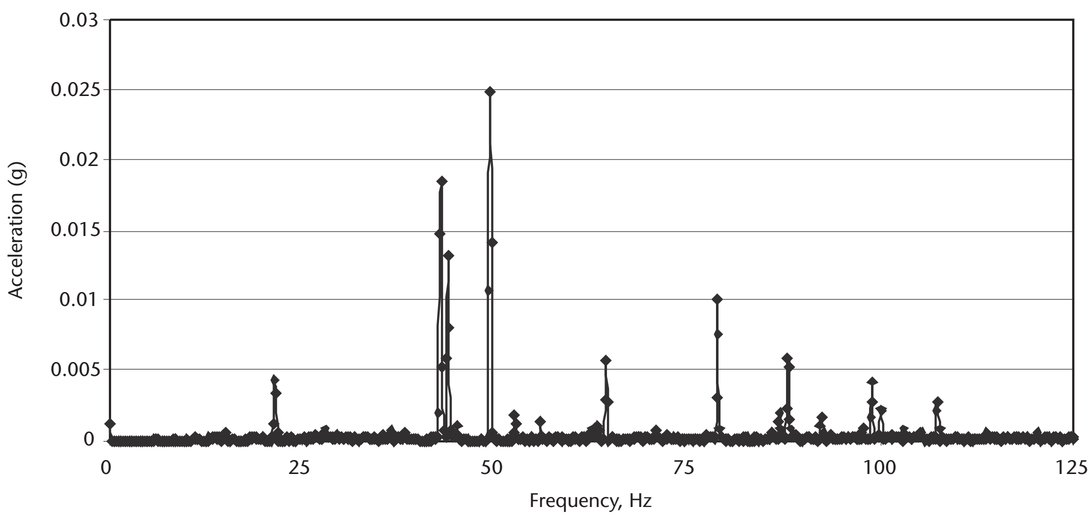

Figure 4.1 Frequency spectrum of a mains powered air compressor.

on the chassis), the type of vehicle, the road conditions, and the speed. Table 4.1 gives some typical data obtained during the VIBES project. It can be seen that the vehicles' suspension systems are very effective at reducing vibration levels (as would be expected) and anything located away from the wheel or suspension elements experiences relatively low levels of vibration at low frequencies. Acceleration levels on the wheel, however, are much greater and hence such locations present a challenging environment in which the generator can operate.

In contrast to the results from a vehicle, the vibration data from a vertical stabilizer on a PZL SW-4 helicopter are shown in Figure 4.2. Vibration frequencies on helicopters are governed by both the rotor speed and number of blades and tend to be relatively consistent. In the example shown, there is a characteristic frequency at 30 Hz, which is repeated around the craft. Vibration levels peak at 19 m/s^2 in this specific location, but can be even greater elsewhere (e.g., on the main rotor gearbox).

4.2.4 Structural

The available kinetic energy in buildings and bridges can be variable for different scenarios. For example, vibrations in buildings can be caused by a variety of effects

Table 4.1 Example Vibration Data from a Range of Vehicles

Type of Car	*Road*	*Location*	*Peak Acceleration (m/s^2)*	*Frequency (Hz)*
Luxury	Highway	Cabin	0.05	40
Luxury	City	Cabin	0.04	30
Small	Highway	Chassis	0.04	23
Small	Highway	Wheel axle	2	16

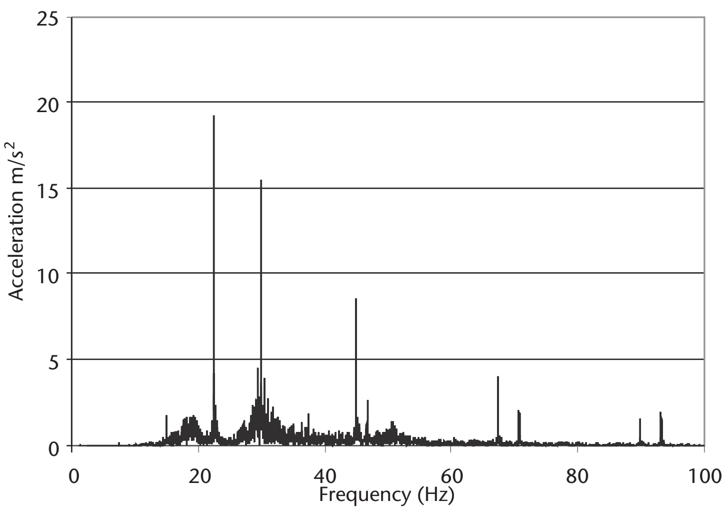

Figure 4.2 Frequency spectrum from the vertical stabilizer of a PZL SW-4 helicopter.

such as seismic activity, subways, road and rail systems, wind, heating, ventilation, air conditioning (HVAC) equipment, elevator/conveyance systems, and fluid pumping equipment. Vibration levels are relatively small, with a maximum of 0.1 m/s^2 exhibited by a range of building types excited by passing traffic at frequencies typically in the range of 10–12.5 Hz, as shown by Hunaidi and Tremblay [6]. Bridges can exhibit vibrations as a result of the traffic flowing over them. The magnitude and frequency will depend upon the nature of the structure and the speed, weight, and number of vehicles traveling over it. According to Yang et al. [7], a 25-m-long bridge with a single vehicle traveling over it at 20 m/s exhibits accelerations of 0.035 m/s^2 at 2 Hz. This can increase to 0.09 m/s^2 for 5 vehicles. Williams et al. [8] also explored the feasibility of vibration energy harvesting on bridges. They found the natural frequency for two different concrete bridges to be 6 Hz and 4.5 Hz, with amplitudes of acceleration less than 0.09 m/s^2 when an articulated lorry crossed one of the bridges.

4.3 Inertial Generators

Inertial generators often have a fixed reference frame to which the vibration energy is applied. The inertial frame transmits the vibrations to a suspended inertial mass, thereby producing a relative displacement between these two components. Such a system will have a fixed resonant frequency, which can be designed to match the characteristic frequency of the applied vibration. This approach effectively "magnifies" the environmental vibration amplitude of by the quality factor of the resonant system. This is discussed further in Section 4.4.

Inertial-based generators are often modeled as second-order spring and mass systems. Figure 4.3 shows a general example of such a system based on a seismic mass, m, on a spring of stiffness, k. Energy losses within the system (comprising parasitic losses, c_p, and electrical energy extracted by the transduction mechanism, c_e) are represented by the damping coefficient, c_T. These components are associated with the inertial frame, which is excited by an external sinusoidal vibration of the form $y(t) = Y\sin(\omega t)$. This external vibration is out of phase with the mass when the structure is vibrated at resonance, thereby resulting in a net displacement, $z(t)$, between the mass and the frame. Assuming (as is almost always the case) that the mass of the vibration source is significantly greater than that of the seismic element and also that the external excitation is harmonic, then the differential equation of motion is described as:

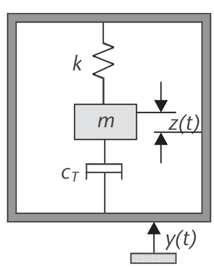

Figure 4.3 Model of a linear, inertial generator.

$$m\ddot{z}(t)+c\dot{z}(t)+kz(t)=-m\ddot{y}(t) \tag{4.1}$$

Since energy is extracted from relative movement between the mass and the inertial frame, the following equations apply. The standard steady-state solution for the displacement of the mass is given by:

$$z(t)=\frac{\omega^2}{\sqrt{\left(\frac{k}{m}-\omega^2\right)^2+\left(\frac{c_T\omega}{m}\right)^2}}Y\sin(\omega t-\phi) \tag{4.2}$$

where ϕ is the phase angle given by:

$$\phi=\tan^{-1}\left(\frac{c_T\omega}{\left(k-\omega^2 m\right)}\right) \tag{4.3}$$

The maximum energy is extracted when the excitation frequency is equal to that of the natural frequency of the system, ω_n, given by:

$$\omega_n=\sqrt{k/m} \tag{4.4}$$

Williams and Yates [9] showed that the power dissipated within the damper (i.e., extracted by both the transduction mechanism and parasitic damping mechanisms) is given by:

$$P_d=\frac{m\zeta_T Y^2\left(\frac{\omega}{\omega_n}\right)^3\omega^3}{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2+\left[2\zeta_T\left(\frac{\omega}{\omega_n}\right)\right]^2} \tag{4.5}$$

where ζ_T is the total damping ratio ($\zeta_T = c_T/2m\omega_n$). The maximum power generated therefore occurs when the device is driven at its natural frequency, ω_n, and hence the output power is given by the following equations:

$$P_d=\frac{mY^2\omega_n^3}{4\zeta_T} \tag{4.6}$$

or, alternatively,

$$P_d=\frac{mA^2}{2\omega_n\zeta_T} \tag{4.7}$$

Equation (4.7) uses the excitation acceleration level, A, in the expression for output power and is simply obtained using $A = \omega_n^2 Y$. It might be tempting here to deduce that the power output will increase towards infinity as the damping ratio decreases toward zero. This is not the case, however, since the equations are steady-state solutions. The maximum power that can be extracted by the transduction mechanism can be predicted by accounting for the parasitic and transducer damping ratios, as shown here:

$$P_e = \frac{m\zeta_e A^2}{4\omega_n \left(\zeta_p + \zeta_e\right)^2} \tag{4.8}$$

P_e is maximized when $\zeta_p = \zeta_e$. Parasitic damping is unavoidable with practical implementations. In some cases it might be useful to have the ability to vary the level of damping. For example, this would allow the displacement of the mass, $z(t)$, to be kept within a fixed limit. Conclusions should not be drawn without first considering the effect of the applied frequency, the magnitude of the excitation vibrations, and the maximum displacement of the mass. If the input acceleration is sufficiently high, then increased damping will result in a broader bandwidth response and hence result in a generator that is less sensitive to variations in excitation frequency, which might be caused by changes in temperature or other environmental parameters or may vary over time. Excessive device amplitude can also lead to nonlinear behavior and introduce difficulties in keeping the generator operating at resonance. It is clear that both the frequency of the generator and the level of damping should be designed to match specific application requirements in order to maximize the power output. It is also worth noting that the output power is proportional to the mass, which should therefore be maximized subject to any given size constraints. It is also important to note that the when a generator is coupled to an electrical circuit, losses (e.g., resistive losses in the coil of an electromagnetic system) will limit the amount of energy available within the electrical domain. Electrical damping coefficients for the three commonly used types of transduction mechanisms are given in Section 4.4.4.

For a fixed level of acceleration, the output power is inversely proportional to the natural frequency of a generator and hence it is generally preferable to operate at the lowest available fundamental frequency within the available vibration spectra. This is compounded by practical observations that acceleration levels associated with environmental vibrations tend to reduce with increasing frequency. The vibration spectra should be carefully studied before designing the generator in order to identify the most appropriate frequency of operation within the design constraints of generator size and maximum permissible displacement.

4.4 Transduction Mechanisms

Some form of transduction mechanism is obviously required to convert the kinetic energy into electrical energy. This mechanism has to be incorporated into the mechanical system that has been designed to maximize the energy coupled from the application environmental to the transducer (e.g., the inertial systems described previously).

The transducer can generate electricity from mechanical strain or the relative displacement present within the system, depending upon the type of transducer. The use of active materials such as piezoelectrics is an obvious example that enables the strain to be directly converted into electrical energy. Electromagnetic and electrostatic transduction exploits the relative velocity or displacement that occurs within a generator. Each transduction mechanism has different characteristics such as damping effects, ease of use, scalability, and effectiveness. The suitability of each mechanism for any particular application depends largely on the practical constraints applied. Assuming no size constraints, electromagnetic harvesting will be the most efficient because the coil can be large, with a high number of turns and low coil resistance (larger diameter wire) providing very high potential coupling factors. The efficiency of piezoelectric generators is fundamentally limited by the piezoelectric properties of the material. The efficiency of electrostatic generators is reduced by technical challenges relating to charging the electrodes, the separation distances, and the amplitudes of displacement.

4.4.1 Piezoelectric Generators

Piezoelectric materials have been used for many years to convert mechanical energy into electrical energy. Piezoelectrics contain dipoles, which cause the material to become electrically polarized when subjected to mechanical force. The degree of polarization is proportional to the applied strain. Conversely, an applied electric field causes the dipoles to rotate, which results in the material deforming. Piezoelectric materials are therefore used in a variety of commercial sensors and actuators and are also a candidate for kinetic energy-harvesting applications. The piezoelectric effect is found in single crystal materials (e.g., quartz), ceramics (known as piezoceramics) [e.g., lead zirconate titanate (PZT)], thin-film materials (e.g., sputtered zinc oxide), screen printable thick films based upon piezoceramic powders, and polymer materials such as polyvinylidene fluoride (PVDF).

Such materials have anisotropic piezoelectric behavior. This means that the properties of the material differ depending upon the direction of the strain and the orientation of the polarization (and therefore the position of the electrodes). The piezoelectric properties of a material are characterized by a series of constants shown in Table 4.2. These are listed with their respective axis notations to fully describe the anisotropy. For example, the 3 direction refers to piezoelectric materials

Table 4.2 Piezoelectric Material Properties

Property	*Constant*	*Definition*	*Units*
Electromechanical coupling coefficient	*k*	√(mechanical energy stored ÷ electrical energy applied) √(electrical energy stored ÷ mechanical energy applied)	—
Piezoelectric constant	*d*	strain ÷ applied field	m/V
		short circuit charge density ÷ applied stress	C/N
Piezoelectric constant	*g*	open circuit field ÷ applied stress	V/N
		strain developed ÷ applied charge density	m/C

that have been polarized along their thickness (i.e., having electrodes on the top and bottom surfaces). If a mechanical strain is applied in the same direction, the constants are denoted with the subscript 33 (e.g., d_{33}). If the strain is applied perpendicular to the direction of polarization (e.g., the 1 direction), the constants are denoted with the subscript 31 (e.g., d_{31}). These are illustrated in Figure 4.4, but for a more complete description, refer to the IEEE standards [10].

In a majority of scenarios, piezoelectric harvesters operate in the lateral 31 mode. This is because the piezoelectric element is often bonded to the surface of a mechanical spring element that converts vertical displacements into a lateral strain across the piezoelectric element. Some designs can operate in the compressive 33 mode and these have the advantage of exploiting the 33 constants, which are typically greater than the 31 equivalents. Compressive strains, however, are typically much lower than the lateral strains occurring when the piezoelectric is bonded onto a flexing structure. One approach, which exploits the high lateral strains and high 33 properties of piezoelectrics, is to use an interdigital electrode arrangement, an example of which is shown in Figure 4.5. Bonded piezoelectric transducers can be used, which convert lateral strains into a compressive mode by using materials that are poled along the length of the element rather than through its thickness. This allows the exploitation of the higher-value 33 properties, but the electrode arrangement has the drawback of reducing the active area of the device.

The piezoelectric constants for quartz, soft and hard lead zirconate titanate piezoceramics (PZT-5H and PZT-5A, respectively), barium titanate ($BaTiO_3$), and polyvinylidene fluoride (PVDF) are given in Table 4.3. These properties typically vary with age, stress, and temperature. The aging rate tends to be logarithmic with time and is dependent on the formation/deposition method and material type. Stressing the material further increases the aging process. Soft piezoceramics (e.g., PZT-5H) are more susceptible to stress-induced changes than the harder compositions such as PZT-5A. Temperature is also a limiting factor with piezoceramics due to the effect of crystal structure changes above the Curie point. Above this limit,

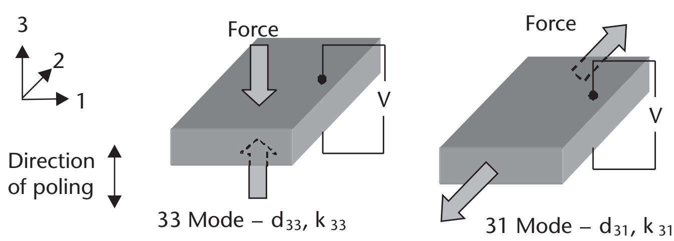

Figure 4.4 Piezoelectric constants in typical energy-harvesting modes.

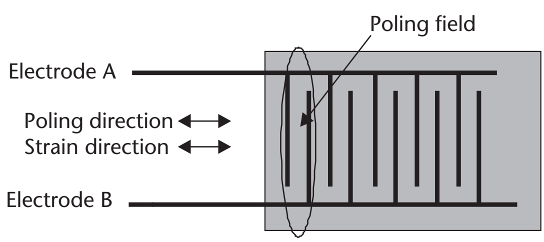

Figure 4.5 Interdigital electrode arrangement.

Table 4.3 Coefficients of Common Piezoelectric Materials

Property	*Quartz*	*PZT-5H*	*PZT-5A*	*$BaTiO_3$*	*PVDF*
Material type	Single crystal	Piezoceramic	Piezoceramic	Piezoceramic	Polymer
d_{33} (10^{12} C/N)	–2.3 (d_{11})	593	374	149	–33
d_{31} (10^{12} C/N)	–0.93 (d_{12})	–274	–171	78	23
g_{33} (10^{3} Vm/N)	–58	19.7	24.8	14.1	330
g_{31} (10^{-3} Vm/N)	—	–9.1	–11.4	5	216
k_{33}	0.07	0.75	0.71	0.48	0.15
k_{31}	—	0.39	0.31	0.21	0.12
Relative permittivity ($\varepsilon/\varepsilon_o$)	4.4	3,400	1,700	1,700	12
Curie temperature (C)	573	195	365	120	~150

Source: [11, 12].

the piezoelectric material will lose it piezoelectric properties, effectively becoming depolarized. The application of stress can also lower the Curie temperature and therefore the maximum practical operating temperature will typically be reduced.

4.4.2 Electromagnetic Transduction

To date, most types of rotating generators are based upon electromagnetic transduction techniques and are used in numerous applications from bicycle dynamos to large-scale power generation. Kinetic energy harvesting can exploit rotary generators, such as those found in the Seiko kinetic watches [13], or linear transducers that are used to harvest power from vibrations. Well-designed generators, which are not constrained in size, can be extremely efficient at converting kinetic energy into electrical energy.

Electromagnetic generators are based on Faraday's law of electromagnetic induction. When an electric conductor is moved through a magnetic field, a potential difference, or electromotive force (emf), is induced between the ends of the conductor. The voltage induced in the conductor (V) is proportional to the time rate of change of the magnetic flux linkage (ϕ) of that circuit, shown in (4.9):

$$V = -\frac{d\phi}{dt} \tag{4.9}$$

In most practical implementations of the generator, the conductor is wound into a multiturn coil and the magnetic field is created by permanent magnets. For a coil with N turns, the voltage generated, V, is given by (4.10), where ϕ becomes the average flux linkage per turn:

$$V = -N\frac{d\phi}{dt} \tag{4.10}$$

In the case of a linear vibration generator, with relative motion between the coil and the magnet in the x-direction, the voltage induced in the coil can be expressed as the product of a flux linkage gradient and the velocity of movement.

$$V = -N \frac{d\phi}{dx} \frac{dx}{dt} \tag{4.11}$$

In the case of a time varying magnetic field, B, where the flux density is uniform over the area, A, of the coil, the induced voltage is given by (4.12), where α is the angle between the coil area and the direction of the flux density.

$$V = -NA \frac{dB}{dt} \sin(\alpha) \tag{4.12}$$

Power is extracted from the generator by connecting the coil to a load resistance, R_L. The resulting current in the coil creates its own magnetic field, which opposes the field giving rise to it. This opposition results in an electromagnetic force, F_{em}, that opposes the generator motion. It is by acting against F_{em} that mechanical energy is transformed into electrical energy. F_{em} is proportional to the current and hence the velocity. It is expressed as the product of an electromagnetic damping, D_{em}, and the velocity.

$$F_{em} = D_{em} \frac{dx}{dt} \tag{4.13}$$

Therefore, for the maximum electrical power output, the generator design must maximize D_{em} and velocity. An expression for electromagnetic damping is given by (4.14), where R_L and R_c are the load and coil resistances, respectively, L_c is the coil inductance, and Φ is the total flux linkage of the coil.

$$D_{em} = \frac{1}{R_L + R_c + j\omega L_c} \left(\frac{d\Phi}{dx} \right)^2 \tag{4.14}$$

Maximizing D_{em} requires the flux linkage gradient to be maximized and the coil impedance to be minimized. The flux linkage gradient is a function of the strength of the magnets, their arrangement with respect to the coil and the direction of movement, and the area and number of turns for the coil. At the low frequencies typically found in application environments, coil inductance can be ignored and its resistance is the dominating factor of the coil impedance. The value of the coil resistance depends on the number of turns and the coil geometry and wire diameter. These factors are discussed in more detail next.

The magnetic field strength is governed by the type of magnetic material used. Permanent magnets are made from ferromagnetic materials. These contain atoms with unpaired electrons and therefore exhibit a net magnetic moment. Magnetic domains are formed when large numbers of atoms are grouped together with their magnetic moments aligned in a particular direction. Initially, the magnet as a whole is unmagnetized (i.e., the domains are randomly aligned). These materials are magnetized by the presence of a strong external magnetic field that aligns the domains and hence produces a net magnetic field after the magnetizing field is removed. Magnetic materials can be "soft," meaning that they can be easily magnetized and demagnetized, or "hard," meaning that they require greater magnetizing fields but are more difficult to demagnetize.

A permanent magnet has a magnetic field (B) and a magnetic flux (ϕ), which is the product of magnetic field multiplied by the area. The magnetizing force is denoted as H, and a figure of merit is the maximum energy product, BH_{MAX}, which can be determined from a material's magnetic hysteresis loop. Finally, another consideration is the Curie temperature, which in the same manner as for piezoelectric materials is the maximum operating temperature that the material can with withstand before becoming demagnetized. Currently, the strongest magnets available are rare earth neodymium iron boron (NdFeB) types, but care has to be taken because of their low operating temperature and poor corrosion resistance. Despite these practical limitations, NdFeB magnets are the preferred option for kinetic energy harvesting since they maximize the magnetic field strength within a given volume, compared to other magnetic materials. Properties of some common magnetic materials are summarized in Table 4.4.

As discussed earlier, the properties of the coil can have a significant effect on the performance of the generator. The number of turns in a coil (N) is related to the geometry of the coil, its wire diameter, and the winding density. The latter is represented by the fill factor, f, which is defined by relating the area of the wire (A_{wire}) to the cross-sectional area of the coil (A_{coil}) where $A_{wire} = f\,A_{coil}/N$. The copper fill factor depends on tightness of winding, insulation thickness, and winding shape. Most coils are scramble wound and have a typical fill factor from 50% to 60% [15]. In the case of small electromagnetic energy harvesters, a copper conductor with a small wire diameter (w_d) is necessary in order to achieve a suitable value of N for a minimal coil size. The resistance of the wire is inversely proportional to its cross-sectional area; so fine wires tend to have increased resistance per unit length.

The properties of a conventionally wound coil are given by the following equations, where r_i = coil inner radius, r_o = coil outer radius, t = coil thickness, L_w = length of wire, V_T = coil volume, ρ = resistivity of coil material, and R_c = coil resistance:

Table 4.4 Properties of Some Common Magnetic Material

Material	*$(BH)_{MAX}$ (kJ/m^3)*	*Flux Density (mT)*	*Max. Working Temperature (C)*	*Curie Temperature (C)*	*Density (kg/m^3)*
Ceramic	26	100	250	460	4,980
Alnico	42	130	550	860	7,200
SmCo (2:17)	208	350	300	750	8,400
NdFeB (N38H)	306	450	120	320	7,470

Source: [14].

$$V_T = \pi\left(r_o^2 - r_i^2\right)t \tag{4.15}$$

$$L_w = \frac{4fV_T}{\pi w_d^2} \tag{4.16}$$

$$R_c = \rho\frac{N^2\pi(r_o + r_i)}{f(r_o - r_i)t} \tag{4.17}$$

$$N = \frac{L_w}{r_i + \frac{\left(r_o - r_i\right)}{2}} \tag{4.18}$$

As stated in Section 4.3, when the generator is limited by parasitic damping, the maximum electrical power can be obtained by making D_{em} equal to the parasitic damping. Assuming that the flux linkage gradient has been maximized, the maximum power can be achieved by choosing an optimum value of load resistance. Equating the expression for D_{em} to parasitic damping and rearranging to obtain the load resistance, the maximum power in the load is a function of the load resistance determined by

$$R_l = R_c + \frac{N^2\left(\frac{d\phi}{dx}\right)^2}{D_p} \tag{4.19}$$

In cases where the parasitic damping is much greater than electromagnetic damping, the optimum load resistance equals the coil resistance.

A comparison of different electromagnetic circuits and their relative merits were discussed by Spreemann et al. [16, 17].

4.4.3 Electrostatic Generators

This section discusses the basic concepts and operating principles of the third most commonly used transduction mechanism: electrostatic. These types of generators consist of a variable capacitor whose two plates are electrically isolated from each other by air, a vacuum, or an insulator. In the simplest case, external mechanical vibrations cause the gap between the plates to vary and hence the capacitance changes. In order to extract energy, the plates must be charged and the mechanical vibrations work against the electrostatic forces present in the device. The fundamental equations for a capacitor are shown next, where C is the capacitance (farads), V is the voltage (volts), Q is the charge (coulombs), A is the area of the plates (m^2), d is the gap between plates (m), ε is the permittivity of the material between the plates (Fm^{-1}), ε_0 is the permittivity of free space, and E is the stored energy (joules).

$$C = Q/V \tag{4.20}$$

$$C = \varepsilon \frac{A}{d} \tag{4.21}$$

$$V = \frac{Qd}{\varepsilon_0 A} \tag{4.22}$$

$$E = 0.5QV = 0.5CV^2 = 0.5Q^2/C \tag{4.23}$$

Electrostatic generators can be either voltage- or charge-constrained. Voltage-constrained devices have a constant voltage applied to the plates, and therefore the charge stored on the plates varies with changing capacitance. This typically involves an operating cycle that starts with the capacitance being at a maximum value (C_{max}) (i.e., the plates being at their closest). At this stage, the capacitor is charged up to a specified voltage (V_{max}) from a reservoir while the capacitance remains constant [18]. The voltage is held constant while the plates move apart until the capacitance is minimized (C_{min}). The excess charge flows back to the reservoir as the plates move apart and the net energy gained is given by:

$$E = \frac{1}{2}\left(C_{max} - C_{min}\right)V_{max}^2 \tag{4.24}$$

Charge-constrained devices use a constant charge on the capacitive plates, and therefore the voltage will vary with changing capacitance. The plates are initially charged when the variable capacitance is at a maximum (plates closest together). As the capacitor plates separate, the capacitance decreases until C_{min} is reached, and, since the amount of charge is fixed, the voltage across the plates increases. The initial charging can originate from a reservoir controlled by the system electronics and the initial charge is returned to the reservoir at the end of the cycle. Alternatively, a fixed charge can be obtained using electret materials, such as Teflon or Parylene. In either case, the mechanical work against the electrostatic forces is converted into electrical energy. The net energy gained in this case is given by:

$$E = \frac{1}{2}\left(C_{max} - C_{min}\right)V_{max}V_{start} \tag{4.25}$$

In either case, V_{max} must be carefully chosen to be compatible with the associated electronics and its associated fabrication technology. These two approaches have different strengths and weaknesses. The constant voltage approach produces greater energy levels but requires the electronics to provide a charging voltage of a different value to that used by the system electronics, which are powered from the reservoir. This requires a dual voltage system, but, since the precharging voltage level affects the damping in the generator, it is possible to use this approach to adjust its dynamics to suit different excitation characteristics [19]. The charge-constrained case produces less power but is simpler to precharge the plates to a voltage less than V_{max}. A third hybrid approach is to operate the generator in the charge-constrained mode but place a fixed capacitance in parallel [18]. In this case, the energy from the charge-constrained system can approach that of the voltage-constrained system because the parallel capacitance can be very large. The drawback of this approach is that more initial charge is required before the conversion process can begin and hence there are potentially more losses.

Electrostatic generators can be broadly classified into three types shown in Figure 4.6: out-of-plane gap varying and in-plane overlap varying and in-plane gap closing [20]. For the constant charge types, the perpendicular force between the plates is given by:

$$F_e = 0.5Q^2 d / \varepsilon A \tag{4.26}$$

The perpendicular force between plates in voltage-constrained types is given by:

$$F_e = 0.5\varepsilon A V^2 / d^2 \tag{4.27}$$

The relationship between the electrostatic force variation and the inertial mass displacement (x) for the three configurations is shown in Table 4.5 [21]. In the case of high damping configurations, the electrostatic damping force has to be counterbalanced almost entirely by the mechanical spring force [21].

4.4.4 Transduction Damping Coefficients

In the case of an electromagnetic generator, El-Hami et al. [22] stated that the damping coefficient arising from electromagnetic transduction c_e is given by:

$$c_e = \frac{(NlB)^2}{R_{load} + R_{coil} + j\omega L_{coil}} \tag{4.28}$$

Table 4.5 Electrostatic Force Variation for the Three Configurations

Structure	*Charge Constrained*	*Voltage Constrained*
Out-of-plane gap varying	F_e constant	$F_e \sim 1/x$
In-plane overlap varying	$F_e \sim 1/x^2$	F_e constant
In-plane gap varying	$F_e \sim x$	$F_e \sim 1/x^2$

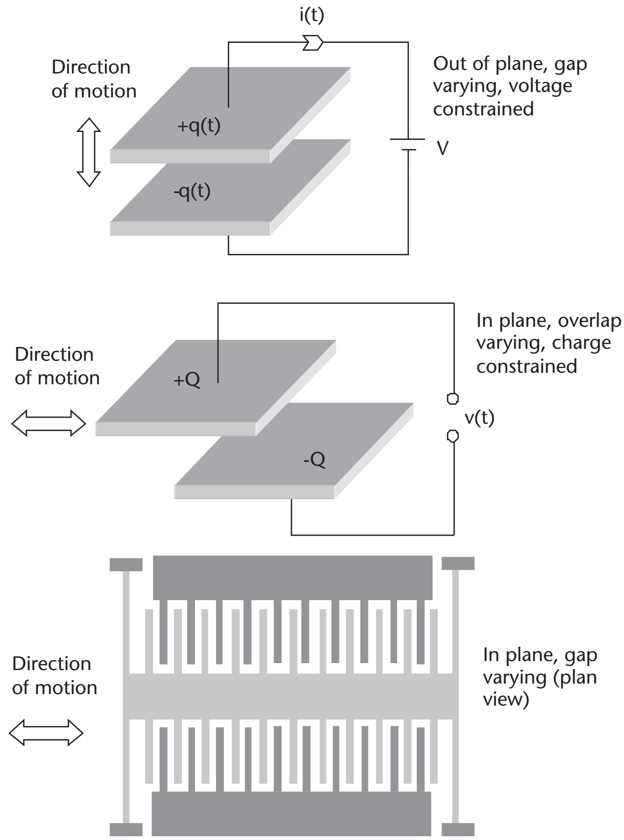

Figure 4.6 Electrostatic generator types.

where N is the number of turns in the generator coil, l is the side length of the coil (assumed square in this case), and B is the flux density to which it is subjected. R_{load}, R_{coil}, and L_{coil} are the load resistance, coil resistance, and coil inductance, respectively. It should be noted that (4.28) is an approximation and only suitable for cases where the coil moves from a high magnetic field region B to a zero field region. A more precise value for the electromagnetic damping can be determined by techniques such as finite element analysis (FEA). Equation (4.28) shows that the value of R_{load} can be used to adjust c_e to match c_p and therefore maximize power within the constraints imposed by the coil wire properties (resistivity, diameter, and overall geometry). Stephen [23] showed that the optimum value of R_{load} can be determined from (4.29) and that the maximum average power delivered to the load can be found from (4.30).

$$R_{\text{load}} = C_{\text{coil}} + \frac{(NlB)^2}{c_p} \tag{4.29}$$

$$P_{\text{eload max}} = \frac{mA^2}{16\zeta_p \omega_n}\left(1 - \frac{R_{\text{coil}}}{R_{\text{load}}}\right) \tag{4.30}$$

For a piezoelectric generator, an expression for the damping coefficient is [24]:

$$c_e = \frac{2m\omega_n^2 k^2}{2\sqrt{\omega_n^2 + 1\big/\left(R_{\text{load}}C_{\text{load}}\right)^2}} \tag{4.31}$$

where k is the electromechanical coupling factor of the piezoelectric material and C_{load} is the load capacitance. The value of the resistive load, R_{load}, can be used to vary ζ_e and the optimum value can be found from (4.32) (noting that the maximum power occurs when $\zeta_e = \zeta_p$).

$$R_{opt} = \frac{1}{\omega_n C}\frac{2\zeta_p}{\sqrt{4\zeta_p^2 + k^4}} \tag{4.32}$$

Electrostatic transduction is characterized by a constant force damping effect, denoted as coulomb damping, and a simple model is shown in Figure 4.7 [25].

The damping terms can vary depending upon the nature of electrostatic generator, which is described in Section 4.4.3. For interdigital electrostatic generators with a variable gap and overlap, the electrical damping coefficient is given by (4.33) and (4.34), respectively [1].

$$c_e = \frac{Q^2}{4N_g\varepsilon_0 tl}z \tag{4.33}$$

$$c_e = \frac{Q^2 d}{2N_g\varepsilon_0 t}\frac{1}{z^2} \tag{4.34}$$

where Q is the charge on the capacitor plates, ε_0 is the permittivity of free space, d is the gap between the electrode fingers, N_g is the number of gaps between fingers, t is the device thickness, and l is the finger length.

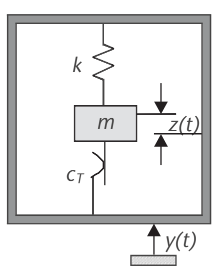

Figure 4.7 Model of an electrostatic resonant generator.

The analysis presented in this section has discussed methods of extracting the maximum electrical power from a resonant mechanical system. The approaches have been based upon a conventional second-order spring and mass system with a linear damper and are better suited to the electromagnetic case, since the damping mechanism is proportional to velocity. The general analysis, however, still provides a valuable insight into resonant generators and highlights some important aspects that are applicable to all transduction mechanisms, the details of which are discussed in the following section.

4.4.5 Microscale Implementations

Microelectromechanical systems (MEMS) comprise mechanical and electrical functionality fabricated at the microscale. They combine microelectronics with micromachining processes to realize microscale mechanical components. MEMS inertial accelerometers are a commercially successful device that, given the similarity between these and vibration energy generators, point to the potential for realizing MEMS vibration energy harvesters. This is attractive from the point of view of simultaneously mass-producing microelectronics, microsensors, and a power supply on a single substrate providing a self-powered wireless sensor chip solution. As discussed previously, however, there is a fundamental relationship between the energy harvested and the size of the generator. Power output is proportional to inertial mass size and the amplitude of displacement, and it is clear that microscale devices will harvest less energy than larger-scale systems. Furthermore, the efficiency of the transduction process is affected by issues with scaling. For example, electromagnetic transduction does not scale down well in size, whereas electrostatic transduction does; the reasons for this fact are discussed next. Finally, as generators are made smaller, the resonant frequency of the system tends to increase. It is therefore challenging to realize a MEMS generator, which is tuned for applications on machinery at around 100 Hz.

Microscale electromagnetic generators are fabricated with planar microcoils and deposited magnetic films. For practical reasons, both of these are inferior to their conventional counterparts (wound coils and bulk magnets) and when combined together, the electromagnetic damping significantly reduces. The reasons for the inferior properties will now be discussed. The planar coil pattern, typically a square or circular spiral, is fabricated using a combination of photolithography, deposition, and etching steps and multiple coil layers can be achieved by building up layers on top of each other. Despite the excellent capabilities of the fabrication processes, it is very challenging to achieve high coil densities and low resistances. There is a trade-off between the resolution of the coil tracks and the thickness of the coils. The thicker the metal film used to realize the coil tracks, the greater the spacing between adjacent turns and the wider the track width. The density of coil winding achievable and the thickness of the conductor layer will depend upon the technology used for the fabrication process. Silicon microfabrication processes can achieve metal films of 1-μm thickness, deposited onto oxidized silicon wafers and patterned with a spacing of 1–2 μm. Advanced photolithographic processes and electroplating techniques can achieve a maximum aspect ratio of metal thickness-to-gap spacing of approximately 10:1.

Electromagnetic damping [given in (4.28)] depends on the coil parameters (R_c and N, ignoring inductance) and the flux linkage. In the case of a wire wound coil, R_c has previously been shown in (4.17) to be proportional to the square of the number of turns (assuming a constant fill factor). Therefore, for wire-wound coils with a constant fill factor, the electromagnetic damping is independent of N. For a planar microcoil, however, the cross-sectional area of the conductor and the length of the track determine the resistance. In order to illustrate the influence of the coil dimensions on resistance, a single-layer planar square spiral coil has been analyzed [26]. The resistance can be expressed in terms of the number of turns (N) and the coil's outer and inner dimensions (d_i and d_o, respectively), as shown in Figure 4.8.

Making the reasonable assumption that the gap between tracks, track width, and thickness are equal, the resistance for a single-layer microcoil is given by [26]:

$$R_c = \frac{2\rho_{cu}(d_o + d_i)}{(d_o - d_i)^2}(4N^3 - 4N^2 + N) \tag{4.35}$$

For a large number of turns, the coil resistance is proportional to N^3. This is because the track's cross-sectional area is given by the width of the track multiplied by the thickness, and, for a fixed d_o, both of these decrease with increasing N. This cubic relationship means that for microcoils the electromagnetic damping is inversely proportional to the number of turns.

Equation (4.28) also shows that electromagnetic damping depends upon the flux linkage. The gradient of the magnetic field and the area of the coil determine the total flux linkage. Considering device scaling alone, for a given magnetic material the B field will remain constant, while reducing the gap spacing between magnets increases the flux density gradient. Flux linkage, however, is dependent upon the area of the coil, which in turn is proportional to the square of the dimension. Therefore, the overall relationship is that the flux linkage gradient is directly proportional to the dimension. For a wire-wound coil with a fixed N, the coil resistance is inversely proportional to the dimension and electromagnetic damping is therefore proportional to the square of the dimension. For planar microcoils the situation is much worse, and in fact the highest EM damping is achieved for a single turn coil. However, in practice it is likely that more turns will be required in order to achieve a sufficient output voltage.

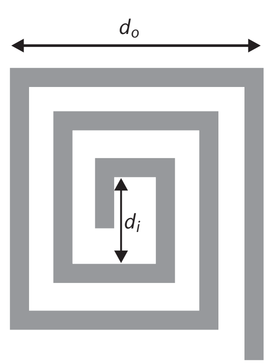

Figure 4.8 Planar coil dimensions.

The situation is further exacerbated by the consideration that true MEMS generators will not use bulk magnetic materials. Rather, the magnetic field will be provided by deposited magnetic materials, which are deposited by techniques such as sputtering and/or electroplating. The sputtering process physically removes material from the sample by firing ions at a target. The removed material then settles on the device substrate. Sputtering can be used for a wide range of materials and achieves good step coverage but limited film thicknesses. Electroplating involves passing an electric current between a metal salt solution and the device substrate, which acts as the cathode. This attracts the metal ions from the solution to the substrate, where they coat the exposed conductive locations. Electroplating is the preferred approach for depositing thicker films. The highest reported magnetic properties from deposited film properties are considerably lower than bulk-sintered rare earth magnets such as samarium-cobalt or neodymium-iron-boron. The highest coercivity and remanence reported for a 90-μm-thick deposited magnet are 160 kA/m and 0.5T [27], compared to around 800 kA/m and 1T for a bulk magnet. It is clear that electromagnetic generators fabricated using deposited magnets will have much lower electromagnetic damping, which when combined to planar microcoils makes MEMS electromagnetic generators unlikely to be successful on a commercial scale.

On the other hand, electrostatic transduction does not deteriorate as the devices shrink in size. The mechanical structures used in electrostatic generators are totally compatible with MEMS (and microelectronics) fabrication processes. MEMS electrostatic generators based upon interdigital finger electrodes, as shown in Figure 4.6, can be used to realize both in-plane gap varying and in-plane overlap varying type generators. Simulations have shown that scaling down dimensions, such as dielectric gaps, can increase the power density of the transduction process for both in-plane types of transducers [20].

Piezoelectric transduction sits somewhere in the middle ground. Good quality piezoelectric materials can be deposited using a range of processes such as sol-gel, sputtering, and screen printing. The piezoelectric properties of these films, however, are not as high as their bulk counterparts; typically they are approximately 50% of the bulk values. Such films can be readily incorporated onto the top surfaces of micromachined structures, but they are not compatible with microelectronics processes. The reduction in size of the piezoelectric element invariably means the capacitance of the piezoelectric transducer is reduced and therefore the piezoelectric damping coefficient given in (4.31) will also be reduced.

4.5 Operating Frequency Range

In applications where inertial-based generators are used to amplify low levels of vibration in the environment, they are designed to operate at a single frequency and will thus have a limited practical bandwidth over which energy can be harvested. Methods of addressing this limitation are currently being widely investigated. Approaches to increasing the frequency range include changing the resonant frequency (or tuning) of a single generator so that it matches the frequency of the ambient vibration at all times. Alternatively, the bandwidth of the generator can be widened by using an array of structures, each with a different resonant frequency, limiting the amplitude of the generator (a physical end stop), exploiting nonlinear (e.g., magnetic) springs, using bistable structures, or simply having a large inertial mass (i.e., large device size with a high degree of damping) [28].

4.5.1 Frequency Tuning

Frequency tuning can be classified as either continuous or intermittent. Continuous tuning includes any approach that is applied constantly to the generator. Intermittent tuning refers to tuning methods that can be periodically activated to adjust the generator frequency and, when the desired value is reached, the tuning mechanism is turned off. Intermittent tuning consumes less energy than continuous tuning and is therefore the preferred option. Roundy [29] concluded that generators using a continuous tuning mechanism can never produce a net increase in power output, as the power required to tune the resonant frequency will always exceed the increase in output power resulting from the frequency tuning. This, however, has been shown not to be the case [28], so both approaches are valid. The suitability of different tuning approaches depends upon the nature of the application. Different approaches can be analyzed by considering:

- The energy consumed by the tuning mechanism (should be minimized and must not exceed the energy produced by the generator);
- The range of frequencies achieved;
- The degree of frequency resolution;
- Its effect on damping levels over the entire operational frequency range (ideally no effect).

Tuning can be achieved by mechanical techniques, which change the mechanical properties of the structure in some manner, or by electrical tuning that exploits the influence of the electrical output load. Some mechanical approaches to frequency tuning will be considered next.

4.5.1.1 Mechanical Tuning

Most vibration energy-harvesting devices are based on a cantilever spring structure, which can be used to highlight the possibilities for mechanical tuning. The principles described are generally applicable to all types of mechanical resonator structures. Mechanical tuning can be achieved by:

- Altering the dimensions of the beam;
- Moving the center of gravity of the proof mass;
- Varying the spring stiffness;
- Straining the structure.

The resonant frequency of a cantilever with an inertial mass, m, at the free end is given by (4.36), where E is Young's modulus of the cantilever material, w, h, and l are the width, thickness, and length of the cantilever, respectively, and m_c is the mass of the cantilever [30]:

$$f_r = \frac{1}{2\pi}\sqrt{\frac{Ewh^3}{4l^3\left(m+0.24m_c\right)}} \tag{4.36}$$

The resonant frequency can be adjusted by altering the dimensions but, in practice, it is difficult to change the width and thickness of the beam. Changing the length is feasible and is suitable for intermittent tuning. This also allows a significant change in the resonant frequency, since this is inversely proportional to $l^{3/2}$. Modifying the length will require the cantilever base clamp to be released and reclamped at a new location along the length of the beam. This can, however, result in inconsistent damping losses through the clamp. Once adjusted, no additional power is required to maintain the new resonant frequency.

Equation (4.36) also shows that by varying the inertial mass, the resonant frequency will change. Once a device has been fabricated, however, it is often difficult to subsequently change m. The resonant frequency of a cantilever structure can be adjusted by moving the center of gravity of the inertial mass (i.e., by moving the mass along cantilever beam length).

Another approach is to vary the spring stiffness by placing an adjustable spring (k_a) in parallel with the mechanical spring (k_m). The effective spring constant of such device, k_{eff}, is given by $k_m + k_a$. The adjustable spring can be achieved by electrostatic, piezoelectric, magnetic, or thermal mechanisms. The majority of variable spring stiffness devices are continuously operated. Many examples of electrostatic tuning have been demonstrated in tuneable micromechanical resonators and have not necessarily been applied to vibration energy generators [31, 32]. Electrostatic generators can be tuned by adjusting the voltage on the plates, as discussed earlier. The presence of the inertial mass in an energy harvester reduces the tuning effectiveness and increases the power required for tuning. Piezoelectric tuning has been demonstrated by using two piezoelectric elements on the energy harvester spring element. One of these elements is used to harvest the energy, while the other element has a tuning bias applied to it [33]. Thermal techniques utilize the variation in Young's modulus of the spring material with the temperature or the thermal expansion of the material. This approach, however, requires relatively high powers and is thus avoided in practical energy harvesting applications.

Magnetics have been used to alter the spring stiffness, by applying external forces to the device. For example, the resonant frequency of a cantilever structure can be tuned by applying an axial load. The resonant frequency of a uniform cantilever with an associated buckling force, F_b, operating in the fundamental flexural mode with an axial load, F (positive for a tensile load and negative in the compressive case), is given by [30]:

$$f_{\text{new}} = f_{\text{noload}} \cdot \sqrt{1+\frac{F}{F_b}} \tag{4.37}$$

$$F_b = \frac{\pi^2 \cdot E \cdot w \cdot h^3}{48 \cdot l^2} \tag{4.38}$$

The change in resonant frequency of a cantilever with an applied axial load is shown in Figure 4.9. The compressive load produces a larger frequency shift than the tensile load. If a large tensile force is applied (i.e., much greater than the buckling load), the resonant frequency will approach that of a straight tensioned cable as the force associated with the tension in the cantilever dominates the beam stiffness. A magnetically applied axial load has been used for energy harvesting, which will be discussed in Section 4.6.7.

4.5.1.2 Electrical Tuning

The basic principle of electrical tuning is to shift the power spectrum of the generator by changing the electrical load and therefore the electrical damping. All reported electrically tuned generators to date are piezoelectric, and capacitive loads are varied in order to realize electrical tuning. A basic bimorph piezoelectric cantilever generator can be represented with an equivalent circuit, as shown in Figure 4.10 where L_m, R_m, and C_m represent the mass, damping, and spring in the mechanical part, respectively, and C_p is the capacitance of the piezoelectric layer. C_L and R_L are the capacitive and resistive load, respectively, and V is the voltage across the resistive load.

The transformer relates the mechanical domain to the electrical domain according to the model of the piezoelectric effect. For a piezoelectric bimorph, which operates in the 31 mode, ε is the dielectric constant of the piezoelectric material and E is the Young's modulus of the piezoelectric material; the transform ratio N is given by:

$$N = -d_{31}E \tag{4.39}$$

Equation (4.39) can be derived to show the mechanical dynamics of the system with electrical coupling [34]. Figure 4.11 compares the resonant frequencies and power output of electrically tuneable piezoelectric generators of different piezoelectric materials (see Table 4.3) with varying load capacitances. Piezoelectric materials with a higher Young's modulus, strain coefficient, and lower permittivity provide a

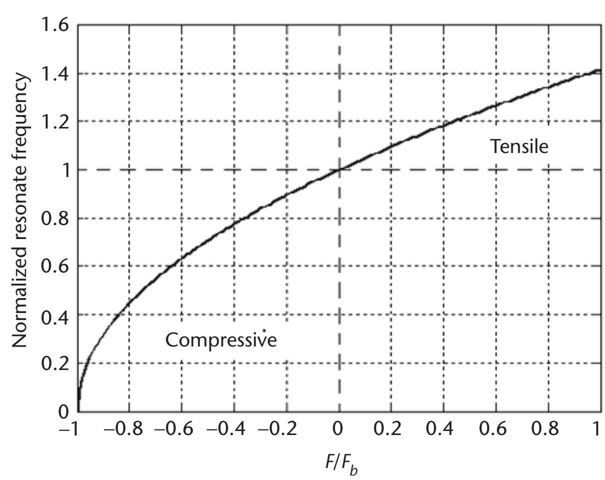

Figure 4.9 Normalized resonant frequency with a variation of axial loads.

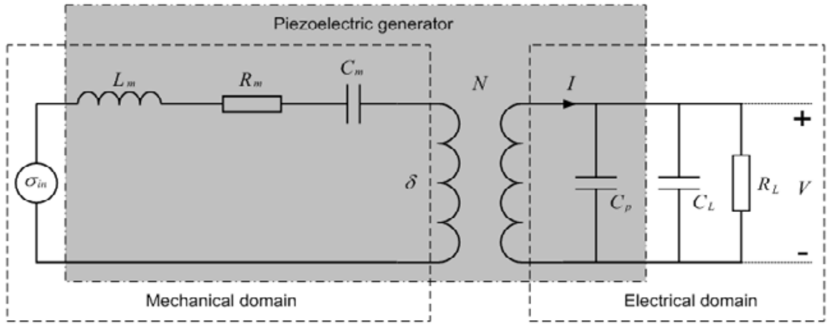

Figure 4.10 Equivalent circuit of the piezoelectric generator with capacitive and resistive loads.

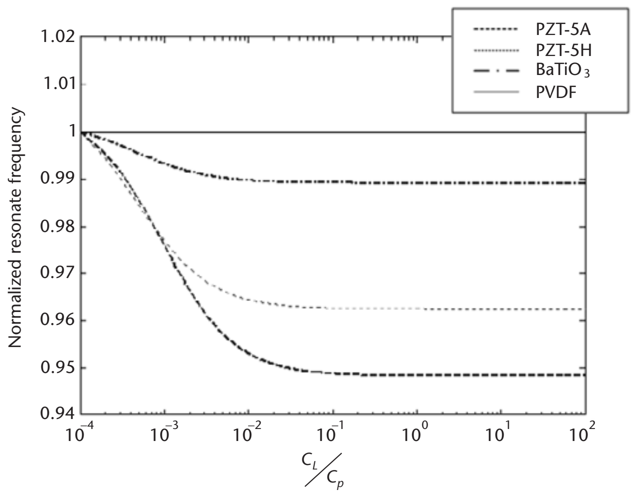

Figure 4.11 Change in the resonant frequency of a piezoelectric generator with different piezoelectric materials.

greater tuning range. Figure 4.11 shows that PZT-5A is the best of the four piezoelectric materials [28].

4.5.2 Strategies to Broaden the Bandwidth

Methods for broadening the effective bandwidth of a generator will now be discussed. The simplest approach is to have a heavily damped inertial generator. This means that the generator will have a lower peak power output but a broader frequency response. Increasing the size of the inertial mass can compensate for the reduction in peak power, but this increases the overall device size. If size is not a major constraint, this approach offers a relatively straightforward process to broaden the bandwidth.

An alternative approach requires the use of an array of individual generator structures, each having a different resonant frequency. By an appropriate design of the array, the individual resonant frequencies can be made to overlap sufficiently, thereby providing a broad operating frequency range made up of discrete individual resonances, as shown in Figure 4.12. The drawbacks of this approach relate to

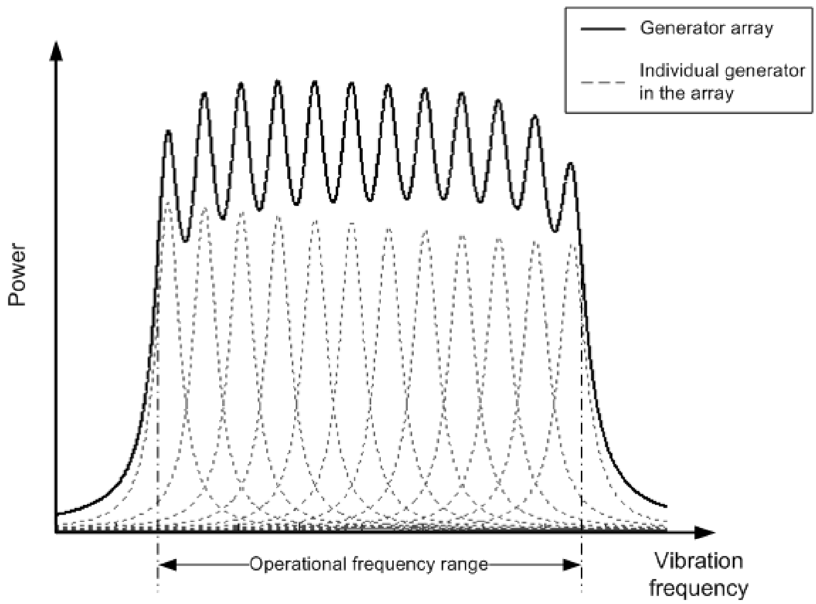

Figure 4.12 Power spectrum of a generator array.

the difficulties associated with reliably fabricating the whole array and the increase in total volume of the device.

A further method for increasing the bandwidth is to use a mechanical stopper to limit the amplitude of the generator as described by Soliman et al. [35]. The generator strikes the end stop, and this is found to increase the bandwidth of the device as the frequency slowly increases. It does not appear to work when the frequency is reduced.

Nonlinear generators utilize the spring stiffening or spring softening effect, whereby the resonant frequency varies as a function of amplitude (see Figure 4.13). The spring stiffening (or hard spring) effect means that the resonant frequency increases with amplitude. Such nonlinear devices have a larger bandwidth over which power can be harvested due to the shift in the resonance frequency. As with the mechanical stopper approach, a generator demonstrating a hard spring nonlinearity

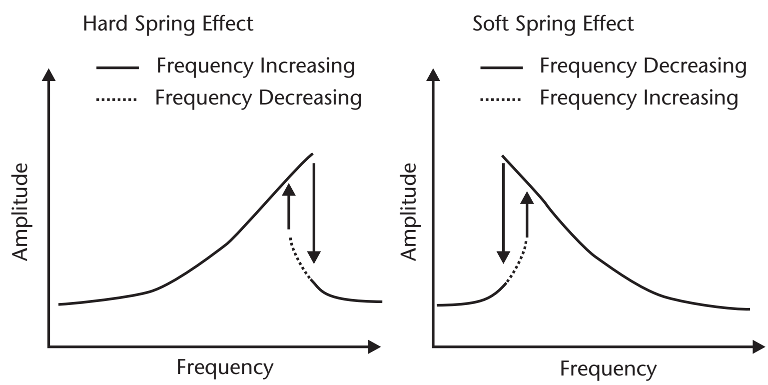

Figure 4.13 Hard and soft spring effects.

will only increase its bandwidth as the frequency increases and will have no effect for decreasing frequencies (vice versa for the soft spring effect).

Bistable structures (known as the snap-through mechanism) can also be used for vibration energy-harvesting applications. Such structures effectively snap backwards and forwards between two stable positions at any frequency, provided that the acceleration stimulus is of sufficient magnitude. The stored elastic energy has the effect of increasing the velocity of the structure for a given input excitation and analysis reveals that the amount of power harvested by a nonlinear device is $4/\pi$ greater than that of the tuned linear device operated out of resonance.

4.6 Rotary Generators

Perhaps the most common example of an energy harvesting application for rotary motion is that of the self-winding watch, which has been in existence commercially since the 1920s, but the concept is thought to date back to the eighteenth century. The concept is depicted in Figure 4.14, which depicts a semicircular mass pivoted around a central axis. The semicircular mass allows both linear and rotational excitation to produce rotating motion in the device. The rotating mass is generally coupled, via a gear train, to a miniature electromagnetic generator, which is often used to charge a secondary storage element such as a battery. The power requirements for watches are generally very low, typically less than 1 μW so the generator only needs to produce a few hundred milli-Joules each day from hand movement.

A review of possible approaches for energy harvesting from rotating motion has been presented by Yeatman [36], who detailed nonresonant devices, resonant devices, and gyroscopic generators. Nonresonant devices, similar to that in Figure 4.14, allow the mass to rotate freely in either direction without any constraints on the displacement. A resonant device requires a spring element (and damper) to couple the mass to the frame and limit the displacement of the mass. In a practical system, this can be a coil spring located underneath the mass.

Kinetron, based in the Netherlands, is a company that specializes in microkinetic energy generators. Kinetron makes a variety of rotary electromagnetic generators. An example of one of its devices is depicted in Figure 4.15.

The generator comprises a multipolar magnet that rotates inside a claw-shaped upper and lower stator. The number of claws is equal to the number of poles on the magnet. The device is similar to that used for automotive alternators. A rotating object is attached to the pinion, and the alternating voltage output is obtained from the coil.

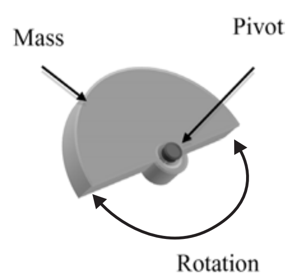

Figure 4.14 Schematic of the concept of a rotating generator as used in a self-winding watch.

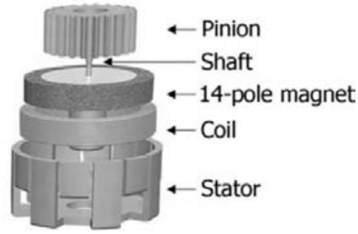

Figure 4.15 A schematic of a multipole miniature rotary generator. (Courtesy of Kinetron, NL.)

4.7 Example Devices

The overview of kinetic energy harvesters described in the literature has been selected to highlight many of the principles discussed previously. It is not meant to be an exhaustive list, as there are too many devices presented in the literature. Devices have been grouped by type: kinetic energy harvesters designed for use on humans, medium to large size conventionally fabricated generators for a variety of applications, MEMS generators, and tuneable/wide bandwidth generators.

4.7.1 Human-Powered Harvesters

Potable electronic devices worn or carried by the user are an obvious target application for energy harvesting. Numerous such devices have become ubiquitous in everyday life (e.g., mobile phone, PDAs, portable media players, and gaming devices), and there is a multitude of portable battery-powered military systems carried by the modern soldier. Powering these from the kinetic energy present in the environment typically requires harvesting energy from the movement of the user. In order for this to be successful, the user should not experience the effects of the harvesting process (i.e., the harvesting should have no notable effect on the activities or effort of the user). In practice, scavenging useful amounts of power without the user noticing is a very challenging problem.

The earliest successful example of a human powered device is the self-powered watch. The first version of this device was the self-winding pocket watch patented in 1770 by Swiss watch manufacturer Abraham-Louis Perrelet. The early versions used an inertial mass oscillating in a linear direction that harvested energy from the user walking. Wristwatches enabled the use of a rotating inertial mass with the movement of the user's arm and numerous examples of self-winding (or automatic) watches have been available during the twentieth century. These have more recently been developed to use the rotating inertial mass to generate electricity and recharge the watch battery. Watches are an ideal candidate for kinetic energy harvesting because of their low energy consumption. The generator to be relatively easily incorporated into the body of the watch and the user will typically not feel any effect from the power harvesting.

Most portable devices, however, require considerably more power than a watch. For example, the average power consumption of a mobile phone, depending upon usage, is of the order of 20 to 50 mW compared to 10 μW for a watch. Numerous different approaches to kinetic energy harvesting these higher levels of power have been investigated and some of the earliest devices were mounted in the shoe. Shoe-mounted piezoelectric materials were investigated at the Massachusetts Institution of Technology (MIT) in the 1990s [37]. An 8-layer laminated PVDF stack was used as an insole in a training shoe locating the piezoelectric materials at the sole. As the user walks, the PDVF stack is strained, producing a charge from the d_{31} mode, by the bending movement of the sole tacks. At around 1 step per second this produced an average power of 1.3 mW into a 250-kΩ load. An alternative approach involved the use of two Thunder TH-6R piezoelectric transducers fabricated into prestressed unimorphs that were positioned in the heel of a Navy work boot [38]. The unimorphs exhibited a naturally curving structure and two of these were mounted together producing a compressible dimorph. As the heel of the shoe hits the ground, the transducers are forced to deform and, as the weight is removed, the transducers spring back into their original shape. This approach produced an average of 8.4-mW power into a 500-kΩ load. The most successful show mounted harvester was demonstrated by researchers from SRI International, who placed flexible electroactive polymers, or dielectric elastomer, in the heel of a shoe. The elastomer material is highly compliant achieving 100% strain and was used in an electrostatic generator that produced 0.8J per step [39].

While harvesting energy from a shoe is an obvious and attractive option, it does highlight the negative effect of the user experiencing the energy harvesting. If too much energy is harvested, it alters the mechanical behavior of the shoe, making it harder to walk (akin to walking on a sandy beach). Other devices overcome this by exploiting the mechanics of the human body during walking. A knee brace developed through a collaboration between Simon Fraser University, University of Pittsburgh, and the University of Michigan extracts power during a particular phase of the gait cycle [40]. The forward movement of the lower half of the leg as it swings into position before being planted on the ground is normally controlled by the leg muscles. This energy is expended without actually achieving any positive work; it is more like a braking exercise. The generator, attached to the knee brace, only extracts energy during the forward swinging phase and the energy harvesting acts to slow the leg movement. The metabolic cost of harvesting the power is described as insignificant with 4.8W of electrical power requiring up to 26W of metabolic power [40].

A low-frequency inertial linear electromagnetic generator designed for human motion has, been described by von Buren et al. [41]. The design consists of a tubular translator, which moves vertically within a series of stator coils. The translator is made up of a number of cylindrical magnets separated by spacers, the dimensions of which were optimized using finite element analysis. The optimum resonant frequency of the generator varies from 5 Hz to 10 Hz depending upon the location and the wearer. The prototype has a total device volume of 30.4 cm^3 and produced an average power output of 35 μW when located just below a subject's knee.

The final type of the human-powered energy-harvesting approach covered here is the energy-harvesting backpack. This approach exploits the relative motion and forces between the wearer and the backpack with the inertial mass being provided by whatever is being carried. This is a logical opportunity for energy harvesting since the mass is already present, provided the energy can be extracted without significant extra effort from the wearer. Two versions have been realized. The first, developed by Rome et al. [42], comprises a backpack with the load supported on a separate frame using a linear bearing and a set of springs. The load is free to move vertically relative to the frame, and a rotary electric generator with a rack and pinion was used to generate electrical energy. This backpack generated a maximum power of approximately 7.37W but at a cost of 19.1-W metabolic power, resulting in increased fatigue for the user. The second approach uses piezoelectric straps made from PDVF to extract the energy [43]. Simulations predicted that the straps could generate 45.6 mW of power while carrying a 45-kg load at a walking speed of 2–3 mph, although this has not been experimentally verified.

4.7.2 Conventional Generators for Industrial and Transport Applications

As highlighted in Section 4.2, the frequency and amplitude spectra of industrial and transport applications are very different to the movements of a human. They are characterized by higher frequency and lower amplitudes. Most types of generators developed for industrial and transport applications employ a cantilever structure as the spring element, an inertial mass and some form of transduction mechanism. The simplest approach is to simply attach piezoelectric material to the top and bottom surfaces, and the electrical charge is generated from the d_{31} effect. Numerous examples of this type of generator have been described in the literature.

An early example is the tapered cantilever beam shown in Figure 4.16, developed at the University of Southampton [44–46]. The tapered shape gives a constant strain in the piezoelectric film along its length for a given displacement. The piezoelectric material was deposited by screen printing onto both sides of a 0.1-mm-thick hardened AISI 316 stainless steel substrate, which acts as the spring element. The screen printable piezoelectric material is based upon PZT-5H and Corning 7575 glass powder and a suitable thick-film vehicle to form a screen printable thixotropic paste [47]. The structure operated in its fundamental bending mode at a frequency of 80.1 Hz and produced up to 3 μW of power into an optimum resistive load of 333 kΩ. The thick-film printing process is an attractive, low-cost, batch fabrication process for depositing the piezoelectric material. In this case, however, the power output is limited by the reduced piezoelectric properties of the screen printable material. Furthermore, the inertial mass is manually attached, which negates some of the benefits of screen printing.

This device has more recently been improved upon during the initial investigations into the design of an energy harvester for aeronautical applications. The latest device exploits recent advances in the film properties [48] and the development of a screen printed inertial mass deposited using a paste based upon tungsten powders [49]. The improvements have increased the power output to 117 μW and an output voltage of 4.1V into optimum loads of 140 kΩ from a similarly sized cantilever

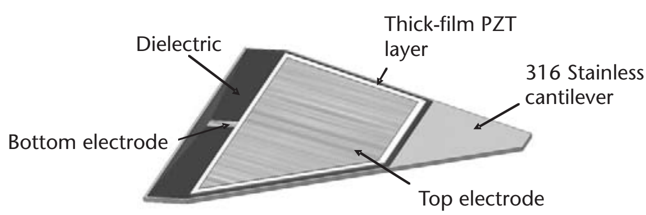

Figure 4.16 Tapered thick-film PZT generator (*After:* [45].)

beam driven at 0.7g and 70 Hz. The device is entirely screen printable, making it suitable for batch fabrication and therefore potentially relatively low cost.

Roundy and Wright demonstrated a composite bimorph piezoelectric cantilever (see Figure 4.17) fabricated by attaching a PZT-5A shim to each side of a steel center beam [1, 50]. The inertial mass was made from an alloy of tin and bismuth and was glued to the end of the generator, which exhibited a fundamental flexural mode at 120 Hz. The generator produced a maximum power output of nearly 80 μW into a 250-kΩ load resistance with 2.5 m/s^2 input acceleration. A later design used a 0.28-mm-thick layer of PZT-5H on a cantilever beam length of 11 mm and a tungsten proof mass, which produced 375 μW with an input acceleration of 2.5 m/s^2 at 120 Hz.

The University of Southampton has also developed electromagnetic generators based upon a cantilever spring arrangement. The earliest device comprised a pair of NdFeB magnets on a C-shaped core at the free end of the cantilever beam located either side of a coil wound from enameled copper wire [22]. This early device produced >1 mW from a volume of 240 mm^3 at a vibration frequency of 320 Hz. This generator was subsequently improved by adding a second pair of rare earth magnets forming the magnetic circuit shown in Figure 4.18 [51]. The improved flux linkage produced, for the same input vibration, more than twice the output voltage and hence more than four times the instantaneous power.

A smaller-scale version of this generator was developed during an EU-funded Framework 6 project VIBES, and a cutaway of the optimized device is shown in Figure 4.19. This device is probably the smallest version of the device that can be practically assembled using conventional fabrication techniques (i.e., non-MEMS) and achieves a total packaged device volume of 850 mm^3 [52]. The final version of this device uses a 50-μm-thick beryllium copper beam and a coil with 2,800 turns wound from a 12-μm diameter wire with a coil resistance of 2,323Ω. The generator was designed to enable a manual frequency adjustment by altering the cantilever beam length. The power output into a 15-kΩ resistive load was 50 μW at 1.1V from 0.6 ms^{-2} vibrations at 50 Hz. This device was demonstrated powering an autonomous wireless condition monitoring sensor system (ACMS) shown in Figure 4.20 [53]. The generator was coupled to a voltage step-up circuit, the output of which was used to charge a 0.047F capacitor. A low-power microcontroller was used to monitor the capacitor voltage and turn on the sensor system when sufficient energy has been stored. The operating duty cycle of the system varies depending upon the magnitude of the vibrations driving the generator. The sensor system was designed to sample external vibration data and transmit the peak acceleration

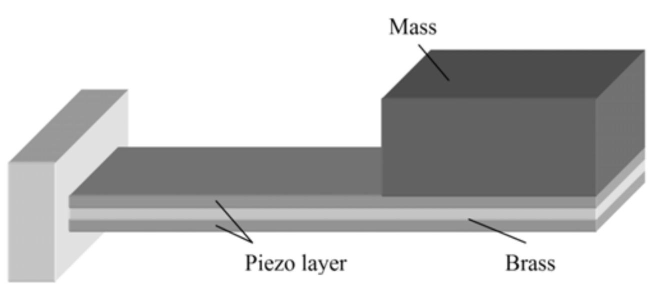

Figure 4.17 Cantilever piezoelectric generator developed by Roundy et al. [50].

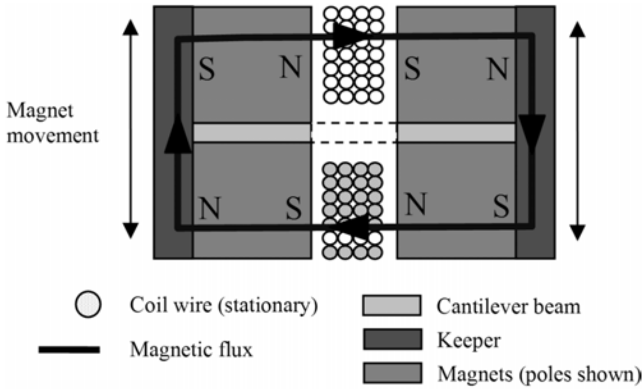

Figure 4.18 Cross-section through the four-magnet arrangement.

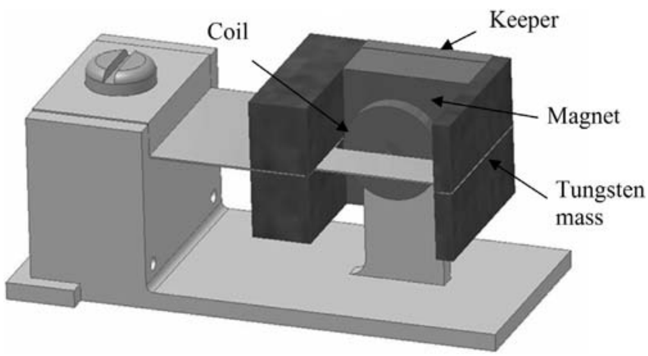

Figure 4.19 Cutaway (two magnets and keeper removed) of the mini-sized generator design.

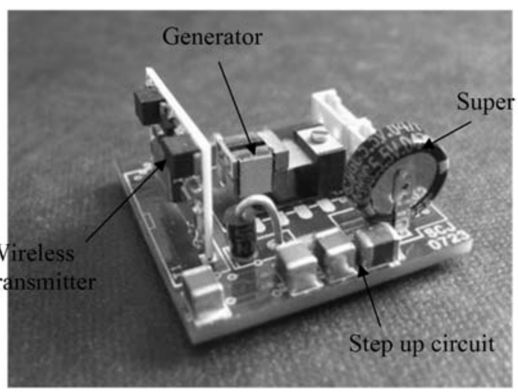

Figure 4.20 Self-powered AWCS. Microcontroller circuit on underside.

data. The system can function from vibration levels as low as 0.2 ms^{-2} transmitting data every 12 minutes, while at 0.85 ms^{-2} the transmission rate is every 3 seconds.

Several companies have begun marketing vibration energy harvesters with PMG Perpetuum Ltd. in the United Kingdom offering the highest performance device available to date [54]. They have developed a series of vibration-powered electromagnetic generators, the most popular variant being the PMG17 tuned to work at 100 or 120 Hz. The output of this device ranges from 1 mW at 0.25 ms^{-2} to over 50 mW at 1*g* (9.81 ms^{-2}). It is interesting to note that the device has a variable damping level, which increases with vibration amplitude as shown in Figure 4.21. This means that the bandwidth of the device widens with acceleration, allowing it to harvest power over a broader spectrum, making it a very useful practical device. It is also an intrinsically safe, ATEX-certified generator and its mean time to failure (MTTF) is estimated to be 440 years, which means that only 2% of devices are expected fail in 10 years. Its volume, including power conditioning electronics, is 130 cm^3 in size and the moving mass is 450g.

Ferro Solutions Inc., Mide Technology Corporation, and AdaptivEnergy offer other commercially available generators. None of these are as well characterized or documented as the PMG-17. The Ferro Solutions VEH-360 is another electromagnetic generator that, coupled with its evaluation circuit, produces 1 mW from 0.5 ms^{-2} at 60 Hz. Its volume, including the power conditioning electronics and space for end user electronics and sensors, is 165 cm^3. The Mide Technology Corporation offers a piezoelectric energy-harvesting device called the Volture PEH25W. It is a cantilever piezoelectric device, based upon Quickpack laminated piezoelectric elements, enclosed in a case with a volume of 40.3 cm^3. The generator produces approximately 1 mW of electrical power from an acceleration of 3.5 ms^{-2} at 100 Hz, which is much less than the devices described above. AdaptivEnergy's Joule Thief is another piezoelectric generator based upon a tapered cantilever beam. The power output is approximately 250 mW from 2 ms^{-2} at 60 Hz from a packaged generator of 35 cm^3 including power conditioning electronics.

One interesting development is the coupling of a cantilevered piezoelectric generator to a radioactive source to produce an energy harvester that does not rely on environmental vibrations [55, 56]. Radiated β particles electrostatically charge a conductive plate on the underside of a piezoelectric unimorphs, causing an electrostatic field to build up. This field attracts the beam to the source until contact is made, the field is dissipated, and the beam is released to vibrate at its natural frequency. The piezoelectric film on the beam harvests the kinetic energy stored

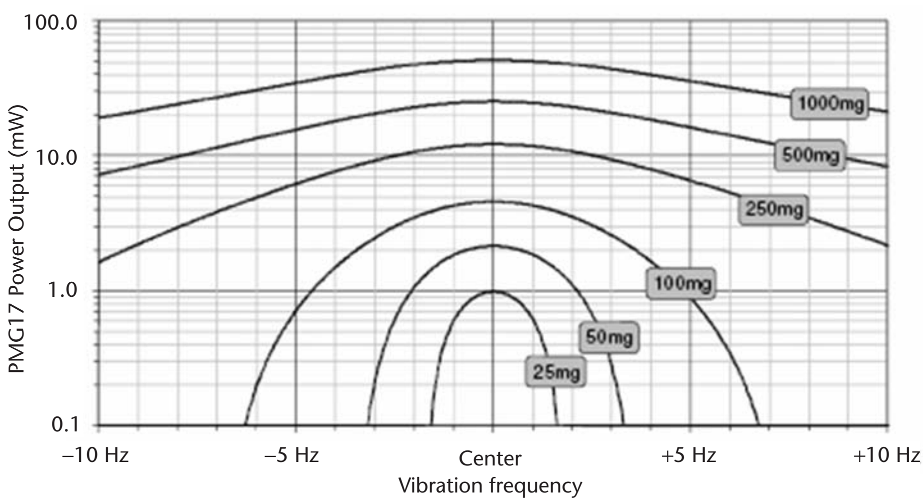

Figure 4.21 PMG17 power output versus acceleration.

in the cantilever. It produces a periodic power burst, which can be stored or used to perform some limited functionality. While the energy supplied with each bust is limited, the device will function for the duration of the radiation source, which means that it is an energy-harvesting technique that can be embedded anywhere and used for many years.

4.7.3 Microscale Generators

Despite the inherent reduction in power output as generators shrink in size, there has been considerable interest in researching MEMS implementations. All transduction types have been demonstrated, and many of the issues discussed in Section 4.4.4 have been highlighted.

Piezoelectric generators are probably the easiest to realize in MEMS. A micromachined silicon cantilever mass geometry was developed during the VIBES project. The device shown in Figure 4.22 consists of an inertial mass, Deep Reactive Ion Etched (DRIE), from an SOI wafer with a 525-μm-thick handle wafer. The 1,200-μm-long, 800-μm-wide supporting cantilever is fabricated from the 5-μm-thick top silicon layer. The cantilever has been coated with a 1-μm-thick layer of aluminium nitride (AlN), which produces 0.8 mW from 2g vibrations at 1,495 Hz [57]. A similar structure with sol gel PZT was also demonstrated during the VIBES project and this produced 1 μW at 2g vibrations at 885 Hz. Interestingly, IDT electrodes, as described in Section 4.4.1, were also evaluated with the PZT film. The electrodes had 4-μm-wide electrodes separated by 6 μm, and this arrangement produced 1.4 mW at the same excitation as in the planar electrodes case (i.e., a 40% improvement [58]). Jeon et al. also fabricated a MEMS piezoelectric generator with sol-gel PZT and an IDT electrode [59]. This device delivered 1.01 μW at 10 ms^{-2} acceleration at an impractically high fundamental resonant frequency of 13.9 kHz.

The earliest microscale device was an electromagnetic generator shown in Figure 4.23 reported by researchers from the University of Sheffield, United Kingdom [60–62]. The circular polyimide membrane, 7-μm-thick spring and inertial mass, was fabricated in an upper gallium arsenide (GaAs) wafer. The planar integrated coil made from a 2.5-μm-thick gold (Au) with 13 turns of a 20-μm line width and 5-μm spacing, was located on a lower wafer. The inertial mass was a vertically

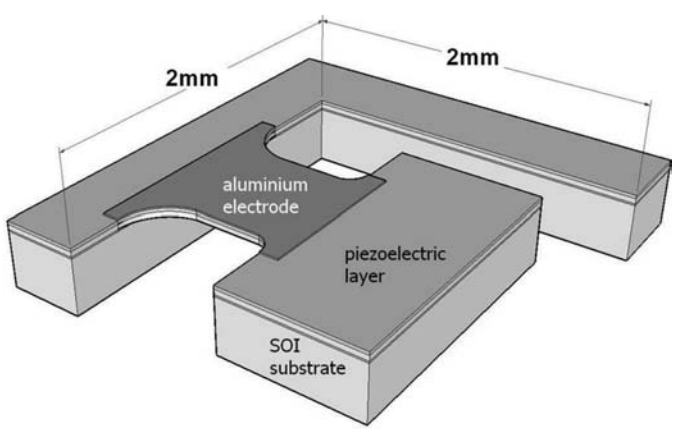

Figure 4.22 Micromachined silicon cantilever mass piezoelectric generator.

polarized 1 mm × 1 mm × 0.3 mm samarium-cobalt magnet of mass 2.4×10^{-3} kg. The two wafers were bonded together using silver epoxy, and the overall size of the electromagnetic transducer is around 5 mm × 5 mm × 1 mm. The mass moves vertically out of phase with the generator housing producing 0.3 μW at 4.4 kHz and an acceleration of 382 m/s^2. A very similar generator was demonstrated by researchers from the University of Barcelona [63]. They bonded a neodymium iron boron (NdFeB) magnet to a polyimide membrane attached to a 1-mm-thick silicon (Si) (100) wafer, which contained the 52 turns coil with 15-μm-thick, 20-μm-wide electroplated copper tracks (coil resistance of 100 ohms). Results demonstrated that 55 μW at 380 Hz from an acceleration of 29 m/s^2 could be obtained.

A similar electromagnetic circuit, described by Wang et al. [64], combines an electroplated copper planar spring with an integrated coil. A NeFeB permanent magnet was manually bonded to the center of the spring, which had been formed on a silicon substrate with the silicon being etched away in KOH to leave the free-standing copper structure. The resonant frequencies were around 55, 121, and 122 Hz. A copper electroplated two-layer coil was fabricated on a glass wafer, and the two wafers were simply glued together to assemble the device, which produced around 60 mV generated from 14.7 m/s^2.

The devices described above all exhibit very low levels of electromagnetic coupling. One method of increasing the electromagnetic coupling by increasing the number of coil turns has been demonstrated using low temperature cofired ceramics (LTCC) to fabricate a multilayer screen-printed coil [65]. The conductive tracks were simply fabricated on the layers using conventional screen printing. The layers are then stacked and compressed to form a laminated structure, which is finally fired in a furnace resulting in a rigid, multilayer device, which comprised a total of 576 turns. Practical results for the generator were not given. In another attempt to overcome the limitations of integrated coils, Beeby et al. integrated a traditionally wound coil within a micromachined silicon resonant structure [66]. A 600-turn coil wound with a 25-mm-thick enameled copper wire wound coil was located in a cantilevered silicon paddle structure shown in Figure 4.24. The silicon/coil paddle was designed to vibrate laterally in the plane of the paddle coil layer between four NeFeB magnets mounted in two Perspex chips. The device had a resonant frequency of 9.5 kHz and generated 21 nW of electrical power from an acceleration level of 1.92 m/s^2. Locating a coil on the moving part of the structure was found not to

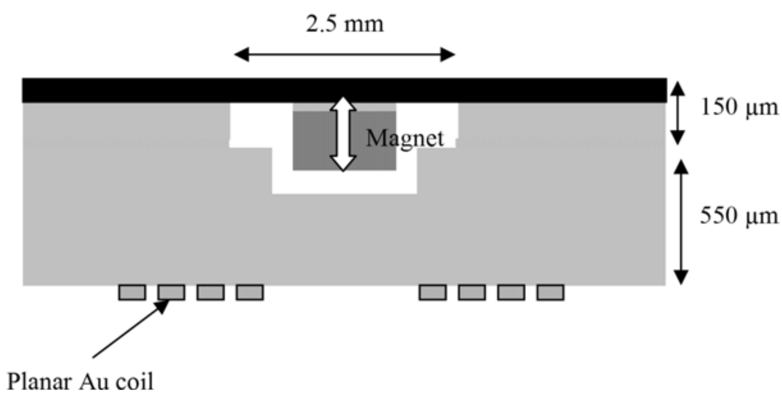

Figure 4.23 Cross-section of the electromagnetic generator proposed by Williams et al. [61].

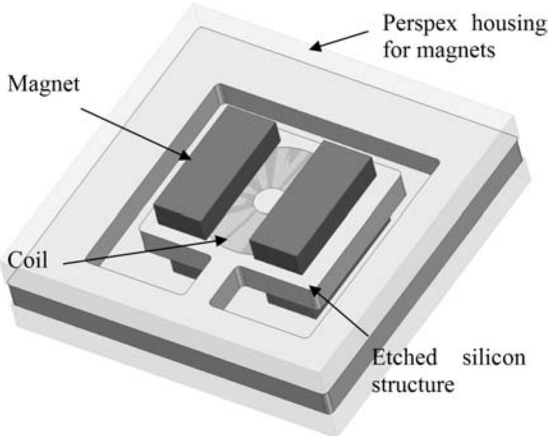

Figure 4.24 A silicon electromagnetic generator with a discrete coil. (*After:* [66].)

be the best approach; it resulted in the low-power output due to the frictional losses from the loose wires that come out from the coil to the edge of device.

Electrostatic transduction is the best suited to MEMS technologies; however, while there are many papers with simulated results, there are not too many published experimental results. Arakawa [67] presented an in-plane, overlap-varying, voltage-constrained, variable capacitor polarized with a fluorocarbon polymer electret. For a 1-mm displacement amplitude at 10 Hz, corresponding to an acceleration of 3.94 ms^{-2}, a 20 mm × 20 mm × 2 mm two-wafer glass device produced 6 μW with a 200-V peak-to-peak output into an optimized external load of 108Ω.

Mitcheson et al. [68, 69] described a nonresonant MEMS electrostatic generator designed to operate when the amplitude of the external motion exceeds the maximum internal displacement of the proof mass (e.g., human applications). The device has four phases of operation: prime, wait, flight, and discharge with the 10^{-4} kg proof mass only moving when the acceleration is at a maximum. In the priming phase the proof mass is located against the charging studs, being held in place by the externally supplied priming voltage of the order of 100V. The priming voltage is chosen so that the holding force is just below the inertial force produced by the maximum acceleration in the chosen application. The inertial mass waits until the inertial force exceeds the holding force, at which point the flight phase begins with the mass separating from the charging plate and moving towards the discharge contacts. The mass holds a constant charge, and as the gap separation increases, the voltage across the capacitor rises until a maximum is reached when the moving mass reaches the discharging contacts. The stored energy increases by the ratio of the initial to the final capacitance, and therefore the initial gap between the mass and the counterelectrode is minimized, which also has the effect of decreasing the required priming voltage. The generator was fabricated from a glass-silicon-glass assembly and consisted of a silicon-proof mass attached to a silicon frame by a polyimide suspension. The bottom plate contained the counterelectrode and discharging studs and the top plate contained the discharge contacts. The overall generator is 20 mm × 25 mm × 1.5 mm thick with a charge capacitor spacing of 6 μm, giving a capacitance of 150 pF while the discharge capacitance is 5.5 pF. The generator produces a net power of 120 nJ per cycle for accelerations of 50 ms^{-2}. The power is limited by the tilting motion of the inertial mass and parasitic capacitances.

4.7.4 Tuneable Generators

This section describes devices that have been designed to maximize the operating bandwidth over which the device can operate. The following generators demonstrate the principles discussed in Section 4.5.

An example of changing dimensions in order to vary the resonant frequency was described in a patent by Gieras et al. [70], where a mechanism was implemented that changes the effective length of a cantilever-based generator. The patent describes a electromagnetic device with a "slider" connected to a linear actuator, which moves the slider along the cantilever, thereby adjusting the effective length.

A technique to move the center of gravity of the inertial mass was demonstrated by Wu et al. [71], who reported a cantilever-based piezoelectric generator having a fixed proof mass with a movable screw within it. The position of the center of gravity of the proof mass could be adjusted by changing the position of the movable screw, which was then fixed in place by a fastening stud (see Figure 4.25). The fixed mass is 10 mm × 12 mm × 38 mm, the movable mass is an M6 screw that is 30-mm long, and the resonant frequency of the device can be adjusted from 180 Hz to 130 Hz. This is an example of manual tuning, but it could be automated.

Electrostatic tuning has been demonstrated in a MEMS resonator by Adam [72]. The resonator is not designed for energy harvesting but demonstrates the feasibility of electrostatic tuning with a resonant frequency of 25 kHz and a tuning range from 7.7% to 146%. It uses the single comb structure shown in Figure 4.26 and a tuning voltage ranging from 0V to 50V. This is an example of continuous tuning and compatible with energy harvesters employing comb drive structures. It should be noted, however, that the greater the mass of the resonator as would typically be the case in an energy harvester, the less the frequency range can be adjusted.

The use of external forces, applied by external magnets, was demonstrated by Challa et al., who reported an intermittently tuned piezoelectric microgenerator [73]. The device has an untuned frequency of 26 Hz and a frequency range of 22–32 Hz. The tuning was realized by applying an attractive magnetic force perpendicular to the cantilever generator, as shown in Figure 4.27. The magnitude of the force can be altered by varying the distance between the two sets of tuning magnets, but it should be noted that the tuning mechanism had the unwanted side effect of also varying the parasitic damping over the frequency range.

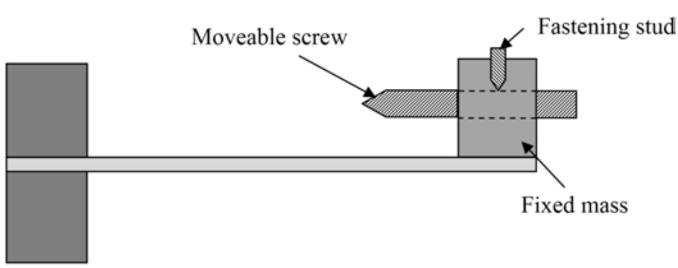

Figure 4.25 Piezoelectric cantilever prototype with a movable mass. (*After:* [71].)

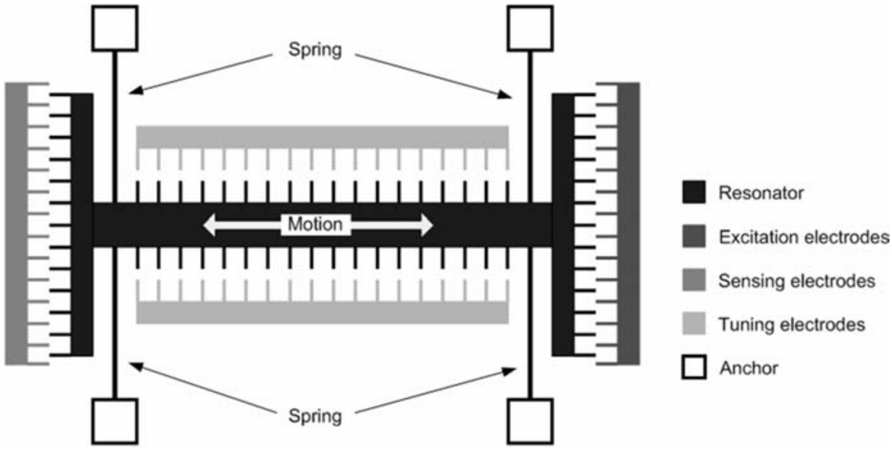

Figure 4.26 Schematic diagram of a single comb structure. (*After:* [72].)

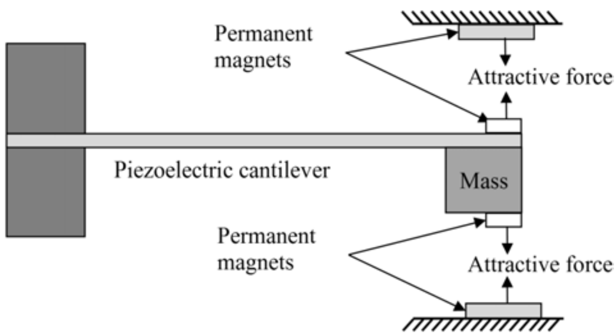

Figure 4.27 Schematic of the tuneable piezoelectric generator. (*After:* [73].)

Another approach using external magnets, in this case to apply an axial load to a cantilever generator, was reported by Zhu et al. [74]. The tuning configuration is shown in Figure 4.28. The axial force is applied by two magnets, one located on the end of the cantilever and one aligned next to it. The tuning magnets can be arranged to provide either a compressive force (magnets repelling) or tensile force (magnets attracting). The force is altered by varying the distance between the two tuning magnets using a linear actuator. The tuning range of the microgenerator was 67.6–98 Hz by changing the distance between two tuning magnets from 5 mm to 1.2 mm. More importantly, when used in the tensile mode, damping levels are unaffected.

Electrical tuning was demonstrated by Wu et al. [75] with a piezoelectric bimorph cantilever. One piezoelectric layer was used for frequency tuning while the other layer was used for energy harvesting. Varying the load capacitance achieves a frequency variation of 3 Hz. varying from 91.5 Hz to 94.5 Hz. A similar approach was demonstrated by Charnegie [33], and again one piezoelectric layer was used for energy harvesting, while the other is used for frequency tuning, as shown in Figure 4.29. Test results show that the resonant frequency can be tuned by 4 Hz with an untuned frequency of 350 Hz by adjusting the load capacitance from 0 to 10 mF if only one layer is used for frequency tuning. In this case, the output power

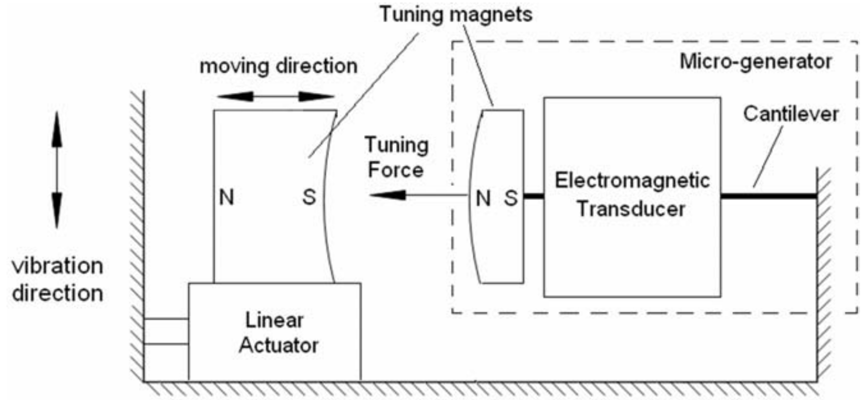

Figure 4.28 Schematic diagram of the tuning mechanism [74].

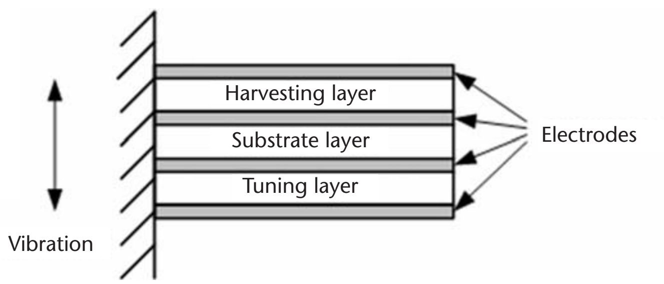

Figure 4.29 Piezoelectric bimorph used for electrical frequency tuning.

remains constant, irrespective of the load conditions. If both layers are used for frequency tuning, then the same range of load capacitance was found to achieve a tuning range of 6.5 Hz. In this case, however, the output power decreased with increasing load capacitance.

Wide bandwidth devices, using multiple generator structures, were demonstrated by Sari et al. [76] and are shown in Figure 4.30. The micromachined electromagnetic generator contains a series of cantilevers with gradually increasing lengths and hence reducing resonant frequencies. The cantilevers are distributed in a 12.5 mm × 14 mm area. The resulting frequency range of the generator is 3.3–3.6 kHz. The cantilevers contain thin-film coils with a central magnet, and the generator produces around 0.5 μW over this range.

4.8 Conclusions and Future Possibilities

The field of vibration energy harvesting continues to expand, both in terms of numbers of academic research papers and also new commercial systems in the market place. A key application area is for powering nodes in wireless sensor networks. The level of ambient vibration only needs to be a few milli *g* to generate a useful amount of electrical power (a few milliwatts). There are three main approaches that can be used to implement a vibration-powered generator. Each of the technologies

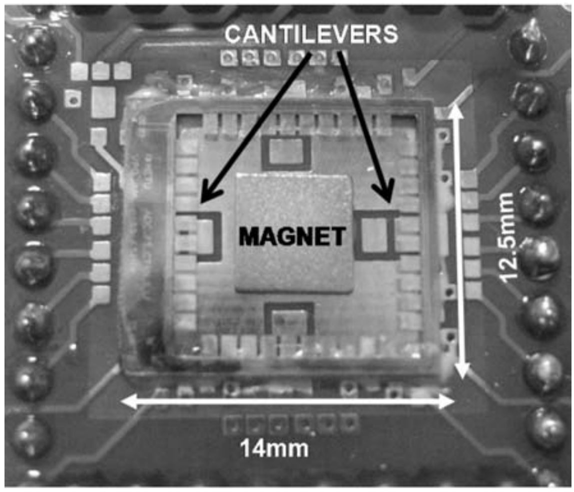

Figure 4.30 Photograph of a wideband electromagnetic generator [76].

described in this chapter have their own advantages and disadvantages, which are summarized in the following sections.

4.8.1 Piezoelectric Generators

Piezoelectric materials offer a simple approach to kinetic energy harvesting, whereby structural vibrations are directly converted into a voltage output by using an appropriate type of piezoelectric material and associated electrodes. Piezoelectric generators are relatively easy to fabricate and can be used in both direct force and impact-coupled harvesting applications. There is a wide choice of piezoelectric materials available for different application environments. One particular advantage of this transduction principle is that piezoelectrics are well suited to microengineering and several processes exist for depositing piezoelectric films in thin- and thick-film forms. Piezoelectric generators are characterized by their ability to produce a relatively high voltage output but only at low electrical currents. The output impedance of piezoelectric generators is typically very high (> 100 kΩ).

The piezoelectric materials need to be strained directly, and therefore their mechanical properties will limit their overall performance and lifetime. Additionally, the transduction efficiency is ultimately limited by the piezoelectric properties of chosen material.

4.8.2 Electromagnetic Generators

Electromagnetic generators are based on a well-established technique of electrical power generation. The effect has been used for many years in a variety of electrical generators. There is a wide variety of spring/mass configurations, which can be used with various types of materials that are well suited and proven in cyclically stressed applications. Comparatively high output current levels are achievable at the expense of low voltages (typically <1V). High-performance bulk magnets and multiturn, macroscale coils are readily available.

Wafer-scale systems, however, are somewhat difficult to achieve because of the relatively poor properties of planar magnets, the limitations on the number of turns achievable with planar coils, and the restricted amplitude of vibration (hence magnet/coil velocity). Inevitably, there are also problems associated with mass production and assembly and the alignment of submillimeter-scale electromagnetic systems.

4.8.3 Electrostatic Generators

A major advantage of the electrostatic device is that they are realizable as microengineered devices. There is also a wealth of processing know-how for the realization of both in-plane and out-of-plane capacitor arrangements. The energy density of the generator can be increased by decreasing the capacitor spacing, thereby facilitating miniaturization. Energy density can, however, be decreased by reducing the capacitor surface area.

Unfortunately, one of the drawbacks of electrostatic generators is that they require an initial polarizing voltage (or charge). This is not an issue for applications that use the generator to charge a battery or other type of secondary storage component, as an initialization voltage will be available. Electrostatic generators sometimes utilize electrets to provide the initial charge, and these are capable of storing the charge for many years. The output impedance of the devices is often very high, which makes them less suitable as a power supply. The output voltages produced by the devices are relatively high (>100V), and, as with piezoelectric generators, they often have a limited current-supplying capability. Parasitic capacitances within the structure can sometimes lead to poor generator efficiencies. There is also the risk of capacitor electrodes short-circuiting or suffering from static friction (stiction) in wafer-scale implementations.

4.8.4 Summary

The three fundamental techniques used for harvesting energy from ambient vibrations have been shown to be capable of generating typical output power levels in the range of a few microwatts to several milliwatts. Only a few years ago, such power levels were considered unusable and the concept of vibration harvesting would have been ridiculed. Modern-day VLSI circuit designs, however, are being built with ultralow-power consumption in mind, and many commercial circuits can now be used with many types of energy-harvesting solutions. Consider, for example, the advances made with the electronic calculator, whose early form required several AA-sized cells, but can now be powered off a single solar cell.

We earlier listed a number of scenarios where vibration-powered, wireless sensor systems can be used. Many academic and industrial research groups across the world are assessing possible uses in ambient intelligence, medical implants, and smart clothing. Wireless, battery-less industrial condition monitoring systems have already been demonstrated, as will be discussed in Chapter 8.

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CHAPTER 5

Thermoelectric Energy Harvesting

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5.1 Introduction

Thermal energy is ubiquitous and found in almost any environment. A vast amount fades away unused. Typical examples of those include waste heat from vehicle exhausts and radiators, geothermal from undergrounds, cooling water of steel plants and other industrial processes, and temperature difference between the surface and the bottom of oceans. Thermoelectric devices, which are capable of converting heat into electricity, have potential for thermal energy harvesting. Success in this endeavor will have a wide implication in both energy supply and environment. Thermoelectric devices can help to improve energy efficiency and reduce CO_2 emissions of fossil fuel systems through waste heat recovery. They can also be integrated into autonomous systems to enhance the capability and lifetime of self-power by harvesting thermal energy from their environment, or even charging wireless sensors and mobile devices from human body heat.

This chapter begins with a brief introduction to the thermoelectric effects and the principles of thermoelectric devices, followed by a discussion of the key factors and challenges that are crucial to the design and fabrication of suitable thermoelectric devices for energy harvesting applications. The purpose is to provide basic design principles of thermoelectric generators, which facilitate realistic estimates of the power level that could be produced and the geometric requirements that would be involved. The understanding of the unique characteristics, as well as drawbacks, of thermoelectric generators holds the key to successful exploitation of the technology for energy harvesting. A number of potential applications are presented and their future prospects will be discussed.

5.2 Principles of Thermoelectric Devices

Thermoelectric phenomena describe the interaction and conversion between heat and electricity in solids, which can be summarized by three thermoelectric effects.

Based on those effects, thermoelectric devices have been developed and employed for power generation, refrigeration, and temperature sensing. This section outlines the operating principles and relevant theories for design and evaluation of thermoelectric devices.

5.2.1 Thermoelectric Effects

Thermoelectric effects include: the Seebeck effect, the Peltier effect, and the Thomson effect. These three effects are related by the Kelvin relationships. They are the foundation for thermoelectrics [1–5].

5.2.1.1 The Seebeck Effect

The Seebeck effect describes a phenomenon that produces a voltage by a temperature gradient. Figure 5.1 shows a circuit that consists of two dissimilar metals or semiconductors joining together. By applying a temperature difference across two junctions, a voltage V will be generated in the circuit,

$$V = \alpha_{ab}\Delta T \tag{5.1}$$

where $\Delta T = (T_H - T_C)$ is the temperature difference across the two junctions and α_{ab} is referred to as the Seebeck coefficient. For the majority of metals or metal alloys, α_{ab} remains approximately a constant over a certain temperature range, which can be determined by

$$\alpha_{ab} \cong \frac{V}{\Delta T} \quad (\text{at } \Delta T \to 0) \tag{5.2}$$

α_{ab} is a relative quantity that is associated with the properties of materials a and b. Therefore, it is called the *relative* Seebeck coefficient. The unit for α_{ab} is $\mathrm{V \cdot K^{-1}}$ and its value can be positive or negative depending on the type of conducting charges [2, 3]. The Seebeck coefficient in metals and metal alloys is usually very small, in a range of a few to tens of $\mu\mathrm{V \cdot K^{-1}}$. It is much larger in semiconductors, which can be up to 1,000 $\mu\mathrm{V \cdot K^{-1}}$. A thermoelectric device operated in the Seebeck mode converts heat into electricity and is a generator.

5.2.1.2 The Peltier Effect

Instead of applying a temperature difference across the junctions in Figure 5.1, we can apply a voltage to the circuit as shown in Figure 5.2. Electric current that flows around the circuit will result in heat absorption at one junction and heat dissipation at another due to thermal transport by moving electrons. Consequently, one junction will become cold and the other junction will become hot. Furthermore, if the electric current changes the direction, the heat absorption and dissipation at the two junctions will be reversed. The amount of heat removed per unit time from one junction to another junction is given by

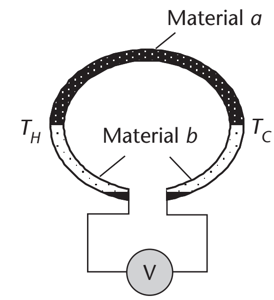

Figure 5.1 The Seebeck effect: a voltage generated by the temperature difference across the junctions.

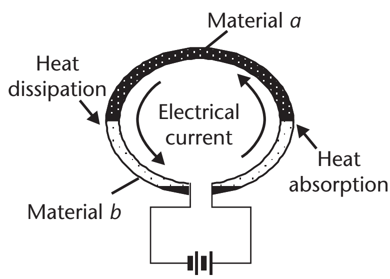

Figure 5.2 The Peltier effect: heat absorption (or dissipation) at junctions due to electrical current.

$$\dot{Q} = \pi_{ab} \cdot I \tag{5.3}$$

where I is the electric current in the circuit and π_{ab} is referred to as the Peltier coefficient,

$$\pi_{ab} = \frac{\dot{Q}}{I} \quad (\text{at } \Delta T = 0) \tag{5.4}$$

It is to be noted that when using (5.4) to determine the Peltier coefficient, the quantity $\dot{Q}$ must be measured under the isothermal condition (i.e., both junctions are kept at the same temperature). Similarly, the Peltier coefficient is a relative quantity. The unit for the Peltier coefficient is $W \cdot I^{-1}$, which is equivalent to volts. A thermoelectric device operated in the Peltier mode pumps heat from one junction to another and is a refrigerator.

5.2.1.3 The Thomson Effect

Both the Seebeck and Peltier effects can only be observed in a system that consists of at least two different materials. However, the absorption (or dissipation) of heat along a single material can occur when the material is subjected to a temperature difference and electric current simultaneously as shown in Figure 5.3. The total heat absorption (or dissipation) is given by

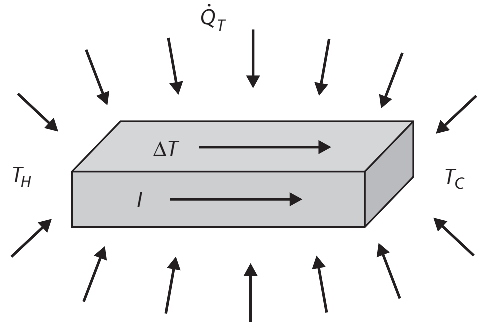

Figure 5.3 The Thomson effect: heat absorption (or dissipation) by a material when subjected to temperature difference and electrical current.

$$\dot{Q}_T = \beta I \Delta T \tag{5.5}$$

where β is referred to as the Thomson coefficient. The unit for the Thomson effect is $W \cdot I^{-1}K^{-1}$, which is equivalent to $V \cdot K^{-1}$. To date, no thermoelectric device is designed to operate in the Thomson mode. However, this effect exists in all thermoelectric conversion devices and its influence can be significant when temperature difference across the device is large [6].

5.2.1.4 The Kelvin Relationships

The three thermoelectric coefficients are not independent of each other, but are related by the Kelvin relationships [1–3],

$$\pi_{ab} = \alpha_{ab} T \tag{5.6}$$

$$\frac{d\alpha_{ab}}{dT} = \frac{\beta_a - \beta_b}{T} \tag{5.7}$$

Equation (5.6) describes the link between the Seebeck and Peltier effect. It indicates that the materials that are suitable for thermoelectric power generation are also suitable for thermoelectric refrigeration. They are, in fact, the reversible effects [7]. In addition, (5.6) provides a simple way to determine the Peltier coefficient. Direct determination of the Peltier coefficient requires the accurate measurements of heat dissipation and absorption at junctions, which are more difficult than the measurements of voltages and temperatures required in the Seebeck measurement.

Equation (5.7) describes the link between the Seebeck and Thomson effect. This relationship enables the definition to be derived for the Seebeck coefficient of a single material ($\alpha = \int(\beta/T)dT$)—the *absolute* Seebeck coefficient. Furthermore, it can be shown that the Seebeck coefficient of the junctions between two materials a and b is the same as the difference between the two absolute coefficients (i.e., $\alpha_{ab} = \alpha_a - \alpha_b$) [8]. Experimental data shows that the lead possesses a very small Seebeck coefficient and has been used frequently as the "zero" reference [1, 8].

Another important implication of the Kelvin relationships is that all three thermoelectric effects exist simultaneously in any thermoelectric device when operated with an electrical current and temperature difference, no matter whether it is in the generating or refrigerating mode.

5.2.2 Thermoelectric Devices

Thermoelectric devices can convert heat into electricity when operated in the Seebeck mode (generation) or pump heat from one junction to another when operated in the Peltier mode (refrigeration). Both modes have been found in many applications with unique characteristics [4, 5]. However, this chapter is concerned with the application of thermoelectric devices to energy harvesting and thus the discussions will only focus on the structures, theories, and performance parameters of thermoelectric generators.

5.2.2.1 Basic Structure

Modern thermoelectric converters are fabricated using semiconductors that offer higher conversion efficiency and larger power output than metal alloys. Since semiconductors are mostly nonductile crystalline solids, the implementation of the thermocouple junction based on Figure 5.1 is rather difficult. A suitable thermocouple structure for semiconductors is shown in Figure 5.4. It consists of an n-type and a p-type semiconductor thermoelement connected electrically in series by a conducting strip (usually copper or aluminium). A load resistor R_L is also shown in Figure 5.4, but it is not part of thermocouple structure. The structure shown in Figure 5.4 is the basic building block for almost all modern thermoelectric generators and refrigerators. A practical thermoelectric device is essentially a matrix constructed by repeating this basic building block (see Section 5.3.2 for details).

The power output and conversion efficiency are two key parameters to evaluate the performance of a thermoelectric generator. Based on the basic building block shown in Figure 5.4, a theoretical model of thermoelectric generators has been developed [1] that provides a basic framework to study the performance and characteristics of thermoelectric generators. In this simplified model, the influence of metal strips on the generator's performance is neglected due to the fact that the thermal and electrical resistances of metal strips are usually much smaller than those of semiconductor thermoelements.

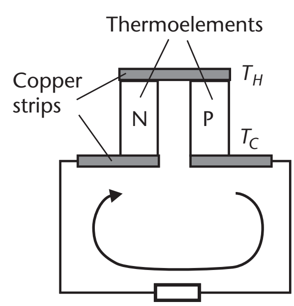

Figure 5.4 The basic building block of thermoelectric devices, which consists of n- and p-type semiconductor thermoelements connected in series by conducting strips.

5.2.2.2 Power Output

A thermoelectric generator can be viewed as a thermal battery. The electromotive force of this thermal battery is the Seebeck voltage $V_o = \alpha_{np}\Delta T$ (where the subscripts n and p represent n-type and p-type thermoelements that form the thermocouple junctions). The internal resistance R is the total series resistance of n- and p-type thermoelements. An equivalent electrical circuit for Figure 5.4 is shown in Figure 5.5, where R_L is the resistance of the load connected to the thermoelectric generator. It can be seen from Figure 5.5 that the voltage across the load can be written as

$$V = \alpha_{np}\Delta T \frac{R_L}{R + R_L} \tag{5.8}$$

The electric current flowing through the load is given by

$$I = \frac{\alpha_{np}\Delta T}{R + R_L} \tag{5.9}$$

Consequently, the electrical power delivered to the load can be expressed as

$$P = \frac{s}{(1+s)^2} \frac{(\alpha_{np}\Delta T)^2}{R} \tag{5.10}$$

where $s = R_L/R$ is the ratio of the load resistance to the device internal resistance. The power output depends on the ratio s, and the maximum power output is obtained at the matched load (i.e., when $R_L = R$)

$$P_{\max} = \frac{(\alpha_{np}\Delta T)^2}{4R} \tag{5.11}$$

For given α_{np} and R values, the maximum power output of a thermoelectric generator increases parabolically with an increase in temperature difference. In

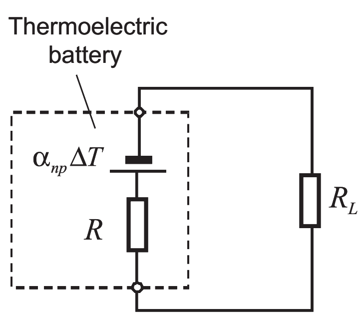

Figure 5.5 An equivalent electrical circuit for the thermoelectric battery of Figure 5.4.

practice, the $P_{max} - \Delta T$ plot deviates slightly from the parabolic relationship because both α_{np} and R change slightly with temperature. The maximum power output for a given temperature difference can be determined experimentally by measuring the power output as a function of resistances using a variable load or, alternatively, by measuring voltages at open- and short-circuit conditions [9].

5.2.2.3 Conversion Efficiency

The conversion efficiency of a thermoelectric generator is defined as

$$\phi = \frac{\text{power delivered to the load}(P)}{\text{heat flux absorbed at hot junction}(\dot{Q}_h)} \tag{5.12}$$

The power delivered to the load is given by (5.10) and heat absorption at the hot junction can be determined by considering heat balance at the hot junction with the Peltier heat, Joule heat, and heat conduction,

$$\dot{Q}_h = \alpha_{np} T_H I - \frac{1}{2} I^2 R + K(T_H - T_C) \tag{5.13}$$

where K is the total thermal conductance of both n- and p-type thermoelements. The conversion efficiency of a thermoelectric generator can be expressed as

$$\phi = \left(\frac{T_H - T_C}{T_H}\right)\left[\frac{s}{(1+s) - (T_H - T_C)/2T_H + (1+s)^2/ZT_H}\right] \tag{5.14}$$

where $Z = \alpha_{np}^2/(R \cdot K)$ is referred to as the thermoelectric figure of merit, which is a measure of suitability of the materials for thermoelectric applications (see Section 5.3.1 for details). The conversion efficiency is also dependent on the ratio s, and the maximum conversion efficiency can be obtained when the ratio s is optimized at $s = \sqrt{1 + Z\overline{T}}$ [1, 3]

$$\phi = \left(\frac{T_H - T_C}{T_H}\right)\frac{\sqrt{1 + Z\overline{T}} - 1}{\sqrt{1 + Z\overline{T}} + T_C/T_H} \tag{5.15}$$

where $\overline{T} = (T_H + T_C)/2$ is the mean temperature across the thermoelements. It can be seen from (5.15) that the conversion efficiency of a thermoelectric generator is the Carnot efficiency $(T_H - T_C)/T_H$, reduced by a factor that depends on the thermoelectric figure of merit. Figure 5.6 shows the maximum conversion efficiency plotted against the hot junction temperature for different thermoelectric figures of merit, assuming that the cold junction temperature remains at 300K. To date, all established thermoelectric materials have a thermoelectric figure of merit in a range of 2 to $3 \times 10^{-3}\ K^{-1}$. This indicates that the conversion efficiency is approximately

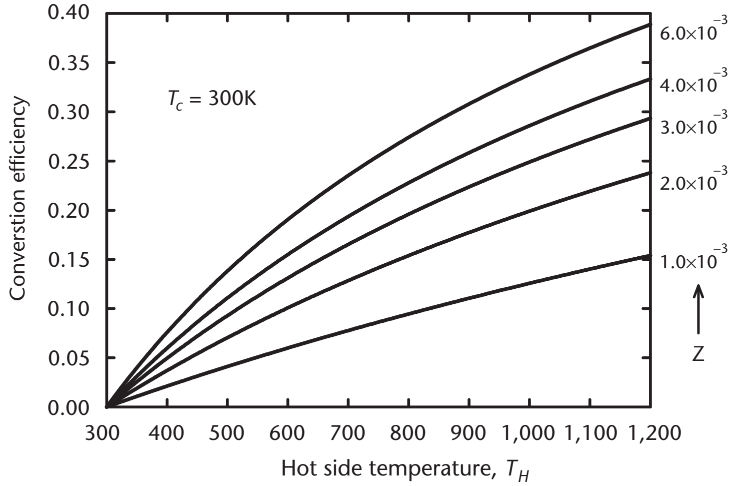

Figure 5.6 Conversion efficiency against hot-side temperatures for a range of Z values.

5% for a temperature difference of about 100K. The value increases to about 20% if the temperature difference is increased to around 800K.

5.3 Influence of Materials, Contacts, and Geometry

The theory outlined in previous section will be expanded to provide useful insights and guidelines for design and optimization of thermoelectric devices that meet the specific requirements, particularly energy harvesting applications. The principal design issues include the selection of appropriate materials, the formation of high-quality electrical and thermal contacts, and the optimization of device geometry.

5.3.1 Selection of Thermoelectric Materials

For given operating temperatures (i.e., T_H and T_C), the conversion efficiency of thermoelectric devices is dependent on the thermoelectric figure of merit,

$$Z = \frac{\alpha_{np}^2}{R \cdot K} \tag{5.16}$$

where α_{np}, R, and K are device parameters that consist of material properties, interface properties, and geometrical influence (see Sections 5.3.2 and 5.3.3 for details). Assuming (this can be readily achieved by using the same semiconductors with opposite doping) that both n- and p-type thermoelements have the same electrical resistivity $\rho_n = \rho_p = \rho$, the same thermal conductivity $\lambda_n = \lambda_p = \lambda$, the opposite Seebeck coefficient $\alpha_n = -\alpha_p = \alpha$, and an identical ratio of length to cross-sectional area $l_n/A_n = l_p/A_p = l/A$, the thermoelectric figure of merit of devices given in (5.16) can be simplified to an expression for a single material,

$$Z = \frac{\alpha^2 \cdot \sigma}{\lambda} \tag{5.17}$$

where $\sigma = 1/\rho$ is the electrical conductivity. The unit for Z is K^{-1}. It is to be noted that (5.17) involves fundamental properties of a single material only. Clearly, in order to obtain a large figure of merit for a thermoelectric device (5.16), it is necessary to find two materials that have large figures of merit (5.17) but opposite Seebeck coefficients. Equation (5.17) provides an important criterion for the selection of suitable materials for thermoelectric applications: a good thermoelectric material should possess a large Seebeck coefficient to produce high-voltage, large electrical conductivity to minimize Joule heating and low thermal conductivity to retain the heat at the hot junction. Figure 5.7 shows schematically the Seebeck coefficient α, the electrical conductivity σ, and the thermal conductivity λ as functions of the charge carrier concentration of a material. An increase in σ will be accompanied by an adverse decrease in α and an increase in λ. Obtaining the maximum Z requires compromise among three parameters and the optimization is achieved at carrier concentration values in the range of 10^{23} to 10^{26} m^{-3}, which correspond to a class of materials termed heavily doped semiconductors. Typically, the optimal values for the Seebeck coefficient, electrical resistivity, and thermal conductivity are in a range of 150 ~ 230 $\mu V \cdot K^{-1}$, 1 ~ 3 × 10^{-3} $\Omega \cdot$ cm, and 1.5 ~ 3 $W \cdot m^{-1} \cdot K^{-1}$, respectively [1–6].

A thermoelectric figure of merit varies with temperature. Since the unit for Z is K^{-1}, a dimensionless figure of merit ZT can be defined that plays the same role as Z. Figure 5.8 shows the dimensionless figures of merits as a function of absolute temperature T for a number of established thermoelectric materials. Currently, all established thermoelectric materials appear to have a maximum ZT value of around unity. The maximum ZT value for Bi_2Te_3 alloys is obtained at around 300K, which makes it suitable for room applications. The maximum ZT values for

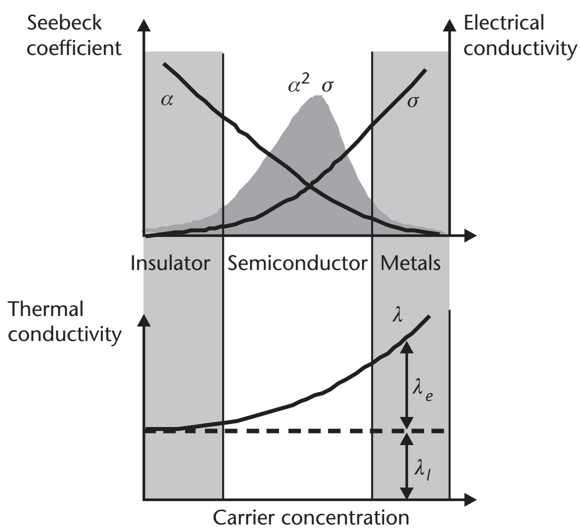

Figure 5.7 Schematic dependence of the Seebeck coefficient, electrical conductivity, and thermal conductivity on charge carrier concentration [2].

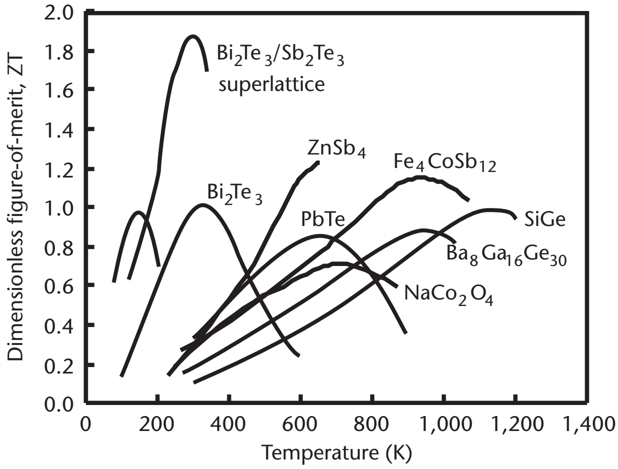

Figure 5.8 Dimensionless thermoelectric figure of merit for a number of thermoelectric materials.

PbTe and SiGe are obtained at ~ 650K and ~ 1,000K, respectively. They are suited to for medium and high temperature applications.

ZT is a main factor that limits the conversion efficiency of thermoelectric devices and the search for high ZT materials is the key challenge in thermoelectric research. Success in this endeavor will lead to a breakthrough in thermoelectric technology, which can have a significant impact on power generation and refrigeration. With improved understanding of thermal and electrical transports over the past 50 years, a plethora of strategies and approaches have been developed in attempts to increase the thermoelectric figure of merit. Recently, it has been reported that ZT values larger than 2 were obtained in nanostructures [10, 11]. Readers who are interested in details of this aspect of research can consult [12–14].

5.3.2 Thermal and Electrical Contacts

Although the device theory described in Section 5.2.2 is adequate for the analysis of large dimension thermoelectric devices, it becomes inaccurate for small dimension devices because it neglects thermal and electrical contact effects. Nowadays, many modern thermoelectric systems are usually constructed using off-the-shelf thermoelectric "modules" whose structure is shown in Figure 5.9. The module consists of a number of the basic building blocks (Figure 5.4) connected electrically in series but thermally in parallel and sandwiched between two ceramic plates. In such a structure, the influence of thermal and electrical contacts can no longer be neglected. In order to provide realistic estimates, an improved device model has been developed based on a module structure shown in Figure 5.9 taking into account thermal and electrical contact resistances. It can be shown [15, 16] that when the module operates with the matched load, the output voltage V and the current I are given by

$$V = \frac{N \cdot \alpha \cdot (T_H - T_C)}{1 + 2rl_c/l} \tag{5.18}$$

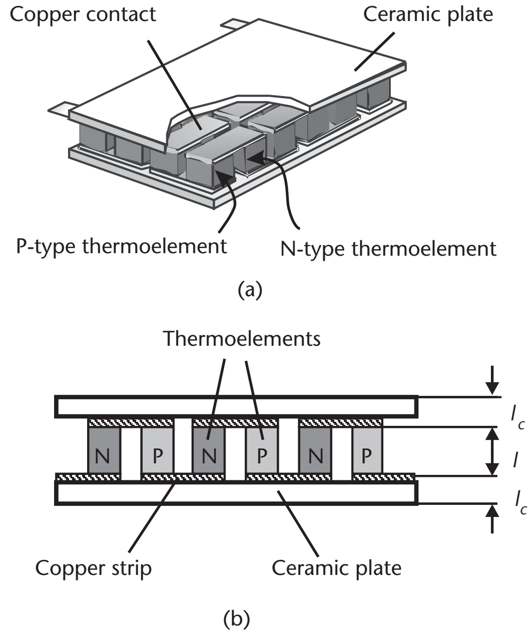

Figure 5.9 Thermoelectric module. (a) Schematic diagram of a thermoelectric module. (b) Cross-sectional view of a thermoelectric module which consists of a number of n- and p-type thermocouples connected electrically in series but thermally in parallel and sandwiched between two ceramic plates.

$$I = \frac{A \cdot \alpha \cdot (T_H - T_C)}{2\rho(n+l)(1+2rl_c/l)} \tag{5.19}$$

where N is the number of thermocouples in a module, A and l are the cross-sectional area and length of the thermoelement, respectively, and l_c is the thickness of the contact layer. $n = 2\rho_c/\rho$ and $r = \lambda/\lambda_c$ (where ρ_c is the electrical contact resistivity and λ_c is the thermal contact conductivity) are referred to as the electrical and thermal contact parameters. The values of n and r can be determined experimentally using the method described in [17]. For commercial off-the-shelf thermoelectric modules, typical values for n and r are ~ 0.1 mm and ~ 0.2, respectively.

The maximum power output $P_{\max}$ and conversion efficiency ϕ of a thermoelectric module, when operated with a matched load, can be expressed as [16]

$$P_{\max} = \frac{\alpha^2}{2\rho} \frac{A \cdot N \cdot (T_H - T_C)^2}{(n+l)(1+2rl_c/l)^2} \tag{5.20}$$

$$\phi = \frac{\left(\dfrac{T_H - T_C}{T_H}\right)}{\left(1 + 2r\dfrac{l_c}{l}\right)^2 \left(2 - \dfrac{1}{2}\left[\dfrac{T_H - T_C}{T_H}\right] + \left[\dfrac{4}{ZT_H}\right]\left[\dfrac{l+n}{l+2rl_c}\right]\right)} \tag{5.21}$$

The effect of thermal contact resistance on the conversion efficiency of thermoelectric modules is shown in Figure 5.10. Based on the state-of-the-art Bi_2Te_3 technology, the thermal contact resistance corresponds to $r = 0.2$, which has only a minor influence on the conversion efficiency of the modules, which have long thermoelements (>3 mm). However, a significant reduction in the conversion efficiency occurs for the modules with a short thermoelement of <1 mm. In order to increase the device efficiency to approach the materials efficiency, it is necessary to minimize the contact resistances. This is particularly important for microdevices, whose thermoelements are very short. The electrical contact resistance exhibits a similar effect on the conversion efficiency of thermoelectric modules. For a detailed study of thermal and electrical contact effects, readers can consult [18].

5.3.3 Geometry Optimization

The parameters embodied in (5.18) through (5.21) can be classified into three groups:

- *Specifications:* The operating temperatures T_H and T_C, the required output voltage V, current I, and power output P;
- *Material parameters:* The thermoelectric properties α, σ, λ, and the contact parameters n and r;
- *Design parameters:* The thermoelement length l, the cross-sectional area A, and the number of the thermocouples N.

The specifications are the requirements provided by customers for a given application. The material parameters are restricted by current fabrication technologies of materials and devices. Consequently, the thermoelectric module design involves the determination of a set of design parameters to meet the required specifications

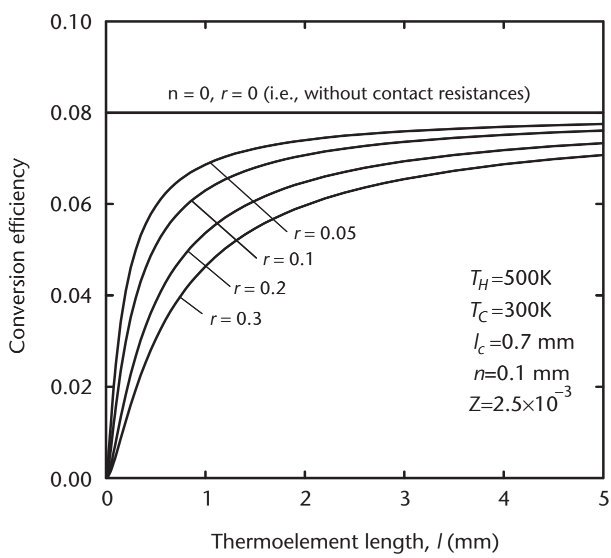

Figure 5.10 The effect of thermal contact resistance on the conversion efficiency of the thermoelectric module.

at a minimum cost. The number of the thermocouples N can be determined using (5.18), while the cross-sectional area, A, can be obtained from (5.19). The determination of both N and A is usually a straightforward calculation. However, the determination of thermoelement length involves optimization between two conflicting requirements.

The existence of thermal and electrical contact resistances results in the conversion efficiency changing with the length of thermoelements. The length dependence of the power output embodied in (5.11) is also modified which leads to occurrence of a maximum value. Figure 5.11 shows the power-per-area p ($=P/AN$) and the conversion efficiency of a thermoelectric module versus the length of thermoelements for different temperature differences. A long thermoelement is required in order to obtain higher conversion efficiency, while a relatively shorter thermoelement is required to obtain a larger power output. Consequently, the selection of thermoelement length is inevitably a trade-off between the conversion efficiency and power output. The ultimate task of the length design is to achieve maximum performance at a minimum cost. In practice, the cost-per-kilowatt-hour (\$/kW·h) is generally used as a yardstick to measure the economic viability of a power system. Using (5.20) and (5.21), the cost per-kilowatt-hour, C, of electricity generated using a thermoelectric module can be estimated using [16, 19]

$$C = \frac{c_f}{\phi} + \frac{c_m}{P \cdot \Delta t} \tag{5.22}$$

where c_f is the cost-per-kilowatt-hour (e.g., \$/kW·h) of input thermal energy, c_m is the fabrication cost of a thermoelectric module, Δt is the operation period, and P and ϕ are the power output and conversion efficiency of a thermoelectric module, respectively. Clearly, if the heat source is expensive, the thermoelectric module design should be guided by obtaining a high-conversion efficiency. On the other

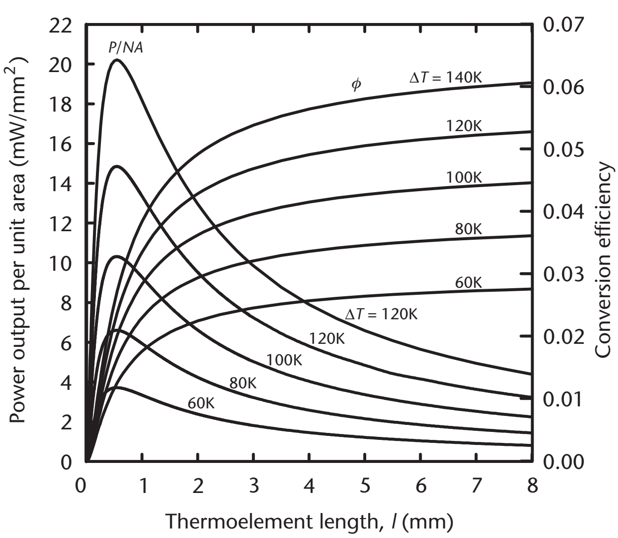

Figure 5.11 The power-per-area and conversion efficiency of thermoelectric module versus the length of thermoelements for different temperature differences.

hand, the module design should be aimed at achieving a maximum power output if the heat source is inexpensive or essentially free, such as in the case of waste heat recovery or energy harvesting.

5.3.4 Heat Exchangers

In order to establish a temperature difference across a thermoelectric device, heat exchangers are needed to extract heat from heat sources and then to dissipate it into ambient [20]. Both the power output and the conversion efficiency of a thermoelectric device increase considerably with increasing temperature difference across it. Efficient heat exchangers are vital to ensure the maximum temperature difference established across the thermoelectric device. Assuming that similar heat exchangers are employed for both the hot and cold sides of the module, the ratio of the temperature drop on the module ΔT_{TE} to that on heat exchangers ΔT_{HE} can be expressed as

$$\frac{\Delta T_{TE}}{\Delta T_{HE}} = \frac{K_{HE}}{2\left(1+ZT_M\right)K} \tag{5.23}$$

where K_{HE} is the heat transfer coefficient of heat exchangers and T_M is a parameter associated with T_H and T_C across the module [21]. Clearly, the design of heat exchangers should be aimed at achieving maximum K_{HE}. In the field of heat transfer, the effectiveness of heat exchangers is usually measured by its reciprocal quantity: thermal resistance ($\Re = 1/K_{HE}$ with a unit: K/W). Table 5.1 shows the thermal resistance of different types of heat exchangers.

In practice, the size, cost, and availability of heat exchangers also need to be considered. Furthermore, design of thermoelectric devices to match the effectiveness of heat exchangers may sometimes be necessary, particularly in the case of energy harvesting from small temperature difference or modules with short thermoelements. In such applications, ΔT_{HE} may still be very large even the available heat exchangers have reached their performance limit. An alternative is to design modules with a relative large $(1 + ZT_M)K$.

5.4 Existing and Future Capabilities

Thermoelectric devices are essentially heat engines but use electrons (or holes) as the working fluid. They are solid-state energy converters that have no mechanical

Table 5.1 Thermal Resistance of Different Heat Exchangers

Type of heat exchanger	*Thermal resistance (W/K)*
Natural convection	2.0 ~ 0.5
Forced convection	0.5 ~ 0.02
Liquid convection	0.02 ~ 0.005

moving parts and hence are silent, reliable, scalable, and environmentally friendly. However, they have a relatively low conversion efficiency compared to conventional heat engines. As a result, thermoelectric devices are usually employed in applications where their desirable features outweigh the low efficiency drawback. Thermal energy harvesting is such a case well suited for thermoelectric applications, particularly in the three aspects listed in the following sections.

5.4.1 Low Power Systems

Advance in modern microelectronics has led to a significant reduction in the power level requirement. Modern-day remote wireless sensors can operate at a power level of about 130 μW. A quartz digital wristwatch requires merely 20 to 40 μW. At such a power level, it is possible to use thermoelectric devices to harvest ambient heat for powering remote sensor networks or mobile devices. This eliminates the needs for replacing batteries or for lengthy cabling from central power sources. A particularly attractive feature of thermoelectric devices is their ability to generate electricity from body heat that could power medical implants, personal wireless devices, or other consumer electronics. A successful example in this attempt is the thermoelectric wristwatch developed by Seiko [22] and Citizen [23]. Figure 5.12 shows a schematic cross-section of a thermoelectric watch. A miniature thermoelectric converter that consists of 2,268 pairs of Bi_2Te_3 thermocouples is mounted on the bottom case of the watch. It produces on average 25 μW of electricity from a temperature difference of 2–3K generated by body heat. The conversion efficiency is about 0.1% (compared with the corresponding Carnot efficiency of 0.66%). Although it is technologically successful, its commercialization has been restricted mainly due to the cost of the thermoelectric converter employed.

In a normal environment, the temperature difference between human body and ambient is around 5–10K. The rate of heat generation for an average human body is typically around 100W [24]. Using such data, the power output that can be harvested using a thermoelectric device is estimated to be 20–50 μW/cm^2 [25]. This indicates that the power of about 2–5 mW may be obtained with a realistic surface area of 0.1 m^2 (equivalent to an area of 40 cm × 25 cm) that enables heat

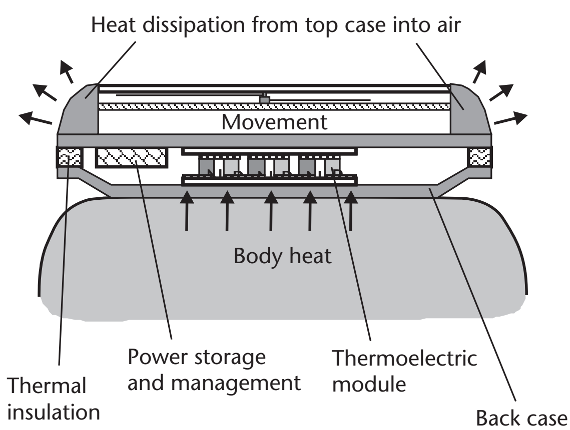

Figure 5.12 A schematic diagram of thermoelectric wristwatch. A miniature thermoelectric converter is placed on the bottom case of the watch.

extraction from a body without causing significant inconvenience or discomfort. Such a power level is sufficient for some low-power applications. An example is a recent project at EADS to develop energy harvesting aircraft seat. The next generation of aircraft seats will have wireless sensors embedded in the seats that report information such as occupancy, backrest, armrest, tray table position and seatbelt status to a remote flight attendance panel. While several seats sensors are already available, a major challenge in achieving a self-sufficient operation of such embedded wireless networks is local power supply at every seat. The battery is an obvious choice; however, it requires replacement and maintenance on a regular basis. Thermoelectric energy harvesting uses the heat dissipated from the passenger's body to generate electricity for powering the wireless sensors, providing an autonomous solution. A preliminary system has been constructed which achieved the target power output of 1 mW under an optimum operating condition. Further work will aim at developing advanced energy systems and reducing the weight-to-power ratio (currently at ~0.6 kg/mW).

The feasibility of integrating thermoelectric devices into smart textiles has been considered for powering medical sensors on smart clothes for the remote health care of patients [25]. Commercially available thermoelectric devices are bulky and rigid. They are not suitable for wearable applications. Recent progress in developing micro/nanothermoelectric devices provides an embedded solution. However, a more adventurous attempt is to develop flexible thermoelectric thin films or polymers that can be woven into fabrics. Prototypes of flexible thermoelectric converters have also been reported by several research groups [26, 27]. Success in this attempt will provide a truly compatible power source for smart textiles.

Another challenge in energy harvesting from body heat is to ensure sufficient heat transfer from the human body to thermoelectric devices without causing any discomfort. Although the human body on average generates 100-W thermal power, only a small proportion of it (possibly limited to 5–10W) can be conveniently collected due to evaporative heat loss and inconvenient locations. Furthermore, the skin's response to ambient temperature can reduce the heat generation at the location where thermoelectric devices are attached. When the skin detects a temperature drop as a result of heat transfer, a constriction of the blood vessels in the skin may occur, leading to a decrease in the skin temperature. Consequently, the power output and conversion efficiency decreases with reducing the temperature difference across the thermoelectric devices. A thermal model of the human body is needed in order to estimate available thermal energy that can be harvested from a human body without any discomfort.

In applications where thermal energy is not continuous or not sufficient enough to power devices, energy storage and a power management system are needed to extend the capacity of thermoelectric energy harvesting. For example, the power generated by thermoelectric energy harvesting from high-voltage power lines is not sufficient for continuous monitoring and data transmission. This shortcoming can be overcome by operating the remote sensor networks intermittently. A supercapacitor is used to accumulate the electrical energy over a period of time and then deliver it in a short burst with large power. Figure 5.13 shows a block diagram of a typical energy storage and power management system. A voltage step-up circuit is often required because the output voltage of thermoelectric devices is usually very low due to small temperature differences in these operating conditions.

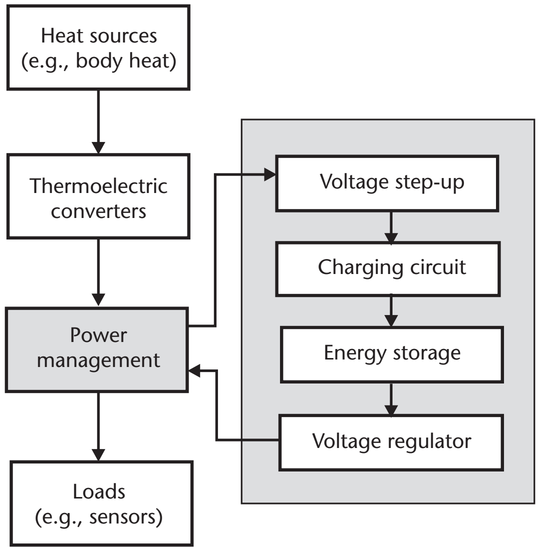

Figure 5.13 The block diagram of energy storage and power management system for thermoelectric energy harvesting devices.

5.4.2 Waste Heat Recovery

Many industrial and transport processes are accompanied by generation of waste heat from burning fossil fuels. Insecurity and continuous price increase of fossil fuels, coupled with their damaging effects to the environment, have spurred significant efforts to improve energy efficiency and reduce the CO_2 emission of fossil fuel systems. About 90% of the world's useful power (approximately 10 TW) is generated by heat engines that convert heat to electrical or mechanical power, but they also dissipate about 15 TW of heat to the environment. The ability to harvest even a small fraction of this low-grade waste heat could have a huge impact on energy efficiency, leading to savings in fuel and reductions in CO_2 emission. For very large scale waste heat recovery, traditional mechanical heat engines such as Rankine, Brayton, or Stirling engines can be used; these offer higher conversion efficiency than thermoelectric devices. However, these systems do not scale easily for small system applications and do not function well at "low" temperatures (<140°C). Thermoelectric devices can be competitive in the case of small systems where the size, reliability, and maintenance-free aspects are major factors.

Thermoelectric waste heat recovery for automobiles is a typical case that is being explored by several major car manufacturers. In a typical vehicle engine with a brake output power of 50 kW, an additional 35 kW of heat is generated and released to the atmosphere in the form of hot exhaust gases and another 35 kW is generated and released to the atmosphere through the engine cooling system. The exhaust flow is at a high temperature and represents a substantial loss of available or useful energy that could be harnessed. The techniques available to convert heat into shaft work include expansion devices, vapor power cycles, and solid state and chemical regenerative techniques. However, in automotive applications, the simplicity of construction and light weight are important parameters. These criteria point to a solid-state solution and thermoelectric devices in particular.

Over the past 10 years, with multimillion-dollar/per year funding support from governments of major car manufacturing nations, a number of large R&D consortia have been formed to develop a thermoelectric recovery system from vehicle exhaust. These consortia involve national research institutes, universities, vehicle manufacturers, and related industries [28]. Recently, BMW has successfully developed a demonstration system that generates about 200W under optimum operation conditions [29]. Figure 5.14 shows a schematic of the thermoelectric system for vehicle waste heat recovery from exhaust. Waste heat collected from the exhaust pipe (after the catalysts converter) is fed to the hot side of thermoelectric modules, which are made from Bi_2Te_3 materials. Heat coming out from the cold side of the thermoelectric modules is dissipated through the radiator. The ultimate goal of such projects as being mentioned by a number of consortia is to develop a system that is capable of generating around 1-kW electricity with 10% fuel savings [28].

Despite significant progress being made in vehicle waste heat recovery applications, a considerable effort is still needed in order to make this technology economically viable. The key challenges to push forward this technology include: high ZT materials, high temperature modules, and effective heat exchangers. It is estimated that ZT ~2 is necessary in order to obtain a system efficiency of ~10% (which corresponds to a materials efficiency of ~15%). In addition, material cost, availability, and nontoxicity are important considerations. Commercially available modules are made of Bi_2Te_3-based materials with a relatively low operating temperature (usually < 450K). On the other hand, since the temperature of waste heat from vehicle exhausts can be up to 850K, materials with large ZT values at high temperatures are required. Furthermore, it can be seen from Figure 5.8 that large ZT values for any given material are limited in a very narrow temperature range. In order to achieve a large average ZT value over a wide temperature range, a module should consists of two or even three materials with different peak ZT values in cascade. This concept can be implemented through advanced structures such as segmented, cascaded [30], and recently proposed tubular [31] configurations.

Thermoelectric technology for waste heat recovery is envisaged as a promising future energy technology that can contribute to: energy security by creation of energy resources from waste heat, environmental protection by reducing CO_2 emission, and wealth creation by initiating a new energy technology sector. Thermoelectric power generation systems can be used for a wide range of waste heat

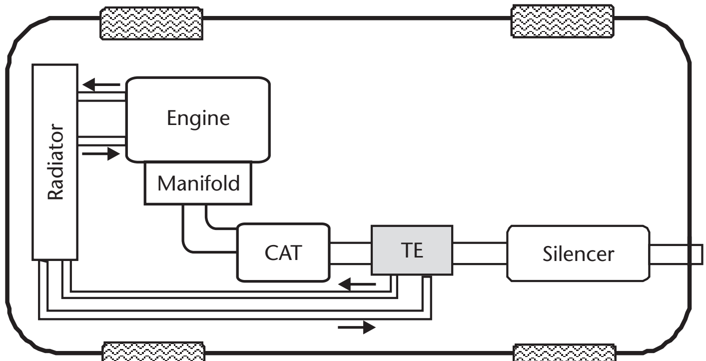

Figure 5.14 Schematic diagram of thermoelectric waste heat recovery from vehicle exhaust.

resources, such as hot water waste from steel plants, geothermal energy from hot springs, hot surfaces of furnaces, incinerators and other industrial processes, and remote subsea oil wells [32, 33].

5.4.3 Symbiotic Cogeneration System

Thermoelectric devices have a relatively low conversion efficiency. This indicates that a significant amount of heat is dissipated from the cold side of thermoelectric devices. An efficient way of using thermoelectric devices is to make use of the heat dissipated from thermoelectric devices [34]. This analysis leads to a proposal of thermoelectric heat and power cogeneration or combined heat and power (CHP) systems. In such case, a thermoelectric device is used as a dual-function component for both heat transfer and power generation. There are many situations where the production of heat is a primary requirement while electricity is also needed. A typical example is domestic central heating systems. Heat is produced by combustion of natural gas or oils, but electricity is required for powering a pump to circulate hot water in radiators. Hence, such systems need to be connected to both gas and electricity networks. Incorporating thermoelectric devices into a central heating system (as shown in Figure 5.15) eliminates the need for mains power [35]. Heat generated from combustion transfers into the water via thermoelectric devices. A major proportion of thermal energy produced goes into the water, with a small proportion converting into electricity to power the water pump. A similar design is also investigated for incorporating thermoelectric devices with solar water heating systems to provide electrical power for water circulation, control, or lights to households [36].

A potential use of thermoelectric devices in a combustion system for heat recirculation has been investigated theoretically [37–39]. Figure 5.16 shows a schematic diagram of a thermoelectric combustion system. Thermoelectric devices are

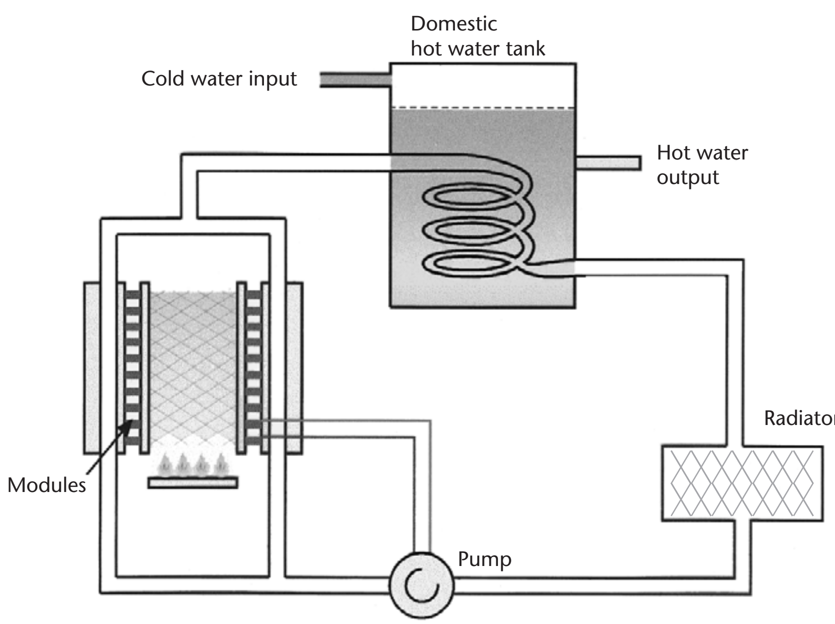

Figure 5.15 A self-powered central heating system using a thermoelectric module as a heat exchanger and a power converter.

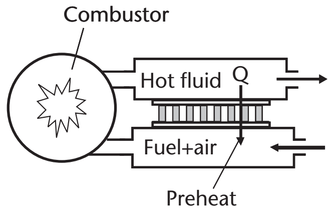

Figure 5.16 Schematic diagram of a thermoelectric combustion system. In addition to power generation, combustion efficiency is improved by preheating the inlet fuel mixtures via a thermoelectric converter.

sandwiched between the inlet and outlet channels of a combustor. When hot flames flow out along the outlet channel, a fraction of heat transfers via thermoelectric devices into the inlet channel, which preheats the inlet reactant mixture (or air alone) to the combustion chamber. In addition to electricity generation, an increase in the temperature of inlet reactant mixtures improves combustion efficiency. It also facilitates the combustion of lean-fuels, which are not combustible under normal circumstances. Furthermore, theoretical calculation shows that with an appropriate preheating temperature of inlet reactant mixtures, the system efficiency for electrical power generation can surpass the maximum module efficiency for the same temperature difference [37, 38].

The thermoelectric kerosene lamp is probably the first example of energy conversion using thermoelectric devices [1]. It is also a good example of another type of symbiotic applications, where the kerosene lamp provides light as its primary purpose, while the thermoelectric devices convert the heat from burning of kerosene into electricity for powering a radio receiver. With currently available ZT values, thermoelectric devices are unlikely to be a mainstream energy technology. However, their applications in energy harvesting, in particular, in symbiotic systems, offer many opportunities to enhance power capability and energy efficiency of autonomous systems.

5.4.4 Commercial Thermoelectric Module Suppliers

There are several international companies that specialize in making thermoelectric modules for power generation. The following list provides a selection of online links to a variety of available products, together with a few comments on the performance of the devices:

1. Marlow Industries Inc., United States (www.marlow.com/power-generators). Supply a variety of modules that can generate up to 8W of electrical power from a temperature difference of 180K.
2. Micropelt, Germany (www.micropelt.com/applications/energy-harvesting.php). Produce a selection of units, some with built-in wireless sensor nodes, in a variety of different packages for various scenarios.
3. Nextreme, United States (www.nextreme.com/pages/power_gen.shtml). Produce a series of thin-film thermoelectric modules, having a low profile (0.65 mm) and a small footprint (2.5 mm × 5.1 mm) capable of producing up to 236 mW from a temperature difference of 120K.

4. Tellurex, United States (www.tellurex.com/products/power.php). Manufacture power generation modules in a variety of sizes, which can produce from 1.3W to 7.5W of electrical power from a temperature difference of 100K.

5.5 Summary

Thermoelectric devices are promising energy converters with unique features in many aspects. The wide-scale application of thermoelectric devices relies on the discovery of high *ZT* materials. Nevertheless, with the state-of-the-art thermoelectric materials and devices available to date, thermoelectric devices have proved to be a very promising technique that is well suited for thermal energy harvesting. The three aspects discussed in this chapter are of particular interest. Low power applications take advantage of the totally scalable feature of thermoelectric devices. They can be fabricated on microscales and integrated with microelectronic systems or woven into smart clothes. Waste heat recovery offers a favored opportunity for current thermoelectric applications where the simplicity, reliability, and cost-per-watt overweigh the conversion efficiency. Finally, from technical point of view, the best application scenario is symbiotic systems where thermoelectric devices are used as dual-function components for heat transfer and power generation simultaneously, facilitating heat and power cogeneration.

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CHAPTER 6

Power Management Electronics

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6.1 Introduction

In most low-power systems, power management is generally thought of as being an ability to switch certain parts of a system off or put them in a low-power state when they are not required, and to manage the charging of a battery. While these are important aspects of low-power electronics powered by energy harvesters, there are much more fundamental reasons for requiring power electronics in an energy-harvesting system than simply managing a battery and conserving energy:

- In order to achieve high-power density from the energy harvester, there should be some form of impedance match between the energy source and transducer and the electrical system. This requires control of the input impedance of the circuit that interfaces to the transducer.
- The output voltage and current from the energy harvester are rarely directly compatible with load electronics and thus some form of voltage regulation is required.
- As discussed in Chapter 7, some form of energy storage is almost certainly required so that the intermittency of the energy-harvesting source does not have a detrimental effect on the continuous operation of the system.

Therefore, the basic power electronics topology for an energy-harvesting system often follows that shown in Figure 6.1.

6.1.1 Interface Circuit Impedance Matching

In a large-scale electrical energy generation plant such as a coal-fired power station, where large amounts of power are produced and where fuel must be purchased, it is important that as much of the energy contained in the original fuel source as possible is converted into useful electrical power. This requires first a high efficiency of conversion of the energy stored in the fuel to a mechanical form, second a high-conversion efficiency of that mechanical energy to electrical energy, and finally a

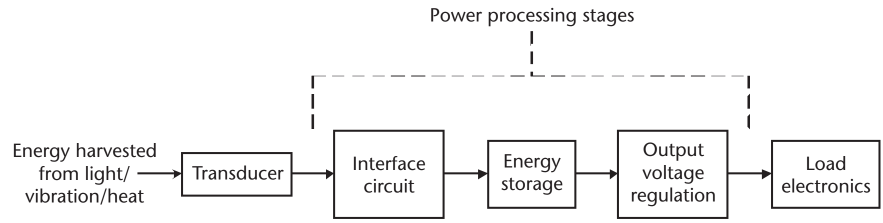


Figure 6.1 Power electronics topology for energy-harvesting systems.

high efficiency of power transfer from the electrical generator to a load. In order to ensure that the energy produced in the electrical generator is efficiently transferred to the load, there is a well-known and fundamental requirement that the impedance of the load should be significantly larger than the impedance of the generator. However, while this arrangement [Figure 6.2(a)] achieves the maximum electrical efficiency (and prevents the generator from thermal destruction), it does not achieve the maximum power transfer from source to load. Maximum power transfer occurs in the case where the load impedance is equal to the source impedance, as illustrated in Figure 6.2(b). In the case of an AC energy source, the load should provide a conjugate match to the source. If the diagrams of Figure 6.2 were taken as a very basic representation of a conventional electromagnetic electrical generator supplying a load resistance, R_{Source} would represent the generator winding resistance and V_{Source} would represent the EMF produced by time-varying flux linkage with those windings.

In the case of energy-harvesting systems, the fuel supply is effectively free, and this leads to the desire to be able to transfer the maximum power into the load, rather than to accomplish this at high efficiency. In addition, the quantities of power generated are low enough that an impedance match rarely has any thermal implications on the system.

In an energy-harvesting generator, the definition of the impedance of the source to which the load should be matched is not generally as trivial as matching the load to a single electrical impedance. The source impedance will be dependent upon the type of energy harvester used and the conditions under which the harvester

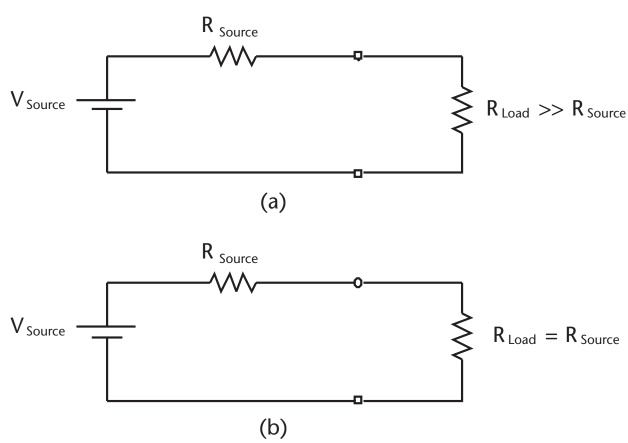


Figure 6.2 (a) Maximum efficiency of energy transfer to load and (b) maximum power transfer to load.

is operating. In some circumstances and harvester operating modes, it may not be optimal to match the impedance of the load to that of the source due to other constraints; however, for energy harvesters studied in this chapter, there is always a clearly defined transducer load impedance that results in maximum power extraction from the transducer. It may therefore be more accurate to specify that the input impedance of the interface circuit to the transducer must be controllable, rather than always matched to the source, although in many cases the input impedance of the interface circuit will be set to match that of the source.

The details of source impedance modeling will be discussed in this chapter for each harvester type considered. The source impedance will always be shown as an electrical circuit that will often contain components that represent quantities other than pure electrical ones. As an example, vibration-driven harvesters, discussed in detail in Chapter 4, have a source model that takes into account the mechanical properties of the system such as the mass, the spring, and the vibration characteristics in addition to including the expected electrical resistance of the generator's windings or capacitance. All of these aspects must be included in the source model so that a suitable interface circuit can be designed; otherwise, global system optimization cannot be achieved [1].

6.1.2 Energy Storage

The vast majority of energy-harvesting transducers will not be able to supply energy at a constant rate over long periods of time. Clearly, a solar cell can only produce electrical energy when illuminated and a vibration harvester only when it is subjected to acceleration. However, many applications of energy-harvesting technology may require a constant source of electrical energy to supply the load. If the average power consumption of the load is greater than the average power generated by the harvester, it is not possible to provide power continually to the load. However, if the average power generated is equal to or exceeds the average consumption by the load, it is possible to run the load continually. However, in order to achieve this, the addition of a storage device, very likely electrical storage in the form of a battery or capacitor as discussed in Chapter 7, may be required.

6.1.3 Output Voltage Regulation

The many different types of energy harvesters produce power at different combinations of voltage and current. Photovoltaic cells and electromagnetic transduction kinetic harvesters tend to produce very low voltages (sometimes significantly less than 1V) while electrostatic devices may produce their output power at over 100V and potentially approaching 1 kV if operated optimally [2]. The output voltage from such devices must therefore be processed before being presented to the load electronics. In addition, if an energy storage element is included in the system, the voltage across that element may fluctuate depending on its state of charge. This effect may be negligible in the case of a storage battery, but may be significant if a capacitor is used as the storage component.

6.1.4 Overview

Often, the most difficult part of the harvester power electronics system to realize is the part that directly interfaces with the transducer (i.e., the part of the system that allows the generator to perform optimally through input impedance control). The implementation of this circuit is the part of the electronics that is most specific to each transducer technology used due to vastly differing voltage and current output combinations provided by the different transduction mechanisms.

The choice of storage, discussed in Chapter 7, and the output voltage regulation circuitry are generally common across all harvester systems with few characteristics being specific to the particular harvester type used. Therefore, the most harvester-specific part of the electronics, the interface circuits with controllable input impedances, will now be discussed.

6.2 Interface Electronics for Kinetic Energy Harvesters

In order to determine an optimal electrical load for a motion-driven harvester, a suitable source model must be developed (i.e., the impedance and output voltage characteristics of the source must be known). All aspects of the energy transfer (from the vibration energy source through to the mass and spring and the transduction mechanism) must be taken into account in the source model. As the overall aim is to provide an optimal electrical load to the system, it is sensible to construct an electrical equivalent model of the generator that takes into account the mechanics of the system as electrical components. Two generic examples of such models are shown in Figure 6.3. A detailed explanation of the construction of these equivalent models is given in [3], and therefore only an overview will be given here.

The circuits of Figure 6.3 show the equivalent circuit models for vibration-driven harvesters using electromagnetic damping and electrostatic damping. The part of the circuit connected to the primary side of the transformer models the mechanical components. In Figure 6.3(a), the current source represents the input energy to the system (i.e., the mechanical vibration), the capacitor, m, represents the mass, the inductor, $1/k$, represents the spring, and the resistor, $1/D_p$, represents the parasitic damping. In Figure 6.3(b), the voltage source represents the vibration source, the inductor represents the mass, the capacitor represents the spring, and the resistor represents the parasitic damping. In both cases the transformer represents the coupling from the mechanical domain to the electrical domain through the transducer. In Figure 6.3(a), voltages across components on the left of the transformer represent the velocity of those components, and currents through them represent forces applied to them. The opposite is true for Figure 6.3(b). In both cases, the terminals on the secondary of the transformer represent the physical electrical connections of the transducer to which the interface circuit can be connected (in this case shown as a simple load resistor). The inductor, L_T, represents the self-inductance of the coil in an electromagnetic device and C_T represents the terminal capacitance of either the piezoelectric material or the moving capacitor in the electrostatic device. It is important to note that the fundamental requirement for stored energy in these transducers places a limit on the maximum real power that can be transferred to a load resistor (in other words, energy stored in the inductance L_T

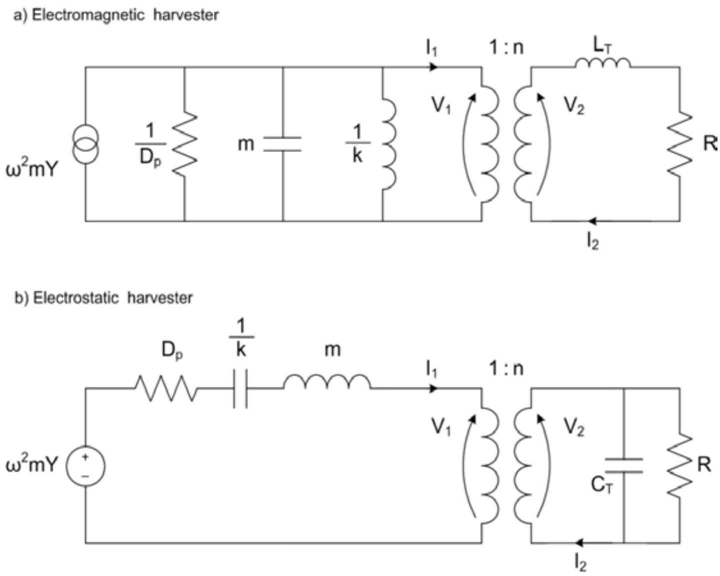


Figure 6.3 Equivalent circuit for motion driven harvester using: (a) electromagnetic force and (b) electrostatic force.

or capacitance C_T). While Figure 6.3(a) is a good model of an electromagnetic harvester and Figure 6.3(b) is a good model of a piezoelectric harvester, neither model is perfect for the electrostatic moving capacitor transducer. This is because Figure 6.3 is a linear circuit and electrostatic transducers are inherently nonlinear systems; their capacitance is nonconstant.

The task, then, in the case of a motion-driven inertial generator, is to connect a value of a load resistance (or much better, a power conditioning circuit feeding a storage element that together emulate a load resistance) that can absorb the maximum amount of energy from the energy source on the left of the transformer.

If we first assume that the storage elements C_T and L_T associated with the transducer have a negligible effect, it is clear from Figure 6.3 that the maximum power can be extracted from the source into the load (shown here as R) if the circuit is operated at a frequency where the inductor and capacitor resonate and if the load resistance equals the equivalent resistance of the parasitic damping when referred through the turns ratio. These models are therefore coherent with the analysis presented in Chapter 4, where it was concluded that the maximum power is transferred to the load at resonance and when the electrical and parasitic dampings are equal.

Therefore, in the case of our impedance match for a load to a motion-driven microgenerator, the aim is often to produce a power converter that can feed energy into a storage element while maintaining an input impedance of a resistance $1/D_p$. It should be noted that operating conditions exist where the optimal load resistance that should be presented by the interface circuit is not simply given by $1/D_p$. A different optimal resistance exists if the generator is operating off resonance, and still a different expression can be found for the optimal resistance if the generator's proof mass becomes displacement limited, which may be the case if the parasitic damping can be made small. A comprehensive derivation of these different constraints is presented in [4]. However, while the optimal load resistance may change depending on the operating condition, in all these cases we conclude that there is an optimal impedance that should be presented by the power electronics interface circuit (Figure 6.4) to the electrical terminals of the microgenerator's transducer.

We are now in a position to discuss the specific implementations of electronics to interface with the three different transducer types for kinetic energy harvesters (i.e., electromagnetic transducers, electrostatic transducers, and piezoelectric transducers).

6.2.1 Electromagnetic Harvesters

The general requirements for interfacing to an electromagnetic transducer on a vibration-driven microgenerator are:

- Rectification;
- Voltage step-up capability;
- Emulatation of a resistive load for the impedance match/impedance control.

The simplest electrical interface for an electromagnetic harvester consists of a step-up transformer that feeds two Schottky diodes (D1 and D2) and a capacitor (C) that acts as a storage component, as shown in Figure 6.5 [5]. Due to the sinusoidal nature of the input vibrations, the output voltage from the electromagnetic harvester is AC. Using the transformer, the typically low transducer output voltage (tens or hundreds of millivolts) is upconverted through the use of the appropriate transformer turns ratio. Rectification of the stepped-up voltage is achieved by diode D1 which conducts during one-half of the AC output voltage, followed by D2 in the other half. This technique of using diodes to rectify the AC voltages from vibration-based energy harvesters is quite common [6–8]. In the configuration shown in Figure 6.5, only one diode conducts during each half cycle of the input vibration when compared to a standard diode bridge, thus minimizing the effect of the diode voltage drop, although this can still pose a problem. This configuration does not perform an impedance match between the electromagnetic harvester's source impedance and the interface electronics, and therefore the maximum power is not transferred from the harvester to the load. However, the simplicity of the arrangement in achieving rectification and voltage step-up is an advantage of this method.

Alternatively, voltage multipliers such as the Villard multiplier (Figure 6.6) and the Dickson multiplier have been used to boost the voltage from the transducer.

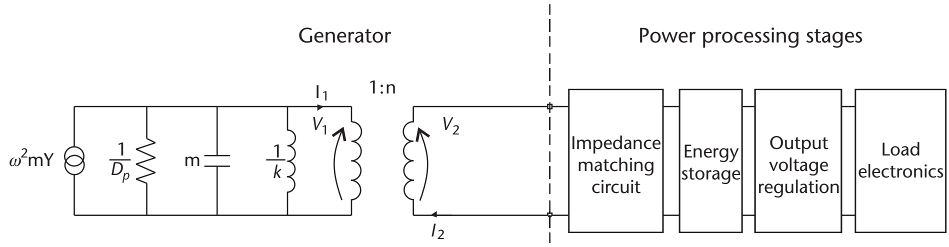


Figure 6.4 Connection of power electronics to the electromagnetic generator model.

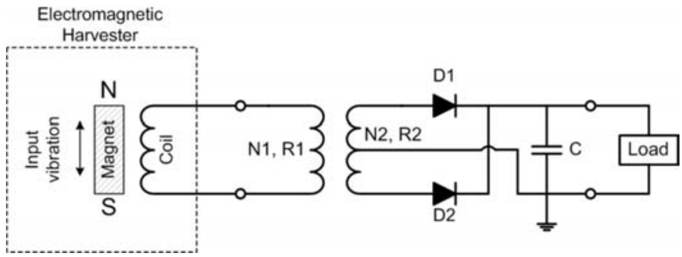


Figure 6.5 A simple electrical interface circuit, which performs rectification and voltage step-up [5].

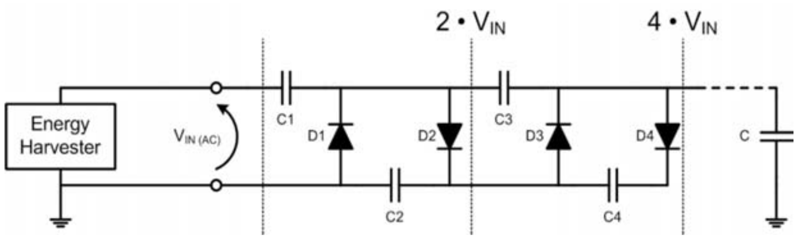


Figure 6.6 Using a Villard voltage multiplier for voltage upconversion [9].

Cascading multiple stages of the Villard multiplier will result in greater step-up ratios on the voltage from the transducer. One benefit of this approach over the previous arrangement is the ability to step up without using magnetic components, which favors integrated fabrication techniques. Again, such an approach fails to provide an impedance match.

Mitcheson et al. proposed a dual-polarity boost converter that interfaces an electromagnetic generator in [1] as a potential solution to provide rectification, an impedance match, and voltage step-up in one circuit, while minimizing diode voltage drops. This converter provides low-voltage rectification of the positive and negative half-cycles of the generated voltage. Two boost converters are activated alternatively to rectify the AC voltage from the harvester's output. The dual-polarity nature of the converter removes the need for a diode bridge rectifier. Additionally, the circuit fulfills the step-up conversion requirements inherent on the output voltage of electromagnetic energy harvesters. Within the boost converter, the authors recommend the use of synchronously switched MOSFETs or Schottky diodes to reduce the effects of power losses in the converter.

In [10], Maurath et al. reported an adaptive impedance matching technique utilizing switched capacitor arrays. The proposed circuit consumed less than 50 μW (simulated) and was geared towards self-powered applications for energy harvesters. Typically, output currents from microgenerators are quite low (less than 1 mA), which was why an on-chip capacitor-based impedance matching circuit was chosen to interface the generator. If the voltage across the switched capacitor array is half that of the generator's voltage, an impedance match exists between the generator's internal resistance and the load. This is an attractive impedance matching technique because it negates the need for current sensing within the power converter. The capacitors in the switched-array are charged to $(0.5V_{gen} + \Delta V_{charge})$ during a charging time period, and then the switch toggles to the other state whereby the capacitors will then discharge to a storage capacitor, which feeds a boost converter. At the end of the discharge cycle, the voltage across the capacitor array will decrease to $(0.5V_{gen} - \Delta V_{discharge})$. The switching frequency for these capacitor arrays depends on how small the ΔVs are required to be and hence is closely linked to the efficiency of the circuit. The control of the circuit is not described in [10] in detail, but it is likely that some open circuit measurement of the transducer open circuit voltage would need to be made during operation as the operating conditions change.

6.2.2 Example of a Complete Power Electronics System for a Continually Rotating Energy Harvester

Many examples have been presented in the literature and, indeed, earlier in this book, about vibration powered harvesters. High-performance power electronics with all the functionality of optimal damping control (the impedance match), energy storage, and output voltage regulation have yet to be demonstrated for such systems, mainly because of the difficulty of achieving these functions with such low-power generation capability and the need that these functions must be powered

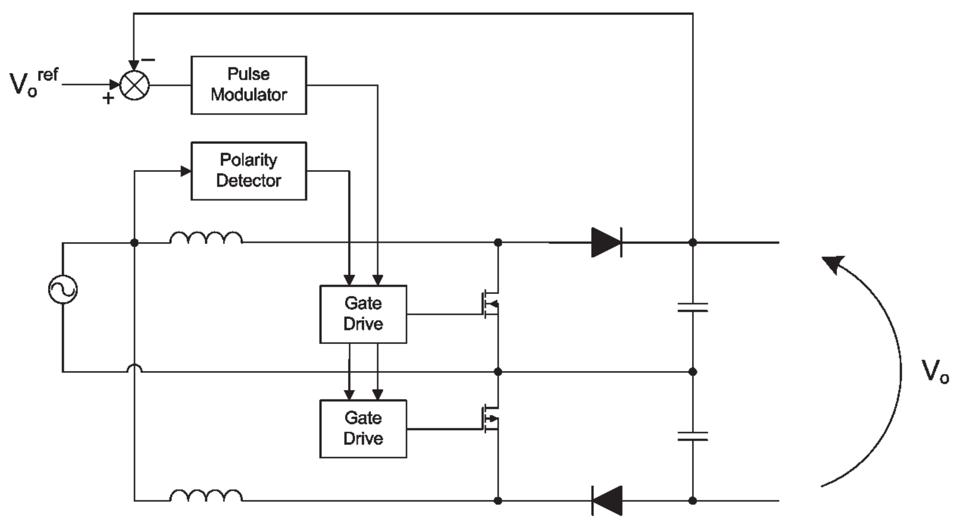


Figure 6.7 Dual-polarity boost converter [1].

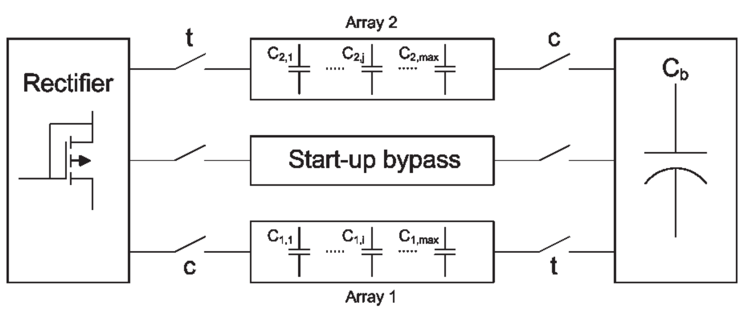


Figure 6.8 Adaptive impedance matching technique using switched capacitor arrays from [10].

from the energy generated (although simulations of some or all of these aspects have been demonstrated). However, all of these functions have already been practically demonstrated for a different type of energy harvesting device: the rotational harvester based on the gravitational torque. This harvester is implemented with an electromagnetic transducer, and therefore many of the features required for the vibration case are shared with the rotational case. Here we will look in some detail about the design and realization of the complete power electronic system, described in Figure 6.1, for this kind of harvester.

The operation of the gravitational torque harvester is as follows. The rotor of a conventional electrical generator is connected to a rotational host source from which energy is being harvested. As the rotor spins, the stator is held in position by the force of gravity acting on an offset counterweight on the stator, as shown in Figure 6.9(a). As a current is drawn from the generator, the torque between the rotor and stator is counteracted by the gravitational torque on the offset mass, and power is generated. Another possibility for configuring the generator is shown in Figure 6.9(b), where the stator of the generator is connected to the host and the offset mass is attached to the rotor of the generator. A detailed operation of these devices is described in [11, 12].

As the current is drawn from the rotational harvester, a torque causes the proof mass to rotate so that the torque from gravity, $T_g = mgL\sin(\theta)$, counteracts the motor torque, as shown in Figure 6.10. For a given rotation speed ω of the host, the limit on the electrical power that can be generated is given by $T_g\omega$, assuming that the mass is held at 90° to the vertical. If the angle of the offset mass exceeds 90°, the rotor and stator of the generator will start to synchronize and power generation will be substantially reduced. From this basic argument it seems that a current should be drawn from the generator such that the angle of the mass is held at 90°. However, when we consider the amount of that power that can be dissipated into a load or pushed into a storage element (in other words, the useful electrical power), we must consider the electrical equivalent circuit of the generator and load as shown in Figure 6.11, while also considering the constraints of the mechanical system. It is clear that, for a given rotation speed and therefore value of an open-circuit generator voltage E_G, the maximum power will be transferred to a load that is matched to the impedance of the armature, R_{ARM}. There are therefore two operating modes for this system to ensure maximum power is generated:

- At low rotation speeds, the impedance of the load should be equal to the generator armature resistance. In this mode the load resistance is constant.

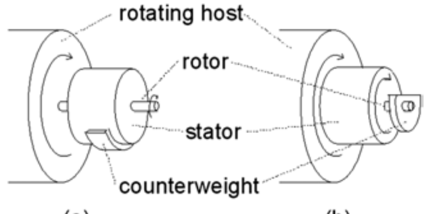


Figure 6.9 Two possible configurations of a rotational harvester constructed from a DC motor: (a) the offset mass is attached to the stator and the rotation is coupled to the rotor, or (b) the offset mass is attached to the rotor with the rotation coupled to the stator

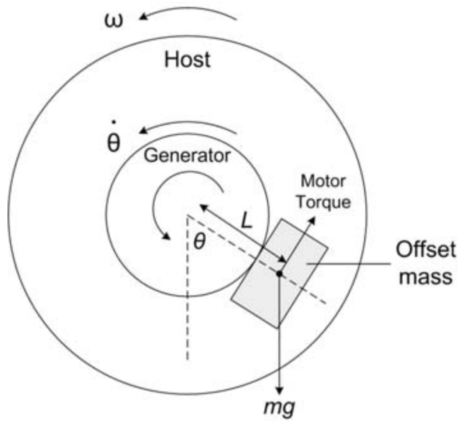


Figure 6.10 End view of rotational torque harvester.

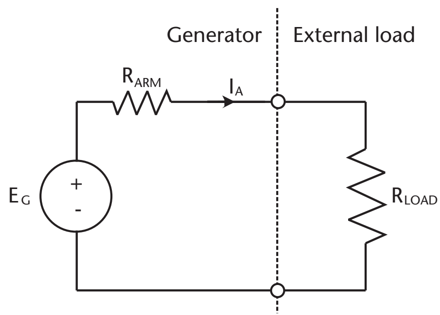


Figure 6.11 Simple DC model of the generator.

- As the rotation speed of the host increases under matched conditions, eventually the offset mass will reach 90°. At this point the load impedance should be increased to prevent the mass flipping and the synchronization of the generator's rotor and stator. Therefore, in this operating mode the generator current should be held constant.

The input impedance of the interface circuit must be controllable to ensure that maximum energy can be harvested under all operating conditions.

As explained previously, we do not want to simply dissipate power in a load resistor, but to supply power to charge a storage element and to power useful loads. Consequently, a power electronic system must be designed that is able to charge a storage element and to present either a constant impedance (at low rotation speeds) or constant current sink interface (at high rotation speeds) to the generator.

The overall topology for the power electronics is as shown in Figure 6.12.

A boost converter was chosen as the interface to the generator because it is able to provide smooth input currents (and thus emulate a resistive input impedance) and step up the relatively low voltages from the generator to push energy into a capacitor, which acts as an energy store able to supply current to the load and smooth out the intermittency of the generation of harvested energy. The voltage on the capacitor will rise if the rate of generation exceeds consumption by the load

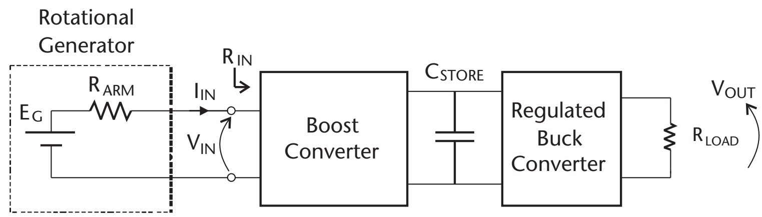


Figure 6.12 Power processing topology for the rotational harvester.

and vice versa. The final stage is a step-down converter that regulates the voltage for use by the load circuit.

A typical rotational harvester may be able to generate around 100 mW, depending on its size and the rotation speed of the host. At these power levels, wide input voltage encapsulated switch mode converters with output voltage regulation are available off the shelf at a low cost and with high efficiency. Therefore, the final stage of the system shown in Figure 6.12 is readily available for this system. The storage element can simply comprise supercapacitors. However, a boost converter with the right characteristics (i.e., input impedance control or input current control) is not readily available and must be designed. The design, construction, and test of this converter will now be discussed.

6.2.2.1 Boost Converter Design

The design of power converters that process power in the range of a few watts would normally involve a relatively standard procedure of choosing a switching frequency and inductor combination that would give an adequately low current ripple, choosing a large enough output capacitor to reduce output voltage ripple and then a diode and MOSFET with suitable voltage and current ratings and switching speeds [13]. However, in the design of a power converter for processing small amounts of power, the overhead of the control circuitry must be taken into account. In such a converter it is also desirable to reduce the component count for simplification in an attempt to reduce power consumption; therefore, the use of components such as a separate gate drive and active filtering of feedback signals should be minimized. In addition, at these low power levels, the energy required to charge the gate capacitance of the MOSFET should be taken into account, as it may constitute a significant proportion of the energy loss in the converter. These additional issues make the optimization of the converter more complicated. The design steps presented in this section assume that the reader has a basic knowledge of operation of switch mode power converters; this section does not cover the mathematical analysis of the basic boost converter operation. A detailed introduction and analysis of the switch mode power converters described in this chapter can be referred to in [13]. Here, we focus on the exact component choices in order to maximize the efficiency of the converter for an energy-harvesting system and to allow the converter to work at a low input voltage (see Figure 6.13).

The approach taken for this design was to optimize the boost converter around what we considered to be a likely operating point for the system, shown in Table 6.1.

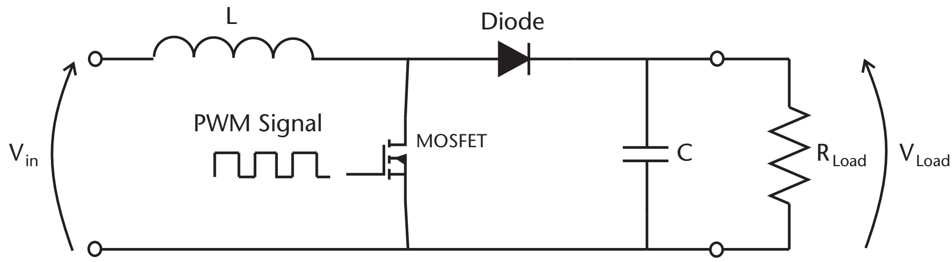


Figure 6.13 Boost converter used as an interface circuit to the transducer.

Table 6.1 Operating Point for Optimization

Generator output impedance	9.1Ω
Generated EMF from transducer	4.4V
Capacitance of energy storage	2 mF
Storage capacitor nominal voltage	15V

The individual power losses in the circuit, whose sum should be minimized, are given in Table 6.2.

There are several free parameters that can be chosen in order to attempt to minimize energy loss in the circuit. These are listed in Table 6.3.

Unfortunately, changing one parameter to reduce one of the losses can cause an increase in other losses. For example, increasing the diode current rating in order to reduce diode conduction loss will almost certainly increase diode reverse recovery losses, and therefore a complete system optimization (accounting for all the parameters at the same time) must be performed.

Table 6.2 Loss Mechanisms in Boost Converter

Inductor conduction loss
Diode conduction loss
Diode reverse recovery loss
MOSFET conduction loss
MOSFET switching loss
MOSFET gate charge energy loss

Table 6.3 Design Parameters

PWM switching frequency
Inductor current rating
Inductor inductance
MOSFET voltage and current rating
Diode current and voltage rating

Expressions for the power losses shown in Table 6.2 were derived in terms of the operating point of the converter and the design parameters of Table 6.3. As an example, the derivation of formulae for the transistor's conduction loss, switching loss, and gate charge energy loss is now described.

6.2.2.2 Conduction Losses

Conduction losses are dependent on the drain-source resistance RDS of the transistor and are proportional to the square of the boost converter's input current multiplied by the duty cycle. The two free design parameters for the MOSFET are the current rating and voltage rating. The maximum voltage blocking capability required by the MOSFET in this case was 40V, as this was the breakdown voltage of the storage capacitance. Since, under a given operating current, conduction loss in a MOSFET is approximately proportional to the square root of the maximum blocking voltage [14], it makes sense to use a MOSFET with the rating required by the application without overrating the device's voltage. This means that the best device for the application is a 40-V MOSFET whose current rating must be determined. Initially, RDS values were gathered for a range of 40-V MOSFETS as a function of their rated operating current, as shown in Figure 6.14.

By applying a curve fit to the points, a relationship between RDS and rated current was obtained as:

$$R_{DS} = 2.56 \cdot \left(I_{rated}\right)^{-2.08} \tag{6.1}$$

The conduction loss can then be expressed as:

$$P_{cond} = \delta I_{in}^{2} \cdot \left[2.56\left(I_{rated}\right)^{-2.08}\right] \tag{6.2}$$

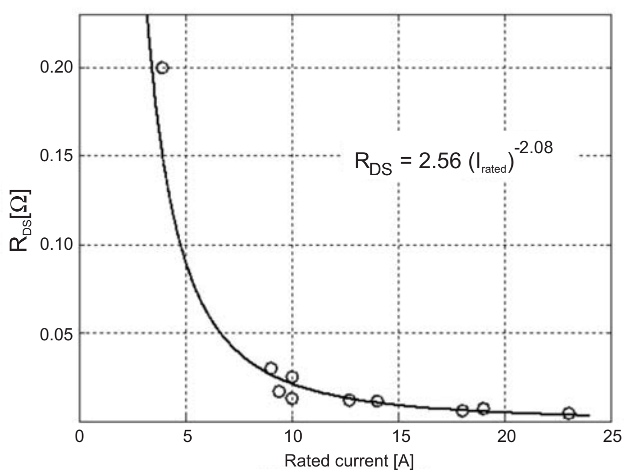


Figure 6.14 Relationship between the drain-source resistance, R_{DS}, and the transistor rated current.

6.2.2.3 Switching Losses

Switching losses arise from the fact that the MOSFET takes time to switch on or off, fundamentally because it takes time to push the charge on and off its gate. The time taken to switch between these two states depends on the stray capacitances at the gate-source and gate-drain junctions, C_{GS} and C_{GD}, respectively, and the current drive capability of the gate drive circuitry. Values of these capacitances are always provided in the datasheets, but as they are voltage dependent, it is better to perform calculations based on gate charge (Q_{GS} and Q_{GD}) instead of capacitance.

Figure 6.15 shows the typical waveforms of a MOSFET switching an inductive load, as is the case in this circuit. This waveform is described in some detail in [15]. When the gate drive source of the FET is initially set high, V_G begins to increase until it reaches the threshold voltage V_{th} of the FET at time t_1. At this point, the drain current I_{DS} starts to increase. C_{GS} continues to charge until the drain current is equal to the inductor current at t_2. At time t_2, V_G and I_{DS} remain constant as the Miller capacitance, C_{GD}, is charged. At t_3, the FET is fully switched on and the voltage drop across the drain-source region is almost negligible. V_{GS} then stabilizes at its final value.

Power loss due to switching occurs in the period between t_1 and t_3, where there is both a nonnegligible current through the MOSFET and a nonnegligible voltage across it. The instantaneous power loss is shown in Figure 6.16.

$$P_{SW} = \frac{1}{2}\left(V_{DD} \cdot I_{DS,\max}\right) \cdot \left(t_3 - t_1\right) \cdot f_{sw} \qquad (6.3)$$

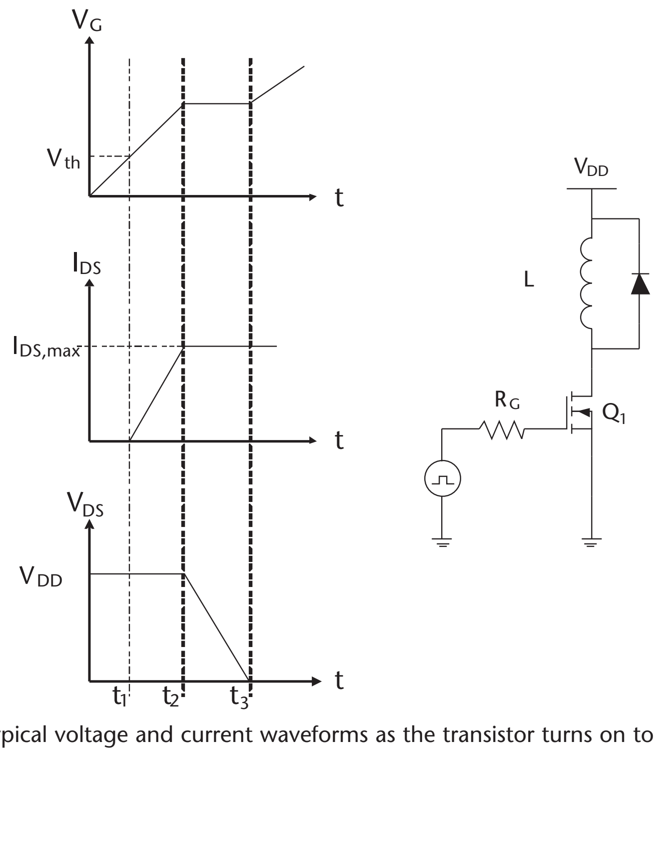


Figure 6.15 Typical voltage and current waveforms as the transistor turns on to switch an inductive load.

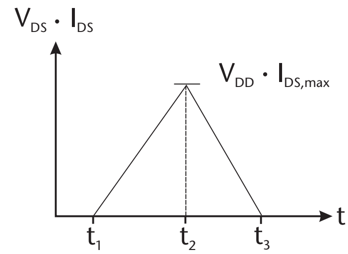


Figure 6.16 Switching power loss waveform.

V_{DD} and $I_{DS,\max}$ are known operating conditions for the converter, and so in order to calculate switching loss, only t_1 and t_3 must be found. In our example, the gate drive for the MOSFET is an output pin on a PIC18F1320 microcontroller [16]. As discussed earlier, the time taken for switching is the time taken to charge C_{GS} and C_{GD}. The current to do this is supplied by the PIC and the output pin on the 18F-series is capable of driving 25 mA.

Therefore, the switching times can be estimated from:

$$t_1 = \frac{Q_{GS}}{I_{PIC}} = \frac{Q_{GS}}{25mA} \tag{6.4}$$

$$t_3 = \frac{Q_{GD}}{I_{PIC}} + t_1 \tag{6.5}$$

The values of Q_{GD} and Q_{GS} can be estimated from the plots of gate-source voltage against the total gate charge given in the datasheets (Figure 6.17). It is possible to correlate the individual gate charges to the time instances t_1 to t_3. For example, t_1 is the time required to raise the gate voltage to the threshold voltage, t_2 is the time at which C_{GS} is sufficiently charged to support the drain current set by the inductor, and t_2 to t_3 is the time taken to charge the Miller capacitance, C_{GD}.

By inspecting the plots of gate voltage against total gate charge, the values of $Q_{G(th)}$, Q_{GS} and Q_{GD} were estimated for each transistor, along with their respective rated currents (Figure 6.18).

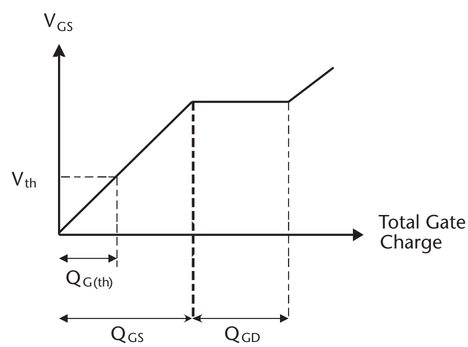


Figure 6.17 The charging of C_{GS} and C_{GD} depends on the applied V_{GS}.

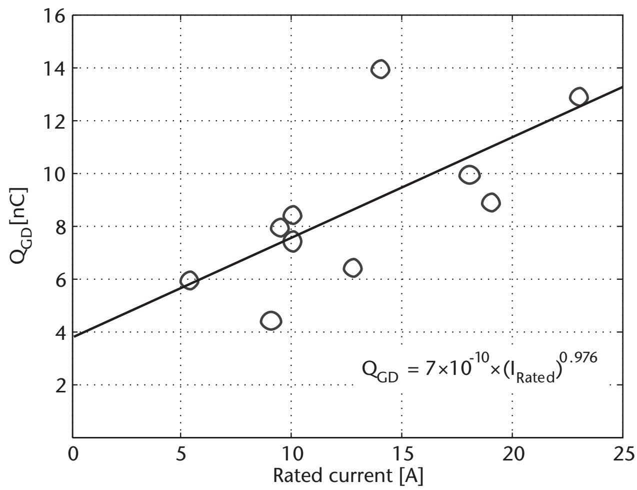


Figure 6.18 Estimated Q_{GD} values as a function of the rated current.

The expression relating Q_{GD} and the rated current was found to be:

$$Q_{GD} = 7\times10^{-10}\cdot\left(I_{Rated}\right)^{0.976} \tag{6.6}$$

The relationship between Q_{GS} and the rated current is:

$$Q_{GS} = 4\times10^{-10}\cdot\left(I_{Rated}\right)^{1.158} \tag{6.7}$$

Finally, the analytical expression for switching power loss is given by:

$$P_{SW} = \frac{1}{2}\left(V_{DD}\cdot I_{DS,\max}\right)\cdot\left[4\times10^{-10}\cdot\left(I_{Rated}\right)^{0.976}\right] \tag{6.8}$$

6.2.2.4 Gate Charge Losses

The stray capacitances C_{GD} and C_{GS} are repeatedly charged and discharged during the turn-on and turn-off transients when switching transistor Q_1 (Figure 6.20). This causes energy loss, as none of the energy placed on these capacitors is ever recovered. Figure 6.20(a) shows a switching circuit with gate drive from which the flow of charge through these stray capacitances can be analyzed and thus the energy losses are calculated. Transistors T_1 (PMOS) and T_2 (NMOS) were assumed to be ideal switches in a gate drive, V_G is the gate drive power supply voltage, and R_G is the output resistance of the gate drive.

The gate charge energy loss occurs in two instances: when the transistor is being switched on and when it is being switched off. Power loss due to the gate capacitance occurs when the charge is taken from V_G or V_{DD} to bias C_{GS} or C_{GD}. Consider the turn-on scenario in Figure 6.20(b) where T_1 becomes a short circuit and T_2 is open-circuited. Energy is transferred from the gate drive supply, V_G, to C_{GD} and C_{GS} as indicated by the flow of currents as shown by the arrows. The

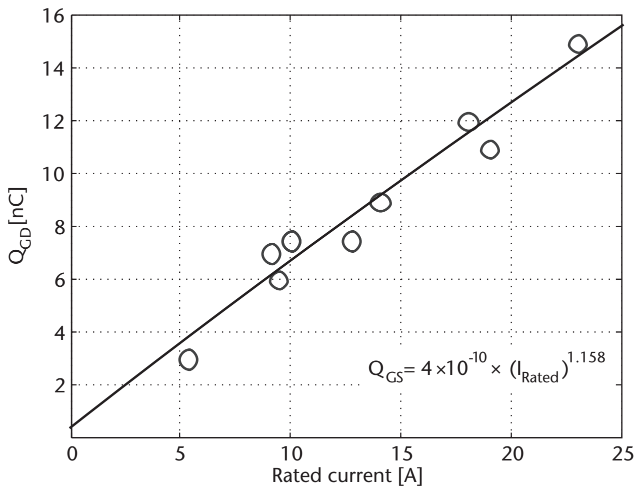


Figure 6.19 Estimated Q_{GS} values as a function of the rated current.

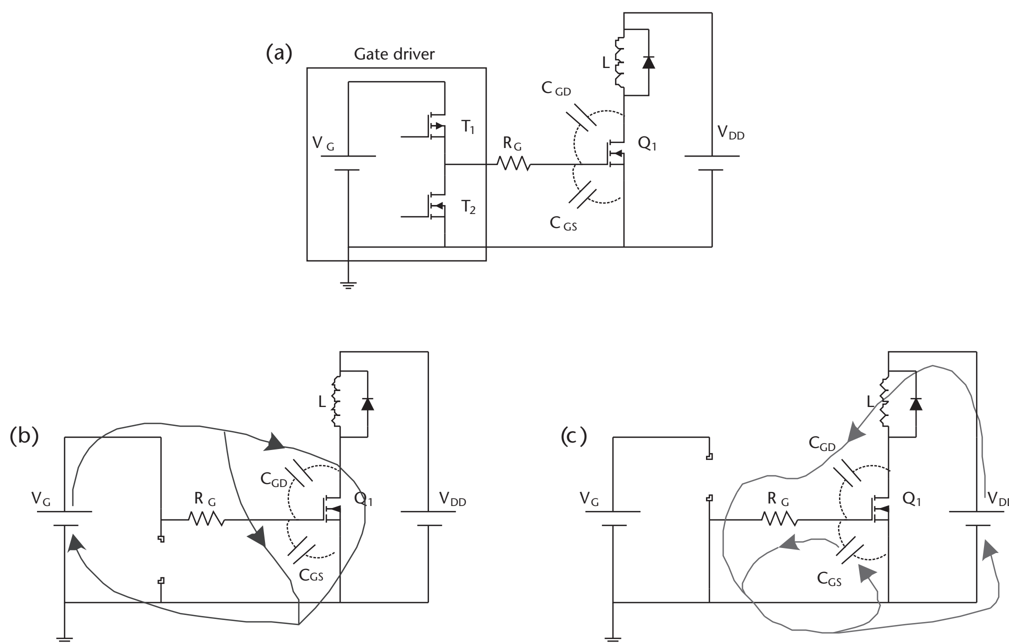


Figure 6.20 (a) Current flow through C_{GD} and C_{GS} during the (b) turn-on and (c) turn-off transients.

capacitor C_{GS} is charged by the gate driver from zero volts to V_G. Also, when Q_1 turns on, its drain voltage must fall from V_{DD} to the ground. To achieve this, C_{GD} would have had to accumulate a charge from V_G, and this amounts to an energy of $(Q_{GD} \cdot V_{GS})$. The amount of power lost in the stray capacitances is therefore given by:

$$P_{Gate(ON)} = f_{SW} \times \left(Q_{GS} \cdot V_{GS} + Q_{GD} \cdot V_{GS}\right) \tag{6.9}$$

As the transistor is switched off, both capacitances will discharge according to the path shown by the arrows in Figure 6.20(c). Here, T_1 is open-circuited and T_2 is shorted to ground. The current from C_{GS} will flow directly to ground (and thus no further energy is taken from the voltage source) whereas V_{DD} supplies the energy to bias C_{GD} in a direction opposite to that in Figure 6.20(b). Therefore, work is done to raise the voltage on the drain from zero to V_{DD}. This gives a turn-off power loss of:

$$P_{Gate(OFF)} = f_{SW} \times \left(Q_{GS} \cdot V_{GS}\right) \tag{6.10}$$

Consequently, the total power loss due to the gate charges is the sum of (6.9) and (6.10):

$$P_{Gate} = f_{SW} \times \left[Q_{GS} \cdot V_{GS} + Q_{GD} \cdot \left(V_{GS} + V_{DS}\right)\right] \tag{6.11}$$

Total Transistor Power Loss Adding all the power loss expressions together gives the total power loss in the transistor as a function of the device's rated current and switching frequency:

$$P_{FET} = P_{Cond} + P_{SW} + P_{Gate} \tag{6.12}$$

6.2.2.5 Optimization in MATLAB

By applying the same approach to finding expressions for losses in the diode and the inductor, an analytic expression for the complete power loss in the circuit was found as a function of the variables in Table 6.3. Then, the MATLAB function *fmincon* was used to find the minimum value of the total power loss. The results are shown in Table 6.4.

These results were validated by sweeping each variable to ensure that the minimum power loss (and thus the maximum useful output power) resulted from these specific values. The graphs in Figure 6.21 confirm that the minimization process was accurate in that it had found a minimum power loss for the system. As a result, the boost converter interface circuit was built using components that were chosen based on the values given in Table 6.4.

Table 6.4 Results from the Minimization Process

Variable	*Value*
FET rated current	4.52A
Diode rated current	0.56A
Switching frequency	36.2 kHz
Inductance	0.8 mH

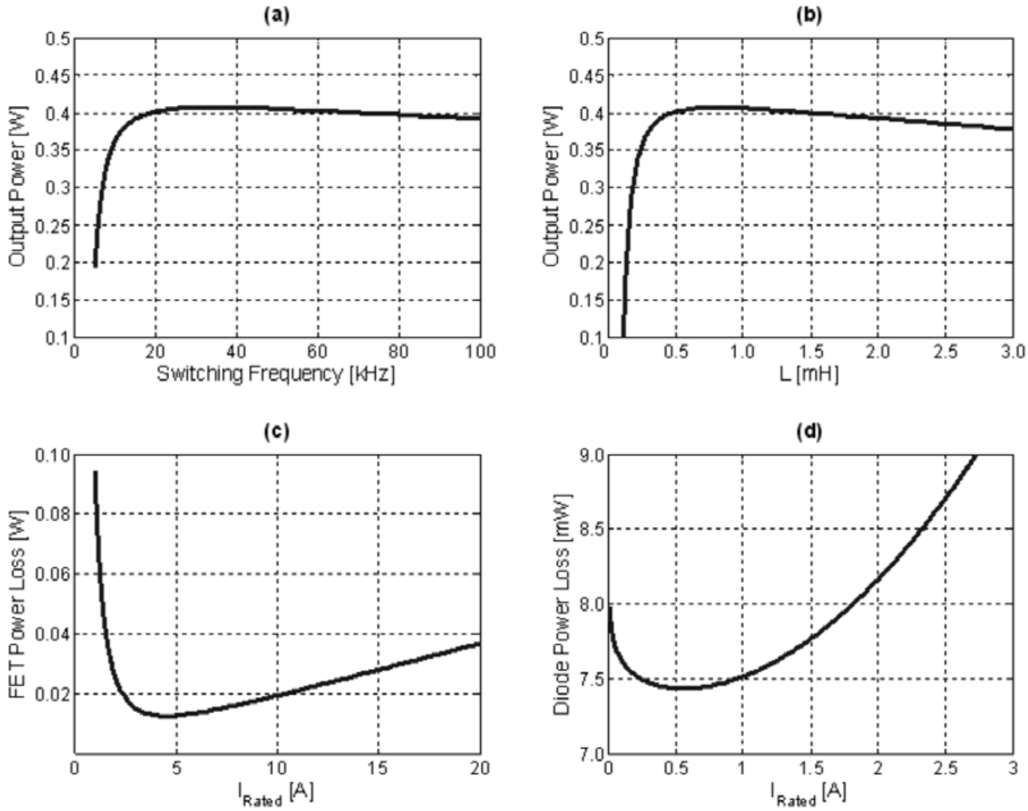


Figure 6.21 (a–d) Validation of the optimization process.

6.2.2.6 Performance of the Boost Converter

The efficiency of the prototype boost converter was found to have peak values of approximately 96% for duty cycle values less than 0.80. The pulse-width-modulated (PWM) signal was provided by an external signal generator and was set up such that the PWM frequency (36.2 kHz) and peak-to-peak (3.3V) values were the same as the microcontroller would provide. Note that, although the optimization sets a switching frequency of 36 kHz, the nearest frequency microcontroller that can be generated is 50 kHz. In addition to that, a 500Ω load resistor was connected to the output. At higher duty cycle values, more current flows in the boost converter, causing an increase in I^2R losses, which degrade the efficiency of the converter as shown in Figure 6.22.

The characterization of the boost converter DC transfer characteristic was performed by applying various input voltages (0.2V, 0.5V, 1.0V, and 2.0V) with a 500Ω load resistor connected at the output. A maximum voltage gain of 11.1 was achieved at a duty cycle of 0.95 for an input voltage of 0.5V. The experimental results follow the ideal voltage gain (gray crosses) very closely, as shown in Figure 6.23.

6.2.2.7 Input Impedance Control

As previously discussed, the optimal power transfer from the generator to a load requires that the load resistance, R_{LOAD}, match the generator's armature resistance,

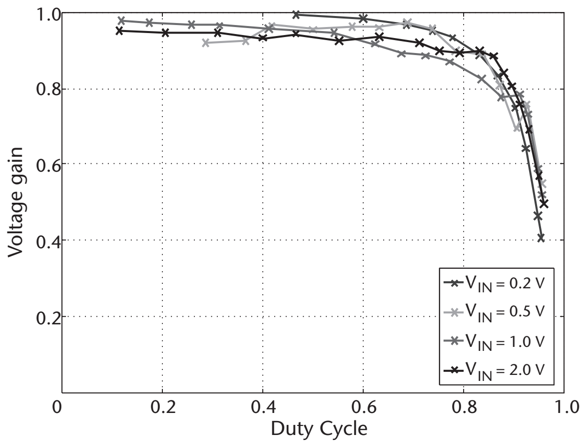


Figure 6.22 Efficiency of the boost converter at various duty cycle values.

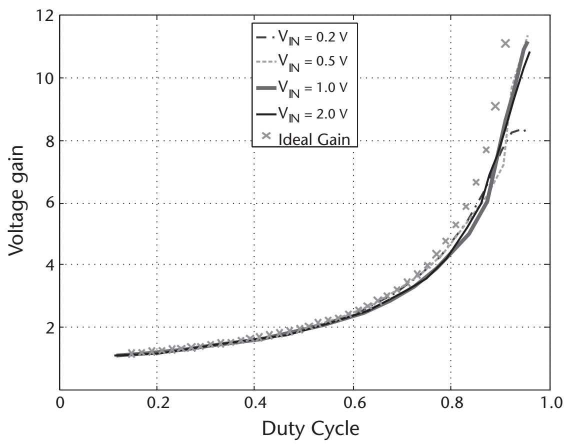


Figure 6.23 Voltage gain characteristics of the boost converter at different input voltages.

R_{ARM}, when the generator's offset mass is held at less than 90° to the vertical and that the current be controlled to a maximum value when the offset mass reaches 90° (Figure 6.10). The input impedance, R_{IN}, of a boost converter can be altered to be less than its load impedance R_{LOAD} by varying its duty cycle, δ. It was assumed that the value of R_{ARM} would be relatively small compared to the input impedance of a device that would potentially be powered by this generator.

$$R_{IN} = R_{LOAD} \cdot \left(1 - \delta^2\right) \tag{6.13}$$

The flow chart in Figure 6.24 demonstrates a conceptual implementation of a boost converter to perform this impedance match.

The boost converter inductor current can be measured using a sense resistor R_{Sense} and a current sense amplifier. Since the inductor current is the armature current from the generator, the online optimization procedure will match this inductor current to a demand value. This current demand value is obtained from the boost converter's input voltage, divided by the armature resistance, which is measured

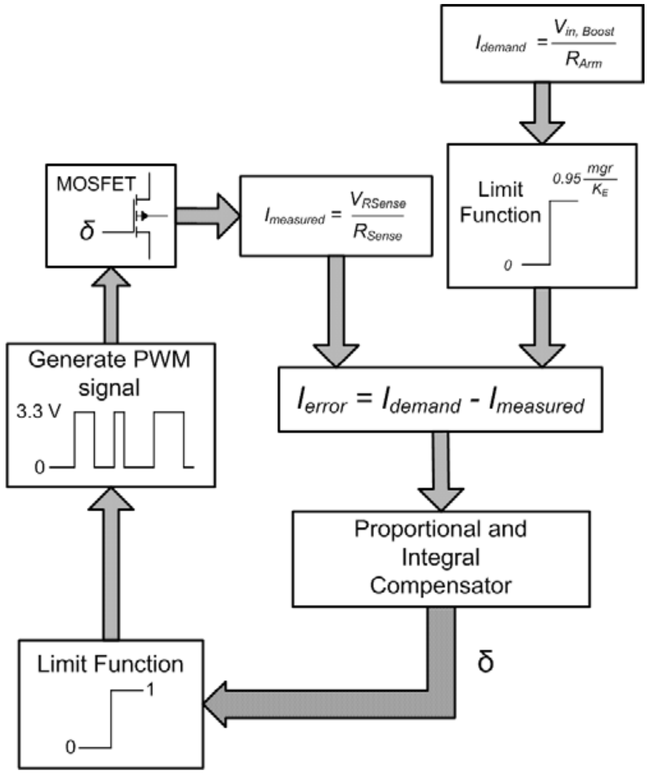


Figure 6.24 Flow chart of the boost converter input impedance matching procedure.

offline. This gives an indication of how much inductor current should be flowing in the circuit in order to present a near perfect impedance match between the generator's armature resistance and the load resistance that the generator sees. The error between the two currents is sent to a proportional and integral (PI) compensator, which calculates the duty cycle required to match the measured current as close as possible to the demand current.

6.2.2.8 Circuit Implementation

Figure 6.25 shows a block diagram of the power processing and control circuitry that implements the impedance match. A storage capacitor C_{STORE} was placed between the boost converter and an off-the-shelf RECOM regulated buck converter, allowing an accumulation of energy and output voltage regulation, respectively. C_{STORE} consists of three series-connected 6-mF supercapacitors rated at 15V, from AVX. The buck converter has a wide input range (4.75V–34V) and a regulated output (3.3V) so that an external device can be powered at a fixed voltage of 3.3V. The microcontroller samples the boost converter's input voltage, inductor current, and the voltage across C_{STORE} while generating the required duty cycle to perform an impedance match. An AM transmitter from RF Solutions, operating at a bandwidth of 433 MHz, was used to transmit the voltage levels of the storage capacitor to a PC, thus implementing a self-powered wireless sensor node.

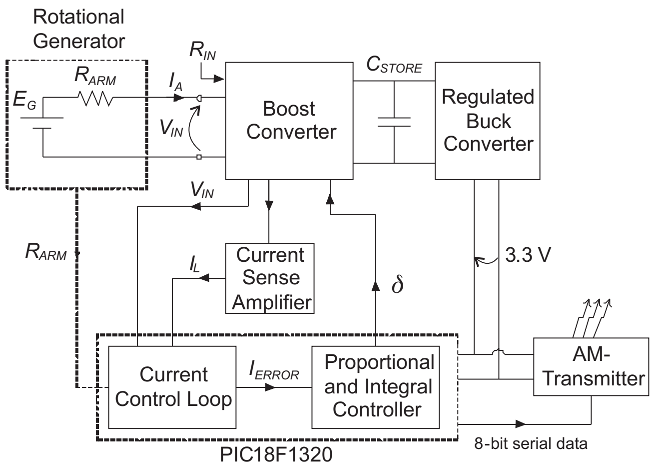


Figure 6.25 Schematic of the power processing and control circuitry.

6.2.2.9 Impedance Matching Results

The control loop outlined in Figure 6.24 was verified using a power supply to mimic the input voltage and current to the boost converter while a series connected resistance (9.1Ω) was used to simulate the armature resistance of the generator. Two load resistance values (50Ω and 100Ω) were connected in parallel with the storage capacitor while the boost converter's input voltage was varied from 0.3V to 2.0V. The input current changes proportionally with the variations in input voltages. The gradient of the graph in Figure 6.26 shows that the input impedance was held at 9.1Ω, for both load resistances.

Figure 6.27 shows the results obtained when two different load resistances (50Ω and 100Ω) were connected to the output of the boost converter. The graphs illustrate changes in duty cycle and correspondingly the storage capacitor voltage, while the input impedance of the boost converter was continuously matched to the target armature resistance of 9.1Ω.

In Figure 6.28, a varying input voltage was applied to simulate a condition where the rotation speed of the generator changes. It was observed that the input current changes proportionally to the input voltage in order to maintain a fixed

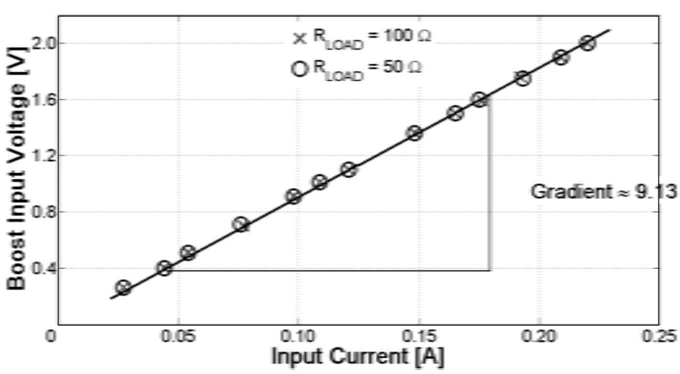


Figure 6.26 Impedance matching performance of the current control loop.

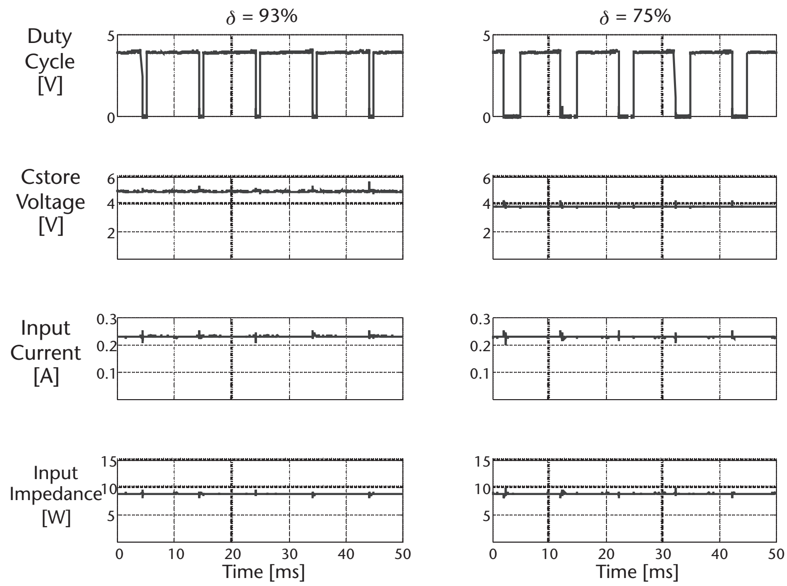


Figure 6.27 Variations in duty cycle under different load resistances to achieve an input impedance of 9.1Ω.

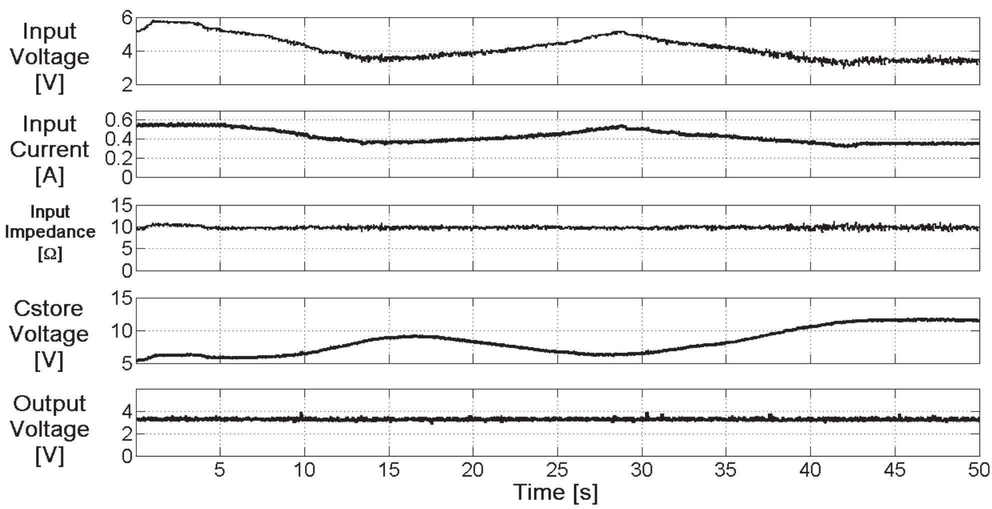


Figure 6.28 Performance of the impedance matching circuit for a varying input voltage and fixed load. The input impedance remains matched to R_{ARM}, 9.1Ω.

input impedance of 9.1Ω. When the generator's speed increases, more power is generated than is consumed by the load, leading to an increase in the voltage across the storage capacitor. When the contrary happens, the storage capacitor will discharge to maintain the operation of the impedance matching circuit. For the whole time, the output voltage from the Buck regulator stays at the predetermined value of 3.3V, and of equal importance, the input impedance stays matched to R_{ARM}—an essential requirement for harvesting energy optimally from a rotational source under practical situations.

6.2.2.10 Conclusions for Power Electronics System for Continually Rotating Harvester

A power electronics system for an energy harvester that includes a transducer interface circuit, energy storage, and output voltage regulation has been developed and demonstrated. The main difficulty in the design is that the circuit must be efficient and operate over wide voltage ranges and the control circuit must consume very little energy so that the system is capable of being self-sustaining from the harvested energy while still being able to supply power to a load. An end-to-end system optimization was described for a boost converter interface circuit; this minimized the losses in the converter, resulting in an efficiency of 96%. The overall aim was to provide an impedance match to the generator's armature resistance and at the same time supply a regulated output voltage from which a load can be powered while storing energy to allow the system to maintain operation when the energy-harvesting source is intermittent. Therefore, all three functions required for an energy harvesting system (i.e., transducer interfacing for maximum power extraction, energy storage, and output voltage regulation) have been demonstrated in the above example.

6.2.3 Piezoelectric Harvesters

The typical electrical equivalent circuit of a vibration-driven piezoelectric harvester is shown in Figure 6.3(b). When previously considering the design of interface circuits for electromagnetic devices shown in Figure 6.3(a), we noted that in order to maximize power extraction from the transducer, we should set the interface circuit to have an input impedance of $\frac{1}{D_p}$, assuming that the generator was operating at resonance and that no other constraints (such as the displacement limit of the mass) were in operation. This argument is valid as long as the inductance of the transducer is negligible; this is frequently the case for the electromagnetic harvester (although not always). However, for piezoelectric transducers, the shunt capacitance can never be neglected because of the low coupling coefficient of the piezoelectric material.

A poor coupling between the mechanical and electrical domains of the piezoelectric material means that the transformer component in Figure 6.3(b) is a step-up transformer with a high turn ratio. This means that very little voltage is developed across the primary side of the transducer. Therefore, at resonance, the mechanical motion of the transducer (i.e., its maximum displacement) is set almost entirely by the mechanical parasitic damping on the primary side of the transformer rather than the electrical loading. As a consequence, the piezoelectric current generated is almost independent of the electric loading on the generator and the equivalent circuit can be replaced with a much simpler model as shown in Figure 6.29, where the current source frequency is the same as the mechanical vibration and the magnitude is set by the properties of the piezoelectric material (which determines the capacitance) and the parasitic damping (which determines the amplitude of the mechanical motion).

As a consequence, it can be shown [17] that the maximum power that can be dissipated in a linear load resistance (or into an interface circuit with an equivalent input impedance) occurs when the load resistance is given by:

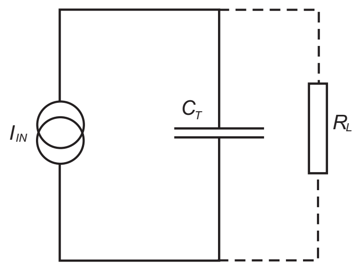


Figure 6.29 Simplified model of piezoelectric generator (assuming poor electromechanical coupling).

$$R_L = \frac{1}{\omega C_T} \tag{6.14}$$

It is clear that in this case the power that can be extracted from the circuit is limited by the intrinsic shunt capacitance of the piezoelectric material. However, if an impedance match as per (6.14) was presented to the piezoelectric harvester, the mass could potentially hit the end-stops of the harvester. This is because the electrical damping force from an optimal load resistance is not large enough to damp the motion of the mass when the displacement of the harvester is significantly larger than the maximum displacement limit of the proof mass. Unlike the power processing circuits presented earlier, a conventional impedance match would not be the best method to use in order to prevent the proof mass from needlessly dissipating energy at the end-stops of the harvester.

Early work on piezoelectric harvesters made use of this resistive match to maximize power output by measuring the power dissipated in a simple load resistor [18, 19], although more recent work has attempted to overcome this limitation by using timed switching elements instead of optimized linear resistive loads.

To increase the power output over what can be achieved with a linear resistive load, two steps can be taken:

1. Prebiasing the piezoelectric material before mechanical work is done against it;
2. Synchronously extracting the charge from the piezoelectric element rather than continuous extracting into a linear resistive circuit.

The first idea of prebiasing can allow a stronger coupling between the electrical and mechanical systems. The second idea of synchronous discharge overcomes the limitation of real power transfer due to the presence of the intrinsic capacitance. When a piezoelectric material is strained in one direction in an open circuit, the resulting charge displacement causes a force that tries to move the material back to an unstrained state, and some work is done in straining the material. If a charge is placed onto the material forcing it to become strained in one direction before the material is forced to move in the other direction by an external force, more mechanical work can be done as the force presented by the piezoelectric material is increased. Therefore, more electrical energy can be generated. This is illustrated in Figure 6.30. When the piezocantilever is strained upward at a maximum displacement such that a positive charge would be generated by the deflection of the

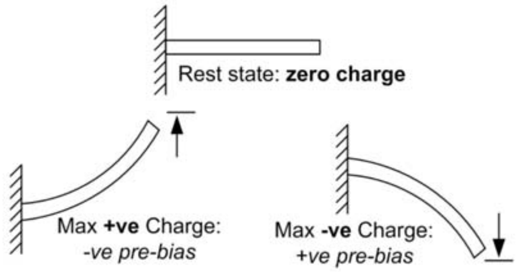


Figure 6.30 Prebiasing of the piezoelectric increases the damping force.

material if in an open circuit, a negative prebias voltage is applied to the material, allowing increased mechanical work to be done as the cantilever's free end moves downwards.

The opposite applies when the free end of the piezocantilever is at the maximum downwards position. If the applied bias V_B is large compared to the piezoelectrically induced voltage change ΔV_p, the force magnitude will now be constant at $\approx \alpha V_B$, rather than oscillating in the range $\pm \alpha \Delta V_p$. The voltage on the piezoelectric material is then as sketched in Figure 6.31.

The first of these techniques, that of prebiasing, was originally proposed by Taylor et al. in [21]; however, Guyomar et al. were the first to apply the technique to the low power energy-harvesting domain in [22]. An increased power output was demonstrated by inverting the charge from the piezoelectric material at the extremes of the motion. The piezoelectric transducer terminals were also connected to a bridge rectifier and smoothing capacitor, allowing the extraction of power in a useful stable DC form. In [22] the explanation of improved power output is given in terms of the nonlinear functioning of the circuit, but it is the increased mechanical force due to the resultant cell biasing that is the essential origin of the increased output power. The disadvantage of this technique is that the charge extraction from the piezoelectric material cannot be controlled independently of the voltage on the output side storage capacitor. Ultimately, this means that the precharge bias cannot be optimized for the particular vibration source and mechanical generator characteristics as it is dependent on the storage capacitor voltage and loads resistance. In other words, the optimal electrical damping, detailed in Chapter 4, cannot be set independently of the capacitor voltage.

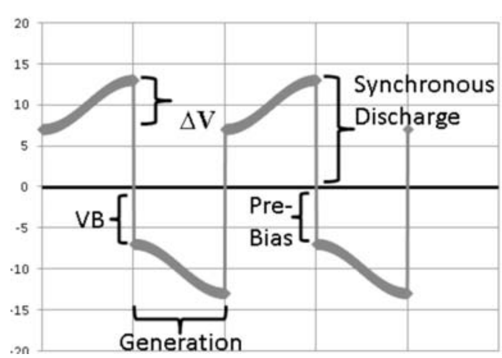


Figure 6.31 Piezoelectric voltage when operated with prebias and synchronous discharge. (*From:* [20]. Reprinted with permission.)

Their latest results are presented in [23], where they propose a synchronized switch harvesting on inductor circuit with magnetic rectifier (SSHI-MR). This circuit, shown in Figure 6.32, utilizes a transformer with a turn ratio that is much greater than one. The transformer, with two antiparallel primary windings, allows the conversion of the AC piezoelectric voltage to DC. Switches S1 and S1' (serially connected to a primary winding) are closed when the displacement of the piezoelectric element reaches its maximum and minimum points, respectively. These switches are alternatively opened at half the resonating time period of $\sqrt{LC_0}$, which arises from the series combination of L and C_0. With the transformer in place, the threshold at which the diode conducts is lowered to $\frac{V_D}{m}$. This could potentially give a significant reduction in the diode conduction losses when compared with a full diode bridge directly connected to the piezoelectric material. At a displacement amplitude of 23 μm and a vibration frequency of 1 kHz, the SSHI-MR technique resulted in a harvested power of approximately 400 μW when an optimal load resistor is used. This harvested power is 56 times greater than when a conventional diode bridge rectifier was used in place of the transformer, signifying the importance of reducing the power losses inherent in discrete power electronics components such as diodes.

In an attempt to allow optimal prebiasing without dependence on the status of the load circuit (i.e., capacitor voltage or load resistance), Dicken et al. presented a new approach to increasing the output power from piezoelectric energy harvesters by prebiasing combined with a synchronous charge extraction circuit.

The key potential improvement of this approach over the techniques presented by Guyomar is that the precharge bias circuit and piezoelectric generation cycle can be completely isolated from the output side circuitry and therefore there is no such thing as an optimal load resistance, only an optimal prebias voltage. The optimization of the energy capture by this circuit therefore only depends on the prebias voltage applied to the piezoelectric device. The prototyped circuit is shown in Figure 6.33.

MOSFETs 1 to 4 are used to prebias the piezoelectric material at the extremes of the cycle. MOSFETs 5 and 6 are used to extract the energy from the piezo to the output stage just before prebiasing occurs. Diodes are present to allow the recovery of energy stored in inductors to the power supply.

The energy stored in the piezoelectric material's intrinsic capacitance is proportional to the square of the voltage generated by its deflection. If an additional

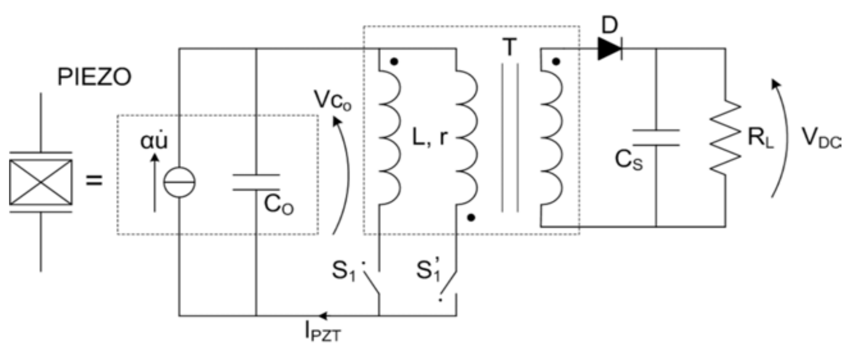


Figure 6.32 Synchronized switch harvesting on inductor (SSHI) with magnetic rectifier circuit as proposed by Garbuio et al. [23].

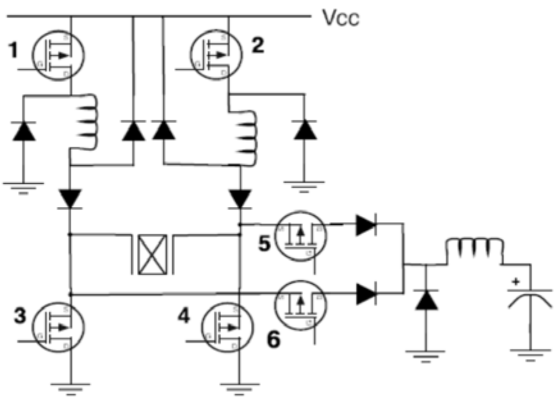


Figure 6.33 Piezoelectric prebiasing circuit with a synchronous charge extraction. (*From:* [20]. Reprinted with permission.)

charge was added to the piezoelectric material prior to the generation of the charge due to mechanical deflections, more work is required to charge the intrinsic capacitance. This is because the voltage on the charge will be higher when compared to the situation where no initial charge was present (no prebiasing). Once the energy generated from the previous half-cycle of the mechanical deflection is discharged, the piezoelectric material will be prebiased at its maximum and minimum deflection positions before the material deflects in the opposite direction. Calculating the gain in energy due to the precharging condition requires the energy used in charging and discharging the piezoelectric material. Defining the efficiencies of the charging and discharging steps as η_c and η_d, respectively, the energy supplied to charge the piezoelectric material to a voltage, V is $\frac{CV^2}{2\eta_c}$, while the useful energy obtained at discharge is $\frac{1}{2}C\eta_d(V+\Delta V)^2$. Variables C and ΔV represent the intrinsic capacitance and the voltage change due to the mechanical deflection of the piezoelectric material. Thus, the net output energy is:

$$E_{out} = \frac{1}{2}C\left[\eta_d\left(V+\Delta V\right)^2 - \frac{V^2}{\eta_c}\right] \tag{6.15}$$

By setting $\frac{dE_{out}}{dV} = 0$, the optimum V in terms of ΔV can be found.

$$V = \frac{\eta_c\eta_d}{1-\eta_c\eta_d}\Delta V \tag{6.16}$$

Using (6.15) and (6.16), an expression for the optimum energy gain in terms of the efficiency can be obtained. Assuming that $\eta_c = \eta_d = \eta$, the energy gain factor, f_E, the ratio of energy generated for synchronous extraction with zero prebias to energy generated with the optimal prebias for a given efficiency, is:

$$f_E = \frac{E(\eta)}{E(V=0)} = \eta + \frac{3\eta^3}{1-\eta^2} \tag{6.17}$$

The energy gain factor in (6.17) is plotted in Figure 6.34. A high output gain is obtainable at efficiencies greater than 90%.

Results presented in [20] showed that the prebiasing technique produced a net output power of about 110 μW at a prebias voltage of 12.5V (Figure 6.35). This is an increase of approximately 10 times the output power compared to that using a simple optimal load resistance. At the moment, this technique has not shown as much increase in power over a simple optimal resistor as that shown by Guyomar, although in the experimental results shown in Figure 6.35, the breakdown of the semiconductors was the limiting factor.

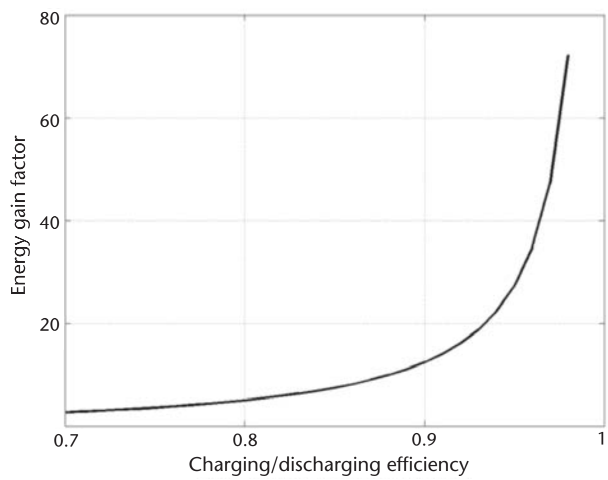


Figure 6.34 Theoretical power enhancement relative to conventional piezoelectric cell versus efficiency of prebiasing [20].

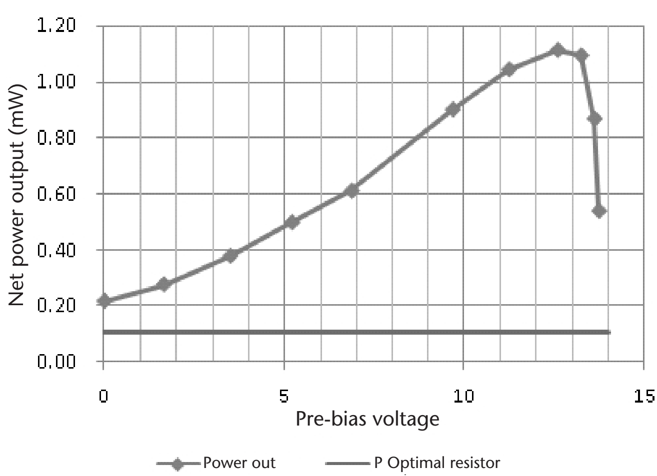


Figure 6.35 Improvement in net output power with prebiasing compared to that using only an optimal resistive load [20].

6.2.4 Electrostatic Harvesters

As discussed earlier in this book, the electrostatic harvester generally uses a moving plate capacitor in order to convert kinetic energy into electrical energy. The existence of the nonconstant valued capacitor makes it difficult to model an electrostatic generator using linear circuit components. Only an approximation is possible [Figure 6.3(b)] and such an equivalent model does not necessarily give insight into the device operation. It is, however, possible to derive the optimal operation of an electrostatic generator in terms of capacitor voltage and thus to determine the optimal operation of the interface circuitry so as to realize the equivalent of an impedance match for the electrostatic case.

Among all the energy harvesters reported to date, the miniaturization of electrostatic harvesters has been more promising than the other transducer technologies in terms of the creation of true MEMS devices utilizing MEMS fabrication techniques at typical MEMS device scales. Consequently, the power electronic circuits presented in this section have generally been designed with a view to the fact that the harvester output powers are very low, in the 1–100-μW range. This minimal power output and the high voltages generated place very difficult constraints on the power electronics in terms of minimizing off-state conductance and parasitic capacitance, and an example of custom semiconductor device design for an electrostatic harvester is discussed.

There are two main techniques that have been used to realize the electrostatic transducer mechanism. These are switched systems and the continuous systems [24], with switched systems being the most studied.

6.2.4.1 Switched Systems

The switched type of connection between the transducer and the circuitry involves a reconfiguration of the system, through the operation of switches, at different parts of the generation cycle. Switched transducers can further be split into two main types: constant charge and constant voltage.

When the transducer is operated under a constant charge, the plates are separated away from one another with a fixed overlap area. However, under constant voltage operation, the plates are moved relative to one another while maintaining a fixed gap between them. The conditions that the interface electronics must present to the harvester in order to extract power optimally can be found using the forces present on the plates of the capacitor as shown in Figure 6.36. The rate of change of capacitance with respect to distance differs depending on the axis of the relative motion of the two plates: x_{perp} for perpendicular motions and x_{par} for parallel motions. Consequently, for a given electric field strength and plate area, the force between the plates not only depends on the distance between the plates, but also on the axis of relative motion. These forces are indicated as F_{perp} (perpendicular force) and F_{par} (parallel force) in Figure 6.36. Using the principle of virtual work, the perpendicular and parallel forces acting on the capacitor plates can be found, depending on whether the charge or the voltage across the capacitor plates is held constant.

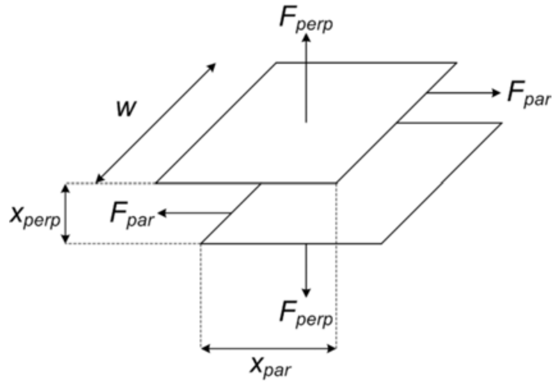


Figure 6.36 Forces acting on charged capacitor plates.

Perpendicular Force The energy stored in the parallel plate capacitor in Figure 6.36 is:

$$Energy = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}Q^2\left(\frac{x_{perp}}{\varepsilon w x_{par}}\right) \tag{6.18}$$

When the plates of the capacitor experience a change in the perpendicular direction (x_{perp}) with the plates having a fixed amount of charge, work is done against the electric field between the plates and electrical energy will be generated. As the plate separation increases, additional potential energy is stored in the increased volume of electric field. The perpendicular force acting on the plates can be found by differentiating the equation for energy with respect to the perpendicular separation of the plates (x_{perp}).

$$F_{perp} = \frac{1}{2}\left(\frac{Q^2}{\varepsilon w x_{par}}\right) \tag{6.19}$$

Parallel Sliding Force Moving the relative positions of the plates such that the overlapping area between them varies with time will change the capacitance between the electrodes. Using the principle of virtual work,

$$Energy = \frac{1}{2}V^2\left(\frac{\varepsilon w x_{par}}{x_{perp}}\right) \tag{6.20}$$

If the capacitor plates have a fixed voltage across them and are moved relative to one another but with a constant separation distance, the electric field strength remains constant, but current is forced to flow because the volume of the electric field decreases.

$$F_{par} = \frac{1}{2}V^2\left(\frac{\varepsilon w}{x_{perp}}\right) \tag{6.21}$$

The expressions of perpendicular and parallel forces acting on the plates of the electrostatic harvester provides an indication of how much electrical damping should be applied to the harvester for it to operate optimally and not hit the end-stops. For both the constant charge and voltage cases, the optimal electrical damping force that results in an optimized power output from the harvester is given by:

$$F_{opt} = \frac{Y_0 m\omega^2}{\sqrt{2}}\frac{\omega_c}{\left|\left(1-\omega_c^2\right)\cdot U\right|} \tag{6.22}$$

where Y_0 is the displacement of the electrostatic harvester, m is the proof mass, ω is the frequency of vibration, and ω_c is the frequency of vibration normalized to the resonant frequency of the harvester. The variable U is defined as:

$$U = \frac{\sin\left(\frac{\pi}{\omega_c}\right)}{1+\cos\left(\frac{\pi}{\omega_c}\right)} \tag{6.23}$$

Each variable in (6.22) and (6.23) has a specific value depending on the operation of the electrostatic harvester. Thus, to extract power optimally from the harvester, the interface electronics has to provide an electrical damping force equivalent to (6.22) by delivering the correct amount of charge or voltage to the transducer. This is equivalent to an impedance match for the electrostatic case. If the applied electrical damping force is greater than the sum of the inertial and spring force (a harvester modeled as a mass-spring-damper system), the mass will cease to move relative to the harvester's frame and no energy is generated.

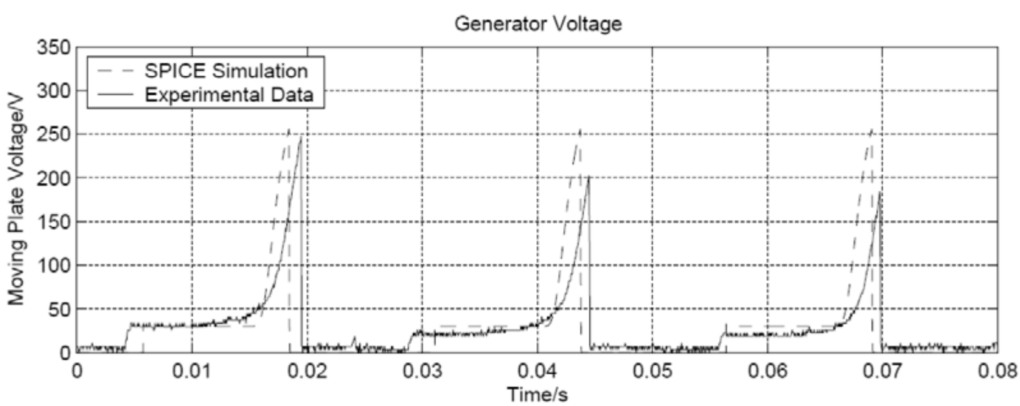


Figure 6.37 Simulated and experimental data for an electrostatic generator operating under a constant charge. (*From:* [25]. Reprinted with permission.)

6.2.4.2 Examples of Interface Electronics for Constant Charge Operation

This type of electrostatic harvester operation was reported in [25] for a MEMS-fabricated energy harvester. The prototype was fabricated using techniques such as DRIE and the movable capacitor plate had an active area of approximately 200 mm^2. In [25], at around 50 ms, the capacitor is precharged, at maximum capacitance, to around 30V. After some time, the source motion causes the plates to separate. This operation is done under constant charge and so a large increase in voltage can be seen. Once the electrodes reach maximum separation, the capacitor is discharged. This generator was shown to generate around 12 μJ from an input motion of 40 Hz and a 6-mm amplitude.

A suitable power conversion circuit for the output side of the aforementioned generator in [25] is the half-bridge, step-down circuit shown in Figure 6.38. The half-bridge has been chosen so that a boot-strap drive can be used to turn on the high-side semiconductor switch, in this case a MOSFET. Although the generation cycle time is long (circa 10 ms) and unpredictable, the power converter need only operate for less than 1 ms to completely discharge the capacitor and so the boot-strap technique is viable. It is desirable to use an integrated inductor, and inductance values in the range 1–10 μH appear to be achievable [26]. The discharge of the generator will occur in a short current pulse; controlling this current through chopping would require a high switching frequency, and consequently the associated power losses will be undesirable.

It is convenient to split the operation of the circuit in Figure 6.39 into three phases, as shown in [27]. The converter is used in single-pulse mode and the source is weak enough to be completely discharged within a few nanoseconds. In the first phase, during the turn-on of the MOSFET, current flows into the diode to establish a reverse bias and to allow the voltage over the MOSFET to reduce. This current is supplied by the generator, and this is an unwanted loss of charge. During the second phase, the inductor current increases and the generator voltage falls until the generator is completely discharged. At this point the inductor current is at its maximum. Then the longest phase begins in which the current freewheels through the diode until the inductor is demagnetized.

As a first step to designing the circuit in Figure 6.38 for the MEMS harvester, an assessment was made of the input resistance and capacitance that the circuit

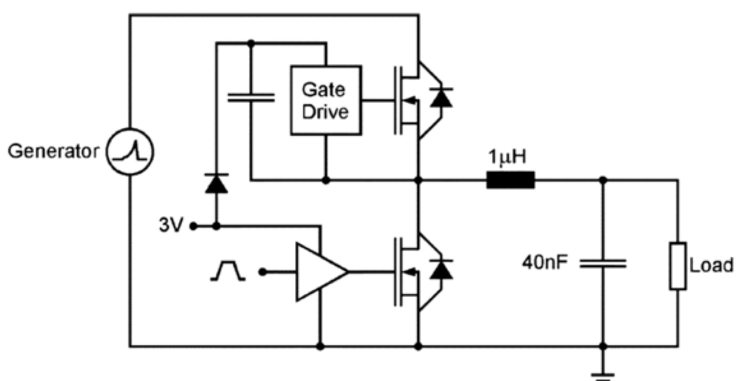


Figure 6.38 Half-bridge converter proposed in [27]; the low side MOSFET is only required for bootstrap gate drive.

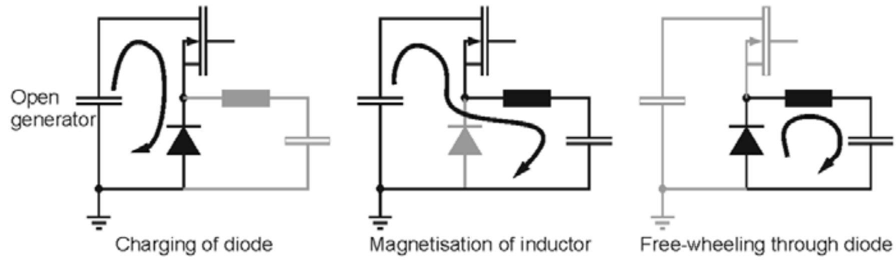


Figure 6.39 Three phases of conversion with distinct current patterns.

must present in the off-state at the maximum generator voltage in order not to compromise generation. The generator's electromechanical system was simulated numerically using MATLAB for a range of static impedances on the generator outputs, assuming a 20-ms flight time. The requirements are unusually strict. To maintain 80% of the generated energy, the off-state loading should be more than $10^{12}\Omega$ and less than 1 pF [4]. These values are not available with standard discrete MOSFETs rated for 300-V blocking. By assuming that the parasitic components of the converter are constant, their effect on the energy generation is analyzed and plotted in Figure 6.40.

To achieve this high level of impedance, thin-layer silicon on insulator technology-based semiconductors must be designed. In [27], in depth simulation studies were carried out to optimize the MOSFET and diode device areas to optimize the energy generated from the system, taking into account conduction loss and charge-sharing effects. A cross-section through the custom MOSFET is shown in Figure 6.41. It was found that the on-state voltage drop of the MOSFET predominantly affects the conversion efficiency because of high peak currents, which are due to the low inductance used in the circuit in order that the inductor could be integrated on a chip.

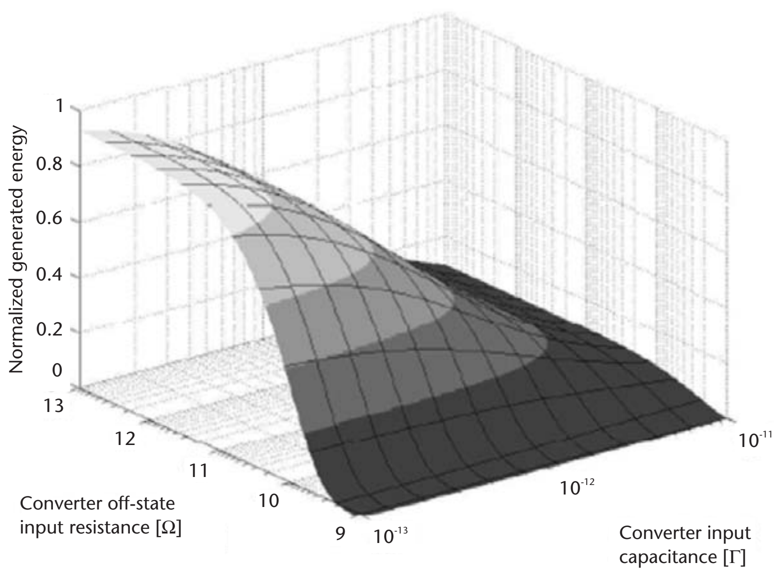


Figure 6.40 Dependence of generated energy on converter impedance.

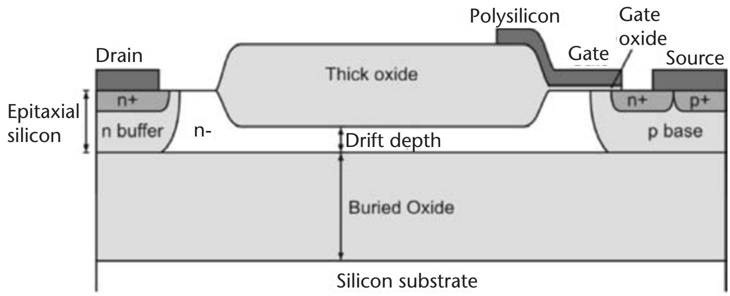


Figure 6.41 Custom-designed silicon on insulator MOSFET for MEMS electrostatic harvester. (*From:* [27]. Reprinted with permission.)

In the above example, interface electronics would be required to charge the variable capacitor through an external precharge power supply (probably battery) at maximum capacitance and to discharge the variable capacitor through a load (or to recharge the battery) at a minimum capacitance. The discharge circuitry alone is not sufficient to make a working energy harvester system. An example of a more complete system with both input and output side electronics for the electrostatic transducer is shown in [27]. A charge pump circuit is used to charge and discharge the variable capacitor. Diode D1 will be on when the variable capacitor is at a minimum position (i.e., capacitance is maximum). Diode D2 will be on when the voltage at node A is more than the load voltage. Both the diodes will be off during the rest of the vibration cycle period. Diodes with a low reverse leakage current are suitable for this application to reduce the leakage power loss. JFETs working in a diode mode have been used in [28] to reduce the reverse leakage current.

The basic circuit of Figure 6.42 will eventually discharge the energy in the precharge source; to avoid this, a flyback inductor was used as shown in Figure 6.43. Charging and discharging of the variable capacitor are done using the charge pump circuit, and the flyback inductor was used to transfer the energy from the temporary storage capacitor (C_{store}). Energy will be stored in the inductor by turning on the MOSFET, and when the MOSFET is turned off, the inductor current will freewheel through diode D_{FLY}. The MOSFET gate pulse need not be synchronized with the vibration cycle, which is the case of modified charge pump circuit, hence reducing the complexity of the circuit. Detailed analysis of calculating the efficiency of power conversion is given in [29].

6.2.4.3 Examples of Interface Electronics for Constant Voltage Operation

To further reduce losses such as forward conduction losses, active switches are used instead of diodes in [30], and the modified charge pump circuit is shown in Figure 6.44. Energy conversion from the mechanical to electrical domain was implemented using low-power digital control circuitry consisting of a delay locked loop (DLL) capable of synchronizing the energy extraction mechanism to the source vibration frequency ω in (6.22). Upon achieving this phase lock, the reference clock in the digital circuitry will be in phase with the motion of the generator's moving plate. This enables the generation of the timing pulses for the gates of SW1 and SW2. During the precharge condition, SW2 will be switched on to store energy in inductor L. The stored inductor energy will be used to charge the variable capacitor C_{var} by turning

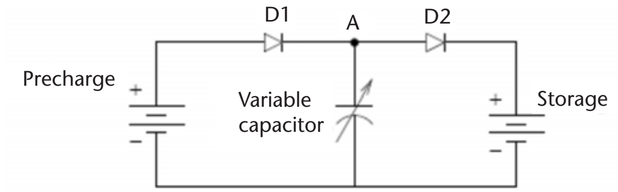


Figure 6.42 Basic charge pump circuit.

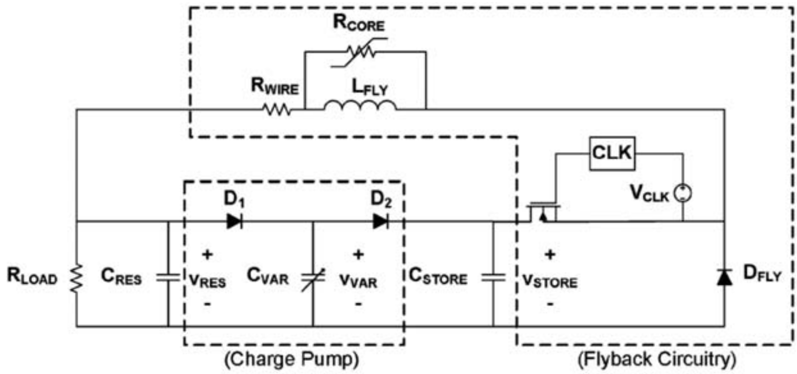


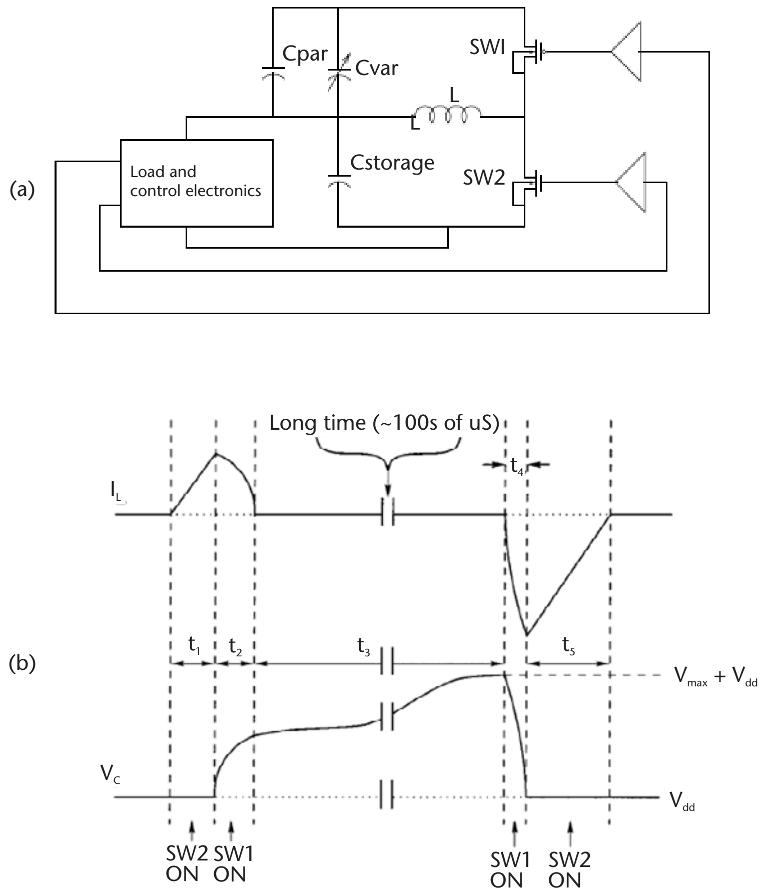
Figure 6.43 Capacitive energy harvester with source-referenced clock controlling the flyback switch [29].

on and off SW1 and SW2, respectively. During the discharge period, the opposite switching sequence of the precharging condition will be implemented to discharge C_{var}. Simulation results of the digital control circuit in HSPICE predicted a control overhead of around 3 μW. The electrostatic generator was predicted to produce 8.6 μW of power, leaving 5.6 μW of electrical power for the load electronics.

Another example of a power processing circuit for a voltage constrained electrostatic microgenerator is shown in [30]. During the precharge condition, SW2 and SW5 will be switched on to store energy in the inductor L. Switches SW3 and SW4 will be turned on by simultaneously turning off SW2 and SW5 to charge the variable capacitor C_{var}. The unidirectional switch SW1 will be turned on to allow the current to flow from the variable capacitor C_{var} to the battery. When the variable capacitor has reached its minimum value, SW1 will be turned off. In order to completely recover the charge across the variable capacitor, a reverse switching sequence of the precharge condition is used. A complete description of the circuit with waveforms is discussed in [31].

6.2.4.4 Continuous Systems

A third mode of operation exists when the variable capacitor is continuously connected to the load circuitry, and this load circuitry provides the capacitor with a polarization voltage. A simple example of this is a voltage source, a resistor, and a variable capacitor wired in series. A change in capacitance will always result in a charge transfer in between the electrodes through the load resistance causing work to be done in the load.



Note: Timing waveforms denote when a switch is on. If a switch is not denoted as on, then it is off.

Figure 6.44 (a) Modified basic charge pump circuit and (b) waveforms [30].

The switched generators previously discussed are special cases of this continuous mode generator. A constant charge generator is equivalent to a continuous generator operated with infinitely high load impedance, while the constant voltage generator corresponds to a continuous generator that is short-circuited. Because no work can be done when either the generated current or the generated voltage is zero, these extremes of operation require a switching circuit to make them operate. The use of controlled switches complicates the implementation of the generator and the circuitry required to control them consumes a minimum amount of the generated power and so in some circumstances the use of a continuous system is preferred.

Electrets are often used in combination with a variable capacitance to make a continuous mode generator. The fixed charges of the electret induce an electric field between the electrodes of the capacitor, corresponding to a potential of several tens of volts. Three possible Q-V diagrams showing the operation of a continuous electret generator are shown in Figure 6.46(a). If the capacitor is operated in a constant voltage mode, a change in capacitance will result in a current through the load circuitry along curve (1-3-1). A high impedance load forces the generator to operate

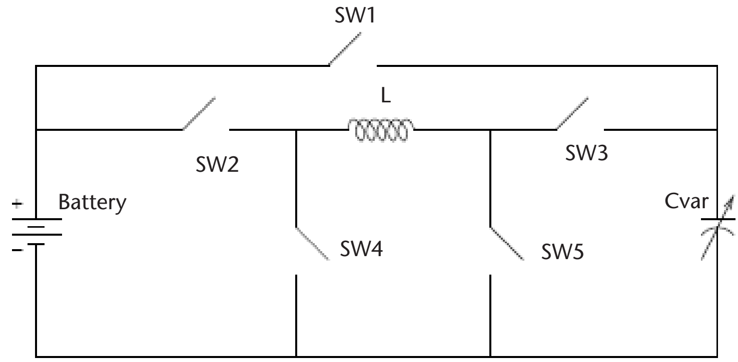
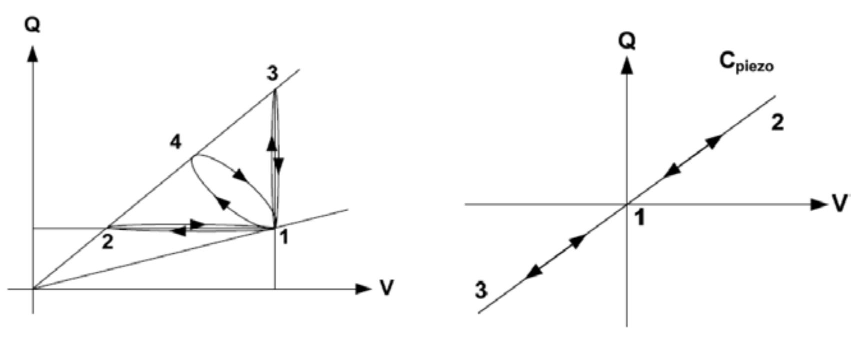


Figure 6.45 Constant voltage based electrostatic microgenerator for battery-charging applications [31].

in constant charge as the high impedance obstructs the charge transport between the electrodes (1-2-1). In both of these cases, the area of the Q-V loop integral is zero as the transition from maximum to minimum capacitance occurs on the same trajectory. An optimized load for a continuous generator will operate the generator in between these extremes along (1-4-1), and as can be seen, work is now done and the loop integral has a finite value. This class of generators is referred to as velocity damped generators because the damping force is approximately proportional to the relative velocity between the proof mass and the frame.

6.2.4.5 Examples of Interface Electronics for Continuous Mode Operation

Sterken et al. micromachined a prototype of a 0.1 cm^2 electrostatic microgenerator using silicon-on-insulator (SOI) methods. This comb-like structure was predicted to be able to generate 50 μW. An electret precharges the moving plate of the capacitor (up to a limit of 50V to prevent clamp-down), which is suspended by meandered beams that function as springs. The lateral displacement of the moving plate changes its capacitance and charge, thus causing a current to flow through the load resistor, *R*, as indicated on the left side of Figure 6.47(a). This load resistor is representative of prospective power management electronics to condition the power from the electrostatic generator.



(a) (b)

Figure 6.46 Operation of an electrostatic generator in (a) continuous mode or (b) using piezoelectric polarization.

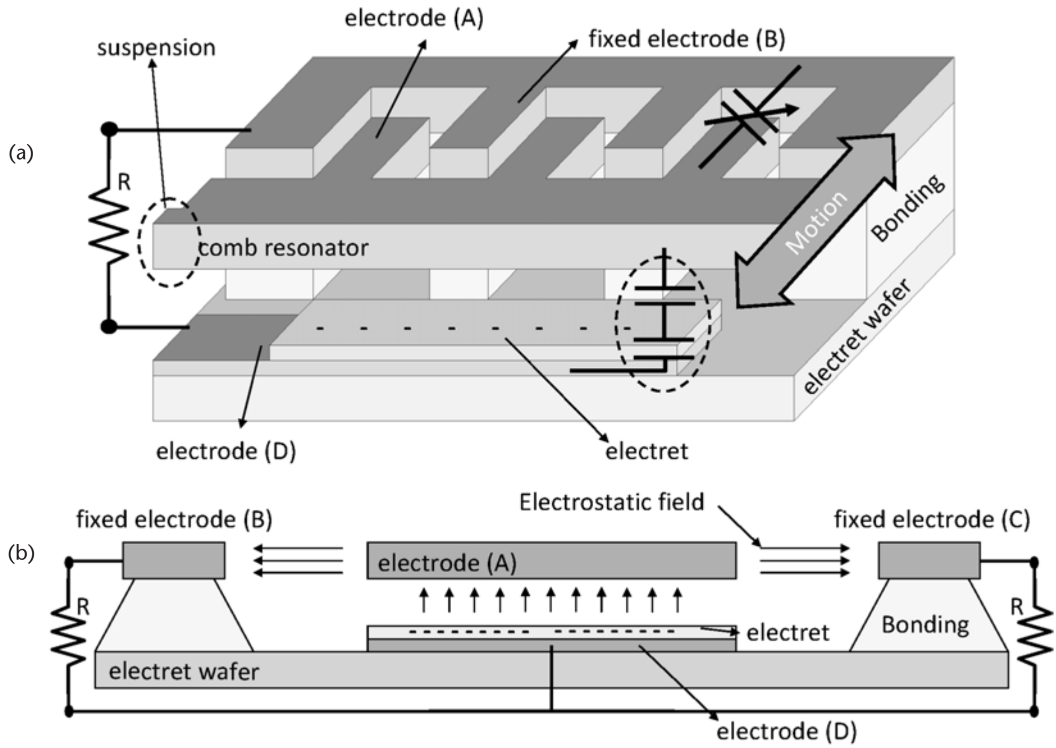


Figure 6.47 (a) Cross-section and (b) side view of a micromachined electrostatic generator [32].

6.3 Interface Circuits for Thermal and Solar Harvesters

Now that the electronics for motion-driven harvesters have been described in some detail, we turn our attention to interface circuits for nonkinetic energy harvesters, namely, thermoelectric generators and solar cells. We must first determine a suitable model of the source to which our electronic interface must connect. The main difference between these harvesting methods and the kinetic devices is that there is almost no frequency dependence in these models. As such, the dynamics of the energy source can effectively be ignored and the system can be analyzed at DC.

6.3.1 Thermal

A structure of a thermoelectric generator (TEG) is shown in Figure 6.48 [33]. The thermoelectric circuit is formed by using two types of semiconductor material, p-type and n-type, which are connected electrically in series and thermally in parallel. A ceramic plate (electrically insulating but thermally conducting) forms a connection between the heat source (heat sink) and the hot side (cold side) of the thermocouple. The rate of heat exchange is denoted by Q_H and Q_C where the subscripts represent hot and cold temperatures, respectively. ΔT_{TEG} is the temperature difference between the hot (T_{HJ}) and cold junctions (T_{CJ}) of the TEG, whereas ΔT is the temperature gradient on the exterior of the generator.

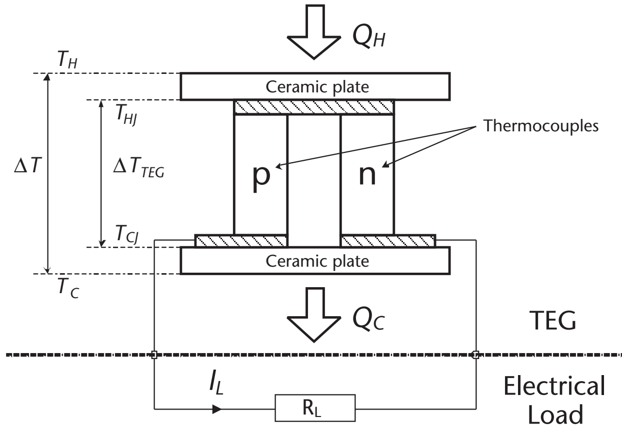


Figure 6.48 Structure of a thermoelectric generator (TEG) connected to a load resistor (R_L).

The thermoelectric effect resulting from a temperature difference between two conductors depends on the Seebeck coefficient of the two materials (α_p and α_n). Equation (6.24) defines the open-circuit voltage generated from a TEG.

$$V_G = \alpha_{pn} \cdot \left(T_{HJ} - T_{CJ}\right) \tag{6.24}$$

When a load is connected to the TEG, a current I_L flows as per (6.25). R_{int} is the internal electrical series resistance of the TEG.

$$I_L = \frac{V_G}{R_{\text{int}} + R_L} \tag{6.25}$$

R_{int} is a function of the height (h) and cross-sectional area (A) of the thermocouple and the resistivity (ρ) of the material.

$$R_{\text{int}} = \frac{2\rho h}{A} \tag{6.26}$$

Figure 6.49 shows the electrical equivalent circuit of the TEG connected to a resistive load based on (6.24) and (6.25). This seems to be a very simple model of the device and shows that the main requirement for the interface circuitry is to set its input impedance to the electrical resistance of the TEG. However, before making this simple conclusion, we must first determine any dependence that a load current may have on the temperature across the device and therefore on the thermoelectric voltage, V_G.

With reference to Figure 6.49, the rate of heat exchange between the hot and cold junctions of the TEG (i.e., Q_H and Q_C) is given by (6.27) and (6.28). K is the thermal conductance of the ceramic plates.

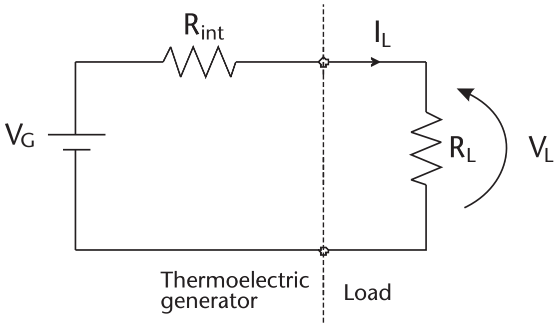


Figure 6.49 Electrical equivalent circuit of the thermoelectric generator connected to a resistive load.

$$Q_H = K \cdot \left(T_H - T_{HJ}\right) \tag{6.27}$$

$$Q_H = K \cdot \left(T_{CJ} - T_C\right) \tag{6.28}$$

Both Q_H and Q_C can be described as the sum of the Peltier effect, the thermal conduction through the p- and n-thermocouples, and the heat loss in the internal series resistance of the TEG as:

$$Q_H = \alpha_{pn} T_{HJ} I_L + K_{\text{int}} \cdot \left(T_{HJ} - T_{CJ}\right) - \frac{1}{2} I_L^2 R_{\text{int}} \tag{6.29}$$

$$Q_C = \alpha_{pn} T_{CJ} I_L + K_{\text{int}} \cdot \left(T_{HJ} - T_{CJ}\right) + \frac{1}{2} I_L^2 R_{\text{int}} \tag{6.30}$$

In (6.29) and (6.30), K_{int} is the internal thermal conductance of the thermocouples and can be expressed as (6.31), where λ is the thermal conductivity of the thermocouple material.

$$K_{\text{int}} = \frac{2\lambda A}{h} \tag{6.31}$$

Equating (6.27) to (6.29) and (6.28) to (6.30), the effective temperature gradient across the hot and cold junctions, ΔT_{TEG}, can be found.

$$\Delta T_{TEG} = \Delta T \frac{K}{K + 2K_{\text{int}} + \dfrac{2\alpha_{pn}^2 \cdot \left(T_H + T_C\right)}{R_{\text{int}} + R_L}} \tag{6.32}$$

The third term in the denominator that involves both R_{int} and R_L is a consequence of the Peltier effect of the load resistance. If the load resistance was reduced, the current flowing from the TEG (I_L) will increase. As a result, the hot junction of the TEG will experience a loss in heat and the cold junction will become hotter due to the Peltier effect. In general, ΔT_{TEG} is affected whenever R_L is changed. However, the term $\frac{a_{pn}^2 \left(T_H + T_C\right)}{R_{\text{int}} + R_L}$ is typically small (α is within the mV/K range) compared to the other terms in the denominator and is usually neglected. This greatly simplifies the source model of the TEG.

The output power (P_{out}) from the TEG is then

$$P_{out} = I_L \cdot \left(\alpha_{pn} \Delta T_{TEG} - I_L R_{\text{int}}\right) \tag{6.33}$$

Substituting (6.25) into (6.33) gives

$$P_{out} = \left(\alpha_{pn} \Delta T_{TEG}\right)^2 \frac{R_L}{\left(R_{\text{int}} + R_L\right)^2} \tag{6.34}$$

Clearly in this case, P_{out} is maximized when R_L is matched to R_{int}, assuming that the Peltier effect that the load resistance has on the TEG is negligible. Therefore, the optimal value of R_L is simply equal to the measured electrical resistance between the terminals of the TEG.

It is possible to increase the thermoelectrically generated voltage by connecting N-thermocouples electrically in series (thermally in parallel). V_G, R_{in}, and K_{int} will increase proportionally to N, whereas the output power increases by N^2. Intuitively, cascading multiple thermocouples seems like an attractive solution to overcome the low voltage levels from TEGs. Limitations of size are generally important in wireless sensor nodes, and this constrains the total number of thermocouples that can be used in a TEG.

6.3.1.1 Example Interface Circuits for Thermoelectric Generators

As an example, consider the work reported in [34] where approximately 100 μW of usable electrical power was extracted from a TEG attached to the human body when the ambient temperature is 22°. The reported open-circuit voltage of the TEG under matched electrical loads was 0.6–1.0V. These low voltage levels are generally insufficient to power a sensor or load electronics. In addition to that, if the ambient temperature changes, the voltage levels from the TEG vary and consequently some form of voltage regulation is needed to supply a load with a constant voltage. By choosing a regulated DC converter with a shutdown input (Figure 6.50), a start-up circuit can be used to keep the converter in shutdown mode until the storage capacitor has accumulated a sufficient charge to overcome the minimum input voltage of the converter. When this happens, the start-up circuit will disable the shutdown mode and the converter can begin regulating the voltage across the load. Moreover, the storage capacitor can discharge in order to supply additional power to the converter in the event of a surge in the load current.

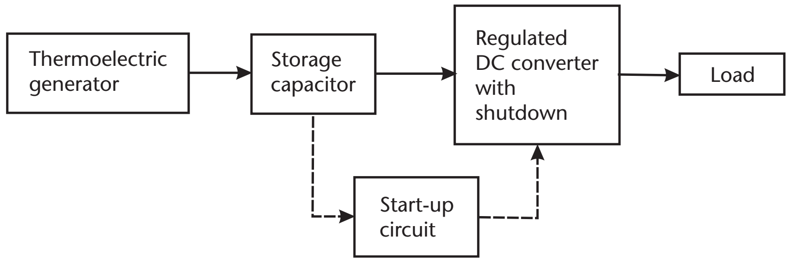


Figure 6.50 Block diagram of a generic power processing setup for the thermoelectric generator.

Mateu et al. reported on a duty cycle controlled maximum power point tracker circuit that was designed for use on a TEG [35]. Due to the low voltage levels from the TEG, a boost converter was chosen to perform the step-up conversion on the voltage. The maximum power point was tracked by changing the duty cycle of the converter so that the output voltage from the TEG was half that of the generated voltage from the TEG. Under such circumstances, the load as seen by the TEG would be equal to that of the TEG's internal series resistance. Hence, an impedance match is present between the TEG's output terminals and its immediate interface, the boost converter.

6.3.2 Power Electronics for Photovoltaics

At the scale of renewable power generation, a typical configuration of the power processing circuits for photovoltaic cells consists of a boost converter interfacing to a DC link and an inverter making a grid connection. In an energy-harvesting device the interface circuit should have the same configuration and function as the boost converter, and of course the inverter for the grid connection is not required.

6.3.2.1 MPPT for PV Arrays

For a set level of illumination, photovoltaic (PV) arrays generate different amounts of electrical energy depending on the current being drawn from them. Typical voltage/current profiles from a PV cell are shown in Figure 6.51, where each curve represents a particular combination of light intensity and temperature. Clearly, the maximum power is extracted from the cell when the product of cell current and voltage is maximized. For each curve, these points are labeled P_{MPP}.

Therefore, as has been the case with other energy-harvesting methods, in order to extract maximum energy from the transducer, an optimal load resistance must be connected in the form of a power converter emulating that resistive load as its input impedance. Setting the load impedance is relatively simple as long as the optimal load is known. The difficulty with PV, and perhaps all harvester technologies, is that finding the optimal load as operating conditions change can be difficult. This difficulty arises from the fact that the maximum power point is dependent on the temperature, the irradiation level, and the age of the solar cell. Some estimation could be made by measuring the temperature and irradiation level, but the usual method employed in large-scale PV installations is to continually hunt for the maximum power point, modifying the duty cycle of a power converter to ensure

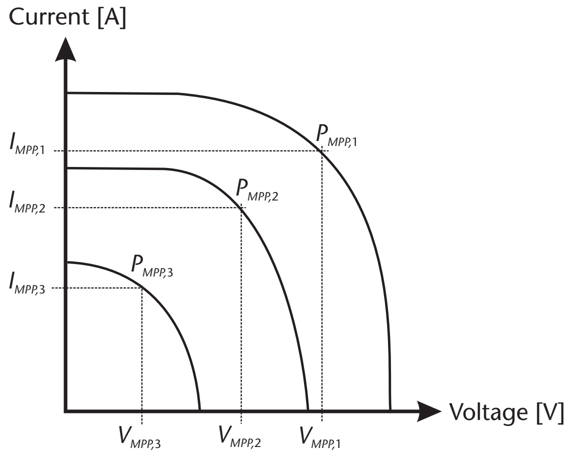


Figure 6.51 Characteristic PV array I-V curve at different operating conditions.

that the maximum power is extracted. The perturb and observe (P&O) method has to date been the preferred technique for hunting for the MPP. The duty cycle of a power converter is continually perturbed, and if a change in duty cycle increases the power output, the duty cycle is again changed in the same direction. If the change in duty cycle produces a reduction in output power, the duty cycle is next perturbed in the opposite way and as such the maximum power point of the solar array is attained.

Figure 6.52 shows a typical output power versus output voltage curve of a PV array when operated under static conditions. If the initial output voltage of the array results in an output power at point A, an increment of Δ1 will move the operating point to the MPP position. However, at point B, decreasing the output voltage by Δ2 will result in optimal operation of the PV array.

6.3.2.2 Power Electronics for Energy-Harvesting PVs

At the time of this writing, little attention has been given to the MPPT control of the interface circuitry for energy-harvesting sized PVs, although miniature boost

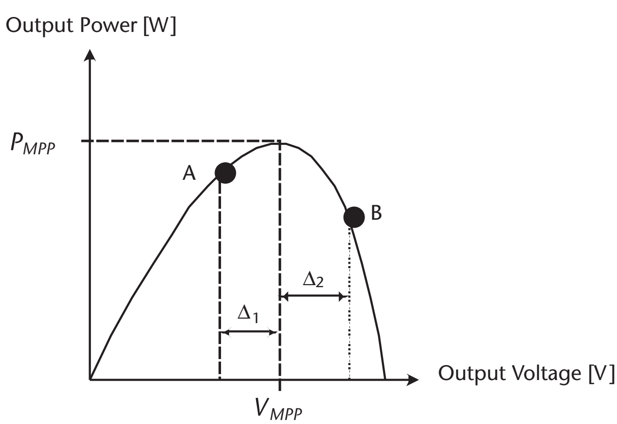


Figure 6.52 Output power versus output voltage characteristic of a PV array when operated under static conditions.

DC-DC converters, suitable for interfacing to PVs, have been designed, such as [36]. In such cases, the main difficulty comes from the very low input voltage produced from individual solar cells. In [36], the authors considered the issue of low voltage start-up; it is not possible to run the converter's control circuitry from a few hundred millivolts available from the energy-harvesting transducer, but it is possible to draw energy from the supply, given that the control circuit for the power converter is operational. In order to achieve this, the authors used a secondary winding on the input inductor with a resonant capacitor to drive the gate of a JFET. Once the circuit has started up, the JFET is no longer used and a parallel MOSFET allows normal controlled operation.

As stated previously, very little attention has been given on the energy-harvesting scale to the maximum power point tracking control for low-power solar harvesting. Figure 6.54 shows the configuration of a typical larger solar cell arrangement with MPPT tracking. Here, the MPPT circuit generates the pulse-width-modulated (PWM) signal that drives the switching transistor in the boost converter. The duty cycle of the PWM signal will depend on the real-time values of the PV array's output voltage and current (V_{PV} and I_{PV}). In this example, the boost converter is the immediate interface between the PV array and the load. This is one of the preferred interfaces to a PV array because the boost converter is able to step up the output voltage from the PV array. Furthermore, the combination of inductor and output capacitor of the boost converter has a smoothing effect on the output current, resulting in smaller output voltage ripples relative to the average output voltage.

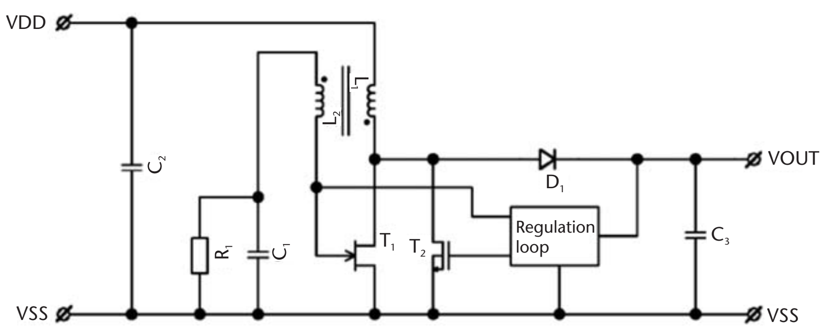


Figure 6.53 Low-voltage start-up DC-DC converter for thermoelectric or solar harvesting applications. (*From:* [36]. Reprinted with permission.)

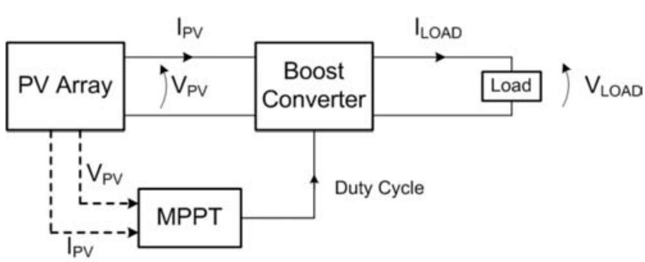


Figure 6.54 A simplified schematic showing the power processing circuitry.

The constraints of implementing such MPPT circuitry on the scale of an energy-harvesting device is simply that when taking into account the power consumption overhead of the control circuitry, the net power output from the system should be greater than the output where MPPT is not applied. Therefore, the success of such circuitry is dependent on both the design of a low-power MPPT implementation and the benefit in increased power extraction from the PV cell, which is highly dependent on the degree of variability of light levels irradiating the device.

6.4 Energy Storage Interfaces

The vast majority of energy harvesting transducers will not be able to supply energy at a constant rate over long periods of time. Clearly, a solar cell can only produce electrical energy when illuminated and a vibration harvester can only produce electrical energy when it is subjected to acceleration. However, many applications of energy-harvesting technology may require a constant source of electrical energy to supply the load. Clearly, if the average power consumption of the load is greater than the average power generated by the harvester, it is not possible to provide power continually to the load. However, if the average power generated is equal to or exceeds the average consumption by the load, it is possible to run the load continually. When excess power is harvested, it is stored in the storage component and when there is insufficient power from the harvester, the storage component can be discharged to supply the load electronics. Besides that, the energy storage component is capable of handling surges in load currents during events like a turn-on transient of the load electronics. As discussed in Chapter 7, electrical storage in the form of a battery or capacitor is generally used.

Using a supercapacitor, as shown in Figure 6.55, for the storage element has the advantage that pushing energy into it is a relatively simple task with few constraints on how the capacitor is charged. In the systems described previously in this chapter, the interface circuit between the energy-harvesting transducer and the storage capacitor only needed to have a controlled input; the interaction between that circuit and the storage capacitor was ignored. In other words, we were free to alter the operating mode of the interface circuit to optimize the operation of the transducer without taking into account how this affected the storage element. Fundamentally, this is because a storage capacitor is very tolerant to the rate at which energy is transferred into and out of it. There is, of course, one constraint that must be taken into account for interfacing with a storage capacitor: its breakdown

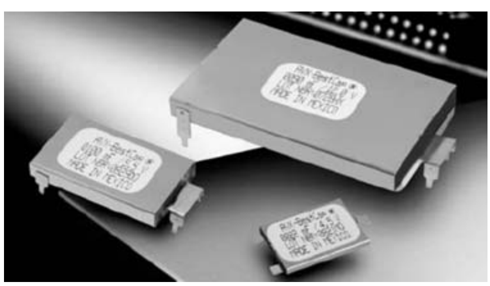


Figure 6.55 Supercapacitors from AVX [37].

voltage. When the capacitor has reached its maximum voltage, the interface circuit must stop transferring energy from transducer to storage to prevent breakdown (i.e., harvesting must stop). However, while the use of a storage capacitor makes the design of the interface circuit simpler, the disadvantage of using capacitive storage is the wide voltage range over which it operates. This, in turn, means that greater difficulty is encountered in regulating the harvester system's output voltage for the load electronics, as a very wide-input power converter is required.

When using batteries as the energy storage element, the opposite is true. The rate and way in which the battery is charged can significantly influence the lifetime of the cells. However, as the battery voltage is relatively constant, load voltage regulation is much simpler than in the capacitive storage case, and indeed output regulation may not even be required as long as the cells are carefully chosen. Limits such as available cell voltages must be taken into account. As an example, lithium-ion cells have nominal voltages of around 3.7V [38], and therefore it is not possible to power lower voltage circuitry from Li-ion cells without some form of output voltage regulation.

6.4.1 Output Voltage Regulation

As we have discussed, fluctuations in the voltage across the energy storage component means that the system may require some form of load voltage regulation. The fluctuation may be negligible if a battery is used, but may be significant if a capacitor is used as the storage component.

Output voltage regulation can in some cases be achieved by using off-the-shelf linear or switching voltage converters. However, the inefficiency of a linear regulator makes them unsuitable for wide-input fixed output conversion. Therefore, when using capacitive storage, a wide-input switching regulator is the preferred interface between the energy storage component and the load electronics. Presently, commercial off-the-shelf switching regulators at capacities of tens to hundreds of milliwatts have reported efficiencies of around 90% [39].

Given that such high efficiencies exist for commercial switching regulators, it may be more convenient to search for one that meets the design requirements,

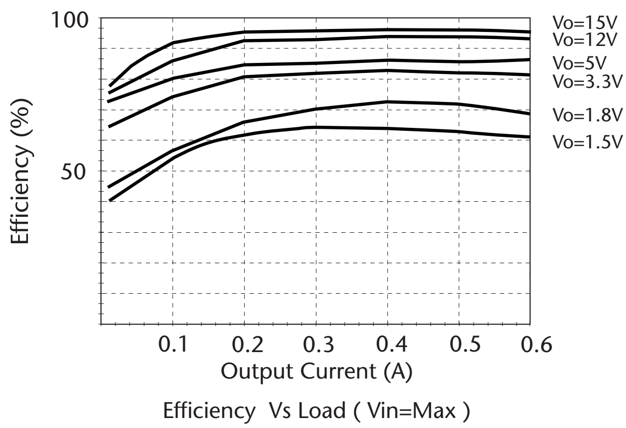


Figure 6.56 Plots of efficiency against output current under different regulated output voltages for the R-78XX-0.5 series from RECOM International.

rather than to design from scratch, depending on the power levels in the system. As an example, consider the R-783.3-0.5 from Recom International [39]. It is small in size at 0.89 cm^3 and has a wide-input range of 4.75–34V, making it suitable for regulating voltage output with significant voltage fluctuations expected from a storage capacitor.

A linear regulator is likely to only be suitable if voltage fluctuations are minimal; otherwise, the efficiency will be very low over at least part of the operating range. However, if a battery is used in the energy storage stage, then a simpler and more efficient solution is probably to store the energy at a voltage that is suitable to run the load electronics, directly avoiding the need for further processing and energy loss.

In summary, a wide-input switching regulator is almost certainly required for an energy-harvesting system utilizing capacitive storage, and a battery storage system should if possible be designed so that output voltage regulation is not required.

6.5 Future Outlook

The natural progression of power electronics for energy harvesters will lead towards low-power, self-starting circuitry that would rely only on the energy scavenged from the environment. As is evident from this chapter, the condition for maximum power transfer from the harvester to the load requires continuous control of the input impedance of the interface circuitry. Future developments of the power processing stages should implement dynamic and accurate online tuning of the optimal damping force or adaptation of the load impedance depending on the transduction mechanism of the harvester. This has a direct consequence on the electromechanical coupling of the harvester and hence the effectiveness of the harvester in converting what is deemed to be useless ambient mechanical energy into usable electrical power. Most control algorithms currently use digital signal processing in the form of microcontrollers. With the development of ultralow-power circuitry,

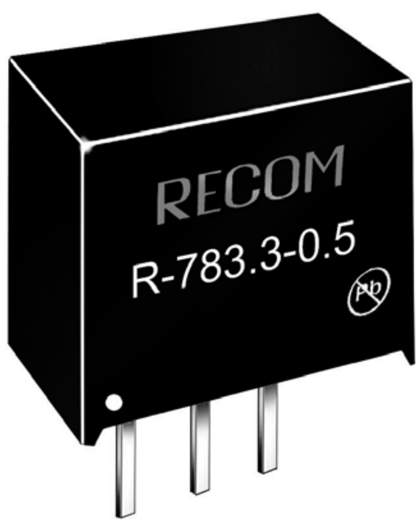


Figure 6.57 Wide-input, output voltage regulator from RECOM International [39].

the options available will be geared towards profoundly customized methods in the control algorithms.

The advent of highly methodical microfabrication techniques will provide a suitable platform allowing for the integration of energy harvesters with their power processing circuitry on a single standalone chip. In other words, energy harvesters that are compatible with MEMS technology can be easily integrated with power electronics. Recent advances in designing smaller magnetic components could significantly reduce the size of DC/DC converter circuits that form the backbone of any adaptive impedance matching or voltage regulation circuit. Due to the high-voltage and low-charge characteristics of electrostatic harvesters, the power electronics design is very difficult and an unsolved problem. It is difficult to see how this will be resolved with existing semiconductor device technologies.

6.6 Conclusions

Optimization of energy harvesters is a system level problem that involves stringent design requirements on the power processing stages. Deploying an energy harvester on its own will yield poor power densities, which is why additional circuitry is needed to implement features such as an impedance match between harvester and load electronics, energy storage capabilities, and output voltage regulation.

Each energy harvester is differentiated by its transduction mechanism, and therefore the equivalent source impedance model must be derived for different harvesters. By matching the source impedance to that of the load or by applying appropriate switching (as is the case for piezoelectric and electrostatic harvesters), the maximum power transfer is achieved from the harvester to the load electronics under optimal conditions. This is even more crucial when energy harvesters are the potential replacements to battery-powered applications. The control overhead of the power processing stages has to be kept as low as possible to place energy harvesters in a viable position in self-powered applications. While the efficiencies of standalone, off-the-shelf power converters can reach almost 90%, this figure reduces to around 50% when the input voltage levels are within the subvolt range. This is where the effects of voltage drop across diodes and the power losses due to the equivalent series resistances of inductors and capacitors can negatively influence the power density of an energy-harvesting system.

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CHAPTER 7

Energy Storage

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For over a century, batteries and electrochemical capacitors have been the mainstay energy storage components of machines, tools, and gadgets. With steady improvements and miniaturization of energy storage technology, the relationship between people and electronic devices has gradually evolved. This evolution has caused modern society to develop a strong dependence on batteries and supercapacitors for the upkeep of both comforts and necessities, from powering our personal mobile phones to large industrial process equipment. Choices for the materials, design, and production of an energy storage device can vary extensively depending on a host of parameters, including an application's performance and lifetime requirements, size, cost, and, of more recent concern, environmental impact. Devising power supplies for small, autonomous wireless systems has been especially onerous, providing engineers with a rich opportunity for innovation because of its stringent demands. This chapter details design considerations, reviews existing technology, and evaluates prospects for microenergy storage devices, in particular batteries and electrochemical capacitors used in conjunction with one or more energy harvesters to provide permanent power to autonomous wireless systems.

7.1 Introduction

The reduction of electronic device form factors and their power demands has augmented the prospects of realizing a fully integrated microdevice platform, one with computation, communication, and sensing capabilities all enabled by a permanent power source [1]. The implications of the widespread deployment of these devices, especially autonomous wireless sensor nodes, are pivotal to a variety of fields including emergency response [2], structural monitoring [3], and the cost- and energy-effective regulation of home, industrial, and office energy use [4]. For wireless microdevices with footprint areas occupying less than 1 cm^2 of a substrate, typical power demands can vary, oftentimes spanning a few orders of magnitude from microwatts (μW) to milliwatts (mW) depending on the application. The need for a micropower source that can satisfy the power requirements of a wireless device within comparable geometric dimensions has incited a surge of research within the fields of microfabrication, energy harvesting, and energy storage [5]. For autonomous wireless sensors, the foremost microenergy storage devices being considered are microbatteries and electrochemical microcapacitors, which are the technologies discussed in this chapter.

7.1.1 Battery Operating Principles

Batteries are galvanic cells in which chemical energy from electrochemical reactions is converted to electrical energy that can be harnessed externally. They contain two chemically different electrodes sandwiching an ion-conducting electrolyte phase. At open circuit, a measurable electrical voltage develops between these electrodes due to the propensity for chemical reactions to occur. Upon electrical connection to a load, chemical reactions occur and ions travel across the electrolyte. A spontaneous flow of electrons occurs in the external circuit from the electrode with the more negative potential to the more positive electrode, and can therefore be used to power a load such as a device (see Figure 7.1). If the battery is rechargeable, the reverse reactions can be induced with an input of energy. Depending on the type of reactions that occur, for example displacement reactions or insertion reactions [6], the discharge behavior of the battery will vary (Figure 7.2), resulting in different cell

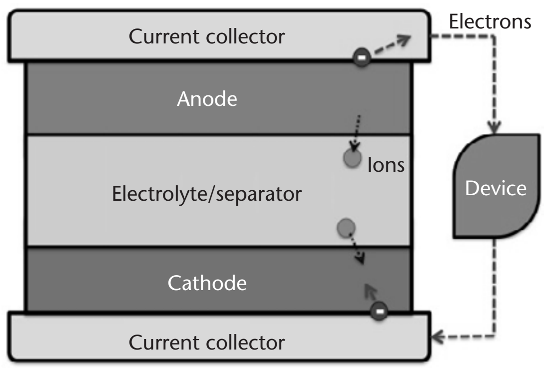

Figure 7.1 Battery schematic.

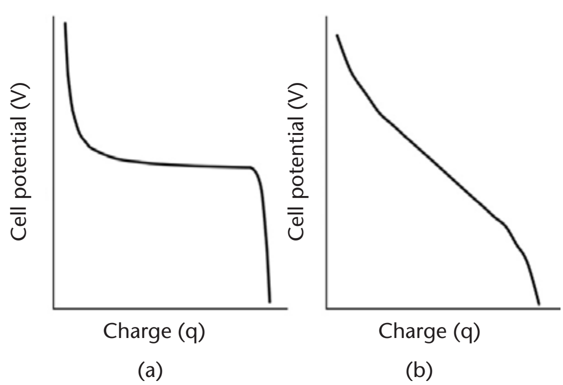

Figure 7.2 Discharge behavior of batteries with (a) invariant cell potential or (b) sloped cell potential.

potential outputs (V) with respect to the amount of charge (q) extracted from the battery, and from this relationship the amount of useful energy (E) obtained from the battery can be determined from the following expression:

$$E = \int V \, dq \tag{7.1}$$

The energy is typically reported in units of joules (J) or watt-hours (Wh). The total amount of charge extracted from the battery, or how much that can be stored in the cell, is known as the capacity typically reported as amp-hours (Ah) or coulombs (C). The maximum power, measured in watts (W), that can be drawn from the battery is dependent on the kinetics of the system. This system can be limited by the summation of impedances in the cell, including the charge transport in the electrodes and electrolyte, the rates at which reactions occur, and interfacial resistances. The power behavior of a battery can also be characterized by its rate performance, and is evaluated by the time (in hours) it takes to deplete a device of its maximum storage capacity (C). For example a battery that takes 10 hours to completely drain was discharged at a $C/10$ rate, while a quick discharge of $2C$ means the battery was depleted in a half hour. Note that all the quantities listed can be normalized with respect to weight, volume, and footprint area. For example, electrochemical storage energy is primarily quantified in terms of the specific energy (Wh/g), the amount of energy stored per unit volume (Wh/cm^3), and the energy stored per unit footprint area occupied on a substrate (Wh/cm^2). Note that when quantifying the theoretical performances of electrochemical materials, metrics normalized with respect to weight and volume are customarily used; however, when comparing microdevices, the constrained unit is most often the footprint area occupied on a substrate, and to compare across many fabrication methods, materials, and device configurations, areal metrics are pertinent.

The chemistries and relative amounts of the battery components will determine its operating voltage and energy storage capacity, and along with the cell geometry and processing, will also influence the maximum power accessible. In addition to these properties, other performance metrics include the battery's lifetime (or for rechargeable batteries, its cycle life) as well as its safety and cost, all of which depend on its inherent materials properties, such as their stability and compatibility. Though batteries are straightforward conceptually, there have been numerous chemistries, geometries, and processing methods proposed.

7.1.2 Electrochemical Capacitor Operating Principles

Electrochemical capacitors, also refered to as supercapacitors, ultracapacitors, or double-layer electrochemical capacitors, are emerging as a practical energy storage device for high power density applications. Electrochemical capacitors are similar in construction to batteries, but are typically symmetric, oftentimes (as seen in commercial cells) having two carbon electrodes. Upon applying an electric field across the capacitor, ions in the electrolyte can dissociate and travel towards oppositely charged electrode surfaces, as illustrated in Figure 7.3. Because the mechanism of energy storage is purely electrostatic, the capacitor voltage scales linearly with charge stored, as shown in Figure 7.4. The resulting capacitance (C), measured in farads (F), of the device can be determined from the following expression:

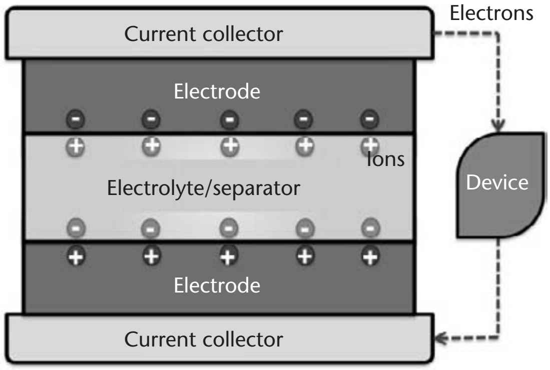

Figure 7.3 Electrochemical capacitor schematic.

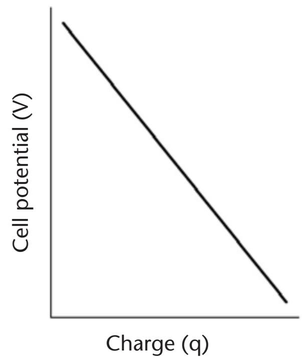

Figure 7.4 Discharge potential of capacitor scales linearly with charge extracted.

$$C = \frac{q}{V} \tag{7.2}$$

Due to the absence of Faradaic or chemical reactions, theoretically no irreversible phenomena occur during standard operation. Therefore, electrochemical capacitors should in theory operate indefinitely, and have been frequently demonstrated to cycle more than one million times.

7.1.3 Comparison of Energy Storage Devices

The energy and power densities of selected energy storage devices are summarized in a Ragone plot shown in Figure 7.5. Though the performance units in the Ragone plot are normalized with respect to weight, the illustrated trends are fairly consistent when normalized with respect to volume. The specific energy of batteries are typically one to two orders of magnitude greater than electrochemical capacitors, which can contain an order of magnitude greater specific energy than ceramic and electrolytic capacitors. The trend for power density is the reverse: ceramic and electrolytic capacitors can theoretically extract at least an order of magnitude greater

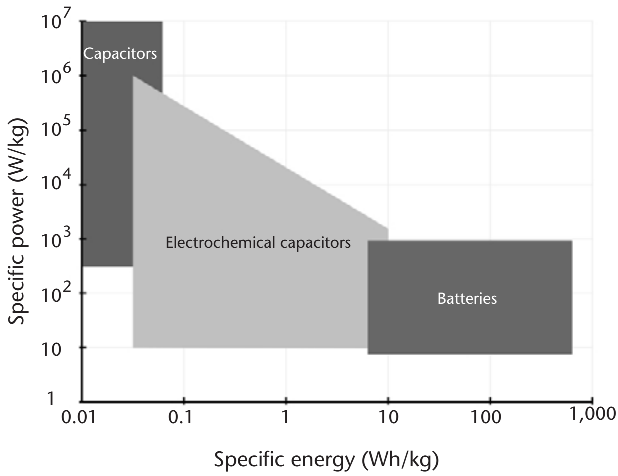

Figure 7.5 Ragone plot comparing energy storage devices. (*From:* [7].)

power density than electrochemical capacitors, while typical batteries report power densities at least one order lower than electrochemical capacitors.

7.2 Micropower Supply for Wireless Sensor Devices

Currently the most prevalently used power sources for wireless sensor platforms are primary macrobatteries (batteries greater than 1 cm^2 in footprint diameter typically assembled separately from the devices they power). Primary macrobatteries are typically assembled in the charged state, utilized until completely discharged, and then disposed of. Today, these batteries are commercially offered in a variety of shapes and sizes, the smallest having a coin or button cell construct (about 10 mm in diameter with a 5-mm thickness). For a device with relaxed size prerequisites, anticipated finite lifetimes, and/or where the replacement of its power supply is unchallenging or inexpensive, primary batteries are simple, hassle-free, and often the most cost-effective choice for a power source. Without the need for special mounting prerequisites or calibration, devices powered by primary batteries can be quickly deployed almost anywhere. Furthermore, most of the battery chemistries offer fairly stable discharge potentials, which can be used as a direct supply voltage to a device without the need for additional regulation. Practical energy densities for various primary battery chemistries are listed in Table 7.1.

For applications that require the ubiquitous and unobtrusive distribution of many autonomous wireless sensors, primary macrobatteries have proven to be unreasonably expensive and difficult to replace once depleted. Additionally, to meet extended lifetime demands, primary batteries would need to be far too large and

Table 7.1 Summary of Primary Battery Chemistries [8]

Chemistry	*Leclanché*	*Alkaline*	*Lithium*	*Zinc-Air*
Energy Density (Wh/kg, mWh/cm^3)	85, 165	145, 400	230, 535	370, 1300

unwieldy, and would overwhelm the total volume of a device. Permanent solutions for power sources have been proposed as viable alternatives, and a large research effort has been directed towards developing microscale ambient energy harvesters and complementary microenergy storage devices. The combination of the two, known as a hybrid micropower supply, incorporates the ability to convert ambient energy (for example, solar, vibrations, and thermal energy) into useful electrical energy, which is then stored in an energy reservoir until needed by the microdevice. In most environments, ambient energy is not always available, and therefore can only be harvested either periodically in recurring timescales or in most cases, intermittently. As a consequence, an energy reservoir is necessary to provide power to the device even when ambient energy is not present, and the size of this reservoir is dictated by the energy source (the magnitude of the harvested energy and the frequency at which ambient energy is available) and on the load (the magnitude of power demanded from the device and the frequency at which this power is demanded). For environments in which energy conversion occurs effectively whenever the microdevice operates, the corresponding energy reservoir storage capacity can be minimal, and is simply limited by the maximum power required. For this case, a standalone electrochemical capacitor may suffice. In the event that ambient energy is unavailable, a battery with higher energy storage capacity is necessary; however, a load-leveling electrochemical capacitor may also be used in conjunction to alleviate the battery from the device's exacting power demands, and as a consequence improve the battery health and life span. In this chapter, we examine the unique materials, processing, and integration considerations involved in designing rechargeable microbatteries and electrochemical capacitors to store energy for microenergy harvesting devices, with emphasis on the performance prerequisites of miniature electronic devices such as autonomous wireless sensor platforms.

7.2.1 Microenergy Storage Considerations

In the late 1970s, a growing trend towards wafer-scale fabrication of integrated circuits engendered the development of microelectromechanical systems (MEMS), elaborate devices having microscopic dimensions. The drivers that have made MEMS devices a widespread commercial success were revolutionary: MEMS technology was capable of making small, integrated systems with immense functionality using low-cost and high-throughput fabrication processes. These same drivers and the need for local and efficient energy delivery to these MEMS components have inspired research efforts towards developing power sources of comparable dimensions to MEMS devices. Microbatteries evolved from a burgeoning concept into a conceivable notion as a result of substantial developments in solid-state ion conducting materials [9], and the construction of the first solid-state microbatteries was demonstrated in the early 1980s [10]. Though most of these prototypes exhibited a fairly low capacity, the demonstrations catalyzed more fervent investigations in the microbattery field. There were, however, many practical barriers, as intimated by Owen in 1985: "the geometry, and in particular, the dimensions of this device are alien to conventional battery production" [11]. However, he forecasted that within ten years, because of the important implications, the production of microbatteries would likely become a reality.

7.2.2 Materials Considerations for Microbatteries

As Owen had predicted, materials and processing innovations in the next decade led to a few successful microbattery implementations, and eventually a variety of materials, geometries, and processing strategies had been proposed and explored. In the 1990s, the field was further invigorated by the growing interest in microenergy harvesting devices and their need for secondary energy storage. Typical materials systems that have been incorporated in rechargeable microbatteries are analogous to macrobattery chemistries (batteries with footprint areas greater than 1 cm^2 and thicker than a few millimeters). A few candidate rechargeable microbattery chemistries are summarized in Table 7.2.

Materials efforts that have contributed to microenergy storage developments and are particularly worth discussing in detail include research on solid electrolyte and nanodimensioned electrode materials. To avoid manufacturing and packaging complexities and to increase resiliency against extreme conditions such as shock, vibrations, and extreme temperatures, it is likely that a solid electrolyte will need to be incorporated into microbatteries; developments of solid-state ionic and polymer electrolyte materials have been underway. Some of the first solid-state batteries incorporated solid, lithium-ion conducting glasses that could be deposited using physical vapor technology to form pinhole-free films with thicknesses as low as 1 μm. Most solid-state electrolyte materials are oxide-based such as lithium phosphorus oxynitride (Lipon). The glassy electrolyte is electrochemically, mechanically, and thermally very robust: it has a large potential stability between 0–5.5V versus Li^+/Li, the mechanical strength to resist lithium dendrite propagation, and

Table 7.2 Summary of Selected Rechargeable Battery Chemistries [8]

Battery Chemistry	*Energy Density (Wh/kg, mWh/cm³)*	*Operating Voltage (V)*	*Advantages*	*Disadvantages*
Nickel-zinc (Ni-Zn)	60, 120	1.6	Environmentally friendly and recyclable. High charge and discharge rates possible.	Zinc dendrite propagation and electrode shape change reduce cycle life.
Rechargeable alkaline manganese (RAM)	145, 400	1.5	Environmentally friendly. Cheap material costs.	Zinc dendrite propagation and electrode shape change reduce cycle life.
Zinc-silver oxide ($Zn\text{-}Ag_2O$)	105, 180	1.5	High power density. Flat discharge potential.	Expensive silver. Zinc dendrite propagation and electrode shape change reduce cycle life.
Lithium-ion (Li-ion)	150, 400*	3–4.2*	Flat discharge potential. No memory effect.	Strict charge and discharge regulation needed. Moisture sensitivity and flammability risk.
Lithium polymer	130, 300*	3–4.2*	Reduced manufacturing complexity and costs. Increased safety.	High internal resistance. Strict charge and discharge regulation needed.

* Values depend on electrode chemistries. Lithium-ion anodes include C_6, Si. Typical cathodes: $LiCoO_2$, $LiMnO_2$, $LiFePO_4$.

exhibits little degradation in temperatures ranging from –40°C to 150°C [12]. The ability to deposit thin films of these solid-state electrolytes compensates for their low lithium-ion conductivity (< 5 μS/cm) so that moderate to high rate performance is possible. Alternative solid-state electrolyte materials such as other oxide and sulfide glasses and glassy ceramics have also been explored.

Solid polymer materials are also being considered as potential electrolytes for a variety of microbattery systems. In addition to the advantages of being solid-state, polymer electrolytes are attractive because of their processing simplicity. There have been two principle material directions in solid polymer electrolyte research, the first being the development of gel electrolytes, which swell salt solutions within amorphous regions formed by crosslinked polymer chains. Gel electrolytes tend to have moderate room temperature ionic conductivities (1 mS/cm^2 [13]), but poor mechanical and electrochemical stability. Typical liquid components incorporated into gel electrolytes for batteries include organic solutions of ethylene or propylene carbonate, and of recent interest, ionic liquids [14]. Ionic liquids are molten salts with high ionic conductivity, negligible vapor pressure, are nonflammable, and will become strong candidates for safe, plastic batteries if cost and large-batch processing considerations can be addressed [15]. The second thrust of solid polymer electrolyte research has concentrated on the development of dry polymer electrolytes, where the polymer acts both as the ion conducting phase as well as the separator. These materials tend to be mechanically robust and can be cast very thin, but have fairly poor room temperature ionic conductivities (0.01 mS/cm^2 [16]). Dry polymer electrolytes made of nanostructured block copolymers have recently been shown to exhibit improved performance [17]. The block copolymers are able to self-organize into discrete rigid or highly ionic conductive regions, effectively decoupling the two properties while significantly increasing their properties relative to other solid polymer electrolytes.

The significant and sometimes unexpected enhancements in performance observed in block copolymer electrolytes due to their nanostructuring have also manifested in nanodimensioned electrode materials. Electrode nanomaterials have oftentimes exhibited higher reaction rates due to increased electrode/electrolyte interfacial areas and shorter charge transport distances [18], and further performance enhancements may be reached with various nanoparticle morphologies, architectures, and composite materials. For example, Nam et al. demonstrated that with the deliberate texturing of dispersed gold nanoparticles along the length of virus-templated, Co_3O_4 electrode nanowires for lithium-ion batteries, a significant capacity increase from 700 mAh/g to 1,000 mAh/g was observed due to the gold's enhanced electrical conductivity as well as its catalytic affect on lithium-ion insertion [19, 20]. This work combines two exciting fields in nanomaterial research with application to electrochemistry: first, the opportunity to explore nanoscale arrangements of materials and their performance effects, and second, it proposes a novel and simple method for using biological systems to program and self-assemble functional nanomaterials. Difficulties packing nanoparticles and the higher propensity for unwanted side reactions as a result of increased electrode/electrolyte contact areas have inhibited the adoption of some nanostructured electrode materials; however, methods to curtail these problems have been proposed [21].

7.2.3 Geometry and Processing Considerations for Microbatteries

Though the various battery chemistries listed in Table 7.2 differ in performance and processing, most of these chemistries have been considered or even implemented in rechargeable micobatteries largely because of the varying materials compatibilities associated with different fabrication processes. In addition, though microbattery electrode and electrolyte chemistries are in most cases indistinguishable from those of macrobatteries, the resulting microbattery performance does not usually scale proportionally because the configurations, materials deposition methods, postprocessing, and packaging methods used to assemble macrobatteries and are oftentimes not feasible below the centimeter scale. For example, many lithium-ion commercial macrobattery electrodes are cast onto large substrates and then calendared, or compacted so the electrodes are pressed into a desired thickness. They are then cut into appropriate shapes and tightly rolled or folded into rigid canisters that provide both hermetic packaging and compression to the cell. For small microbatteries, which are intended to be deposited directly onto a microdevice, these compression techniques and packaging processes cannot be efficiently implemented. As a result, in addition to materials optimization, microbattery researchers have focused on advancing microbattery fabrication processes.

7.3 Implementations of 2D Microbatteries

Two-dimensional microbatteries generally are cells configured from successive film depositions, either in planar or stacked arrangements as illustrated in Figure 7.6. Planar batteries are one of the simplest configurations to construct, comprising of two electrodes positioned adjacent to each other on a substrate. Since the electrodes are physically detached, a separator is not necessary. Oftentimes, planar cells are simply submerged in a liquid electrolyte or blanketed with a solid or polymer electrolyte film. The electrodes may be patterned into a variety of shapes such as square pads or more elaborate interdigitated structures, but regardless the current density distribution across each electrode varies with respect to its relative distance from the adjacent electrode. This uneven current distribution may result in unfavorable consequences especially during rapid charge and discharge. However, if the internal resistances within the electrodes are minimal, reasonable performance may be achieved.

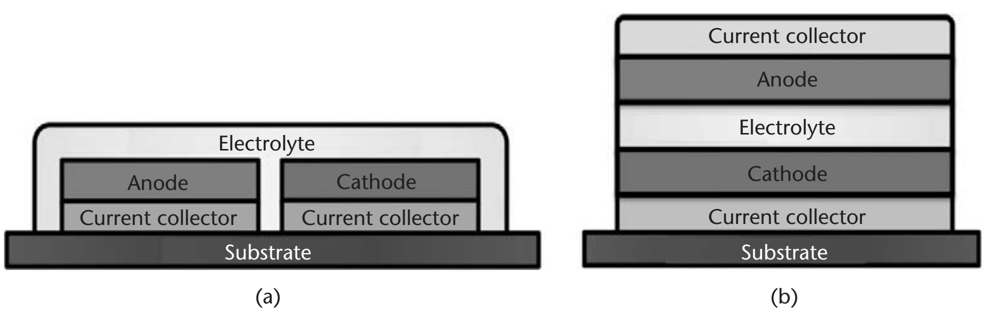

Figure 7.6 (a) Two-dimensional planar and (b) stacked microbattery configurations.

Stacked batteries are constructed by the consecutive deposition of films assembled layer-by-layer, with all films occupying the same approximate footprint area. The current distribution in a stacked cell runs perpendicular to the through thickness of the electrolyte/separator layer and is approximately constant, resulting in uniform performance across both electrodes; it is more suited for high-rate applications in comparison to planar batteries. Furthermore, better areal energy density should be achieved with this configuration in comparison to planar microbatteries. Fabricating pinhole-free thin electrolytes/separators that can structurally support the weight and stress developed upon deposition of electrode and current collector films has not been a trivial feat. However, solid-state ionic materials and polymer electrolytes have been utilized to enable this configuration.

Two-dimensional microbatteries of both planar and stacked configurations have been constructed using a diverse spectrum of processing tools. Two-dimensional microbattery implementations are classified according to two processing strategies, from thin film to thick film microbattery deposition, and are manufactured by a variety (and in some cases a combination) of microfabrication and solutions-based processes. Prototypes of 2D microbatteries have been fabricated with wide-ranging degrees of completion, and notable efforts will be summarized and compared in the following section.

7.3.1 Thin Film Solid-State Microbatteries

Perhaps the most advanced implementations and extensive explorations of microbatteries have been conducted using thin film deposition methods and recently, thin film batteries are being sold commercially [22, 23]. Most thin film, solid-state batteries are fabricated using physical vapor deposition (PVD) tools that deposit materials onto a substrate via condensation from its vapor phase. PVD tools that have been used for thin film batteries include sputtering and pulse laser deposition. Researchers at Oak Ridge National Laboratory (ORNL) have demonstrated stacked lithium-ion thin film microbatteries in which all components, including the current collectors, electrodes, electrolyte, and protective packaging, were deposited using thin-film techniques (see Figure 7.7) [24, 25]. Each of the layers can be successively deposited as conformal films, and depending on the deposition parameters, the film thicknesses range between submicrons to a few micrometers. The deposited films are usually very uniform, and as a result, consistent current and charge distributions occur across the whole device. Commercial thin film batteries are typically a few cm^2 in footprint area with capacities ranging from 10 $\mu Ah/cm^2$ to 1 mAh/cm^2.

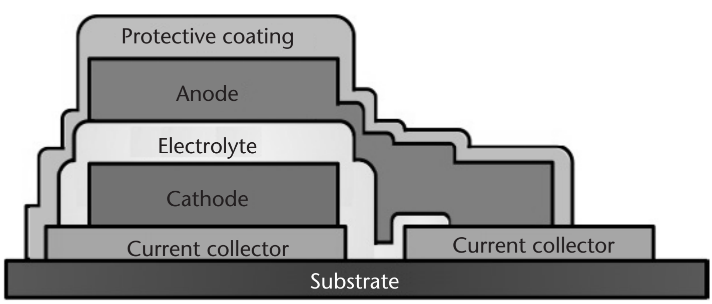

Figure 7.7 Thin film microbattery structure [24].

However, the thin film processes can be modified to deposit batteries with active areas under 1 mm^2 (Figure 7.8) [26, 27]. Because the fabrication tools used to build these microbatteries are compatible with standard microfabrication methods, it is conceivable that thin film microbatteries can be deposited while concurrently also patterning other wafer-scale components onto the same substrate. Multiple thin film microbatteries can be patterned in series and parallel configurations to match voltage and current requirements [28], and the integration of localized energy storage components can be achieved to provide efficient power delivery for discrete on-chip components such as integrated circuits.

Though microbatteries and macrobatteries share common electrode chemistries, electrochemical performance can differ quite significantly depending on the material processing methods and parameters used. In general, the as-deposited PVD electrode films are homogenous in composition, lacking binders and additives. The dense films are amorphous until annealed between 300–1,000°C. The resulting crystalline films are microstructurally robust and exhibit rapid lithium-ion diffusion, leading to excellent cycle lives (> 1,000 deep cycles) and moderate maximum power densities (10 mW/cm^2) [25]. Furthermore, for crystalline electrodes, the discharge voltages are very flat, simplifying supply voltage regulation circuitry. In general, failure and electrode ageing may occur from poor voltage regulation if overcharged, exposure to unsafe operating temperatures, or mechanical volume changes that might occur during electrochemical cycling—all of which can greatly affect the capacity and cycle life of the device [25]. Typical anodes for thin-film lithium and lithium-ion batteries include lithium, silicon-tin oxynitrides, Sn_3N_4, and Zn_3N_2, to name a few, and cathodes such as $LiCoO_2$, $LiMn_2O_4$, and LiV_2O_5 have been thin film deposited. The thicknesses of thin film deposited cathodes and anodes are usually no greater than 5 μm, and are limited by the inherent stress accumulated in the film due to processing, which results in difficulties adhering to substrates. For applications requiring high energy densities, the limited storage capacity of thin film microbatteries has been the greatest barrier to their widespread adoption as an integrated energy storage component.

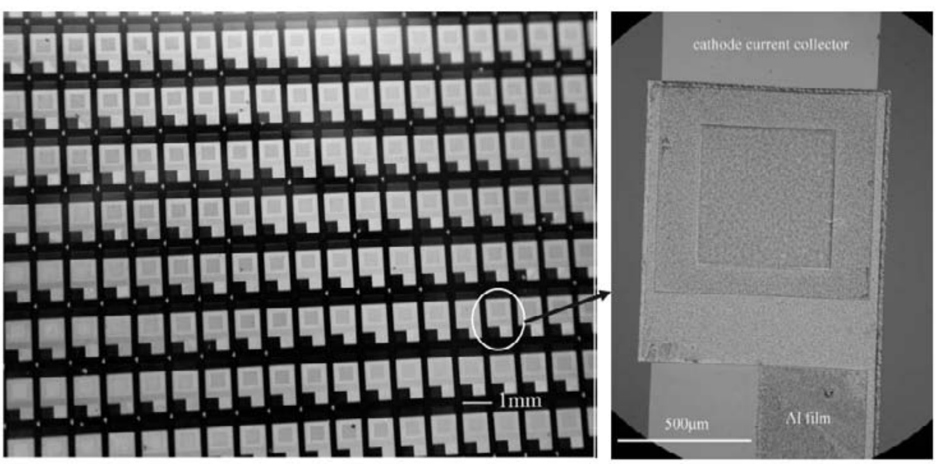

Figure 7.8 Array of thin film microfabricated microbatteries. (*From:* [26]. Reprinted with permission. Copyright 2009, IOP.)

The difficulty in achieving higher energy densities is further compounded by inactive packaging and support components occupying a majority of the total device weight and volume, and this is further exacerbated as the microbattery size decreases. The greatest challenge in lithium-ion microbattery manufacture has been to find equally thin, cost-effective hermetic sealants that can protect the electrodes and electrolyte from exposure to moisture. For lithium-ion batteries, moisture levels must be kept below a few parts per million to avoid detrimental irreversible reactions and maintain prolonged operation [29]. For thin film microbatteries, finding a hermetic material that can be deposited conformably with the same fabrication tools used to deposit the active components of the microbattery would be ideal. Moisture permeability depends heavily on the encapsulating material used, its thickness, and its resilience to mechanical and chemical wear. Researchers have had varying degrees of success with depositing thin films of glass, polymers, and metals. At ORNL, a microbattery packaged with alternating layers of Parylene polymer and sputtered metal films was able to operate for a few months before failing due to moisture exposure [25]. Other necessary improvements include the need for faster, more efficient manufacturing processes, higher device yields, and reduced materials and processing costs. Surprisingly, despite the achievements in thin film microbattery research and development, there has been a scarcity of reported attempts to integrate these microbateries with energy harvesters, though feasibility and system architecture studies have been presented in literature.

7.3.2 Thick Film Microbatteries

Despite significant achievements in the development of thin film microbatteries, high processing costs and limitations in areal capacity motivated the development of alternative microbattery manufacturing strategies. In 1999, Birke et al. [30] proposed the development of miniaturized coin cells of monolithic solid-state battery materials tailored in geometry so that, along with other integrated circuit components and microchips, the battery could be embedded into a substrate. Their microbattery consisted of lithium titanate ($Li_4Ti_5O_{12}$) and lithium manganese oxide ($LiMn_2O_4$) electrodes and a ceramic lithium titanate phosphate based ($Li_{1.3}Al_{0.3}Ti_{1.7}(PO_4)_3$) electrolyte. Two electrode pellets, each no more than a few millimeters thick, and a 200-μm-thick electrolyte were stacked, cold pressed, and sintered at 750°C for 12 hours. Then the electrodes were polished down to tens of micrometers and the entire structure was then cut into a desired shape so that the final monolithic microbattery could be incorporated into a substrate (see Figure 7.9). For a C/2 discharge rate, a microbattery of 1 cm^2 diameter footprint area and 500 μm in thickness had an approximate energy density of 42.5 μWhr/cm^2 when cycled between 1.5–3.2V. Though this assembly method would have been difficult to implement with smaller, more complex geometries, Birke seeded a few important concepts with regard to microbattery fabrication: the need for a method to fabricate thicker electrodes from tens to hundreds of microns thick, and the importance of developing a cost-effective method to tailor both the lateral and horizontal dimensions of all microbattery components. For example, balancing the maximum current density with the desired storage capacity performance for different device applications could change the optimal thicknesses of both microbattery electrodes by orders of magnitude, making methods that allow customization very compelling. Additionally, Birke

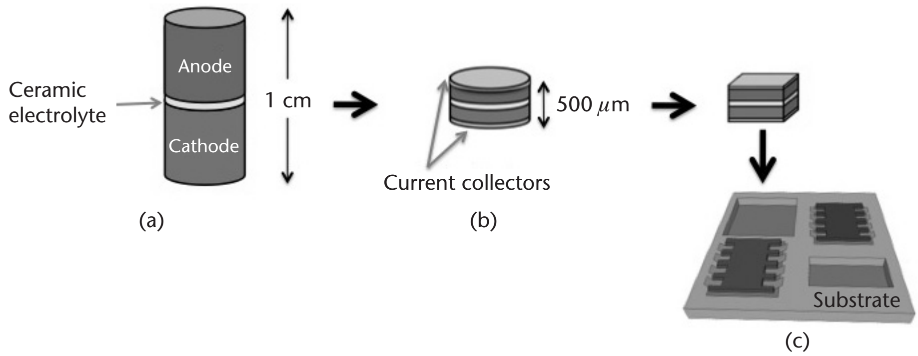

Figure 7.9 A monolithic solid-state battery formed by (a) cold pressing and sintering a stack of electrodes and ceramic electrolyte, (b) polishing the electrodes to the desired thickness and sputtering current collectors at each end, and (c) cutting the monolithic microbattery into a desired shape to be embedded into a substrate. (*From:* [30].)

acknowledged that the integration of a microbattery onto a substrate would create new temperature and stability limitations that might prompt alternative materials and processing choices, which might be considered unconventional compared to macrobattery fabrication. These themes have continued to impel the microbattery research field, and many efforts concentrating on applying thick film, customizable fabrication processes have been explored.

Thick film microbatteries have electrode components with thicknesses that are at least an order of magnitude greater than thin film batteries, but still much thinner than button and coin cell macrobatteries. Therefore, like thin film microbatteries, thick film microbatteries can be embedded into or on substrates in an inconspicuous fashion. Though many thick film fabrication methods vary in actuation, they share a common advantage of being able to deposit thick and oftentimes porous electrodes, in the tens to hundreds of microns. As a consequence, significantly larger energy storage capacities can be achieved for a given footprint area. Generalizing the performance of thick film microbatteries can be quite difficult since the material chemistries, cell configurations, and deposition capabilities are varied, but a few noteworthy efforts are discussed in the next sections.

7.3.2.1 Microfabricated Thick Film Microbatteries

Microfabricated thick film microbatteries employ similar processing tools used to construct integrated circuits and MEMS devices. The microbatteries can be processed in parallel with other microfabricated components on a substrate, and many microbatteries can be made simultaneously with high throughput and yield. Humble et al. used standard microfabrication techniques to construct planar configurations of thin film nickel-zinc microbatteries on silicon wafer substrates [31]. Using a combination of electron-beam evaporation, electroplating, and photolithography, 15–75-μm-thick, porous zinc and NiOOH electrodes were patterned adjacent to each other with a total cell area of 2 mm^2 (see Figure 7.10). The electrodes were blanketed in a liquid alkaline electrolyte solution of potassium hydroxide and zinc oxide, and this electrolyte was retained by 100-μm-tall, spin-coated epoxy sidewalls. Manually applied polymer films were able to adequately seal the cell

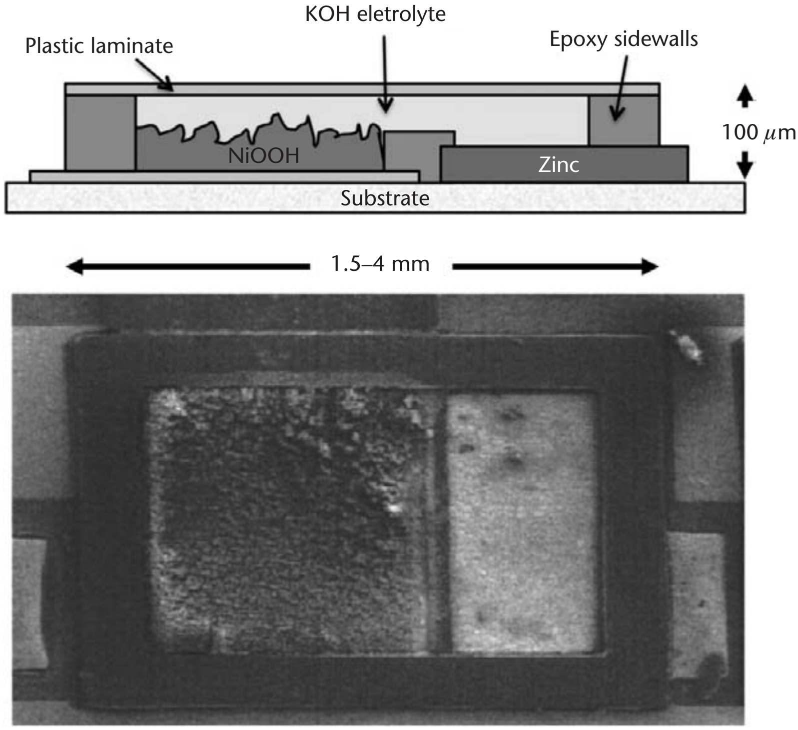

Figure 7.10 Microfabricated Zn-NiOOH planar microbattery with thick film electrodeposited electrodes. (*From:* [31]. Reprinted with permission. Copyright 2001, The Electrochemical Society.)

chambers to prevent evaporation and leakage of the corrosive electrolyte solution over short experimental times, but improved hermetic materials and an improved encapsulation process will be needed for extended device lifetimes. The relative electrode geometries and their separation distances were modeled and optimized for specific power and energy density requirements [32]. The prototypes had operating voltages between 1.2–1.85V, and for a C/1.2 discharge rate, demonstrated energy and power densities as high as 500 μWh/cm^2 and 150 mW/cm^2, respectively. To simulate the power demands of an autonomous microsensor, a commercial solar cell was used to charge the microbattery [33]. More than 2,300 recurring pulse discharges were extracted from the microbattery without performance degradation. Additionally, this was one of the first reported implementations of a hybrid power supply combining an energy harvester with a microbattery.

7.3.2.2 Solutions Processed Microbatteries

The majority of thick film microbattery fabrication procedures are solution-based processes that can deposit porous electrodes with compositions similar to commercial macrobatteries. The materials that can be deposited vary, including metals, ceramics, and polymers, and they can be deposited as suspensions, solutions, or slurries, all of which are referred to as inks. Most thick film electrodes are composites, containing active particles, conductive additives, a polymer binder, and a solvent that adjusts the rheology of the slurry and is eventually evaporated from the film after deposition. The active particles determine the electrochemical properties of the cells. While most conductive additives and polymer binders are electrochemically inactive, their functions are vital to the performance of the electrode, providing enhanced electronic conductivity and mechanical strength to the film, respectively. Adequate electronic conductivity in the electrodes is needed for the battery to provide high discharge rates, and this is achieved when a large enough population of conductive particles forms a contiguous network. Note that for most thick film methods, the electrochemical and materials properties of the deposited films are largely predetermined by the composition of the ink rather than its deposition and postprocessing conditions. Consequently, most solution deposition processes deposit materials at room temperature and ambient conditions, are indiscriminate to the substrate they deposit on, and do not usually require postprocessing steps such as doping or the application of extremely high temperatures. A further understanding of the properties of a solutions deposited film (such as film adhesion, deposited material quality, and interfacial morphology) and effective characterization methods are needed.

Viable solution processes for the fabrication of microbatteries include screen printing and various direct write printing techniques. Though there has been a large amount of literature reporting electrode and electrolyte formulations compatible with screen printing [34], a fully screen-printed microbattery has not yet been demonstrated; however, the successful screen printing of other microdevices suggests that microbatteries made with this process should be feasible. In the following sections, advanced demonstrations of microbatteries fabricated using direct write solution processes will be discussed in detail.

Direct Write Fabricated Microbatteries Direct writing refers to a broad subset of patterning processes that are able to deposit functional materials onto specific locations of a substrate designated by computer controlled translational stages [35]. With these processes, patterns and structures both simple and complex can be fabricated. Direct write processes can be classified according to their ink-writing mechanisms: drop, flow, energy beam, and tip-based direct writing [36]. These categories of material transfer techniques vary greatly in actuation, material compatibility, fabrication capabilities, feature sizes, and writing speeds. Despite their variety, most direct write processes share a few general characteristics. Because materials are patterned additively and often at room temperature and ambient conditions, the amounts of material waste and energy expended are minimal, especially compared to production-scale microfabrication processes that depend on subtractive techniques like lithography and etching. Furthermore, the functionality and structure of the deposited materials are typically independent of the substrate material, its orientation, and its morphology. Because of the versatility of direct writing, there has been a growing interest in applying these manufacturing tools to a variety of fields. Common obstacles preventing the upwards scaling of these processes towards commercialization include necessary improvements in throughput, reliability, resolution, and cost.

The ability of direct write processes to precisely pattern multilayer structures both laterally and vertically in a repeatable and additive fashion is especially beneficial for the fabrication of energy storage devices, which rely on the precise geometry of their active layers as well as the quality of the interfaces between them to achieve good performance. For example, the geometry of a deposited electrode must be optimized to be thick enough to provide adequate energy storage capacity, but thin enough so that charge carrier transport is not impeded, and additionally it requires a high interfacial surface area shared with the electrolyte so that chemical reactions occur rapidly. The fabrication of microbattery components has been demonstrated using flow-based dispenser printing and energy-based laser direct writing, and these processes are discussed in the next few sections.

Flow-Based Direct Write Microbattery Flow-based direct writing refers to a class of tools that, through a positive pressure, are able to dispense a flow of ink through a small orifice such as a syringe needle (Figure 7.11). This is an ambient, room temperature process that is capable of depositing small volumes (tens of picoliters) of inks ranging in viscosities from 10–1 million cP. Line widths and feature sizes depend heavily on the processing parameters employed and ink properties, but typically run between 10 μm–3 mm. Due to the flexibility of this tool to pattern a variety of materials, a few demonstrations of flow-based direct write microbatteries with different configurations and materials have been reported.

Dokko et al. used a pneumatic-controlled microinjector to pattern sol-gel materials into planar microdot [37] and interdigitated lithium manganese oxide ($LiMn_2O_4$) and lithium titanate ($Li_{4/3}Ti_{5/3}O_4$) electrode microarrays [38]. As shown in Figure 7.12, the planar interdigitated electrodes were 100 μm in width and separated by 50-μm gaps. The resulting electrode thicknesses measured between 0.5–1 μm after undergoing high temperature calcination (450°C) and anneal (700°C) processes. Polymer electrolyte films were either manually applied, cast, or printed to cover the electrodes. The interdigitated planar array, occupying 6.6 μm^2 and having an operating voltage of 2.5V, displayed an areal energy density of 11 μWh/cm^2 for a 1C discharge rate. The authors suggested by increasing the thicknesses of the electrodes and decreasing the gaps between them, the performance could be improved.

Steingart et al. developed a similar pneumatic dispenser printing process to fabricate stacked microbatteries with composite slurry electrodes and a gel electrolyte [39, 40]. Because this printing method is gentle (in comparison with other direct write processes, such as inkjet printing, which propels solution droplets or

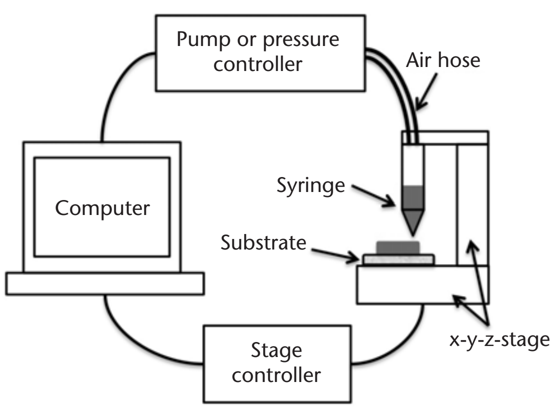

Figure 7.11 Schematic of flow-based direct write printing.

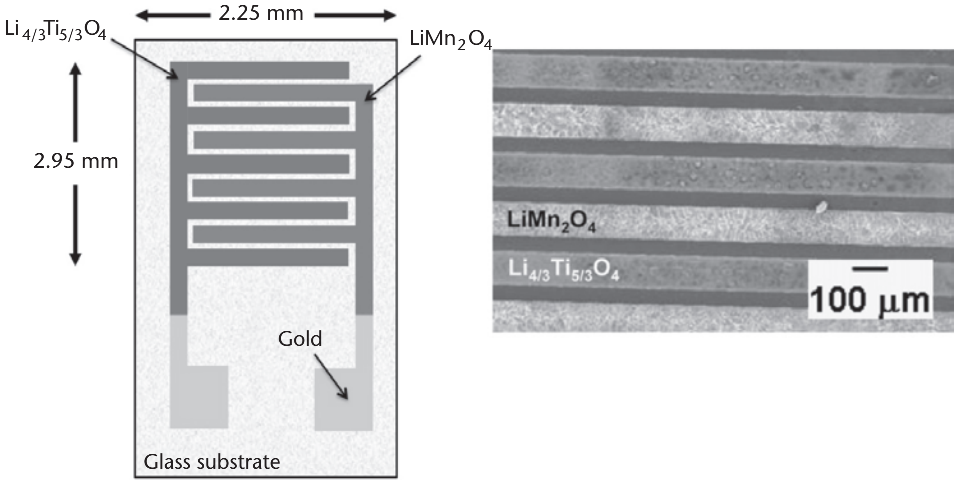

Figure 7.12 Flow-based printed interdigitated, planar lithium-ion electrode array. (*From:* [38]. Reprinted with permission. Copyright 2007, Elsevier.)

films towards a substrate with a large amount of kinetic energy), multiple conformal layers of different inks can be deposited without completely mixing and good interfacial adhesion is achieved [41]. As a result, the precise construction of interesting multilayer devices and structures is possible. Both batteries of lithium-ion and zinc metal-oxide chemistries have been printed; however, the printed lithium-ion battery's performance was hampered by residual moisture contamination. The zinc metal-oxide microbattery consists of electrode slurries of zinc or manganese dioxide (Figure 7.13). Separating the two electrodes is a gel electrolyte of zinc trifluoromethanesulfonate $(Zn^{+}Tf^{-})_2$ salt dissolved in 1-butyl-3-methylimidazolium trifluoromethanesulfonate ($BMIM^{+}Tf^{-}$) ionic liquid and polyvinylidene fluoride–hexafluropropylene (PVDF-HFP) polymer binder. For a stacked microbattery with a 0.25 cm^2 footprint area and 100-μm thickness and operating voltage between 1–2V, an areal capacity and energy density of 1.05 mAh/cm^2 and 0.37 mWh/cm^2 were demonstrated. Further investigations on the cycling and power behavior are underway.

Laser Direct Write Printed Microbatteries Extensive research at the U.S. Naval Research Laboratory on laser-induced forward transfer (LIFT) of electrochemical materials and laser micromachining has inspired the use of direct write technologies in a variety of fields [42, 43]. The variation of LIFT known as matrix-assisted

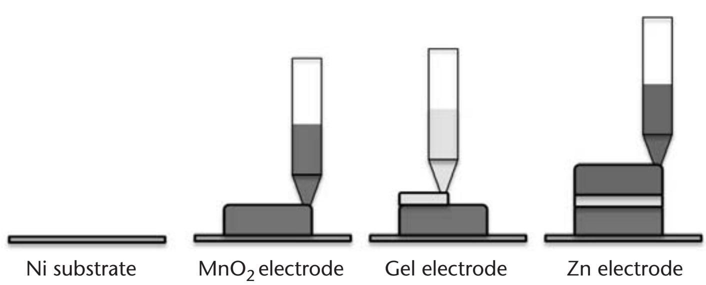

Figure 7.13 Dispenser printed zinc metal-oxide stacked microbattery. (*From:* [40].)

pulsed laser evaporation (MAPLE) employs a laser, which, when directed on an ink mounted on the underside of a laser transparent support material, will propel the ink towards a substrate positioned in parallel (Figure 7.14). The ink is a composite of an active powder, solvent, and a UV-absorbing matrix, which upon interacting with the laser at the interface of the support material, evaporates to release the ink. The MAPLE process is capable of depositing thick films without altering or damaging their properties. This is especially advantageous in electrochemical systems that benefit from maintaining porous, high surface area electrode structures. Both planar and stacked microbattery configurations have been demonstrated using a combination of the MAPLE process along with laser micromachining [44, 45]. For planar microbatteries, laser micromachining was used to create thin electronically isolated gaps between two adjacent electrodes. Typical micromachined line widths of 10–50 μm have been shown. The added capability of laser micromachining allows for the high spatial utilization and intricate patterning of the electrodes with good accuracy and repeatability. Besides the printing of adjacent rectangular electrodes, some complex electrode configurations including interdigitated and concentric circle electrodes were also fabricated (Figure 7.15) [42]. To activate the batteries, drops of alkaline electrolyte solutions were blanketed over the laser-transferred electrodes without any containment. For operation voltages of 1.55V, the typical capacity and energy density measured were 100 μAh/cm^2 and 0.6 mWh/cm^2, respectively. An approximated maximum power density of 0.8 mW/cm^2 was reported.

Similarly, laser micromachining was also used in conjunction with the MAPLE process to fabricate stacked lithium-ion microbatteries embedded into a substrate. Sutto et al. micromachined a 30-μm deep, 3-mm by 3-mm trench into polyimide backed with aluminum, which acts as the cathodic current collector [46]. Laser-transferred inks of $LiMnO_2$, a ceramic-solid polymer ionic liquid (c-SPIL) nanocomposite electrolyte, and carbon electrode slurries were deposited sequentially into the trench, as shown in Figure 7.16. Platinum was sputtered onto the top of the

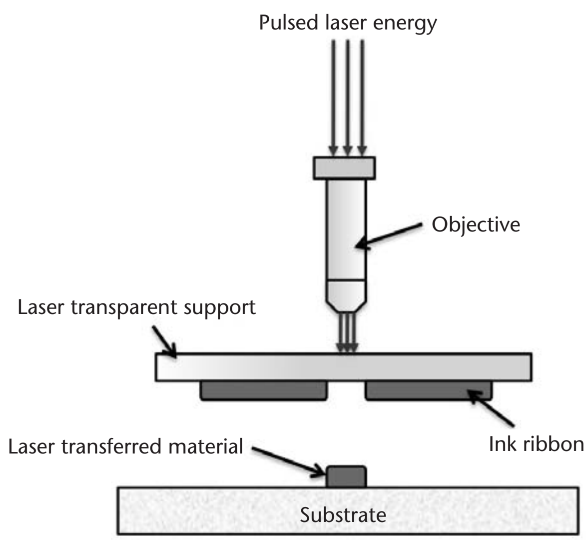

Figure 7.14 Schematic of matrix-assisted pulsed laser evaporation (MAPLE) direct write deposition. (*From:* [43].)

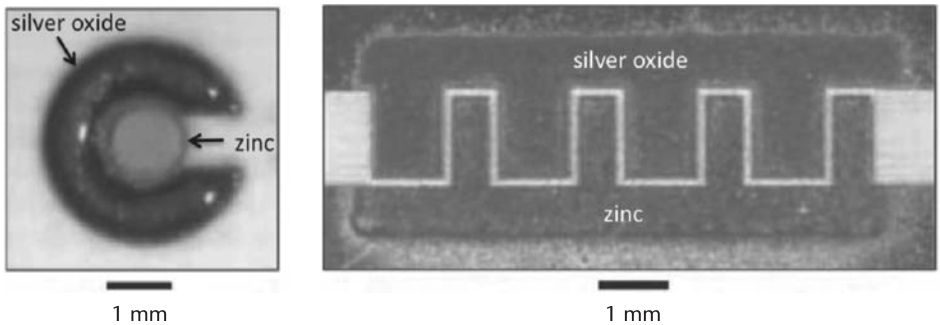

Figure 7.15 Planar (a) concentric circle and (b) interdigitated zinc-silver oxide alkaline microbatteries fabricated using laser direct write. (*From:* [42] Reprinted with permission. Copyright 2007, *MRS Bulletin.*)

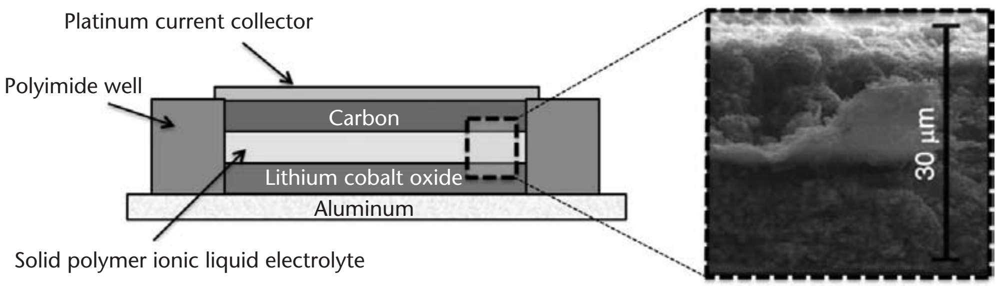

Figure 7.16 Stacked lithium-ion microbattery embedded in a substrate using laser direct write fabrication. (*From:* [46]. Reprinted with permission. Copyright 2006, The Electrochemical Society.)

trench, acting both as the anodic current collector as well as an encapsulant. When cycled at a C/3 discharge rate between 3–4.65V, the stacked microbattery exhibited a capacity and energy density of 110 μAh/cm^2 and 1.32 mWh/cm^2, respectively. By designing the c-SPIL to structurally retain separation between the electrodes during fabrication, Sutto et al. were able to successfully demonstrate the fabrication of a microbattery with a stacked configuration integrated and sealed into a substrate without any manual assembly. Because it is versatile and indiscriminate of materials and substrates, the laser forward-transfer process could feasibly be applied to many other battery chemistry systems.

7.3.3 Concluding Remarks for 2D Microbatteries

The miniaturization of traditional, 2D, rechargeable macrobatteries in all three dimensions has proven to be a fairly challenging task because of the complexities in reconciling materials, geometrical, processing, and performance constraints into a small, integrated electrochemical device. Thin and thick film fabrication strategies have been explored and encouraging strides towards integrating these microbatteries into hybrid micropower supplies have been made. Table 7.3 summarizes the two-dimensional microbatteries discussed in this chapter.

Table 7.3 Two-Dimensional Microbattery Implementation and Performance Summary

2D Microbattery	*Chemistry*	*Geometry**	*Operating Volt (V)*	*Areal Performance** for 1 CM²*
Thin Film				
Physical vapor deposition [24–28]	Li-ion, glass electrolyte	(t) 10-15 μm (f) 1 mm²–25 cm² stacked	3–4.2V	10 μAh–1 mAh/; 50 μWh–5 mWh; 10mW
Thick Film				
Electrode polishing [30]	Li-ion, ceramic electrolyte	(t) 500 μm (f) 1 cm² stacked monolith	1.5–3.2	15 μAh; 42.5 μWh
Microfabrication [31, 32]	Ni-Zn, liquid electrolyte	(t) 50 μm (f) 2 mm² planar	1.2–1.85	305 μAh; 500 μWh; 150 mW
Dispenser printing [38, 40]	Li-ion, polymer electrolyte	(t) 0.5–1 μm (f) 6.6 mm² planar, interdigi-tated electrodes	1.5–3	4.5 μAh; 11 μWh
	$Zn\text{-}MnO_2$, gel electrolyte	(t) 100 μm (f) 0.25 cm² stacked	1-1.9	1.05 mAh; 0.37 mWh
Laser direct write printing [44, 46]	$Zn\text{-}Ag_2O$, liquid electrolyte	(t) 10 – 20 μm (f) 4.9 mm² planar, concentric electrodes	0.8–1.6	450 μAh; 0.6 mWh; 0.8 mW
	Li-ion, gel electrolyte	(t) 30 μm (f) 9 mm² stacked, embedded in substrate	2.9–4.2	110 μAh; 1.32 mWh

* (t): thickness, (f): footprint area; ** areal capacity, energy density, and, if provided, power density are listed.

7.4 Three-Dimensional Microbatteries

Designing small footprint area two-dimensional microbatteries oftentimes leads to a fundamental conundrum: a trade-off between the amount of energy stored (determined by the electrodes and their thicknesses) and the rate at which this energy can be extracted (determined by the contact area between electrodes and electrolytes, their through thicknesses, and charge transport properties). For optimal performance in a 2D microbattery, this manifests as a practical limit to the relative electrode and electrolyte dimensions. As an alternative, nontraditional three-dimensional microbattery architectures have been explored, and initial implementations are discussed in the following section.

Three-dimensional microbatteries function identically to two-dimensional cells, but by increasing the amount of electrode and electrolyte interfaces while maintaining short ion transport distances between electrodes, drastically enhanced electrochemical performance can be derived. This has been a burgeoning field of research with consistent developments in materials, processing, modeling, and characterization. Though configurations and approaches towards fabricating three-dimensional microbatteries may vary, a defining property is that at the microscopic scale, charge transport between the electrodes is essentially one-dimensional; at the macroscopic level, the electrodes exhibit nonplanar geometries [47]. As a consequence, large areal energy densities are achievable without sacrificing high rate power performance.

A variety of three-dimensional architectures are potentially suitable configurations for microbatteries. However, for classification purposes, the structures can be distinguished according to the continuity of its electrode and electrolyte phases. The most commonly examined architectures are cells with interdigitated cylinder or plate electrode arrays [Figure 7.17(a, b)] or arrangements of concentric electrode and electrolyte units surrounded by a matrix of the opposing electrode [Figure 7.17(c)]. In both cases, at least one electrode component is discontinuous. Alternatively, microbatteries can also be constructed from 3D configurations in which all phases (both electrodes and the electrolyte) are continuous regardless of their arrangement, be it periodic or aperiodic. This can be as simple as using a 2D stacked battery motif and folding it into a 3D pattern [Figure 7.18(a)], or creating two interpenetrating, uninterrupted electrode networks, either regular or irregular in morphologies, and separating them by a conformal and continuous electrolyte coating [Figure 7.18(b, c)].

In the last two decades, there have been numerous feasibility studies on the fabrication and electrochemical properties of components that can be applied to

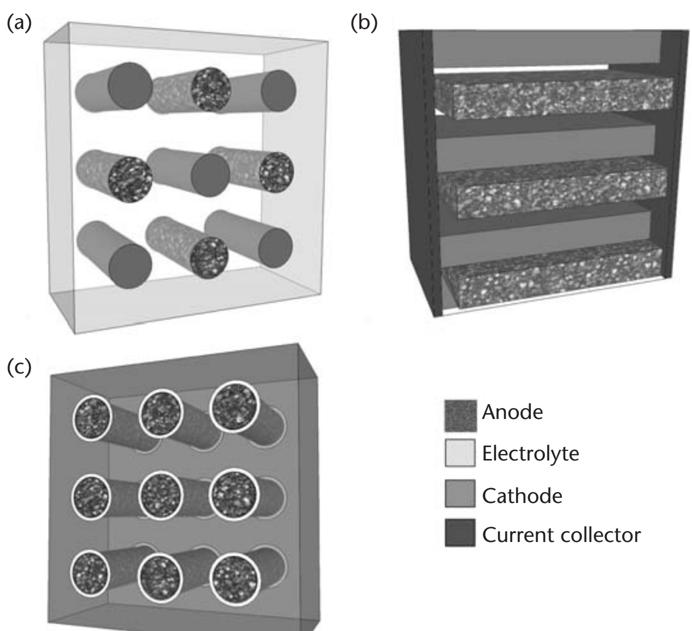

Figure 7.17 Three-dimensional microbattery architectures with at least one discontinuous component: (a) regular arrangements of interdigitated anode and cathode cylinders or (b) interdigitated electrode plate arrays distributed in an electrolyte. (c) Electrolyte-coated anode cylinders dispersed in a continuous phase of cathode material. Note, (a) and (c) do not display current collectors. (*From:* [47].)

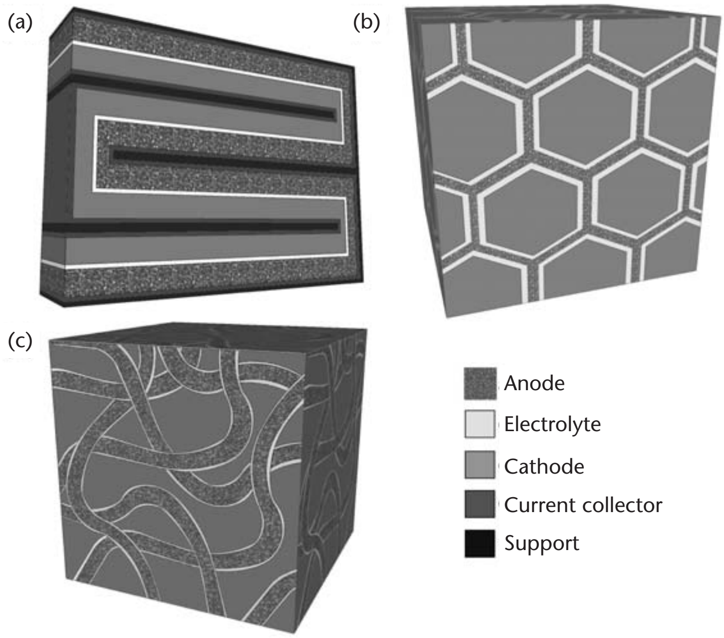

Figure 7.18 Three-dimensional microbattery architectures where the electrodes and electrolyte are each continuous phases: (a) a 2D stacked battery configuration that is folded into a 3D architecture, and (b) a periodic and (c) aperiodic scaffold (composed of cathode material conformably coated with a thin electrolyte) back-filled with anodic material occupying the free volume. Note, (b) and (c) do not display current collectors. (*From:* [47].)

3D microbatteries. This section attempts to highlight those significant efforts. Note that as the fabrication technologies for 3D architectures are far from mature, the comparison of the performance of different 3D microbattery prototypes is difficult because most attempts use test structures that either do not fully exploit their 3D architecture or only successfully demonstrate some but not all necessary components.

7.4.1 3D Microbattery Architectures with a Discontinuous Element

The microbattery architectures discussed in this section, summarized in Figure 7.17, are composed of at least one electrode phase that is made of discrete elements, usually arranged in an array. Each of the electrode components must be electronically connected using a current collector network. As an electrode arrangement becomes more elaborate, the current collector configuration will need to equally become as intricate. Most of the methods described involve constructing a 3D mold and then filling it with the active components by vapor deposition, electrodeposition, or colloidal methods. However, alternate fabrication processes, for example direct-write printing, have also been used to construct three-dimensional geometries.

7.4.1.1 Interdigitated Electrode Array Microbattery

Interdigitated microbattery architectures incorporate regular arrangements of anode and cathode structures surrounded by a continuous electrolyte phase. Potential electrode shapes include plates, rods with triangular or square cross-sections, and most commonly, cylindrical rods. For this microbattery architecture, the relative locations and dimensions of the electrodes will determine current and potential distributions within the cell; therefore, both precise spatial positioning and patterning capabilities are imperative. Dunn et al. fabricated Ni-Zn and lithium-ion microbatteries with square and cylindrical electrode arrays by filling removable silicon molds with electrode materials via colloidal or electrodeposition methods [48, 49]. Electrodes with diameters ranging from tens to hundreds of micrometers and aspect ratios between 10:1 to 50:1 (rod length:diameter) could be achieved by micromachining silicon substrates using a combination of photolithography with either deep reactive ion etching (DRIE) or photo-assisted anodic etching. Alternating rows of nickel hydroxide and zinc posts were fabricated on the same substrate within a 5-mm by 5-mm footprint (Figure 7.19), and when submerged in an alkaline electrolyte, demonstrated an areal capacity of 2.5 μAh/cm^2 for operating potentials of 1.3V–1.75V. The authors suggested that further capacity improvement could be achieved with higher density electrode arrays and alternative electrode arrangements; however, the repeated cycling of the interdigitated Ni-Zn battery was severely impaired by the gradual removal of zinc electrode material by the electrolyte. This process is relatively indiscriminate to the electrode materials deposited;

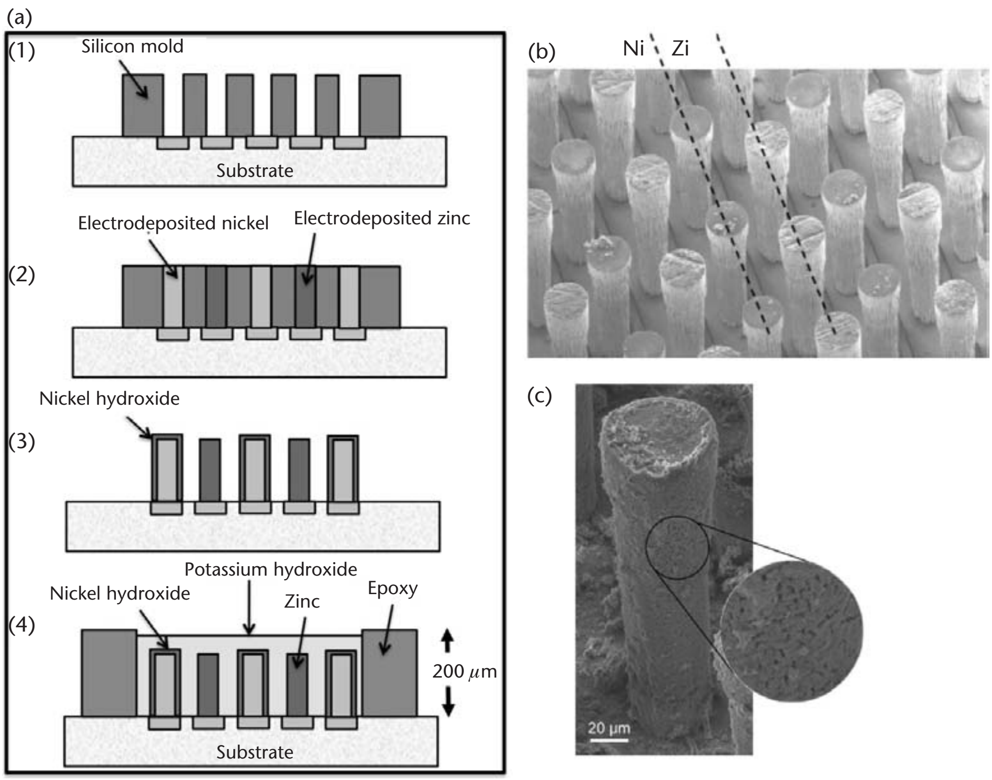

Figure 7.19 (a) Fabrication process of interdigitated Ni-Zn post electrode microbattery. (1) Silicon mold is micromachined with high-aspect ratio trenches; (2) nickel and zinc are selectively electrodeposited into the mold; (3) silicon mold is removed and nickel hydroxide is conformally coated onto the nickel electrodes; (4) microbattery is defined by epoxy walls and filled with aqueous alkaline electrolyte. (b) Micrographs of the nickel and zinc electrodeposited posts; (c) a nickel post coated with nickel hydroxide. (*After:* [49]. Reprinted with permission. Copyright 2007, IEEE.)

arrays of high-aspect ratio lithium-ion and zinc-air electrodes [50] have also been demonstrated.

Similar to Dunn's work, Ripenbein et al. used silicon substrates as molds for the patterning of electrode posts. Trenches were etched onto both sides of a silicon substrate using a deep reactive-ion etching (DRIE) process, and then the mold was made porous and electronically insulating through a metal-assisted etching and oxidation process [51]. The resulting porous silicon partition serves multiple purposes, acting as a mold and physical separator between the anode and cathode structures, and also providing a porous network for a liquid or gel electrolyte (Figure 7.20). By etching trenches on both the top and bottom of the wafer, an interlaced geometry of electrode arrays was formed. By isolating the arrays, elaborate masking processes during electrode deposition can be avoided. The researchers developed a silicon porous partition with 8,265 trenches/cm^2, and plan to fill the trenches with MoO_2, $LiCoO_2$, or $LiFePO_4$ cathode materials and lithiated graphite anodes. Though full microbatteries have not been demonstrated, optimization of the fabrication processes, pore morphology, and compatible materials are underway.

Interdigitated lithium-ion electrode arrays were also fabricated by Min et al. using carbon-microelectromechanical systems (C-MEMS) fabrication, a versatile process for patterning carbon structures and tailoring their material properties [52]. Photoresist is patterned using photolithography, then pyrolyzed under specific conditions [53, 54]. Carbon posts of a 65-μm height and aspect ratios ranging between 1 and 4 were fabricated into 120 by 120 arrays in a 1 cm^2 footprint area (Figure 7.21). Alternating rows of carbon posts became current collectors to electropolymerized coatings of polymer cathode material [dodecylbenzenesulfonate-doped polypyrrole (PPYDBS)] while bare carbon posts were used as anodes. When submerged in a liquid electrolyte, the interdigitated microbattery exhibited a capacity of 10.6 μAh/cm^2 when cycled between 0.7–3.5V. Reported difficulties with the shorting between posts as well as high internal resistances measured in the carbon current collectors may have led to poor cycling behavior and reduced capacity.

By using a unique drop-on-demand ink jet printing system, Ho et al. fabricated interdigitated zinc–silver oxide microelectrode arrays without the use of templates or surface patterning, eliminating the many fabrication steps typically associated with subtractive photolithography and etching processes [55]. The super ink jet (SIJ) printer used in this work is able to eject droplets of nanoparticle solutions

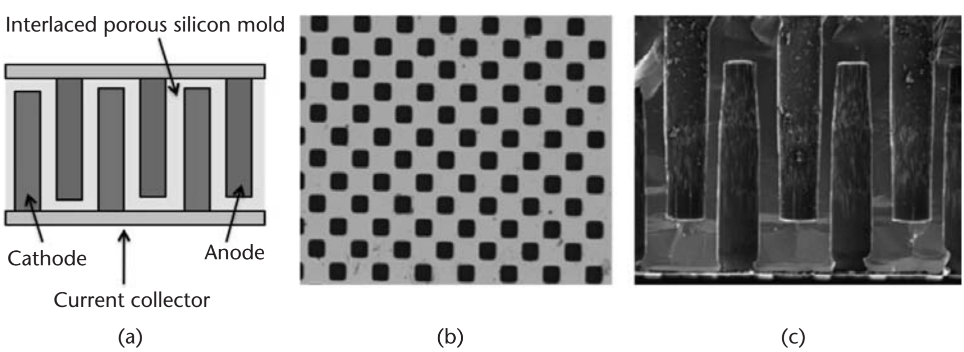

Figure 7.20 (a) Interlaced electrode rods separated by a porous silicon mold. (b) Top view and (c) cross-section view of the silicon mold. (*After:* [51].)

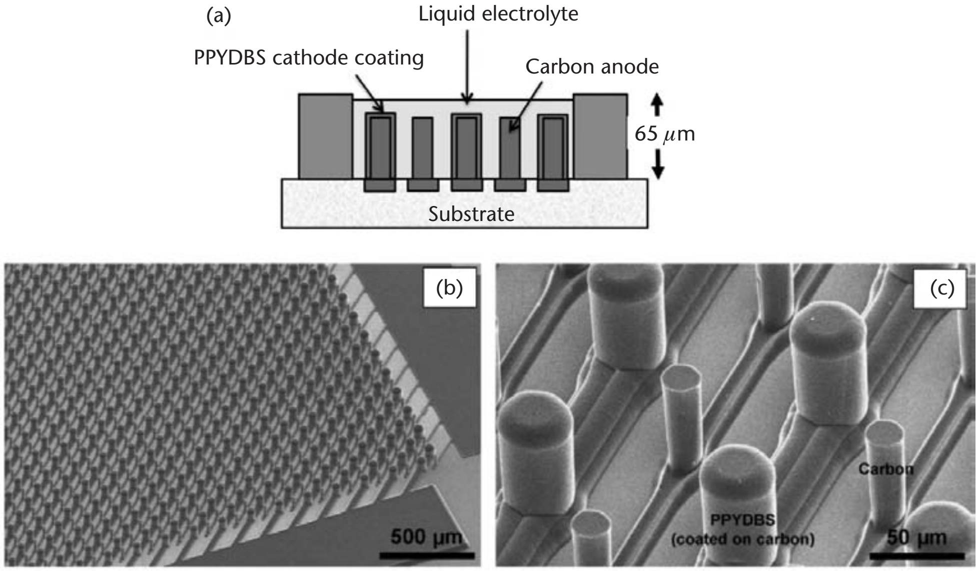

Figure 7.21 (a) Schematic of a C-MEMS fabricated interdigitated electrode post array of pyrolyzed carbon anodes and polymer cathode coatings of dodecylbenzenesulfonate-doped polypyrrole (PPY-DBS). (b, c) Micrographs show an array of interdigitated carbon anodes and thick polymer-coated cathodes. (*After:* [52].)

three orders of magnitude smaller in volume than commercial ink jet printers, allowing the deposition of submicron printed feature sizes [56]. As a SIJ printed drop is ejected, its reduced volume facilitates rapid solvent evaporation so that when deposited, the droplet viscosity is so high that the printed ink is essentially dry, enabling the fabrication of 3D structures [57]. With the precise deposition of a succession of ink droplets, 10-μm diameter, 40-μm length pillars of silver were printed. Pairs of 3-mm by 3-mm electrode arrays, each with 722 pillars each were printed adjacent to each other and then submerged in an alkaline electrolyte solution with dissolved zinc oxide. Upon applying an electric field across the pair of silver arrays, silver oxidizes on the positive electrode while the negative electrode acts as current collectors on which zinc electrodeposits from the electrolyte; essentially a zinc–silver oxide microbattery is assembled in its charged state (Figure 7.22). Though the configuration of the electrodes was not optimal, performance was enhanced significantly by the 3D structure. An areal capacity and energy density of 2.5 mAh/cm^2 and 3.95 mWh/cm^2 were measured, respectively, for operating potentials between 1.2V and 2V. Implementing an interdigitated structure, increasing the density of pillars, and stabilizing the zinc electrode in the electrolyte solution can generate substantial improvements in performance.

Some of the most advanced 3D microbattery research efforts have utilized the interdigitated electrode architectures discussed in this section, and have been very convincing in demonstrating the inherent benefit of implementing nonplanar microbattery configurations. Continued modeling and investigation are needed, including the consequences of the novel geometries on nonlinear current and potential distributions, and an understanding of the materials stability and cycle life of the cells. Such endeavors will ultimately be critical in demonstrating the viability of these new microbattery architectures.

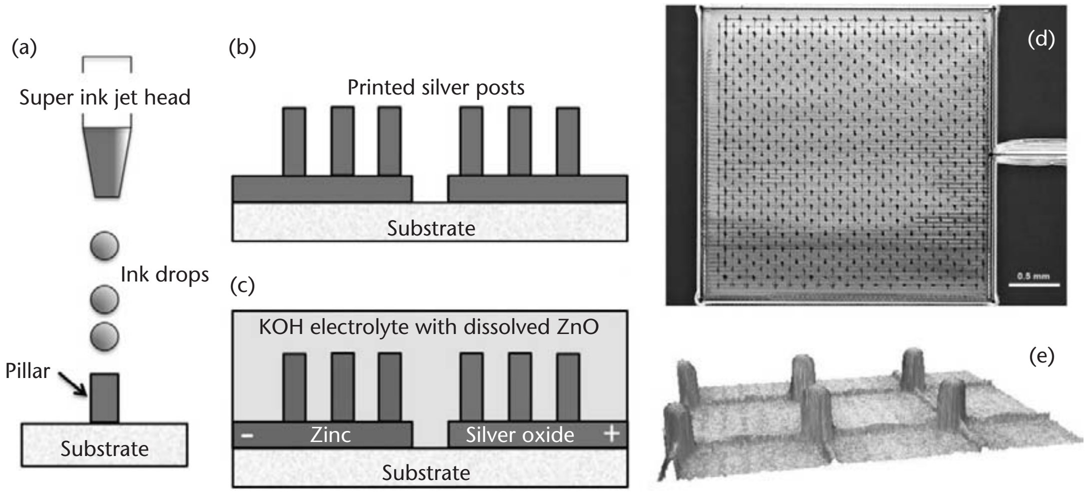

Figure 7.22 Super ink jet printed Zn-Ag_2O post electrode arrays. (a, b) Silver pillars are printed on a substrate. (c) When submerged in an alkaline electrolyte with dissolved zinc oxide, an applied electric field causes silver to oxidize at the positive electrode while zinc electrodeposits on the negative electrode, assembling a microbattery in its charged state. Micrographs of (d) an electrode array and (e) printed silver posts. (*After:* [55].)

7.4.1.2 Concentric Microbattery Array

Conformal-coated electrode arrays and interdigitated electrode architectures appear similar in configuration; however, the former incorporates one electrode phase which is continuous through the structure. By implementing a concentric configuration of electrodes and electrolyte layers, simple 1D current and potential distributions are achieved through the individual concentric structures, and the electrochemical behavior can be assumed to resemble that of stacked 2D battery configurations. As a result, a 3D microbattery with this concentric architecture relies on the composite performance of many small concentric microbatteries. Though essentially simple in concept, the feasibility of fabricating such architectures has been difficult, and demonstration efforts remain in their infancy.

Nathan et al. employed 2D microbattery techniques for desposition onto textured surfaces, sequentially depositing thin conformal films onto substrates modified with 3D features [58]. By perforating silicon or glass substrates with microscale channels, the available surface area was enhanced by an order of magnitude. 0.5-mm-thick silicon substrates with 50-μm diameter microchannels were conformably coated with thin films of a nickel current collector, MoS_2 cathode, and hybrid polymer electrolyte, and then back-filled with a graphite slurry (Figure 7.23). The cells displayed an operating voltage of 1.5V and over more than 200 cycles, capacities of 2 mAh/cm^2 were measured with high Faradaic efficiency, about 30 times greater than a 2D battery with the same chemistry and footprint area. Further improvements in conformal deposition techniques and materials properties have been explored [59, 60]. Despite fabrication difficulties, this architecture offers a few advantages: though the discrete microbatteries were arranged in a periodic array in this work, the periodicity is not required since the individual concentric

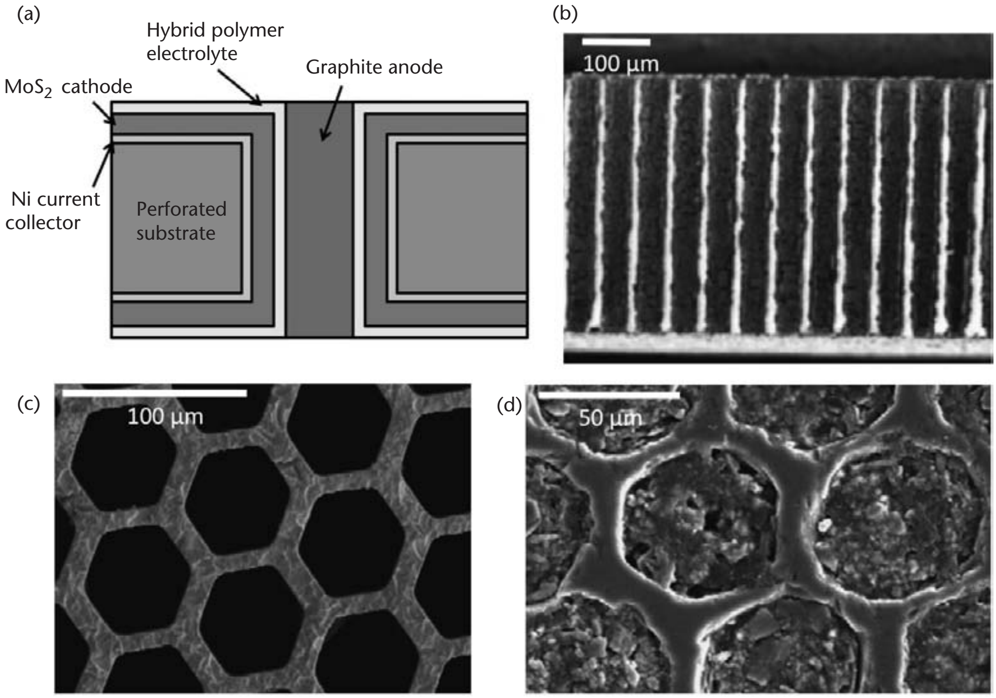

Figure 7.23 Concentric battery (a) schematic of the cross-section of a perforated substrate with high aspect ratio holes that are conformably coated with continuous, thin films of current collector, cathode, and electrolyte layers. The anode is back-filled into the open volume. (b, c) Micrographs of the perforated substrate and (d) 3D enhanced microbattery with graphite filled microchannel volume. (*After:* [58, 60]. Reprinted with permission. Copyright 2006, Elsevier. Copyright 2005, IEEE.)

microbatteries presumably do not interact with each other. Also, the energy density of the microbattery can be optimized by occupying the larger, continuous electrode phase with the electrode material of less theoretical volumetric capacity.

7.4.2 3D Microbattery Architectures with Continuous Elements

All microbattery configurations discussed in the next section have continuous electrode and electrolyte phases. These components may be arranged in regular patterns, examples of which are illustrated in Figure 7.18(a, b), or in aperiodic, undirected configurations, as shown in Figure 7.18(c). By having continuous electrodes, as long as they have sufficient electrical conductivity, applying current collectors to these structures may be less challenging since the electrodes need only be accessed at one point rather than throughout the whole structure.

7.4.2.1 3D Enhanced Thin Film Microbatteries

The prospect of using silicon-compatible fabrication methods to concurrently integrate microbatteries with MEMS sensors or energy harvesting components onto the same substrate has encouraged researchers to extend the capabilities of thin film processes and develop complementary methods to fabricate 3D geometries. Niessen et al. textured silicon substrates with high aspect ratio trenches and pores [61, 62].

By successively depositing thin films of silicon, Lipon glass electrolyte, and $LiCoO_2$ onto patterned silicon substrates with trenches 135-μm deep and 5 μm in width, a 3D microbattery with an operating voltage of 3.5V and projected capacity and energy density of 1.5 mAh/μm-cm^2 and 5 mWh/μm-cm^2, respectively, should be achievable (Figure 7.24). Efforts towards a working demonstration include fabrication and deposition improvements, an understanding of the silicon electrochemical behavior upon repeated cycling, and integration efforts to enable top and bottom substrate texturing to further increase the energy storage capacity. Since 3D silicon morphologies are of interest to a variety of other research fields such as thermoelectric and sensor devices [63], ongoing improvements in 3D micropatterning continue to increase the viability, cost-effectiveness, and appeal of fabricating 3D microbatteries using patterned substrates coupled with thin film deposition processes.

7.4.2.2 Interpenetrating Microbatteries

The performance enhancements gained by incorporating nanostructured materials into electrochemical devices have been profound. Nanostructured materials have already been shown to significantly increase the electrochemical properties of 2D macrobatteries, and Ergang et al. further exploited this phenomenon by patterning nanostructured electrochemical materials into 3D microbattery architectures [64]. Patterned monoliths of carbon anode material were fabricated by dispersing colloidal poly(methyl methacrylate) (PMMA) spheres with 200-nm diameters in a precursor solution, which is then carbonized. The resulting three-dimensional ordered macroporous (3DOM) carbon electrode monolith served as a template for an interpenetrating microbattery. It was conformably coated by an electrodeposited pinhole-free poly(phenylene oxide) polymer electrolyte that was soaked in a liquid

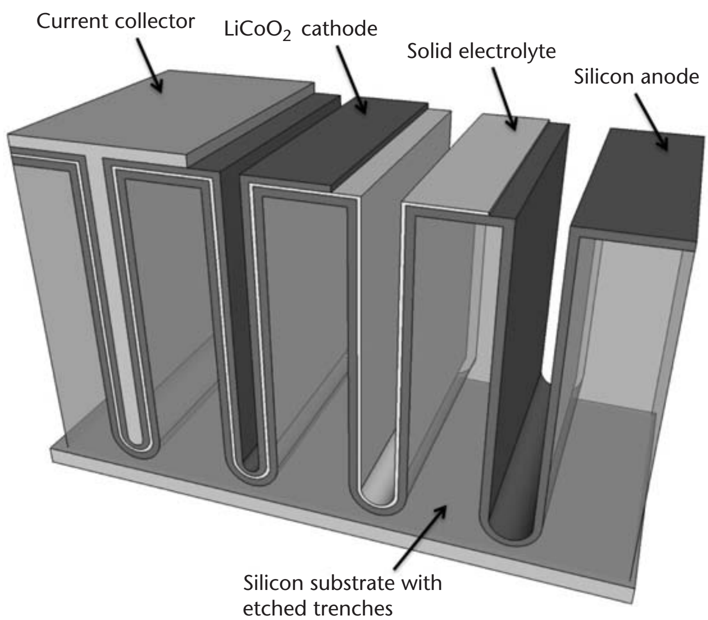

Figure 7.24 Schematic of textured silicon substrate conformably coated with lithium-ion thin film battery layers. (*After:* [61].)

lithium salt solution, and then any free volume was infiltrated by vanadium pentoxide (V_2O_5) ambigel cathode precursors. Subsequent careful ageing and removal of residual solvent was necessary to maintain the nanostructure of the vanadia ambigel. Finally, the interpenetrating 3D microbattery was manually sandwiched between two foil current collectors (Figure 7.25). For a microbattery occupying a 0.26 cm^2 footprint area and a thickness of 0.1343 cm, when cycled between 1.6–3.3V, a gravimetric capacity of 350 μAh/g was reported, resulting in an estimated areal capacity of 25 $\mu Ah/cm^2$. Improved cathode and electrolyte material properties may be achieved with processing improvements; however, the largest difficulty has been increasing the electrical conductivity of the V_2O_5 cathode.

Rolison et al. have been developing a microbattery composed of networks of nanoscale interpenetrating electrodes and electrolyte structures engineered without periodicity constraints [65, 66]. The authors suggest that eliminating the periodic arrangement of battery components will reduce processing complications, especially since the self-assembly of nanodimensioned aperiodic sponge structures has been extensively demonstrated using simple solution processes such as sol-gel synthesis. A sponge structure, in this case an aerogel material, is used as one electrode that provides continuous scaffolding on which a thin electrolyte can be conformably deposited. The remaining open volume defined by the extensive network of pores throughout the electrolyte-covered electrode scaffolding is infiltrated with a continuous phase of the opposing electrode, forming a microbattery (Figure 7.26). So far aerogel electrodes of MnO_2 covered in 10–100-nm electrodeposited poly(phenylene oxide) (PPO) polymer electrolyte were fabricated, and reversible lithium-ion intercalation and deintercalation into and from MnO_2 have been demonstrated through the thin electrolyte layer. Ruthenium oxide (RuO_2) electrode colloids were back-filled into the pore volume of the structure using cryogenic methods; however, difficulties ensuring good interfacial contact between the electrolyte and RuO_2 electrode and electrically accessing this electrode have prevented full battery operation. Progress towards developing a full microbattery will only be

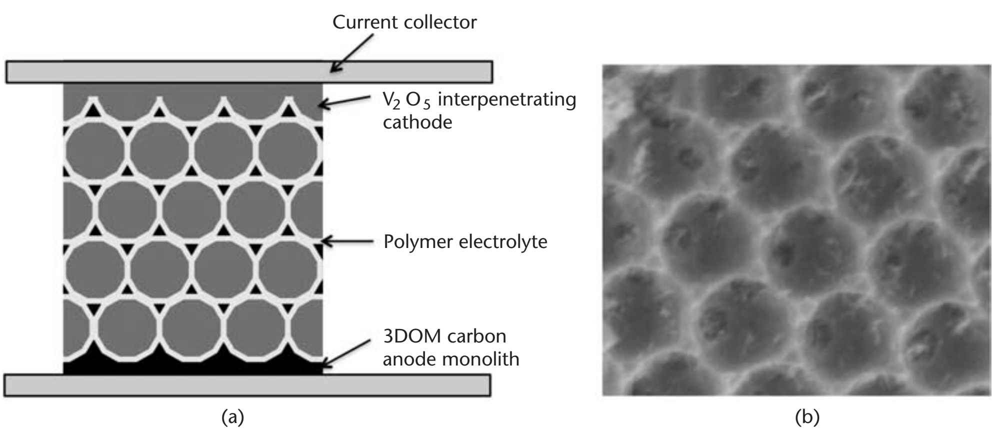

Figure 7.25 (a) Three-dimensional interpenetrating lithium-ion microbattery. A three-dimensional ordered macroporous (3DOM) carbon anode monolith is coated with a polymer electrolyte film then (b) back-filled with a vanadium pentoxide ambigel cathode. (*After:* [64]. Reprinted with permission. Copyright 2007, The Electrochemical Society.)

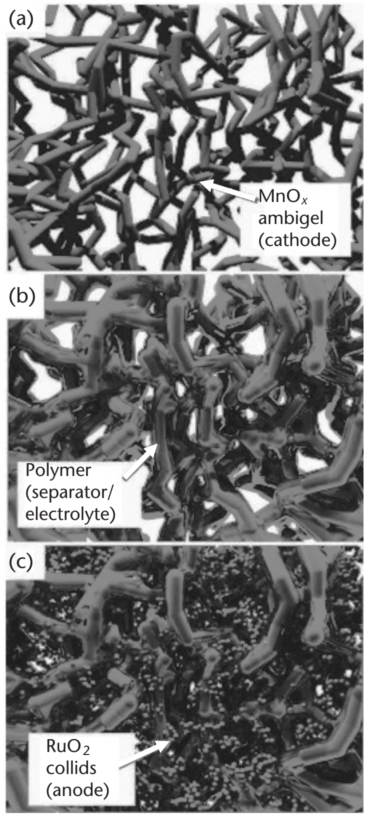

Figure 7.26 Aperiodic sponge microbattery assembled from (a) a manganese oxide cathode ambigel coated with (b) a continuous polymer electrolyte and back-filled with (c) ruthenium oxide anode material. (*After:* [66].)

possible with effective methods to evenly coat pin-hole free polymer films on 3D surfaces without occluding pores, enhanced electronic conductivity of the electrode materials, and successful infiltration of the final electrode within the open volume of the sponge structure without damaging the scaffold. Architectures with interpenetrating tricontinuous components are promising structures for augmenting the areal storage capacity of microbatteries. New compelling concepts to implement percolating 3D battery structures are continually being proposed and demonstrated, such as the self-assembly of batteries from electrode colloids via their attractive and repulsive surface forces in a continuous electrolyte phase [67]. Complementary characterization techniques and device modeling for these structures will be needed for further maturation of the field.

7.4.3 Prospects for Three-Dimensional Microbattery Implementation

The ever-expanding field of three-dimensional microbattery research has provided extensive opportunities for architecting innovative battery structures, material designs, and new fabrication processes. Promising research efforts in the 3D microbattery field are summarized in Table 7.4. As was asserted throughout this section, deeper understanding of the effects of these novel microbattery architectures on

Table 7.4 Three-Dimensional Microbattery Demonstrations and Performance Summary

	Fabrication Methods	*Chemistry*	*Geometry**	*Operating Voltage (V)*	*Performance for 1 CM²***
3D microbattery architectures with at least one discontinuous element	*Interdigitated electrode array microbattery*				
	Removable silicon mold, electrodeposited rods [49]	Ni-Zn, liquid electrolyte	(d) 50 μm (l) 400 μm (f) 0.25 cm^2	1.3–1.8	2.5 μAh; 4 μWh (measured)
	Interlaced trenches in porous silicon mold [51]	Li-ion, liquid electrolyte	(w) 50 μm (l) 290 μm (n) 8,265 (f) 1 cm^2	3–4	Cycling through the porous silicon material has been demonstrated
	C-MEMS fabricated rods [52]	Li-ion, liquid electrolyte	(d) 20 μm (l) 65 μm (n) 14,400 (f) 1 cm^2	0.7–3.5	10.6 μAh; (measured)
	Super ink jet printed rods [55]	Zn-AgO, liquid electrolyte	(d) 10 μm (l) 40 μm (n) 722 (f) 0.9 cm^2	1.2–2	2.5 mAh; 3.95 mWh (measured)
	Concentric microbattery array				
	Thin film-coated concentric rod array [58]	Li-ion, hybrid polymer electrolyte	(d) 50 μm (l) 0.5 mm (f) 5.3 cm^2	1.3–2.2	2 mAh; 3 mWh (measured)
3D microbattery architectures with continuous elements	*3D enhanced thin film microbatteries*				
	Thin film-coated textured substrate [61]	Li-ion, glass electrolyte	(w) 5 μm (l) 130 μm (f) 1 cm^2	3–4	1.5 mAh; 5 mWh; 50 mW (predicted)
	Interpenetrating microbattery structures				
	Templated ordered scaffold [64]	Li-ion, polymer electrolyte	(t) 0.1343 cm (f) 0.26 cm^2	1.6–3.3	25 μAh; 60 μWh (measured)
	Aperiodic "sponge" [66]	Li-ion, polymer electrolyte	10-nm thin electrodeposited polymer	N/A	Inserted/removed Li ions from polymer-coated cathode at 2C rate

*d: element diameter, l: element length, w: element width, n: number of elements, t: thickness of device, f: footprint area of device. **if provided, areal capacity, energy density, and power density.

the current and potential distributions within the cell, interfacial reaction kinetics, modes of failure, and overall device performance will require new research including processing, characterization, and modeling capabilities. Though commercial viability is far from realization, important concerns that are often overlooked regarding packaging, environmental robustness and stability, as well as process scalability and cost should all be thoughtfully considered.

7.5 Electrochemical Microcapacitors

With ongoing improvements in materials performance and processing, electrochemical capacitors have also become viable energy storage buffers for autonomous wireless sensors. Alone, electrochemical capacitors are limited in energy density compared to batteries, but may be appropriate for scenarios requiring frequent, high-power pulse operation such as in emergency response applications, which rely on rapid real-time information. Electrochemical capacitors can also be considered complementary technology when used with batteries for applications requiring larger energy storage capacity, and if used in conjunction, the capacitor can address high power surges demanded from the load, effectively mitigating detrimental battery operation and therefore increasing the battery's and consequently the device's lifetimes.

Largely overshadowed by microbattery research, microcapacitors have until recently been overlooked, except perhaps to vet fabrication processes developed primarily for microbatteries. This processing compatibility is possible due to their similar configurations and in some cases, identical materials. Like microbatteries, microcapacitors use materials similar to macrocapacitor devices, and implementation of these miniature devices has been explored for both two-dimensional and three-dimensional cell configurations.

7.5.1 Electrochemical Capacitor Materials

Electrochemical capacitor research efforts have largely concentrated on the discrete improvements of electrode and electrolyte materials. Typical materials used as electrochemical capacitor electrodes are listed in Table 7.5. Significantly higher capacitance can be achieved by using pseudocapacitive electrode materials. Unlike high surface area carbon electrodes, which derive their storage capacity from the electrostatic arrangement of electrolyte ions along their electrode-electrolyte

Table 7.5 Electrochemical Capacitor Electrode Materials

Mode of Energy Storage	*Capacitor Material*	*Examples*	*Specific Capacitance* * *(F/g)*	*Comments*
Purely electrostatic	Carbon	Activated carbon	150	Stable, long cycle life, inexpensive
	High surface area carbon nanomaterial	Carbon aerogels, fibers, and nanotubes	300	Moderately expensive
Pseudocapacitive	Conductive polymers	Polyaniline, thiophene-based polymers	300	Inexpensive, unpredictable cycle behavior and stability
	Metal oxides	Hydrous ruthenium oxide	1,358	High electrical conductivity, fast charge transport, expensive
		Manganese dioxide, nickel oxide	800	Inexpensive, environmentally friendly, poor electrical conductivity

*Values are maximum reported specific capacitances found in [69, 70].

interfaces, pseudocapacitive materials store energy through Faradaic reactions at these regions. In comparison to battery electrodes, which often undergo irreversible phase changes as a consequence of redox reactions, pseudocapacitive charge and discharge processes are almost reversible within reasonable voltage ranges. In general, pseudocapacitive electrodes provide higher capacitance in exchange for cycle life and power density [68].

To achieve higher cell voltages, complementary electrolytes stable within the operating potential range must be used. Traditionally, most electrochemical capacitors have utilized aqueous electrolytes; however, water degrades at high voltages. Nonaqueous electrolytes such as carbonate-based organics and ionic liquids are able to achieve higher cell potentials of (2–5V), but their ion conductivities are at least an order of magnitude less than aqueous solutions. Similar developments seen in microbatteries in solid-state and polymer electrolytes as well as nanostructured materials [69] have also been applied to microcapacitors as well.

7.5.2 Microcapacitor Prototypes

In the last decade there have been limited demonstrations of capacitors with diminutive geometries (< 1 cm^2 footprint area); however, working prototypes utilizing thin film microfabrication and direct-write fabrication methods have been reported. Configurations described in the microbattery section, including 2D adjacent pads or interdigitated electrodes, stacked cells, and 3D interdispersed cells have been applied to microcapacitor structures as well. For comparison areal capacitances (mF/cm^2) with respect to footprint area occupied will be reported.

7.5.2.1 Thin Film Microcapacitors

Thin film supercapacitor (TFSC) research has largely benefited from the advancements of thin film battery research, so trends in electrode and solid-state electrolyte battery materials have largely translated into the TFSC field. Kim et al. [70] demonstrated some of the first TFSCs by incorporating 5-μm stacked configurations (as illustrated in Figure 7.27) of amorphous Lipon ($Li_xPO_yN_z$) solid-state electrolytes with pairs of ruthenium oxide (RuO_x) electrodes, and later with less costly sputtered cobalt oxide (Co_3O_4) electrodes [72]. Capacitances of 28.5 mF/cm^2 and 13.2 mF/cm^2, respectively, were reported. Both TFSC chemistries experienced problematic capacity fade due to structural changes in the thin film electrodes with extensive cycling.

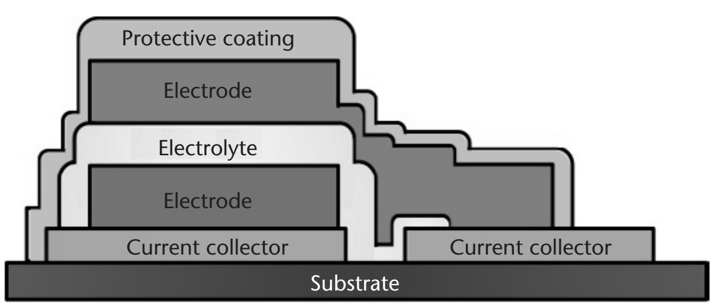

Figure 7.27 Schematic of thin film electrochemical capacitor.

7.5.2.2 Origami Microcapacitors

The Nanostructured Origami process reported by In et al. [73] applies 2D planar microfabrication techniques to fabricate structures that are "automatically folded" via strain and Lorentz force actuation into 3D structures (see Figure 7.28). Carbon slurry electrodes and a droplet of liquid electrolyte were manually deposited onto the microfabricated features and folded into stacked capacitor structures. With an aqueous liquid electrolyte, the microcapacitor showed a capacitance of 0.4 mF/cm^2 when cycled between 0–0.6V. Using this method, a high-density array of capacitors with microdimensions could be fabricated; however, the feasibility of scaling the folding process to large substrate areas will need to be addressed.

7.5.2.3 Direct Write Fabricated Microcapacitors

Because of the versatility and reduced processing complexity required to pattern materials, a few direct write methods have been used to demonstrate both planar and stacked configurations of microcapacitors. Pech et al. [74] used microfabrication methods to pattern interdigitated gold or platinum current collectors on which carbon slurries were ink-jet printed (Figure 7.29). Each interdigitated finger was 40-μm wide and 400-μm long, and spaced 40 µm from each other. The total device occupied a 2 mm^2 footprint area. In an organic electrolyte the cell was cycled to 2.5V and achieved 2.1 mF/cm^2.

Arnold et al. [75] fabricated microcapacitors with planar configurations utilizing both square and interdigitated electrodes. Laser direct writing was used to deposit thick film pseudocapacitive ruthenium oxide capacitor electrode slurries using planar configurations. A laser-deposited capacitor film was bisected with a 10–20-μm microchannel defined using laser micromachining. With an aqueous liquid electrolyte, the capacitors were cycled up to 1V and displayed a high areal

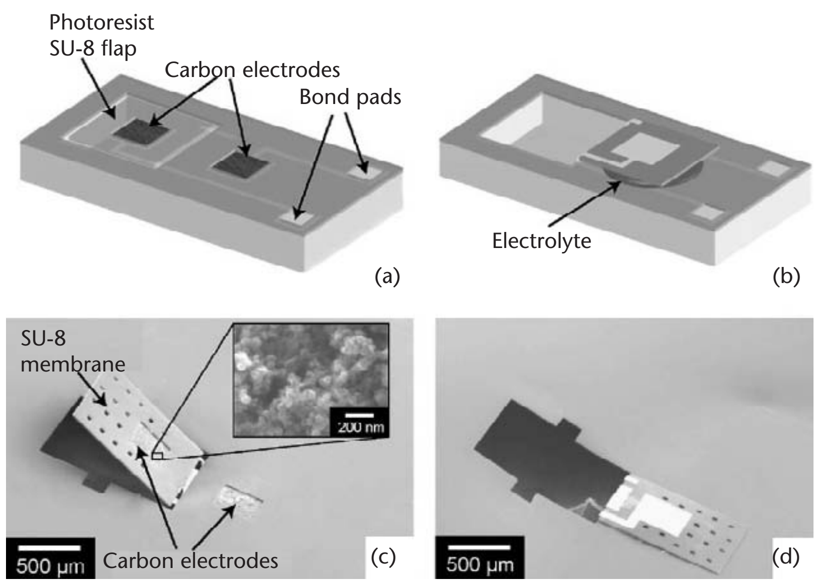

Figure 7.28 (a, b) Schematics and (c, d) micrographs of origami folded supercapacitors. (*From:* [73]. Reprinted with permission. Copyright 2006, American Institute of Physics.)

capacitance of 320 mF/cm^2. Capacitors in series and parallel configurations were also demonstrated to exhibit the ability to achieve higher voltages and currents.

Stacked carbon microcapacitors were fabricated by Ho et al. [76, 77] using direct write dispenser printing. Layers of activated carbon slurry electrodes and an ionic liquid gel electrolyte were deposited in succession within a 1 cm^2 footprint area and 100–150-μm thickness, as shown in Figure 7.30. The gel electrolyte developed was able to provide physical separation between the electrodes even under significant compression, yet maintain good ion transport. The resulting structure was shown to cycle over 120,000 times up to 2V and average capacities ranged from 40–100 mF/cm^2. This corresponded to a maximum areal energy and power density of 10 μWh/cm^2 and 575 mWh/cm^2, respectively. Microcapacitors fabricated using simple direct write processes were able to demonstrate some of the highest capacitance performances, largely due to the ability to pattern thick porous electrodes.

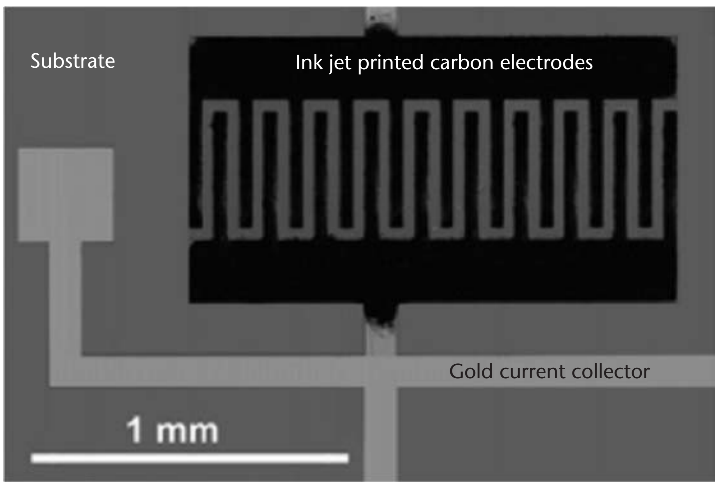

Figure 7.29 Ink jet printed carbon supercapacitor electrodes on gold, interdigitated current collectors [74].

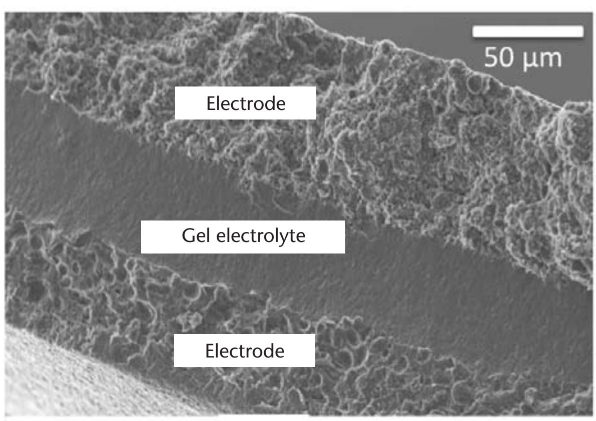

Figure 7.30 Cross-section micrograph of a dispenser printed stacked, carbon electrochemical capacitor [77].

7.5.2.4 Interpenetrating Microcapacitors

The concept of a sponge-like, 3D, interpenetrating energy storage device developed by Rolison et al. [65] has also been applied to asymmetric hybrid electrochemical capacitors. Pairing a nanostructured carbon foam anode with an interpenetrating MnO_2 cathode, electrochemical measurements in an aqueous electrolyte resulted in energy densities of 20 Wh/kg with rapid charge and discharge times of less than 10 seconds (areal capacitance was not reported). Because of the high overpotential of hydrogen evolution achieved at the carbon electrode, the cells were cycled to 2V, well above the typical water breakdown potential.

7.5.3 Conclusions and Prospects for Microcapacitors

The implementations of microcapacitors have for the most part resembled the progress of microbatteries. Continued material improvements have diminished the gap between achievable battery and electrochemical capacitor energy densities, and fabrication advancements have resulted in preliminary demonstrations of microcapacitors for integrated micropower supplies, which are summerized in Table 7.6.

Besides the prospects described in this chapter, there are opportunities for large performance improvements, specifically through accessing higher cell voltages by using two dissimilar electrode materials of either battery or capacitor character. These asymmetric capacitor configurations are known as hybrid electrochemical capacitors [78]. By combining a battery cathode with a pseudocapacitive anode, the hybrid electrochemical device, in theory, offers the "best of both worlds," providing high energy density from its battery electrode, yet maintaining high power

Table 7.6 Electrochemical Microcapacitor Demonstrations and Performance Summary

Fabrication Methods	*Chemistry*	*Geometry**	*Operating Voltage (V)*	*Performance*
Thin film [71, 72]	RuO_x, glass electrolyte	(t) 5 μm, stacked	0–2	28.5 mF/cm^2
	Co_3O_4, glass electrolyte	(t) 5 μm, stacked	0–2	13.2 mF/cm^2
Origami [73]	Carbon, liquid electrolyte	(f) 0.12 mm^2 stacked	0–0.6	0.4 mF/cm^2
Ink jet printing [74]	Carbon, liquid electrolyte	(l) 400 μm (w) 40 μm (f) 2 mm^2 planar interdigitated	0–2.5	2.1 mF/cm^2
Laser direct writing [75]	RuO_x, liquid electrolyte	(t) 15 μm (f) 2 mm^2 planar	0–1	320 mF/cm^2
Dispenser printing [76, 77]	Carbon, gel electrolyte	(t) 100 μm (f) 1 cm^2, stacked	0–2	100 mF/cm^2
Interpenetrating sponge [65]	MnO_2, carbon, liquid electrolyte	N/A	0–2	350 F/g

*l: element length, w: element width, t: thickness of device, f: footprint area of device.

density through its pseudocapacitive electrode. Achievable energy densities are usually an order of magnitude greater than carbon electrochemical capacitors without significant power density reduction (~1 kW/kg). Future work investigating the operating principles and longevity of this device is necessary, but it seems that the eventual miniaturization of this device could be straightforward.

7.6 Conclusion

The need for miniature electrochemical energy storage devices for secondary storage with microenergy harvesting devices has prompted new paradigms in materials, geometry, and processing. As a result, a surge of new research concepts and demonstrations has emerged, especially in the last decade. Both microbatteries and microcapacitors have been fabricated with a variety of two-dimensional and three-dimensional configurations, and depending on the processing methods developed, a wide-range of material chemistries has been implemented. Perhaps the numerous endeavors summarized in this chapter indicate that there will not be a single energy storage solution, but rather many appropriate strategies for a diverse number of applications.

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CHAPTER 8

Case Study: Adaptive Energy-Aware Sensor Networks

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8.1 Introduction

The work presented in this case study was undertaken by two research groups from the School of Electronics and Computer Science at the University of Southampton. Researchers from the Electronic Systems and Devices group were responsible for the design and implementation of the energy-aware, energy-harvesting nodes and associated software. Researchers from the Intelligence, Agents, Multimedia group were responsible for the higher-level coordination functionality that was implemented using the concept of autonomous agents.

This work was undertaken as part of the Data Information Fusion Defence Technology Centre (DIF DTC) Phase II "Adaptive Energy-Aware Sensor Networks" project, which was jointly funded by the UK Ministry of Defence and General Dynamics UK. Additional support was also provided by the Engineering and Physical Science Research Council (EPSRC) UK. Within this chapter details of the energy-aware, energy-harvesting node design will be reported, including the hardware and software design and details of demonstrations of the system capabilities. Details on the agent-based coordination work will not be presented here, but can be found in [1–3].

The aim of this work was to develop and demonstrate an integrated approach to energy management across a network of independent, autonomous, energy-harvesting sensor nodes. Each node was to be able to harvest energy from a range of sources and to be able to monitor the quantity of energy available from each source and from the node's energy store. Based upon this information, the individual node can make intelligent decisions on how to use the available energy, cooperating with other nodes within the network to maximize the coverage and availability of the overall network.

The scenario for which the nodes were designed comprised urban surveillance over a wide area, with the nodes being deployed with no prior knowledge of their location or that of their neighbors. The sensors were to be used to monitor events such as the movement of pedestrians and vehicles, and having formed an ad hoc network to relay this information through the network.

8.2 Requirements

The first requirement of the deployed sensor network was that it would operate for an indefinite length of time and would thus have to be energy neutral. Therefore, each sensor node would need to be able to harvest energy from its local environment, monitor the current availability of harvested energy, predict future energy availability, measure current energy reserves, and manage the consumption of energy. The main means of controlling the rate of energy use is by varying the sensing duty time, increasing the inactive/nonsensing periods to reduce overall energy usage.

As this duty cycling of the individual sensor nodes causes periods during which no sensing is performed, sensor nodes with overlapping sensing zones coordinate at the local level so that overall sensor coverage is maintained as much as possible by staggering the inactive periods of nodes in the same locality. This coordination is performed in a decentralized manner to reduce energy-costly data flow through the network, avoiding overloading any bottlenecks in the data path and removing the reliance on a central coordinating node, the absence or failure of which could cause the network to become unreliable, ineffective, or inoperative. By allowing the nodes to consider their energy status during this decentralized coordination process, nodes with higher energy reserves can be allocated a higher proportion of tasks requiring higher energy consumption, typically tasks such as relaying of radio messages.

For the coordination of sensor nodes with overlapping sensing zones to be successfully achieved, a learning phase has to be performed, during which the nodes can learn which other nodes in their locality can sense the same events as them by comparing the events sensed.

To enable rapid deployment in a range of different energy environments, the central node is not designed to accommodate a particular form of energy harvesting. Instead the design permits a combination of different energy harvesting and storage subsystems to be simply plugged in, with the node able to determine the nature of the attached harvesting or storage device and act accordingly.

8.3 Energy Harvesting Sensor Node Hardware Design

8.3.1 Node Core Design

The core of the node is required to be able to process information acquired from the node's local sensors and information passed to it by other nodes and to act accordingly on this information and to pass information from its own sensors onto other nodes. To accomplish these core tasks, the node core needs to contain a microprocessor with associated support components and a form of a suitable RF transceiver. This node core forms the center of a modular design, with further subsystems being attached to it.

8.3.2 Overview of Modular Design

As stated previously, the node is designed in a modular manner, allowing the functionality of the node to be tailored to the environment in which it is being deployed. A basic node configuration would include the core node containing the microprocessor and RF circuitry. This is connected to an energy multiplexer subsystem, which provides power regulation and multiplexed power and data routing and interconnection connection between harvesting and storage modules. The core node also connects to the selected sensor module, providing power to the sensor and controlling it as necessary. To provide a functioning node, the core node is also required to be connected to at least one energy harvesting module and an energy storage module via the energy multiplexer subsystem to provide a source of energy for the node and a means of storing energy during periods when the rate of harvesting is insufficient for continuous operation. An outline of this basic modular configuration is shown in Figure 8.1.

8.3.3 Choice of Microprocessor

The microprocessor selected for the node design is the CC2430 from Texas Instruments. The key features of this device are the very low-power operation modes and the integrated IEEE 802.15.4 compliant RF transceiver, contained within a 7 × 7 mm package. To achieve the low-power operation, the microprocessor has several sleep modes where the processor speed and functionality are reduced; with corresponding energy savings, the device also operates down to a supply voltage of 2.0V. To further the efficient use of energy while the microprocessor is in an operational mode, the integrated peripherals, including the RF transceiver, can be powered down, and, subject to clock-speed critical modules such as the RF transceiver being inactive, the clock generation can be switched from a 32-MHz crystal to a lower power and frequency RC oscillator. The peripherals and I/O available within the CC2430 device are: a 12-bit ADC with up to 8 inputs, an AES security coprocessor;

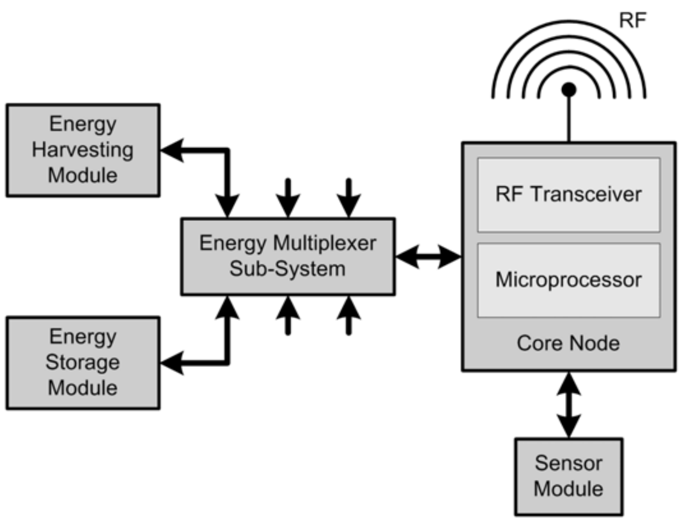

Figure 8.1 Outline of basic modular energy-harvesting node.

two USARTs with multiple protocol support, 21 general purpose I/O pins, a 16-bit timer; and two 8-bit timers, a watchdog timer, a sleep timer, a supply voltage monitor, and an integrated temperature sensor. The current consumption of the device is 27 mA while the radio module is transmitting or receiving, with the current reduced to 0.5 μA when the chip is in a sleep mode from which it can be woken by the internal sleep timer. The lowest consumption of 0.3 μA is achieved when the device is in a sleep mode that requires an external interrupt to wake it. For program storage the CC2430 has 128 kB of flash memory and 8 kB of RAM for program variable storage, 4 kB of which is preserved during the lowest power modes.

8.3.4 Energy Multiplexer Subsystem

The function of the energy multiplexer subsystem is to permit the connection of multiple energy harvesting and storage modules to the core node system, providing a regulated supply voltage to the microprocessor and permitting interrogation and monitoring of the individual modules connected to it. The multiplexer subsystem is designed to support up to six harvesting or storage modules in any combination, including duplicate modules, with each module having the same defined interconnections with the multiplexer subsystem. The multiplexer subsystem permits the node to harvest energy from all of the attached harvesting modules whose energy source is currently available, rather than from just one energy form. The multiplexer subsystem to energy harvesting/storage modules interface supports the following interconnections:

- A 1-wire interface for reading the module's details from the on-board EPROM.
- Four general-purpose bidirectional control and monitoring lines that can be defined as digital I/O or as ADC inputs to the microprocessor depending on the requirements of the module.
- A signal line at the microprocessor supply voltage to enable logic level translation between that on the harvesting and storage modules and the level used by the microprocessor.
- A connection to the energy multiplexer's energy bus that interconnects the output of all the harvesting modules, the storage modules, and the regulator used to provide the microprocessor's supply voltage; this level is defined to be in the range of 0–4.5V and any modules connected must not exceed this range.
- A common ground to which all signals are referenced.

Each of the six module connections is assigned a multiplexer channel number, and the selection of this channel by the microprocessor enables the multiplexer connecting that channel to the microprocessor, thus facilitating communication with the module. The multiplexers used within the energy multiplexer subsystem are Analog Devices' ADG708 low-voltage analog CMOS 8-channel multiplexers. These have an operating supply range of 1.8V–5.5V and low current consumption. Of the eight channels provided by the multiplexers, six are used to communicate with the harvesting and storage modules. One is used to enable the control and monitoring of the energy multiplexer itself. The final channel is used to provide a low power consumption isolated state. When the microprocessor is not communicating with the modules or performing monitoring operations, the multiplexers are set into this state to reduce power loss through such means as pull-up resistors.

A key function of the energy multiplexer subsystem is the regulation of the voltage supplied to the microprocessor. As the microprocessor requires a minimum of 2V to operate correctly and a maximum supply of 3.3V, both under and over voltage protections are required. The under and over voltage protection has been achieved through the use of a Torex XC6215 low power regulator with enable control. In a normal operational mode this device has a quiescent current of 0.8 μA. With the regulator disabled by the enable pin, this falls to less than 0.1 μA. This enable control is operated by a Torex XC61C low power voltage detector, with a 2-V device chosen to prevent the regulator being enabled when the supply voltage is below the operating level of the microprocessor. The current consumption of this detector is of the order of 0.7 μA. Providing under voltage protection prevents the system from wasting energy by supplying power to the microprocessor when it can not be gainfully used. Preventing the bleed-off of energy in this way also reduces the time taken for the voltage on the energy storage to reach the desired operational level. A block diagram of the energy multiplexer subsystem is shown in Figure 8.2.

8.3.5 Supercapacitor Energy Storage Module

The forms of energy storage used within this system are supercapacitors. These have been selected over other forms of storage such as NiCad or NiMH rechargeable cells for several reasons. There are no special charging requirements, other than ensuring that the maximum voltage is not exceeded, so energy-consuming charge control circuitry is not required. Supercapacitors have a low ESR, typically of the order of 30 mΩ, reducing internal losses during charge and discharge cycles and allowing them to handle current surges without the output voltage dropping significantly. The supercapacitor used in the energy storage module is a CAP-XX GS206F, which is a compact flat-packaged device with a nominal capacitance of 600 mF and a maximum voltage of 4.5V.

In addition to the supercapacitor used for energy storage, the module contains a Maxim DS2502 1-wire memory to contain details of the module and circuitry to permit the microprocessor to determine the level of charge within the supercapacitor. This circuitry consists of a high-value potential divider to scale the voltage to the required 0–1-V range for the ADC inputs and a unity gain buffer to isolate the supercapacitor from the microcontroller ADC input. The buffer is based on an ST TS941 micropower rail-to-rail single-rail operational amplifier, which has an operating current of 1.2 μA and is specified to operate down to a supply voltage of 2.7V. One of the multipurpose I/O lines is used to connect the potential divider to the supercapacitor supply voltage through low on-resistance MOSFETs and also the power supply to the buffer when the microcontroller requires to monitor the charge level. At all other times the potential divider is isolated and the buffer unpowered to minimize energy wastage. A block diagram of the module is shown in Figure 8.3.

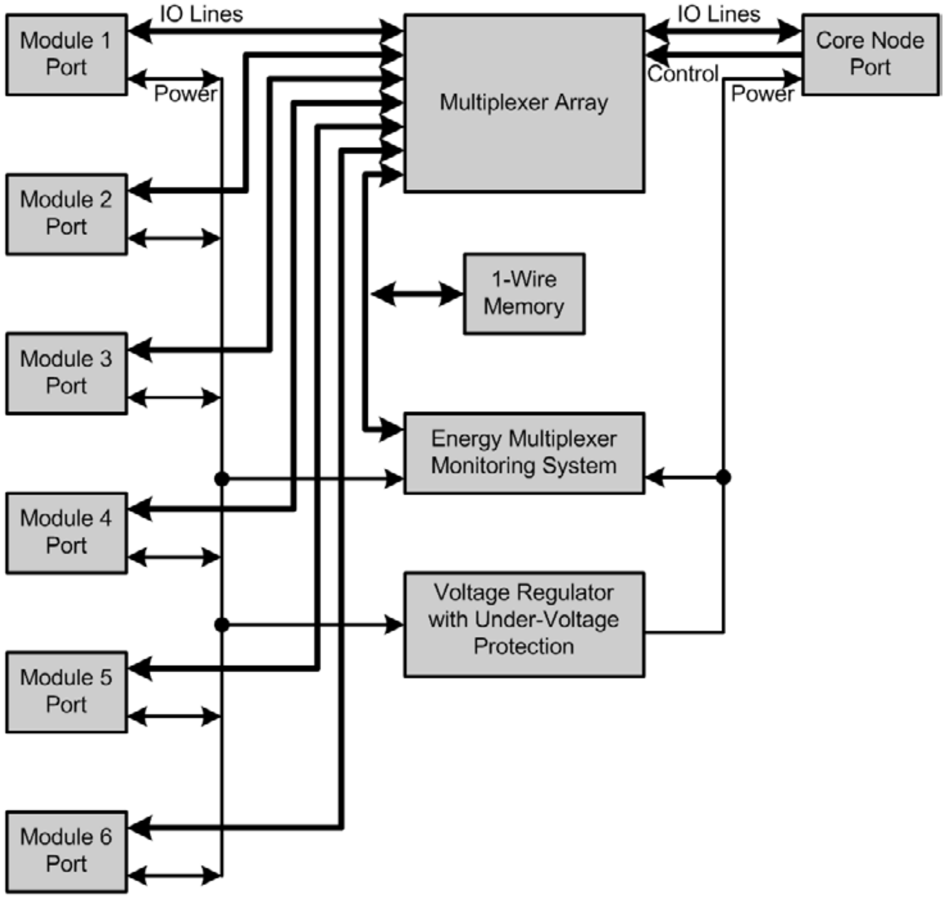

Figure 8.2 Block diagram of the energy multiplexer subsystem.

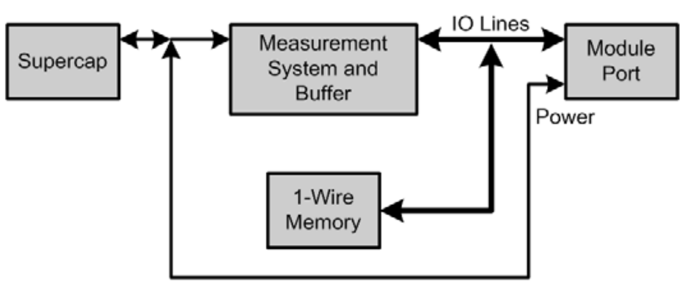

Figure 8.3 Block diagram of the supercapacitor energy storage module.

8.3.6 Solar Energy-Harvesting Module

The solar energy-harvesting module uses a Schott Solar module, type ASI O3i 07/090/072 JJF, as its primary harvester. This module is optimized for use indoors under artificial illumination at normal office light levels (100–1,000 lux).

For solar cells the output power is maximized for a particular value of load. To operate the module at its maximum efficiency, the module must be operated at this load point. With the module panel under normal illumination conditions and under load the output voltage is of the order of 3V, as the energy bus within the multiplexer subsystem can operate at levels of up to 4.5V, it is necessary to step up the output voltage from the solar module to this value. To address both operating the solar module at its maximum efficiency and the requirement for a step-up converter, a buck-boost converter has been used that has a feedback system that keeps its input at a set voltage. This set voltage is selected such that the solar module is operating at or close to its optimum efficiency. A block diagram of the solar energy harvesting module is shown in Figure 8.4.

The buck-boost converter is implemented using a National Semiconductor LMC7215 micropower rail-to-rail CMOS comparator that has a quiescent operating current of 0.7 μA. This comparator has its input level set using an ON Semiconductor LM385Z micropower voltage reference diode to give the desired operation point for the solar module. To increase the start-up speed of the switching converter, under voltage protection is provided by a Torex XC61C micropower voltage detector that isolates the comparator from the supply derived from the solar module until the module's output voltage reaches 2.7V, at which point the converter will start to operate.

In addition to the switching converter circuitry, the module also contains control and monitoring circuitry. The monitoring circuitry includes the ability to isolate the solar module from the switching circuit and to measure the open-circuit voltage, which is used to allow the microprocessor to calculate the instantaneous power available from the solar module. This voltage is buffered using a unity gain buffer based on an ST TS941 micropower rail-to-rail single-rail operational amplifier, which has an operating current of 1.2 μA, the supply to this buffer is controlled by one of the I/O lines from the microprocessor and power is only supplied when a measurement of the open circuit voltage is being performed. To protect the other modules connected to the energy bus, over-voltage protection is provided by a second Torex XC61C micropower voltage detector, which isolates the solar module from the energy multiplexer in the event of its output voltage exceeding 4.5V. The module also includes a Maxim DS2502 1-wire EPROM, which contains the details of the solar energy-harvesting module and which can be read by the microprocessor to allow it to identify the connected module.

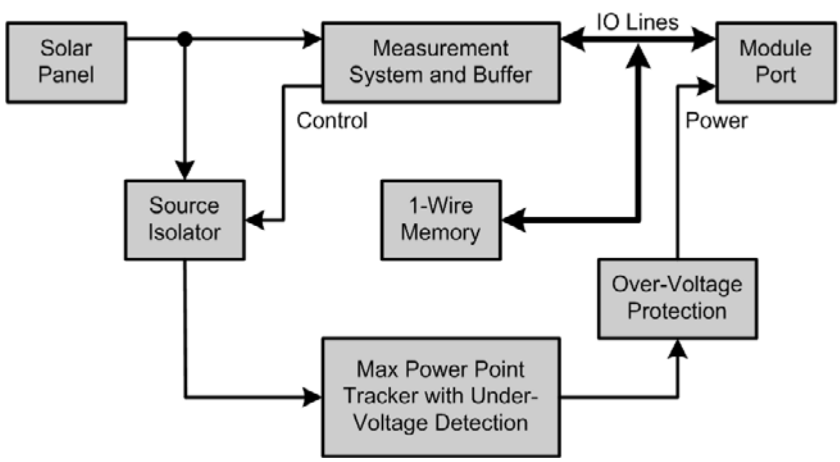

Figure 8.4 Block diagram of the solar energy-harvesting module.

8.3.7 Vibration Energy-Harvesting Module

To harvest vibrational energy, a module incorporating a Perpetuum PMG17-100 energy harvester has been implemented. The PMG17-100 electromagnetic energy harvester is designed to harvest stray vibration from machinery. In this case, it is tuned to operate from vibrations at a frequency of 100 Hz, as typically found in equipment powered from a 50-Hz AC supply. The output of the energy harvester used is a rectified sine wave with a peak of 8V. It is necessary to regulate this down to the 4.5V used by the energy bus within the energy multiplexer subsystem in an efficient manner. This down regulation is accomplished with the use of a Maxim MAX639 high-efficiency, low-quiescent current step-down converter. To speed up the start-up of the converter, the output of the converter circuit is isolated from the energy bus until a threshold voltage is reached on the output capacitor. This isolation is controlled by an ST STM1061 low-power voltage detector with a quiescent current of 0.9 μA. In common with the other energy-harvesting modules, a means of measuring the instantaneous power being harvested by the device is provided. In the case of the PMG17-100, this is realized by shutting down the step-down converter and switching in a known load across the output of the generator and measuring the voltage across this load. The known load is provided by a potential divider that scales the voltage to a suitable range for the microprocessor's ADC input after buffering by a unity gain buffer based on a ST TS941 micropower rail-to-rail single-rail operational amplifier. The test load and buffer amplifier are isolated from the generator and power supplies, respectively, at all times when an instantaneous power measurement is not being made. The electronic data sheet that details the nature of the module and the operations of the interface lines is contained in a Maxim DS2502 1-wire EPROM that can be interrogated by the microprocessor. A block diagram of the module is shown in Figure 8.5.

8.3.8 Thermal Energy-Harvesting Module

To harvest thermal energy, the module utilizes a Peltier-type thermoelectric power generation device, a Tellurex G1-1.4-127-1.14 Power Generation Module. The Tellurex module has an open-circuit voltage of 2.8V, and with the hot side at 150°C and a thermal gradient of 100°C across it, the device can produce 2.3W of power.

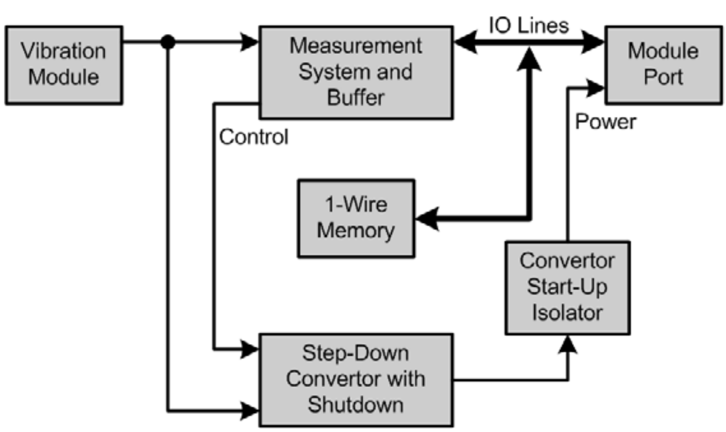

Figure 8.5 Block diagram of the vibration energy-harvesting module.

As the output voltage is lower than the desired voltage for the energy bus, it is necessary to step it up to 4.5V. This has been accomplished using a Maxim MAX1674 high-efficiency, low-supply current, step-up converter. The output of the step-up converter is connected to the energy bus in the multiplexer submodule through an over-voltage protection circuit using a Torex XC61C micropower voltage detector to control the International Rectifier IRLML6401 ultralow on-resistance MOSFETs.

For measurement of the instantaneous power production of the thermoelectric device, control signals from the microprocessor cause the step-up converter to be isolated from the harvester module and a test load to be applied to the output of the thermoelectric device. The test load is implemented as a potential divider and is used to scale the output voltage of the harvester to that of the microprocessor's ADC inputs. While no power measurement is being made, the test load is isolated from the generator to avoid unnecessary energy losses. A block diagram of the module is shown in Figure 8.6. As with other energy-harvesting modules, the details of the nature and interface requirements of the module are contained on a Maxim DS2502 1-wire EPROM, connected to the microprocessor via the energy multiplexer subsystem.

8.3.9 Wind Energy-Harvesting Module

To harvest energy from the wind, a miniature wind turbine has been used to convert the air flow into electrical power. The turbine used with this demonstration system is a miniature device with 60-mm diameter blades. This produces an unrectified output with the RMS voltage varying with applied wind speed. As tests showed that the output voltage of the turbine with varying applied wind speeds produced output levels greater and less than the desired module output voltage, a step-up/step-down switching regulator was required. A block diagram of the wind energy-harvesting module is shown in Figure 8.7. Within the module the output from the turbine is full wave rectified using International Rectifier MBRA120TRPbF Schottky diodes with a low forward voltage. The upconverter/downconverter is realized using a Linear Technology LT1307 micropower high-efficiency DC-DC converter with an automatic operating mode switching depending on the applied load. To protect the

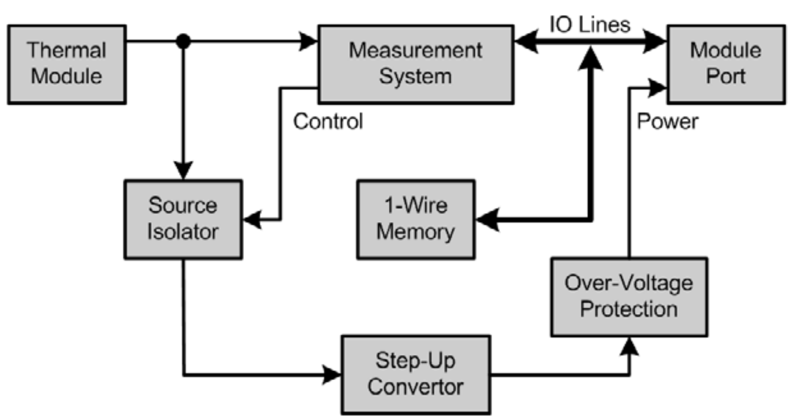

Figure 8.6 Block diagram of the thermal energy-harvesting module.

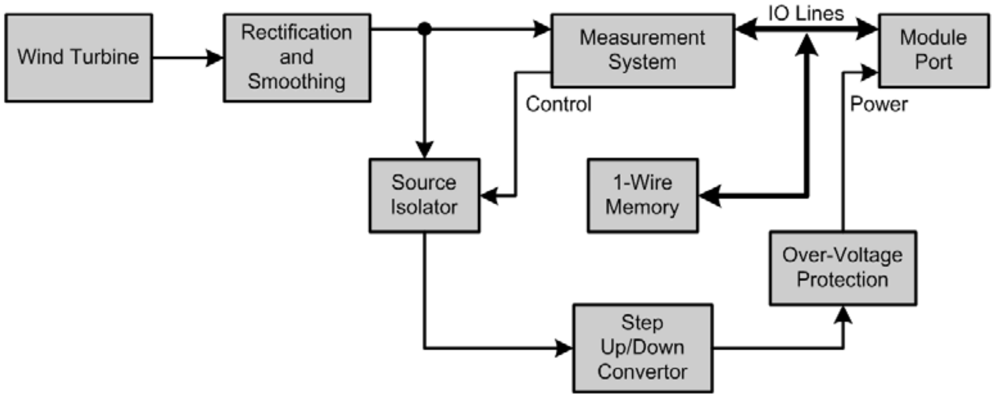

Figure 8.7 Block diagram of the wind energy-harvesting module.

energy-harvesting module and the energy multiplexer subsystem from the risk of over-voltage damage, a protection circuit composed of a Torex XC61C micropower voltage detector and International Rectifier IRLML6401 ultralow on-resistance MOSFETs has been implemented that will isolate the module in the event of an over-voltage condition. To permit the microprocessor to measure the applied wind speed, under the control of one of the interface lines from the microprocessor, the rectified output of the generator is disconnected from the switching converter and connected to a potential divider that serves as a test load and serves to scale the voltage to a range compatible with the microprocessor's ADC input range. During normal operation this test load is isolated from the generator to avoid energy loss. So the microprocessor can interrogate the energy-harvesting module to learn its capabilities and operational parameters, a Maxim DS2505 1-wire EPROM is included that can be read via the energy multiplexer.

8.3.10 Other Energy-Harvesting and Storage Modules

As the energy multiplexer subsystem uses a standardized interface to connect to the modules, other forms of energy sources and stores can also be integrated with the system. In addition to the harvesting and storage techniques described previously, modules have been designed that allow energy to be collected from external DC sources and primary batteries. Also developed is a module that will allow energy storage using rechargeable batteries, although due to the inefficiencies involved in the charging process, this form of storage would only be applicable in a situation when an unconstrained energy source was intermittently available. These modules have been designed to demonstrate the flexibility of the plug-and-play energy multiplexer subsystem, but are not used in the demonstration system applied to the current scenario.

8.3.11 Plug-and-Play Capabilities

To enable the sensor node to be readily configured by unskilled operators, a plug-and-play approach has been taken to the interface between the energy multiplexer subsystem and the energy-harvesting and storage modules. This approach allows the system to be tailored to suit the available ambient energy and the amount of energy storage required for a given deployment. Once deployed, the sensor node will harvest energy from all the attached harvesters at the same time, providing that there is the applicable form of ambient energy available for any given harvester type in use. This ability to use multiple forms of energy harvesting in parallel gives the node a potentially greater supply of energy and also provides alternative sources of energy should one form not be producing to its normal level. For instance, a solar harvester's energy output would fall from its normal levels on an unusually cloudy day. The plug-and-play interface used contains signal lines that enable the node to determine on demand the instantaneous energy output from any one of the individual energy-harvesting modules, allowing for trends in energy availability to be identified and used for energy forecasting in conjunction with knowledge of the type of harvesting modules that are connected.

One of the key enabling technologies that allows the microprocessor to identify which energy harvesting and storage modules have been connected to the energy multiplexer subsystem is the use of Electronic Energy Data Sheets (EEDS). These data sheets are stored in the 1-wire EPROM memory on each module and contain information regarding the nature and functionality of the module and the interface configuration to use to access the module. The electronic energy data sheet is a further development of the IEEE 1451.4 standard, which defines an interface between transducers and application processors and a format for the transducer electronic data sheet (TEDS). This data sheet, which may be stored on an EPROM within the transducer, contains information to identify:

- The manufacturer;
- The type and model of the transducer;
- The transducer's serial number;
- The calibration data and/or curves for the device allowing the attached processor to correctly interpret the readings from the transducer.

Within the EEDS format used in this work, the information contained within the module's EPROM consists of:

- The manufacturer of the device;
- The model number of the module;
- The serial number of the module;
- The type of module (e.g., Vibration Harvester, Supercapacitor Store);
- The configuration and function of I/O lines 1 to 4;
- The device model details to allow the calculation of the current energy status and future energy availability prediction;
- The details of the control functions provided through the four I/O lines and the actions to which they relate.

As each module includes a unique serial number, the microprocessor can distinguish between multiple modules of the same type, as well as between modules of different types. For instance, in an environment with copious solar energy available, it may be desired to deploy some nodes with two solar energy-harvesting modules attached to provide a greater energy budget. Another potential use of a multiple module of the same type would be the use of multiple energy storage modules in a node that is likely to be inactive for longer periods of time, as this will enable the continued storage of harvested energy and reduce the likelihood of the node being unable to store harvested energy due to filling the energy store up. A further advantage of increasing the energy storage capacity of a node with longer inactive periods would be that when it was active, it would be able to perform more energy intensive tasks and/or for longer periods of time, given that it has a greater amount of energy available to use. Due to the presence of the serial number, the microprocessor can determine between the two otherwise identical modules and can act accordingly in its energy monitoring and predicting processes. Further information on the EEDS concept is contained in [4]. The EEDS concept is not just applicable to the energy harvesting and storage modules, but it can also be used within the energy multiplexer subsystem to contain information regarding it, providing information such as the number of modules that it is capable of hosting.

As there is an associated processing and energy cost related to reading the 1-wire memory devices used to store the EEDS, it is not practical to continually reread the EPROMs on the attached modules and refresh the corresponding data tables within the microprocessor's memory. A different strategy is used to track the addition and removal of energy and storage modules. In normal operation a scan of the attached modules is performed during the start-up process of the microprocessor. Once this has been completed, the 1-wire interface is deactivated to remove the energy drain. The microprocessor will periodically rescan the attached devices, with the frequency of this scan being dictated by available energy levels and with a higher availability of energy facilitating more frequent rescans. In the case of a user adding and/or removing a module and wishing to update the microprocessor immediately, this can be achieved by making the desired module changes and then pressing a Rescan button on the energy multiplexer subsystem. This will cause the microprocessor to rescan the modules at the first opportunity and rebuild the configuration tables and not wait for the next scheduled rescan.

8.3.12 Sensor Module

In keeping with the modular design of the node, the desired sensing capabilities are hosted on a separate module to the main node, with a specified interface allowing the ready interchangeability of different sensors. The sensor interface standard consists of four connections: the microprocessor supply voltage, a sensor enable line, an ADC input channel line, and a common ground connection. The demonstration sensor module used in this work consists of a positional sensor, which requires a regulated supply that is provided by a National Semiconductor LP3990 voltage regulator with enable control. The enable input to the regulator is controlled by the sensor enable line from the microprocessor, so the sensor is only powered up while a measurement is being made. When the voltage regulator is not enabled, its quiescent current is less than 10 nA. The output signal from the sensor is connected to the ADC input line.

8.3.13 Built-In Sensing Capabilities

In addition to the sensor module, the energy-harvesting node has further sensing capabilities. The CC2430 has a built-in temperature sensor that can be used to measure the ambient temperature around the microprocessor. Further sensing abilities are provided by virtue of the energy-harvesting modules that have been attached. For example, a variation in the output level from the solar module may indicate the passing of something over the node, causing a shadow to fall on the solar module. Likewise, an increase in the vibration level measured by examining the instantaneous power output of the vibration generator may indicate that the structure to which it is attached is under a greater load, for instance, due to the travel of people or vehicles.

8.3.14 Energy Efficient Hardware Design

To minimize the waste of harvested energy within the node, a number of energy efficiency rules and guidelines have been followed during the design of the modules that make up the node. First, care has been taken with the selection of components. Capacitors have been selected for low leakage and equivalent series resistance values. Diodes have been chosen with low forward voltages and reverse leakage values. Switching components such as MOSFETs have been specified to have very low on-resistances and to operate in a fully-on state. The use of pull-up resistors has been minimized, and values as high as possible are used where they are necessary. Some components may seem overspecified for the currents typically being carried, but this can prove necessary to reach some of the performance goals. This is done as power components may have better performance in key areas. For instance, they typically have lower on-resistances than nonpower components to minimize the power dissipation and heating effects in their intended use. Within our energy efficient designs, these low on-resistances help to minimize losses within switching and routing circuits.

At the circuit level many submodules have been designed so that they are isolated when they are not in use. For instance, potential dividers used to scale sensor voltages to the input range of the ADC are isolated from their input voltage at all times except for when the measurement is actually being made. Where possible, voltage regulation is achieved through switching converters instead of linear regulators and thereby avoiding their associated wastage of energy through thermal emissions.

In a manner similar to the isolation of circuit submodules when not in use, the various peripherals within the microprocessor can be powered down when not in active use. A key item that must be powered down as much as possible is the RF transceiver module, which consumes current in either the transmitting or the receiving mode. Other peripherals only consume power when they are actually processing, for instance, the ADC. Further power savings can be achieved by switching the microprocessor's clock source from a high stability 32-MHz crystal to a lower frequency, less stable RC oscillator, with the crystal being restarted and switched to when operations requiring the stable high-frequency source, such as RF communications, are to be performed. The highest energy savings can be achieved within the microprocessor by switching it into one of its lower power modes. The CC2430 has four power modes: PM0, PM1, PM2 and PM3. In PM0 the microprocessor is fully functional, with the 32-MHz crystal oscillator, the RF transceiver, and all peripherals active. In this mode the clock source can also be switched to a 16-MHz RC oscillator, which disables the RF transceiver and the ADC module, reducing the microprocessor's power consumption both by deactivating these modules and through the reduction in clock speed and maintenance power. In PM1 both the 32-MHz and 16-MHz oscillators are stopped and a shutdown routine is run to power down some of the peripherals; a 32-kHz oscillator is run to enable the timing of the sleep duration before the device wakes up again and restarts clocks and peripherals. This mode is intended for use when the microprocessor is expected to be woken up within 3 ms of being sent into sleep, as it has a fast power-down sequence. In PM2 the high-speed oscillators are again shut down, along with the peripherals and the internal voltage regulator. This mode is intended for use when the microprocessor is expected to be woken up using the internal sleep timer after a delay of greater than 3 ms. The final power mode is PM3. In this mode all of the oscillators are stopped, all peripherals and internal voltage regulators are shut down, and the current consumption is at its minimum. Waking from this mode requires an external interrupt, as all internal oscillators have been stopped.

In the design used for this application, much of the processing is done with the microprocessor in PM0 with the 16-MHz RC oscillator providing the master clock. The 32-MHz crystal oscillator is started when RF communications or ADC conversions are to be made. While the microprocessor is not active, it is put into sleep mode in PM2 and the internal sleep timer is used to wake it up at the desired point in time. In this design there is not an external clock source to wake the node up on a transmission schedule, so the lowest power mode, PM3, is not used.

Within the energy-harvesting modules steps are also taken to operate as efficiently as possible and minimize losses. Included within this energy efficient design strategy are: avoidance of pull-up/down resistors where possible or the use of high values where their use is required; isolation of measuring circuitry at all times except when a measurement is being made, both from the power supplies within the node and from the measurand; selection of components to minimize power dissipation within switching and routing components, for instance, the IRLML6401 MOSFETS used for isolation purposes have an on-resistance of 0.05Ω; and voltage conversion and regulation achieved using switching converters designed for use at low supply currents and having low quiescent current requirements. Depending upon the design of the conversion and regulation circuits within the individual energy-harvesting modules, internal energy buffers may be used to facilitate a faster and more efficient start-up of the converters prior to connecting the power output of the module into the energy bus within the energy multiplexer subsystem. All of the energy-harvesting modules have protection circuitry built in that prevents energy from being drawn from the energy bus within the energy multiplexer subsystem. This prevents a module that is not producing power output due to a lack of its applicable ambient energy from drawing energy from the other sources to charge output capacitances and thus wasting harvested energy.

8.4 Energy-Harvesting Sensor Node Demonstration Overview

To demonstrate the energy aware functionality and capabilities of the energy-harvesting sensor node, a demonstration system was designed. Included in this system are further nodes with which the energy-harvesting node can communicate but which are not powered by energy harvesting and are not energy-aware. The demonstration consists of a four-node network comprising the energy-harvesting node, two battery-powered remote nodes, and a PC-powered sink node. The energy-harvesting and remote nodes each have a sensor module attached, enabling them to make individual measurements. The network is arranged such that the remote and sink nodes can only communicate with the energy-harvesting node; they are not able to communicate directly in this scenario and any messages from the remote nodes must be routed via the energy-harvesting node. This network topology is shown in Figure 8.8.

8.5 Energy-Harvesting Sensor Node Software Design

8.5.1 Node Software

The software used within the node also needs to be written with energy conservation in mind, both in the overall operations being performed and also in the code

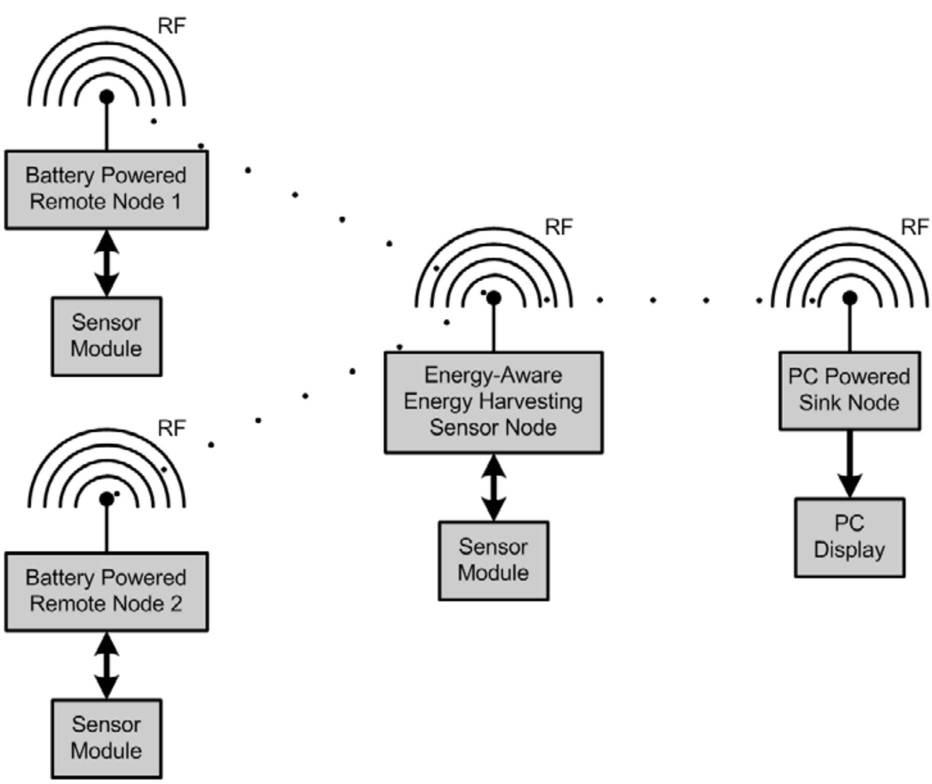

Figure 8.8 Demonstration network topology.

implementation. A key strategy used to operate the node in an energy-neutral manner is to track the level of available energy and to vary the duty cycle of the node accordingly. In this system this is achieved by interrogating the level of charge of the storage module(s) and assigning the node's energy level to one of a set number of levels ranging from zero energy to full. At lower levels the sleep duration of the node will be increased, reducing as the energy store reaches higher levels. These energy levels are also used to make decisions regarding relaying messages from other nodes. The full details of these techniques will be presented in Section 8.5.2.

The basic operation of the energy-harvesting, energy-aware node within the scenario used for the demonstration consists of the node repeating a set core routine:

- The node queries and updates its current energy state and sets its power priority (PP) level.
- The energy multiplexer subsystem is scanned for any changes to the attached modules, either addition or removal.
- Any changes detected to the modules attached to the energy multiplexer subsystem are reported to the sink node using the RF link.
- Measurements are requested from the remote nodes, and responses are awaited for a fixed period of time.
- If a response is received from a remote node, a decision on whether to relay the message onwards is made by the intelligent energy management process, with the message being retransmitted if appropriate.
- The node will then perform a measurement using the sensor module attached to it and send the result of this measurement along with details of its current energy state to the sink node using the RF link.
- The current instantaneous production rates for all harvesting modules and the current amount of energy stored by storage modules that are attached to the energy multiplexer subsystem are measured and details of these reported to the sink node using the RF link.
- Having completed the main processing tasks, the node will update its energy status and power priority level.
- The node uses the power priority level to determine the duration for which it should sleep before starting the cycle again. When the energy available to the node is reduced, the sleep period will be increased, and conversely, as the energy level rises, the sleep period reduces.
- The microprocessor goes into a low-power sleep mode for the duration calculated previously, prior to restarting the cycle.

The software used to control the operation of the node has been written in C and then cross-compiled using the IAR Embedded Workbench development environment for the CC2430 System on Chip. Function libraries supplied for use with the development environment enable the control and operation of the various peripherals within the CC2430.

8.5.2 Intelligent Energy Management

Within the intelligent energy management process there are several techniques to be used in conjunction to optimize the life of the node and its availability with the final goal being an energy-neutral operation. The first process used is to monitor the availability of energy from the different energy sources and the quantity of energy available from the attached energy store(s). This enables decisions to be made regarding how long the node should spend in the low-power sleep mode if the energy used while in its active state is to be replaced and thus yield an energy-neutral operation. During times of lower ambient energy availability, it may be necessary to set a sleep duration that will not allow full replenishment of the energy store between cycles, but will keep the node at the minimum level of availability called for in the scenario. Once more energy becomes available from the energy harvesters, the node can then start to replenish its energy store, and once sufficient energy has been stored, the sleep duration can be reduced, increasing the availability of the node. Steps are also taken to minimize energy usage within the node, with peripherals and modules being powered down when they are not required and any I/O pins not currently in use set as inputs to prevent them from driving the current into the external circuitry.

Using the ability of the node to monitor individual energy-harvesting sources via the energy multiplexer subsystem, we can also attempt to predict the likely availability of future energy. For example, if the node is equipped with a solar energy-harvesting module and deployed in a sunny environment, we may be able to anticipate that the sun is going to rise and set at regular intervals and that the level of energy available from that module is likely to correspond to the time of day. From this we can potentially predict that there will be more energy available from this module after a certain time in the morning, corresponding to when the sun has risen sufficiently.

A further strategy uses the current energy status of the node to make decisions regarding the relaying of messages from other nodes, based on their perceived importance. The current energy status of the node is defined as one of a series of discrete points, ranging between zero energy and sufficient energy for continuous operation. This approach has been described by Merrett et al. [5] and is used in their IDEALS/RMR system. In the implementation used in this demonstration, when the energy status of the node is updated, it is assigned a power priority (PP) level based on the amount available. We use levels PP0, PP1, …, PP5, where PP0 represents no energy in the store(s) and PP5 equates to the energy store(s) being full. When a message is produced by a node, for instance, when reporting a sensor reading, a message priority (MP) level is assigned to it based on the value of the information contained in the message. These priorities range from MP1 to MP5 with MP1 messages carrying the highest value information and MP5 carrying the lowest value information. When a message is received for relaying, the message and power priorities are compared and only if the message priority is high enough in comparison to the power priority is the message relayed. The relationship between the message and power priority levels is shown in Figure 8.9. For example, a message with a priority MP1 would be relayed if the node is in the power priority levels PP1 to PP5. However, a message with priority MP3 will only be relayed if the node has a power priority level of PP3 or above. In the case when the energy level of

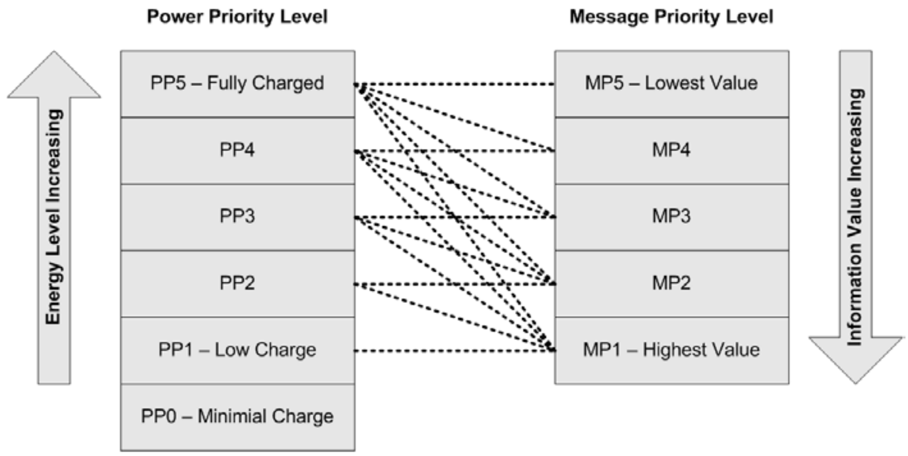

Figure 8.9 Relationship between the message and the power priority levels. (After: [5].)

the node has been assigned a lower level, say, PP2, the message will be dropped, as the value of the information within the message is not considered high enough to justify the energy used in relaying it.

The software that deals with the energy management process has been built in a modular manner and is organized as a three-layer stack. These layers consist of:

- The *physical energy layer* interfaces directly to the energy resources and their associated hardware.
- Above this layer is the *energy analysis layer*, which takes information from the physical layer and uses information about the energy sources to calculate the actual energy produced or stored depending on the type of module.
- The top layer of the stack process is the *energy control layer*, which provides a high-level view of the energy subsystem and reports the status of the energy hardware to the rest of the application software in a manner independent of the actual hardware used.

Further details on this stack approach to energy management can be found in [6].

8.5.3 Information Reported by the Energy-Harvesting Node

The energy-harvesting node relays information to the sink node in the form of set message types. Messages are sent using the RF link to the sink node containing the following information:

- *Sensor readings from the sensor modules attached to the energy-harvesting node and to the remote nodes:* The sensor messages contain the source of the measurement, the measured value, and the message priority level for the message.
- *Energy multiplexer module changes:* These messages contain details of modules added or removed from the energy multiplexer subsystem presented as

module type and physical location on the energy multiplexer subsystem and whether it has been added or removed.

- *Energy module readings:* These messages contain the current energy production rate in the case of harvesting modules or, in the case of storage modules, the amount of energy currently stored, and the physical location on the energy multiplexer subsystem to which the reported module is connected.

8.6 Energy-Aware, Energy-Harvesting Node Demonstration

To demonstrate the energy-aware, energy-harvesting node that has been described here, it has been used with supporting hardware to facilitate data flow and also to provide energy sources for the energy harvesters to utilize.

8.6.1 Supporting Nodes for Demonstration

In the demonstration previously outlined, the energy-harvesting node communicates with two remote nodes and a sink node. The remote nodes consist of CC2430 nodes powered by batteries. The nodes have a control allowing the user to set the message priority (MP) of the readings that it makes from the attached sensor module when a measurement is requested. This control was implemented so that by varying the MP the intelligent routing aspects of the energy-aware node can be demonstrated. The remote nodes are programmed so that they can only communicate directly with the energy-harvesting node and must relay their results to the sink node via it. The sink node consists of a CC2430-based node connected to a PC using a USB connection, which is also used to provide the power to the node. The node is programmed to only receive messages from the energy-harvesting node. On receiving a message, it is decoded and then a text string is displayed on a terminal window on the PC to show the content of the message received.

8.6.2 Energy Sources for Demonstration

As it was required to be able to control the quantity of energy available to the energy harvesters during the demonstration, controllable energy sources were provided for each of the four energy harvester types being utilized. The energy sources used are as follows:

- *Solar energy harvester:* For this module the ambient illumination was used, along with some paper clouds that could be used to partly or fully occlude the active face of the solar module and thus simulate variations in incident brightness.
- *Vibration energy harvester:* As the device used with this module is designed to operate from vibrations from machines powered by the mains electricity supply, a small mains-operated vacuum pump was used as the test source. This has a low level of operational vibration that can be perceived when the pump is touched but is not visible to the naked eye. The amount of vibration

produced can be varied by changing the loading of the pump by constricting the air inlet line.

- *Thermal energy harvester:* To provide heat to the hot side of the thermal energy harvester, a small adjustable hot plate was used. To aid the dissipation of heat from the cold side of the device, a multipin heat sink was attached and arranged so air could flow across it. To allow the measurement of the hot and cold side temperatures, two k-type bead thermistors were mounted with thermally conductive adhesive to the corresponding device faces.
- *Wind energy harvester:* Operation of the miniature wind turbine has been achieved either by simply blowing onto it or by directing the output from a standard small desk fan onto the blades.

The component modules used in the energy-aware, energy-harvesting node demonstration are shown in Figure 8.10. Interconnecting cables between the core node and the energy multiplexer subsystem, between the energy multiplexer subsystem and the energy modules, and between the energy harvesters and their corresponding modules are omitted for clarity. In addition, an energy bleed circuit was connected to the energy bus comprising of a low value resistor in series with a push switch, pressing the switch to make the circuit adds in a further energy drain for demonstration purposes.

8.6.3 Demonstration Sequence

For the demonstration the two remote nodes (RN1 and RN2) are set up with differing message priority levels, with RN1 having a message priority of MP2 and RN2

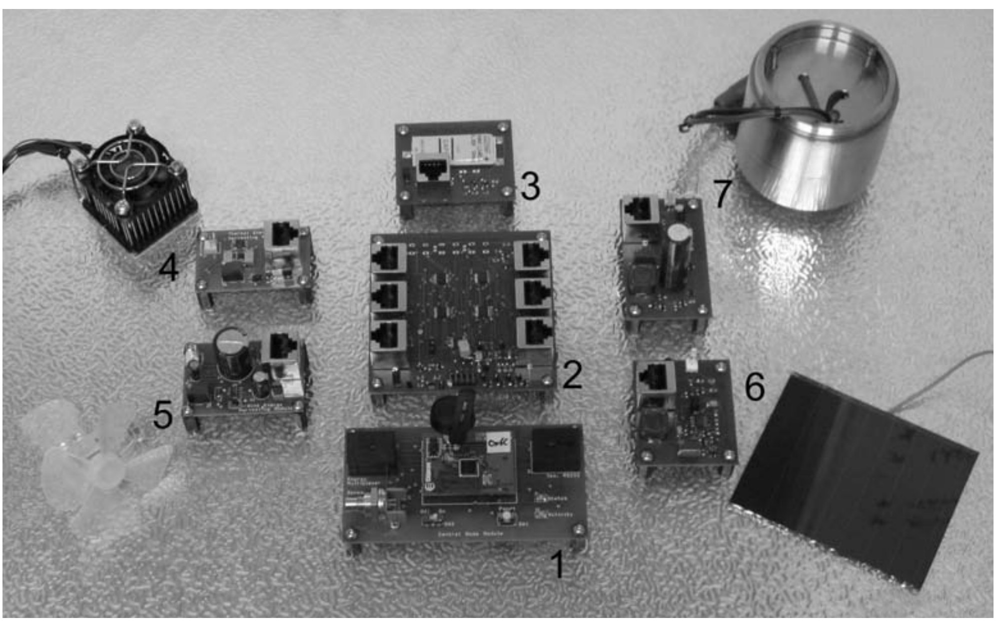

Figure 8.10 Energy–aware, energy-harvesting node components. (1: core node, 2: energy multiplexer subsystem, 3: supercapacitor energy store, 4: thermal energy harvester, 5: wind energy harvester, 6: solar energy harvester, 7: vibration energy harvester.)

having a level of MP4. The sink node is attached to a PC, with the PC configured to display the messages received by the sink node on a terminal window.

In its initial state the energy-aware, energy-harvesting node and the energy multiplexer subsystem are connected to the supercapacitor energy storage module, which is uncharged, and to the vibration and solar harvesting modules. The energy-harvesting modules are initially inactive, with the vibration source switched off and the paper clouds placed over the active surface of the solar module. A typical demonstration then proceeds as follows:

- The vibration source is started and the clouds removed from the solar module. The system then starts to harvest energy from these sources and charge up the energy store.
- Once the energy store has charged up sufficiently to operate the microprocessor node, status messages start to be received and displayed by the sink node. Initially, the node reports that it has the vibration and solar harvesting modules connected and that its energy store is at 2%, giving a power priority level of PP1. As a result of this, none of the messages from the two remote nodes are relayed. The energy-aware node sends a set of reporting messages each cycle reporting the level of the energy store and the production level of the connected harvesting modules.
- The level of energy in the energy stores rises over time to 20%, at which point the node switches to power priority level PP2. This causes the energy-aware node to start to relay messages from remote node RN1 but not from RN2, the node's sleep period decreases, and these changes are reflected in the update messages sent to the sink node.
- The node's energy store continues to increase, and at 40% the node switches to PP3 and the sleep duration is further reduced. This change is again reported by the sink node.
- The thermal harvester is then connected to the energy multiplexer subsystem and its heat source switched on.
- The next node status messages report the addition of the thermal harvesting module, its physical location on the energy multiplexer's ports, and its current rate of harvesting.
- The vibration source is switched off, but the module is not disconnected from the energy multiplexer.
- The subsequent node status messages displayed by the sink node show the rate of generation of the vibration harvester to have dropped to zero, but that the module is still present.
- The wind energy harvester module is connected to the energy multiplexer subsystem, but no wind is provided to the miniature turbine, and the vibration harvesting module is disconnected.
- The next node status message received by the sink node reports that the wind energy harvester module has been connected and that it currently not generating any power. The message also reports that the vibration harvesting module has been disconnected from a specified port on the energy multiplexer subsystem.

- The energy in the supercapacitor store continues to rise, and after a period of time the node energy status messages report that it has reached 60% and that the node has changes to power priority PP4. The results of this are that messages from remote node RN2 start to be relayed and the sleep period of the node is further reduced.
- The source of wind is started to blow onto the miniature turbine.
- Subsequent node energy status messages report the now-nonzero level of energy harvesting from the wind energy harvesting module, along with the production levels of the other connected harvesting modules.
- “Clouds” are placed over 75% of the solar module’s active surface.
- Subsequent energy status messages displayed by the sink node report the drop in output from the solar module and that the level in the energy store has started to fall.
- After the level in the energy store has fallen sufficiently, the energy status message reports that the energy-aware, energy-harvesting node has changed its power priority level to PP3. As a consequence of this, the node’s sleep period is increased and messages from remote node RN2 are dropped instead of being relayed.
- With the node at PP3, the reported energy level in the supercapacitor store starts to rise again.
- The node reaches sufficient energy reserves to switch its power priority level back to PP4, reducing the sleep duration and permitting the relaying of messages from remote node RN2.
- As the level of energy in the energy store varies above and below 60%, the node reports changes to its power priority level between PP4 and PP3. The energy store gains during the lower power priority level and then starts to lose during the higher level. These energy level changes are reflected in the information reported to the sink node and also by the relaying or dropping of messages from remote node RN2.
- The thermal harvesting module is disconnected from the energy multiplexer subsystem and the clouds are removed from the active surface of the solar module.
- The next energy update message received by the sink node reports the removal of the thermal module from a specified port on the energy-multiplexer subsystem and the increased energy production from the solar module.
- Following a reduction in the total harvester power, the node settles into power priority level PP3, which is reported by the sink node.
- The message priority level of messages from remote node RN2 is increased from MP4 to MP3.
- The energy-aware, energy-harvesting node starts to relay messages from remote node RN2 to the sink node as a result of the increased priority of the messages now being sufficient in relation to the node’s current energy status.
- The energy bleed is used to the discharge the energy store until the power priority level reported by the node in its energy status messages to the sink

node drops to PP1. The effects of this drop in level are increased sleep duration and messages from both remote nodes being dropped.

- The message priority level of messages from remote node RN2 is decreased from MP3 to MP4.
- The energy-aware, energy-harvesting node starts to replenish its energy store, reporting the increase in energy level through the energy status messages that it sends to the sink node.
- The energy level in the node reaches 20% and the node switches to power priority level PP2 and reduces its sleep time.
- Messages from remote node RN1 are again being relayed to the sink node by the energy-aware, energy-harvesting node.
- Once the energy-aware, energy-harvesting node reports that it has reached power priority level PP3, the vibration energy harvesting module is reconnected to the energy multiplexer subsystem and the vibration source is restarted.
- The next energy status message displayed by the sink node reports the addition of the vibration harvesting module, the physical location of it on the energy multiplexer subsystem, and the current rate of energy harvesting.
- After a period of harvesting, the reported level in the energy store reaches 60% and the energy-aware, energy-harvesting node goes to power priority level PP4. This causes the sleep duration to be further reduced and the message from remote node RN2 to be again relayed to the sink node.
- With the rates of energy harvesting and consumption balanced, the energy-aware, energy-harvesting node remains at power priority level PP4, reporting the state of the attached energy-harvesting modules and its sensor module's reading while also relaying messages from both remote nodes.

This demonstration demonstrates the active management and reporting of multiple energy-harvesting sources, along with intelligent message prioritization for energy management purposes. This demonstration has been performed at the Data Information Fusion Defence Technology Centre (DIF DTC) Conference 2009 and at the WiSIG Wireless Sensing Showcase 2009 [7].

8.7 Conclusions

The energy-aware, energy-harvesting sensor node described in this chapter has been developed for use within a demonstrator system, simulating use in an urban surveillance scenario. A plug-and-play approach to the configuration of energy-harvesting and storage modules has been adopted. This approach yields several benefits including the ability of configuration by nonskilled operatives and easy customization of the energy-harvesting system to suit the ambient energy available in a given location. Custom hardware has been developed to efficiently harvest, store, and process energy. Modules have been developed that allow the harvesting of solar, thermal, vibrational, and wind energy. Through the use of the energy-efficient hardware and intelligent energy management techniques implemented in software on the node's microprocessor, the node can operate in an energy-neutral manner and device operation has been demonstrated, which achieves this goal.

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CHAPTER 9

Concluding Remarks

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This book has presented an overview of the design and implementation of autonomous systems that are powered using ambient forms of energy within the operating environment. The availability, over recent years, of low-cost and low-power radio frequency devices has led to the development of many commercial wireless sensor nodes. Examples have been given of how such systems can be used in specific application areas. A variety of energy harvesting mechanisms has been described within this text; these include techniques based on kinetic, solar, and thermal energy. Examples have also been given that demonstrate how multiple harvesting sources can be coupled together to increase the available power within a given environment. Owing to the variability in electrical output characteristics between different types of harvester, a wide range of interfacing circuits are required to ensure efficient connectivity with the autonomous system. In broad terms, the amount of electrical energy generated in small (typically milliwatts at best) and careful design of the electronic systems is required in order to maximize the useable quantities of power for the overall system. The ability to store electrical energy is desirable for many applications, and devices such as supercapacitors and planar rechargeable batteries have been described.

The reader should at this point appreciate that the decision to employ energy-harvesting techniques for a given scenario should not be taken lightly. There are many design challenges and technical issues that need to be thoroughly assessed before committing to the energy harvesting route. We hope that many of these issues are addressed within this book and that any reader new to this subject area will not have unrealistic expectations regarding the harvesting approach, as is often the case. Indeed, the whole issue of powering small-scale electronic devices is being addressed from a variety of angles and we must not ignore the fact that almost all of today's portables electronic systems use only a small fraction of the electrical power that their predecessors used and they are more computationally intensive. If the fuel cell technology matures to a point where low-cost, miniaturized, safe, reliable, and high-energy density devices are readily available, then the opportunities for energy harvesting will almost certainly be limited. At this point in time, however, such scenarios appear to be a long way off.

Another key aspect that emerges from this study is that self-powered autonomous systems are rather complex entities comprising many different types of subsystems that can interact with each other. The nature of, say, a solar harvesting system is vastly different from that of a vibration harvesting system, not simply because of the different form of ambient energy, but also because of the requirements for the electronic interfacing between the respective harvester and autonomous system. Traditionally, the design issues for the generator, storage, RF module, sensor, and microprocessor have been addressed in isolation, but the need for a holistic process becomes evident. This will require a standardized approach to the design process, and we have already seen examples of this in several of the main building blocks. Perhaps the most advanced standards to date are in the areas of low-power wireless communications, in particular the IEEE 802.15.4 standard and the high-layer protocols such as ZigBee, 6LoWPAN, and Bluetooth Low Energy. Standards for energy harvesting will undoubtedly emerge since bodies such as the International Society for Automation (ISA) have already formed a working group to this end.

Advances in enabling technologies will continue to have an impact on future harvesting systems. A majority of energy harvesters are, by their nature, currently rather large and bulky devices. In many instances, the power output is a direct function of the physical size and scaling down is not necessarily desirable in all cases. There are, however, many benefits that arise from systems that are inexpensive, lightweight, and flexible, which can be easily integrated into a wide variety of application areas. Advances in microelectromechanical systems and nanoelectromechanical systems (MEMS/NEMS) materials and processes will play a key role in the future of energy harvesting systems, especially as power consumption levels in end user systems continue to fall.

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Index

F

ENERGY Harvesting for AUTONOMOUS SYSTEMS

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